

CYTOKININS

Cytokinins (SIE-toh-KIE-ninz) promote cell division in plants. They are produced in the developing shoots, roots, fruits, and seeds of a plant. Cytokinins are very important in the culturing of plant tissues in the laboratory. A high ratio of auxins to cytokinins in a tissue-culture medium stimulates root formation. A low ratio promotes shoot formation. Cytokinins are also used to promote lateral bud growth of flower crops.

Word Roots and Origins

abscisic

from the Latin *abscisus*,
meaning "to cut off"

ABSCISIC ACID

Abscisic acid (ab-SIS-ik) **acid**, or ABA, generally inhibits other hormones, such as IAA. It was originally thought to promote abscission, hence its name. Botanists now think that ethylene is the main abscission hormone. Absciscic acid helps to bring about dormancy in a plant's buds and maintains dormancy in its seeds. Absciscic acid causes a plant's stomata to close in response to drought. Water-stressed leaves produce large amounts of ABA. It is too costly to synthesize ABA for agricultural use.

OTHER GROWTH REGULATORS

Many growth regulators are used on ornamental plants. These substances do not fit into the five classes of hormones. For example, utility companies in some areas may apply **growth retardants**, chemicals that prevent plant growth, to trees to prevent the trees from interfering with utility lines. It is sometimes less expensive to apply these chemicals than to repeatedly prune the trees. Scientists are still searching for a hormone to retard the growth of lawn grass so that it does not have to be mowed so often.



SECTION 1 REVIEW

1. List the five major types of plant hormones.
2. Which plant hormones generally have stimulatory effects, and which generally have inhibitory effects?
3. What are important commercial uses of auxins, gibberellins, and ethylene?
4. Explain why growth retardants are useful in maintaining ornamental plants.

CRITICAL THINKING

5. **Analyzing Methods** How might a grower use auxin and cytokinin to grow large plants quickly from pieces of stem tissue?
6. **Applying Information** Why is it adaptive for some plants to grow a fleshy fruit even when each seed has its own food source?
7. **Relating Concepts** How might gibberellins cause an embryo to end its dormancy?

SECTION 2

OBJECTIVES

- **List** the environmental stimuli to which plants respond for each type of tropism.
- **Explain** the current hypotheses regarding auxins and their function in phototropism and gravitropism.
- **Describe** two types of nastic movements, and explain how they help a plant survive.

VOCABULARY

tropism
phototropism
solar tracking
thigmotropism
gravitropism
chemotropism
nastic movement
thigmonastic movement
nyctinastic movement

PLANT MOVEMENTS

Plants appear immobile because they are rooted in place.

However, time-lapse photography reveals that parts of plants frequently move. Most plants move too slowly for us to notice.

Plants move in response to several environmental stimuli, such as light, gravity, and mechanical disturbances. Tropisms are growth responses that occur slowly, while nastic movements happen more quickly.

TROPISMS

A **tropism** (TROH-piz-uhm) is a response in which a plant grows either toward or away from an environmental stimulus. Growth toward an environmental stimulus is called a *positive* tropism, and growth away from a stimulus is called a *negative* tropism. Each kind of tropism is named for its stimulus. For example, plant growth in response to light coming from one direction is called **phototropism**. Thus, in Figure 31-5, the shoot tips of a plant that grow toward the light source are positively phototropic. Types of tropisms are summarized in Table 31-2.

Phototropism

Phototropism is illustrated by the appearance of the sprouts in Figure 31-5. Scientists have observed that light causes the hormone auxin to move to the shaded side of the shoot. The auxin is thought to cause the cells on the shaded side to elongate more than the cells on the lighted side. As a result, the shoot grows toward the light and exhibits positive phototropism.



FIGURE 31-5

The way these new sprouts of the flowering shamrock, *Oxalis rubra*, grow toward a light is an example of positive phototropism. The hormone auxin stimulates the cells on the shaded side of the plant stem to elongate.

TABLE 31-2 Various Plant Tropisms

Tropism	Stimulus	Positive example
Phototropism	light	plant grows toward light
Thigmotropism	contact with object	vine twines around a tree
Gravitropism	gravity	roots grow downward
Chemotropism	chemical	pollen tube grows toward ovule

In some plant stems, phototropism is not caused by auxin movement. In these instances, light may cause the production of a growth inhibitor on the lighted side. Negative phototropism is sometimes seen in vines that climb on flat walls where coiling tendrils have nothing to coil around. These vines have stem tips that grow away from light, toward the wall. This growth brings adventitious roots or adhesive discs in contact with the wall on which they can cling.

Solar tracking, also called *heliotropism*, is the motion of leaves or flowers as they follow the sun's movement across the sky, as shown in Figure 31-6. By continuously facing toward a moving light source, the plant maximizes the light available for photosynthesis. These movements are not actually a tropism but a related kind of movement.

Thigmotropism

Thigmotropism is a plant's growth response to touching a solid object. Tendrils and stems of vines, such as morning glories, coil when they touch an object. Thigmotropism allows some vines to climb other plants or objects, increasing the vine's chance of intercepting light for photosynthesis. It is thought that ethylene and a type of auxin are involved in this response.



Quick Lab

Visualizing Phototropism

Materials 2 in. pots (2) containing potting soil, 4 bean seeds, cardboard box

Procedure

1. Plant two bean seeds in each pot. Label each pot for your group. Place one pot in a window or under a plant light.
2. Cut a rectangular window in the box, and place the box over the second pot so that the window faces the light. Place the box in a different location from the open pot. Keep both pots moist for several days.
3. Remove the box 2–3 days after the seedlings have emerged from the soil. Compare the seedlings grown in the light with the ones grown in the box. Sketch your observations.

Analysis How are the seedlings different? How do you account for the difference? Describe or draw what you think the cells inside the curved part of the stem look like.

Word Roots and Origins

thigmotropism

from the Greek *thiga*, meaning "touch," and *tropos*, meaning "turning"

FIGURE 31-6

These sunflower plants face the sun as it sets in the evening. In the morning, the plants face the sun as it rises, then they follow it as it moves across the sky. This movement is called *solar tracking*.



FIGURE 31-7

This photograph was taken seven days after cuttings of the inch plant, *Zebrina pendula*, were placed in growing tubes in different orientations. Notice that the two lower cuttings have started to grow upward. The upward growth of the lower cuttings is caused by the plants' response to gravity and is called *negative gravitropism*. When the roots turn and begin to grow in a downward direction, the growth is called *positive gravitropism*.

Gravitropism

Gravitropism is a plant's growth response to gravity. A root usually grows downward and a stem usually grows upward—that is, roots are positively gravitropic and stems are negatively gravitropic.

Like phototropism, gravitropism appears to be at least partly regulated by auxins. One hypothesis proposes that when a seedling is placed horizontally, auxins accumulate along the lower sides of both the root and the stem. This concentration of auxins stimulates cell elongation along the lower side of the stem, and the stem grows upward. A similar concentration of auxins inhibits cell elongation in the lower side of the root, and the root grows downward, as shown in Figure 31-7.

Chemotropism

Plant growth that occurs in response to a chemical is called **chemotropism**. An example is the growth of a pollen tube after a flower is pollinated. The pollen tube grows out from the pollen grain and down through the stigma and style to the ovule, in the direction of increasing concentrations of chemicals produced by the ovule.

NASTIC MOVEMENTS

Plant movements that occur in response to environmental stimuli but that are independent of the direction of stimuli are called **nastic movements**. These movements are regulated by changes in the water pressure against the cell wall (*turgor pressure*) of certain plant cells.

Thigmonastic Movements

Thigmonastic (THIG-mah-nas-tik) **movements** are nastic movements that occur in response to touching a plant. Many thigmonastic movements are rapid, such as the closing of the leaf trap of a Venus' flytrap around an insect. Figure 31-8 shows how the leaves of a sensitive plant fold within a few seconds after being touched.



(a)



(b)

FIGURE 31-8

(a) *Mimosa pudica*, also called the *sensitive plant*, is a small shrub that has leaflets. (b) When a leaflet is touched, it folds together. This rapid movement is a thigmonastic movement.



(a)



(b)

This movement is caused by the rapid loss of turgor pressure in certain cells, a process similar to that which occurs in guard cells. Physical stimulation of the plant leaf causes potassium ions to be pumped out of the cells at the base of leaflets and petioles. Water then moves out of the cells by osmosis. As the cells shrink, the plant's leaves move.

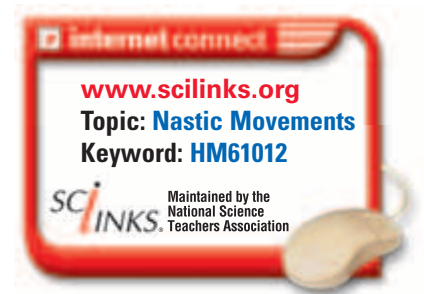
The folding of a plant's leaves in response to touch is thought to discourage insect feeding. In addition, thigmonastic movements may help prevent water loss in plants. When the wind blows on a plant, the rate of transpiration is increased. So if the leaves of a plant fold in response to the "touch" of the wind, water loss is reduced. This could be an important adaptive advantage to a plant.

Nyctinastic Movements

Nyctinastic (NIK-tuh-NAS-tik) **movements** are responses to the daily cycle of light and dark. These movements involve the same type of osmotic mechanism as thigmonastic movements, but the changes in turgor pressure are more gradual. Nyctinastic movements occur in many plants, including bean plants, honeylocust trees, and silk trees. The prayer plant, shown in Figure 31-9, gets its name from the fact that its leaf blades are vertical at night, resembling praying hands. During the day, its leaf blades are horizontal. The botanist Linnaeus planted many plant species with nyctinastic movements in a big circle to make a "flower clock." The nyctinastic movements of each species occurred at a specific time of day.

FIGURE 31-9

A common houseplant is the prayer plant, *Maranta leuconeura*. (a) During the day, the leaf blades of the prayer plant are oriented horizontally in response to light. (b) During the night, the leaf blades are oriented vertically. This movement is called a *nyctinastic movement*.



SECTION 2 REVIEW

1. Give an example of each of the four types of tropisms.
2. What role is auxin thought to play in the growth of plants in response to light and gravity?
3. What adaptive advantages might thigmonastic movements provide a plant?
4. Explain how a tropism and a nastic movement differ.

CRITICAL THINKING

5. **Evaluating Information** Why is it adaptive for plant cells to respond to stimuli received from the environment?
6. **Predicting Results** Would you expect a plant that had its shoot tip removed to exhibit phototropism? Why or why not?
7. **Inferring Relationships** Why aren't nastic movements regulated by hormones?

SECTION 3

OBJECTIVES

- **Define** photoperiodism.
- **Describe** the role of critical night length in flowering.
- **Explain** the process of vernalization.
- **Explain** changing fall colors in leaves.

VOCABULARY

photoperiodism
critical night length
short-day plant
long-day plant
day-neutral plant
phytochrome
vernalization
biennial
bolting
fall color

SEASONAL RESPONSES

In nontropical areas, plant responses are strongly influenced by seasonal changes. For example, many trees shed their leaves in the fall, and most plants flower only at certain times of the year. How do plants “sense” seasonal changes? Although temperature is involved in some cases, plants mark the seasons primarily by sensing changes in night length.

PHOTOPERIODISM

A plant's response to changes in the length of days and nights is called **photoperiodism**. Photoperiodism affects many plant processes, including the formation of storage organs and bud dormancy. However, the most-studied photoperiodic process is flowering. Some plants require a particular night length to flower. In other species, a particular night length merely makes the plant flower sooner than it otherwise would.

Day Length and Night Length

Researchers have found that the important factor in flowering is the amount of darkness, or night length, that a plant receives. Many plant species have specific requirement for darkness, called the **critical night length**. Although scientists now know that night length, and not day length, regulates flowering, the terms *short-day plant* and *long-day plant* are still used. A **short-day plant** flowers when the days are short and the nights are longer than a certain length. Conversely, a **long-day plant** flowers when the days are long and the nights are shorter than a certain length.

Plants can be divided into three groups, depending on their response to the photoperiod, which acts as a season indicator. The three groups of plants are summarized in Table 31-3. The largest group, made up of **day-neutral plants** (DNPs), is not affected by day length. Plants that are DNPs for flowering include tomatoes, dandelions, roses, corn, cotton, and beans.

TABLE 31-3 Flowering Photoperiodism

Type of plant	Conditions needed for flowering	Seasons of flowering
Day-neutral plant (DNP)	not affected by day-night cycle	spring to fall
Short-day plant (SDP)	short days (long nights)	spring, fall
Long-day plant (LDP)	long days (short nights)	summer

Short-day plants (SDPs) flower in the spring or fall, when the day length is short. For example, ragweed flowers when the days are shorter than 14 hours, and poinsettias flower when the days are shorter than 12 hours. Chrysanthemums, goldenrods, strawberries, tulips, and soybeans are typically SDPs for flowering.

Long-day plants (LDPs) flower when days are long, usually in summer. For example, wheat flowers only when the days are longer than about 10 hours. Radishes, asters, irises, spinach, and beets are typically LDPs for flowering.

Adjusting the Flowering Cycles of Plants

Examine Figure 31-10, which compares the flowering of a poinsettia (an SDP) with that of an iris (LDP). With an 8-hour night, the LDP flowers but the SDP does not. With a 16-hour night, the SDP flowers and the LDP does not. However, if a 16-hour night is interrupted in the middle by one hour of light, the LDP flowers and the SDP does not. This response shows that the length of uninterrupted darkness is the important factor. Even though there is a daily total of 15 hours of darkness, the SDP does not flower because of that one hour of light. Flower growers who want to obtain winter flowering of LDPs simply expose them to a low level of incandescent light in the middle of the night. Summer flowering of SDPs is obtained by covering the plants in the late afternoon with an opaque cloth so that the SDPs receive enough darkness.

Regulation by Phytochrome

Plants monitor changes in day length with a bluish, light-sensitive pigment called **phytochrome** (FIET-uh-KROHM). Phytochrome exists in two forms, based on the wavelength of the light that it absorbs. The form that absorbs red rays is called P_r , and the form that absorbs far red (infrared) rays is called P_{fr} . Daylight converts P_r to P_{fr} . In the dark, P_{fr} is converted to P_r . Phytochrome plays a role in bud dormancy and seed germination as well as in flowering.

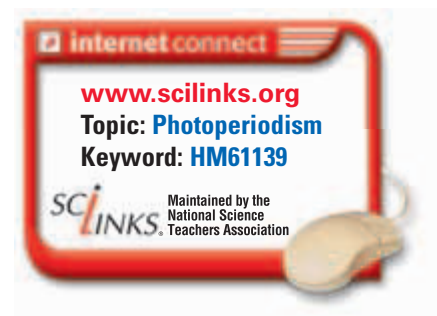
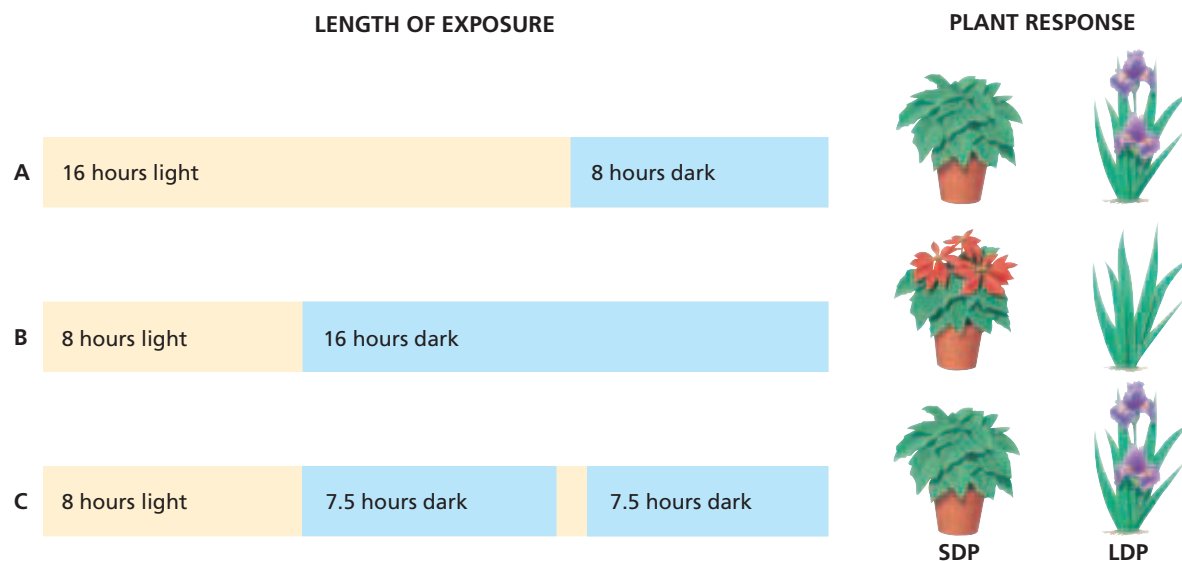


FIGURE 31-10

The figure compares a short-day plant (SDP) with a long-day plant (LDP) in three variations of night length. The SDP has a critical night length of 14 hours. The LDP has a critical night length of 10 hours.



VERNALIZATION

Vernalization is a low-temperature stimulation of flowering. Vernalization is important for fall-sown grain crops, such as winter wheat, barley, and rye. Farmers often take advantage of vernalization to grow and harvest their crops before a summer drought sets in. For example, wheat seeds are sown in the fall and survive the winter as small seedlings. Exposure to cold winter temperatures causes the plants to flower in early spring, and an early crop is produced. If the same wheat is sown in the spring, it will take about two months longer to produce a crop. Thus, cold temperatures are not absolutely required for many grain crops, but they do quicken flowering.

A **biennial** plant is a plant that usually lives for only two years, producing flowers and seeds during the second year. Biennial plants, such as carrots, beets, celery, and foxglove, survive their first winter as large roots with small plants above ground. In the spring, their flowering stem elongates rapidly, a process called **bolting**. Most biennials must undergo vernalization before they flower during the second year. They then die after flowering. Treating a biennial with gibberellin is sometimes used as a substitute for cold temperatures.

FIGURE 31-11

The colors of the carotenoids are visible in these autumn leaves, which have lost most of their chlorophyll.



FALL COLORS

Some trees are noted for their spectacular fall colors, as shown in Figure 31-11. The changing **fall colors** are caused mainly by a photoperiodic response. As nights become longer in the fall, leaves stop producing chlorophyll. As the green pigment degrades, other leaf pigments—the carotenoids (kuh-RAHT'n-OYDZ)—become visible. Carotenoids include the orange carotenes and the yellow xanthophylls (ZAN-thuh-filz). The carotenoids were always in the leaf but were hidden by the chlorophyll. Another group of leaf pigments, the anthocyanins (AN-thoh-SIE-uh-ninz), produce red and purplish red colors in cool, sunny weather.

SECTION 3 REVIEW

1. State the name of a plant's response to changes in the length of days and nights.
2. How do flower growers get short-day plants, such as chrysanthemums, to flower at any time of year?
3. In what ways do farmers use the process of vernalization to their benefit?
4. Explain the process by which tree leaves change color in the fall.

CRITICAL THINKING

5. **Analyzing Concepts** Why might the term *nyctoperiodism* be a better term for describing the process known as *photoperiodism*?
6. **Recognizing Relationships** Explain how the floral business might be affected if the flowering cycle could not be altered artificially.
7. **Evaluating Results** Many plants will flower if just a single leaf is exposed to the appropriate night length. How can you explain this finding?

CHAPTER HIGHLIGHTS

SECTION 1 Plant Hormones

- Plant hormones are formed in many plant parts and regulate many aspects of growth and development. There are five major groups of plant hormones: auxins, gibberellins, ethylene, cytokinins, and abscisic acid.
- Hormonal responses often have adaptive advantages.
- Synthetic auxins are used for killing weeds, stimulating root formation, and stimulating or preventing fruit drop.
- Gibberellins are used to increase the size of fruit, to stimulate seed germination, and to brew beer.
- Ethylene is used to ripen fruit and promote abscission.
- Cytokinins are used to culture plant tissues in the lab and to promote lateral bud growth of flower crops.
- Abscisic acid promotes dormancy in plant buds, maintains dormancy in seeds, and causes stomata to close.
- Growth retardants are widely used to reduce plant height.

Vocabulary

plant hormone (p. 631)
growth regulator (p. 631)
auxin (p. 631)
indoleacetic acid (p. 631)

naphthalene acetic acid
(p. 632)
apical dominance (p. 633)
2,4-D (p. 633)

Agent Orange (p. 633)
gibberellin (p. 633)
ethylene (p. 634)
ethephon (p. 634)

abscission (p. 634)
cytokinin (p. 635)
abscisic acid (p. 635)
growth retardant (p. 635)

SECTION 2 Plant Movements

- Tropisms and nastic movements are plant responses to environmental stimuli. Tropisms occur slowly; nastic movements happen more quickly.
- A tropism is a response in which a plant grows either toward or away from an environmental stimulus. Tropisms can occur in response to light, touch, gravity, and chemicals.
- Phototropism is thought to occur in some plants when auxin moves to the shaded side of a plant and causes cells there to elongate more than the cells on the lighted side.
- Gravitropism is thought to occur when auxin accumulates on the lower sides of a horizontal root and stem. This accumulation causes cell elongation on the lower side of the stem and inhibits cell elongation on the lower side of the root.
- Nastic movements are responses to environmental stimuli but are independent of the direction of the stimuli. Thigmonastic movements occur in response to touch, and nyctinastic movements occur in response to the daily cycle of light and dark.

Vocabulary

tropism (p. 636)
phototropism (p. 636)
solar tracking (p. 637)

thigmotropism (p. 637)
gravitropism (p. 638)
chemotropism (p. 638)

nastic movement (p. 638)
thigmonastic
movement (p. 638)

nyctinastic
movement (p. 639)

SECTION 3 Seasonal Responses

- Photoperiodism is a plant's response to changes in the length of days and nights.
- Plants fit in one of three photoperiodic classes for flowering: day-neutral, short-day, and long-day plants. Short-day and long-day plants have a specific requirement for darkness, called the *critical night length*.
- Vernalization is the promotion of flowering by cold temperatures. Farmers often plant wheat seeds in the fall so that the seedlings can be exposed to winter temperatures and will flower before summer droughts begin.
- Changing fall colors in tree leaves are due to chlorophyll degradation, which reveals other pigments already present.

Vocabulary

photoperiodism (p. 640)
critical night length (p. 640)
short-day plant (p. 640)

long-day plant (p. 640)
day-neutral plant (p. 640)
phytochrome (p. 641)

vernalization (p. 642)
biennial (p. 642)
bolting (p. 642)

fall color (p. 642)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
 - auxin* and *gibberellin*
 - apical dominance* and *abscission*
 - tropism* and *nastic movement*
 - short-day plant* and *long-day plant*
- Use the following terms in the same sentence: *thigmotropism*, *gravitropism*, and *chemotropism*.
- For each pair of terms, explain the relationship between the terms.
 - vernalization* and *biennial*
 - abscisic acid* and *abscission*
 - short-day plant* and *critical night length*
- Word Roots and Origins** The word *auxin* is derived from the Greek *auxein*, which means “to increase.” Using this information, explain why the term *auxin* is a good name for the plant hormone that the term identifies.

UNDERSTANDING KEY CONCEPTS

- Summarize** the observed actions of each of the five types of plant hormones.
- Identify** an adaptive advantage of abscisic acid production.
- Describe** three ways in which a home gardener might use auxin.
- Discuss** the commercial use of gibberellins.
- Describe** how a fruit grower might use ethylene on crops.
- Compare** the functions of cytokinins and abscisic acid.
- Describe** how growth retardants are used on plants.
- Identify** the type of plant movement that is a growth response.
- Identify** the environmental stimulus involved in each of the plant tropisms.
- Explain** how auxins are thought to cause plant growth in response to gravity.
- Summarize** the adaptive advantages of the two types of nastic movements.
- Define** photoperiodism.
- Explain** how critical night length applies to photoperiodism.
- Describe** how a biennial plant lives, and explain why vernalization is important in its life cycle.
- Name** the type of pigment that becomes visible in a leaf when chlorophyll begins to degrade.
- CONCEPT MAPPING** Use the following terms to create a concept map that shows the major effects of some plant hormones: *apical dominance*, *auxin*, *cytokinin*, *gibberellin*, *gravitropism*, *plant hormone*, *phototropism*, and *thigmotropism*.

CRITICAL THINKING

- Predicting Results** Suppose you placed a green banana in each of several plastic bags, placed a ripe pear in half of the bags, and then sealed all of the bags. Which group of bananas do you think would ripen sooner? Justify your answer.
- Evaluating Results** Suppose that a friend who lives in North Dakota gives you a cutting from a flowering plant that you admired when you saw it growing in your friend's yard. You plant the cutting at your home in Georgia. The plant grows but does not produce flowers. Based on your knowledge of plant responses, what might be preventing the plant from flowering?
- Inferring Relationships** The growth of most deciduous trees in the northern United States and Canada, where winters are severe, is regulated strictly by photoperiodism. That is, temperature plays no part in the regulation of their yearly growing cycle. Explain why this is ecologically significant.
- Interpreting Graphics** Suppose you notice that your neighbor has a perfect yard but that some of the plants in your yard have distorted leaves, like those in the photo shown below. What likely caused the leaves' unusual appearance?



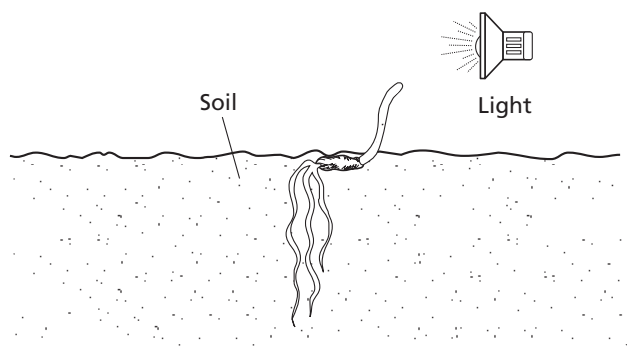


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- Which of the following is the name for the process in which flowering is stimulated by exposure of plants to cold?
 - dormancy
 - vernalization
 - thigmotropism
 - photoperiodism
- Which of the following is a result of abscission?
 - Leaves fall off stems.
 - Cuttings are produced.
 - Plants sense night length.
 - Absciscic acid is produced.
- Which of the following describes the relationship between nastic movements and a stimulus?
 - Nastic movements occur without a stimulus.
 - Nastic movements occur toward a stimulus.
 - Nastic movements occur away from a stimulus.
 - Nastic movements occur independently of the direction of a stimulus.
- What is the response of a plant to the length of days and nights called?
 - gravitropism
 - phototropism
 - photoperiodism
 - thigmotropism

INTERPRETING GRAPHICS: The illustration below shows a growing seedling. Use the illustration to answer the question that follows.

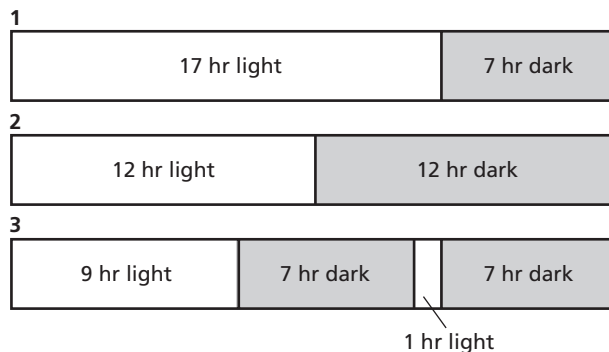


- Which of the following statements about this seedling is true?
 - The seedling is exhibiting abscission.
 - The shoot is exhibiting positive phototropism.
 - The roots are exhibiting negative gravitropism.
 - The seedling is exhibiting negative thigmotropism.

DIRECTIONS: Complete the following analogy.

- cell elongation : gibberellins :: cell division :
 - auxins
 - ethylene
 - cytokinins
 - absciscic acid

INTERPRETING GRAPHICS: The diagrams below represent three different growing conditions. Use the diagrams to answer the question that follows.



- In which of the conditions will a long-day plant with a critical night length of 8 hours flower?
 - 1
 - 2
 - 1 and 3
 - 2 and 3

SHORT RESPONSE

The Venus' flytrap obtains nutrients by closing its leaves around insects and then digesting the insects.

Explain why a thigmonastic movement is a more useful plant response than a thigmotropic response would be in this situation.

EXTENDED RESPONSE

Reproduction is essential for survival of flowering plant species.

Part A Identify two plant hormones that have roles in flowering, fruiting, and/or seed or fruit dispersal, and describe these roles.

Part B Identify the important environmental signals for flowering, and describe how these signals affect flowering.

Test TIP

For a question about a structure or phenomenon that has a complex name, write down the name and review its meaning before answering the question.

Testing the Effect of a Gibberellin on Plant Growth

OBJECTIVES

- Study the effect of a gibberellin on the growth of bean seedlings.

PROCESS SKILLS

- observing
- hypothesizing
- measuring
- comparing and contrasting
- collecting data
- organizing data

MATERIALS

- safety goggles
- lab apron
- disposable gloves
- 2 flowerpots, 20 cm or larger
- marking pen
- labeling tape
- potting-soil mixture
- 10 bean seedlings, at least 3 cm tall
- centimeter ruler
- cotton swabs
- gibberellic acid solution (contains a gibberellin)
- water

Background

1. Gibberellins are found in all parts of plants but are most concentrated in immature seeds. Gibberellins are only one of many kinds of hormones that affect plant functions.
2. What is the major effect of gibberellins on plant tissues?

PART A Testing a Hypothesis

1. Discuss the objective of this investigation with your lab partners. You will apply gibberellic acid solution to the leaves and shoot tips of one group of bean seedlings, and you will apply only water to an identical group of bean seedlings. Develop a hypothesis about the effect of gibberellic acid solution on bean seedlings. Record your hypothesis in your lab report.

2.    **CAUTION** Wear safety goggles, a lab apron, and disposable gloves at all times during this investigation.
3. Obtain two flowerpots, and label them A and B. Also write your initials on the flowerpots.
4. Fill each flowerpot with potting-soil mixture. The pots should be filled to the same level.
5.  **CAUTION** Keep the seeds, which may have been treated with a fungicide, away from your skin. Plant five bean seedlings in each flowerpot, spacing them evenly around the edge. Each seedling's roots should be at least 1 cm below the soil surface.
6. Measure the height of each plant in pot A to the nearest millimeter. Measure from the soil surface to the shoot tip. In your lab report, record your measurements in a data table similar to the table on the facing page.
7. Average the height of the five seedlings. In your lab report, record your average measurement under "Initial height" in your data table.
8. Repeat steps 6 and 7 using the plants in pot B.
9. Wet a cotton swab with water. Rub the water over the leaves and shoot tips of each seedling in pot A.
10.  **CAUTION** Gibberellic acid is an irritant. If you get gibberellic acid solution on your skin or clothing, rinse it off at the sink while calling to your teacher. If you get gibberellic acid solution in your eyes, flush it out immediately at an eyewash station while calling to your teacher. Wet a clean cotton swab with gibberellic acid solution. Apply the solution to the seedlings in pot B by rubbing the cotton swab on the leaves and shoot tips.
11. Place both flowerpots in a window or other area where they are likely to get light. Ensure that each seedling will receive the same amount of water, sunlight, darkness, and air circulation for the next 10 days.
12.   Clean up your materials, and wash your hands before leaving the lab.

PART B Observing the Results

13. Each day over the next 10 days, measure the height of each seedling. Average the height of the seedlings in each flowerpot, and record the average height in your data tables.
14. Calculate the average change in height for each day. For example, the average change in height during the second day is equal to the average height on day 2 minus the average height on day 1.
15. If you are unable to make measurements over the weekend, simply record in your data table the number of days that elapsed since the last measurement was taken. To calculate the average change in height for the days when measurements were not taken, you will need to divide the amount of change by the number of days that have elapsed since your last measurement.
16.   At the end of 10 days, clean up your materials according to your teacher's instructions.

Analysis and Conclusions

1. Use the data you have recorded to calculate the percentage increase—between your first measurement and your last measurement—in the average height of the two groups of plants.
2. Use your data to make a line graph of height as a function of time. Label the *x*-axis of your graph *Time*, and measure time in days. Label the *y*-axis *Height*, and measure height in millimeters.
3. Do your data support your hypothesis? Explain why or why not.
4. What differences, other than height, did you detect in the bean seedlings?
5. What were some of the possible sources of error in this experiment?
6. How might a knowledge of plant regulatory chemicals be of use to farmers and other plant growers?

Further Inquiry

Design an experiment to determine whether gibberellic acid accelerates the germination of bean seeds.

FLOWERPOT A: EFFECTS OF WATER ON BEAN SEEDLINGS

Day	Height (mm)	Total change in height (mm)
Initial		
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

FLOWERPOT B: EFFECTS OF GIBBERELIC ACID ON BEAN SEEDLINGS

Day	Height (mm)	Total change in height (mm)
Initial		
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

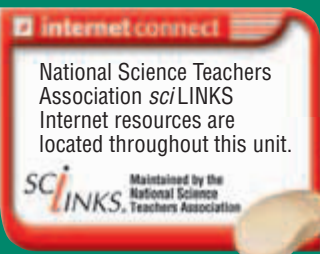
UNIT 8

CHAPTERS

- 32** *Introduction to Animals*
- 33** *Sponges, Cnidarians, and Ctenophores*
- 34** *Flatworms, Roundworms, and Rotifers*
- 35** *Mollusks and Annelids*
- 36** *Arthropods*
- 37** *Insects*
- 38** *Echinoderms and Invertebrate Chordates*



References to *Scientific American* project ideas are located throughout this unit.



INVERTEBRATES

“It has taken biologists some 230 years to identify and describe three quarters of a million insects; if there are indeed at least thirty million, as Erwin (Terry Erwin, The Smithsonian Institution) estimates, then, working as they have in the past, insect taxonomists have ten thousand years of employment ahead of them.”

From “Endless Forms Most Beautiful,” from *The Sixth Extinction: Patterns of Life and the Future of Humankind*, by Richard Leakey and Roger Lewin. Copyright © 1995 by Sherma B. V. Reprinted by permission of Doubleday, a division of Bantam Doubleday Dell Publishing Group, Inc.



This Sally lightfoot crab lives on bare volcanic rock on the Galápagos Islands.

This sea star clearly demonstrates pentamorous radial symmetry.

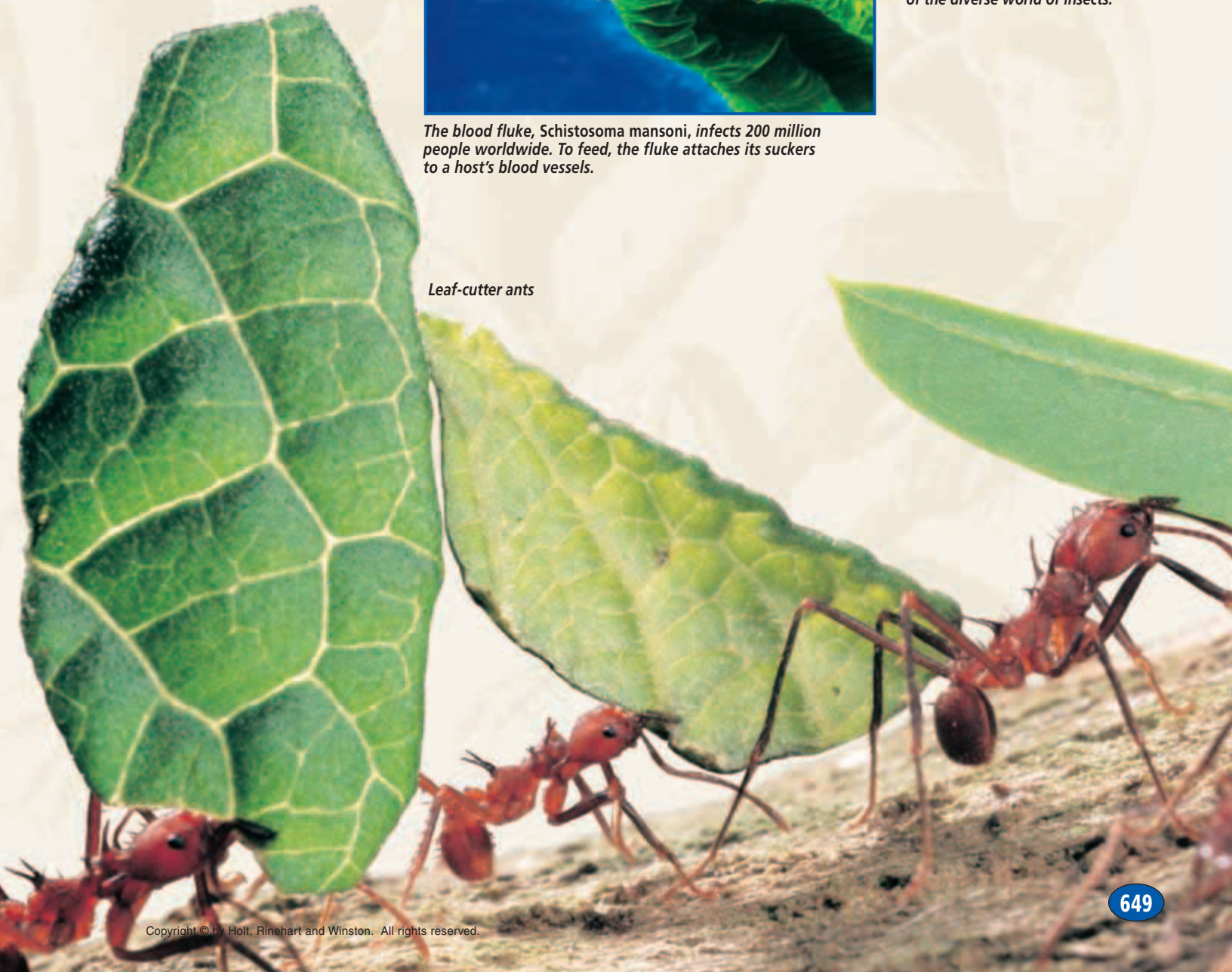


*The leaf-footed bug, *Diactor bilineatus*, is a colorful member of the diverse world of insects.*



*The blood fluke, *Schistosoma mansoni*, infects 200 million people worldwide. To feed, the fluke attaches its suckers to a host's blood vessels.*

Leaf-cutter ants



INTRODUCTION TO ANIMALS



The diversity of animal life is staggering. Animals have adapted to Earth's lushest environments and to its harshest environments. This Sally Lightfoot crab, *Grapsus grapsus*, lives on the bare volcanic rock of the geologically young Galápagos Islands.

SECTION 1 *The Nature of Animals*

SECTION 2 *Invertebrates and Vertebrates*

SECTION 3 *Fertilization and Development*

THE NATURE OF ANIMALS

If you are asked to name an animal, you might respond with the name of a familiar large-bodied animal, such as a horse, a shark, or an eagle. But the kingdom Animalia is much more diverse than many people realize.

CHARACTERISTICS

Animals are multicellular heterotrophic organisms that lack cell walls. Some animals, called **vertebrates**, have a backbone. Other animals, called **invertebrates**, do not have a backbone. Invertebrates account for more than 95 percent of all animal species alive today. Most members of the animal kingdom share other important characteristics, including sexual reproduction and movement.

Multicellular Organization

The bodies of animals are multicellular. Some animals contain large numbers of cells. For example, some scientists estimate that the body of an adult human contains 50 trillion to 100 trillion cells. Unlike the cells of unicellular organisms, the cells of multicellular organisms do not lead independent lives. Each cell depends on the presence and functioning of other cells.

In all but the simplest animal phyla, there is a division of labor among cells. **Specialization** is the evolutionary adaptation of a cell for a particular function. Just as a general contractor makes use of carpenters, electricians, and plumbers to build a house, a multicellular organism makes use of specialized cells to perform particular functions, such as digesting food, removing wastes, or reproducing.

A *tissue* is a group of similar cells that perform a common function. Most animal bodies are composed of combinations of different kinds of tissues. The formation of tissue from many individual cells is made possible by *cell junctions*, connections between cells that hold the cells together as a unit. The members of most animal phyla have *organs*, body structures that are composed of more than one type of tissue and that are specialized for a certain function.

Without multicellularity, the enormous variety found in the animal kingdom would not exist. The size of unicellular organisms is limited. Moreover, all of their functions, such as reproduction and digestion, must be handled within a single cell. Multicellularity and cell specialization have enabled organisms to evolve and adapt to many environments.

OBJECTIVES

- **Identify** four important characteristics of animals.
- **List** two kinds of tissues found only in animals.
- **Explain** how the first animals may have evolved from unicellular organisms.
- **Identify** four features found only in chordates.
- **Identify** two functions of the body cavity.
- **List** the structural features that taxonomists use to classify animals.

VOCABULARY

animal
vertebrate
invertebrate
specialization
ingestion
zygote
differentiation
chordate
notochord
dorsal nerve cord
pharyngeal pouch
symmetry
radial symmetry
dorsal
ventral
anterior
posterior
bilateral symmetry
cephalization
germ layer



Word Roots and Origins

ingestion

from the Latin *ingestus*,
meaning “carry in”

Heterotrophy

Plants and some unicellular organisms are autotrophic. They make food using simple molecules from their environment and an energy source, such as the sun. Animals, on the other hand, are heterotrophic. They must obtain complex organic molecules from other sources. Most animals accomplish this by ingestion. During **ingestion**, an animal takes in organic material or food, usually in the form of other living things. Digestion then occurs within the animal's body, and carbohydrates, lipids, amino acids, and other organic molecules are extracted from the material or cells the animal has ingested.

Sexual Reproduction and Development

Most animals can reproduce sexually, and some can also reproduce asexually. In sexual reproduction, two haploid gametes fuse. The **zygote**, the diploid cell that results from the fusion of the gametes, then undergoes repeated mitotic divisions. Mitotic division of a cell produces two identical offspring cells. How does an adult animal, with its many different organs, tissues, and cell types, arise from a single cell? In the process called *development*, the enlarging mass of dividing cells undergoes differentiation. During **differentiation** (DIF-uhr-EN-shee-AY-shuhn), cells become specialized and therefore different from each other. For example, some cells may become blood cells, and others may become bone cells. The process of differentiation is the path to cell specialization.

Movement

Although some animals, such as barnacles, spend most of their lives attached to a surface, most animals move about in their environment. The ability to move results from the interrelationship of two types of tissue found only in animals: nervous tissue and muscle tissue. Nervous tissue allows an animal to detect stimuli in its environment and within its own body. Cells of nervous tissue, called *neurons*, conduct electrical signals throughout an animal's body. Multiple neurons work together to take in information, transmit and process it, and initiate an appropriate response. Often, this response involves muscle tissue, which can contract and exert a force to move specific parts of the animal's body. The bat shown in Figure 32-1 continuously processes information about its position in space and the position of its prey. It can adjust its muscular responses so rapidly that it can intercept insects in flight.

FIGURE 32-1

Capturing fast-moving prey requires exquisitely timed coordination between the nervous tissue and muscle tissue in the body of this heart-nosed bat, *Cardioderma cor*.



ORIGIN AND CLASSIFICATION

The first animals probably arose in the sea. The structural characteristics of invertebrates suggest that they were the first multicellular animals and that they evolved from protists. Because protists are both heterotrophic and eukaryotic, scientists have inferred that multicellular invertebrates may have developed from colonies of loosely connected, flagellated protists, such as the one shown in Figure 32-2.

What path did cell specialization take in these early organisms? Colonial protists may have lost their flagella over the course of evolution as individual cells in the colony grew more specialized. They may have been similar to modern colonial protists that do show some degree of cell specialization, such as some species of algae. In these species, the gametes are distinct from nonreproductive cells. A similar division of labor in early colonial protists may have been the first step toward multicellularity.

Scientists often use a type of branching diagram called a *phylogenetic diagram*, such as the one in Figure 32-3, to show how animals are related through evolution. Taxonomists have grouped animals into several phyla (singular, *phylum*) by comparing animals' fossils, body symmetry, patterns of embryo development, and ribosomal RNA (rRNA) and other macromolecules. Taxonomy is an ever-changing branch of science; therefore, it should not be surprising that the actual number and names of animal phyla continue to change and be debated. Many taxonomists recognize 30 or more different animal phyla, though some phyla contain a very small number of species.

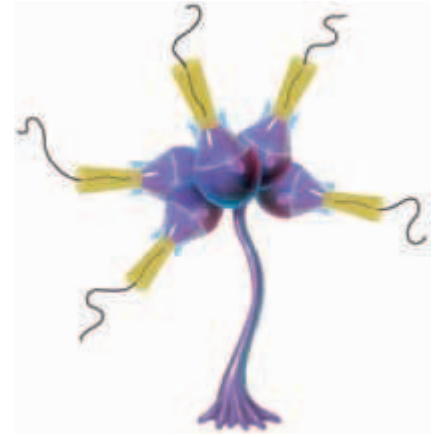


FIGURE 32-2

The first animals may have evolved from colonial protists similar to the one shown in this drawing.

FIGURE 32-3

This phylogenetic diagram represents a hypothesis for the relationship among members of the animal kingdom based on rRNA analysis. Notice on the diagram the locations of similarities in body tissues, body symmetry, and embryo development. For updates on phylogenetic information, visit go.hrw.com. Enter the keyword HM6 Phylo.

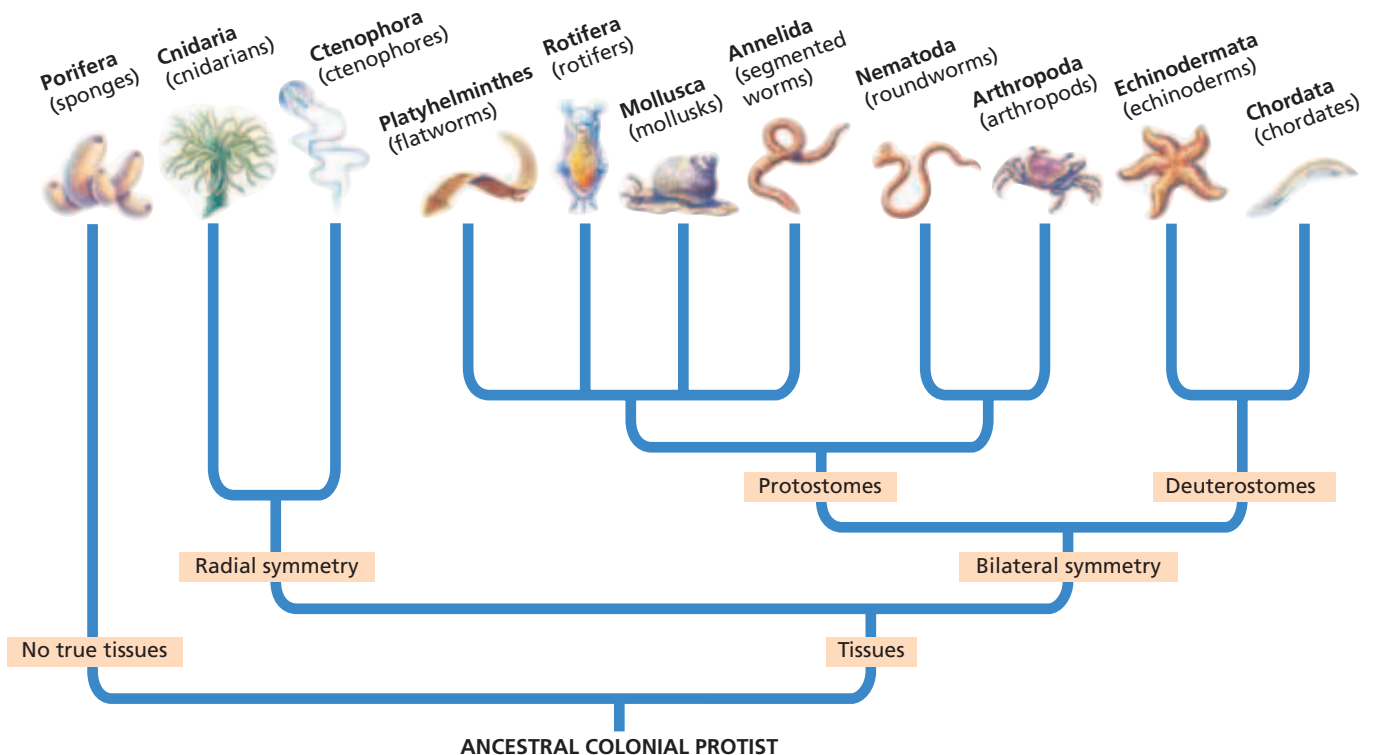




FIGURE 32-4

The palm spider, *Nephila* sp., is an arthropod, with a segmented body and body parts specialized for trapping, killing, and eating its prey.



Invertebrates

Invertebrate body plans range from the absence of body symmetry and true tissues, as is found in sponges, to bilateral symmetry and specialized body parts found in arthropods, such as the spider shown in Figure 32-4. Invertebrates do not have a backbone. Invertebrates make up the greatest number of animal species.

Chordates

The name **chordate** (KAWR-DAYT) refers to animals with a **notochord**, a firm, flexible rod of tissue located in the dorsal part of the body. At some stage in development, all chordates have a notochord, a dorsal nerve cord, pharyngeal pouches, and a postanal tail. The **dorsal nerve cord** is a hollow tube above the notochord. **Pharyngeal** (fuh-RIN-jee-uhl) **pouches** are small outpockets of the anterior digestive tract. The postanal tail consists of muscle tissue and lies behind the posterior opening of the digestive tract. A few chordates retain their early chordate characteristics all their lives. In most vertebrates, a subphylum of the chordates, the dorsal nerve cord develops into the brain and the spinal cord, and the notochord is replaced by the backbone. In aquatic vertebrates, the pharyngeal pouches have evolved into gills for breathing.

Although vertebrates are only one small subphylum of animals, they merit discussion from a human perspective. Humans are vertebrates, and the ecology of humans includes extensive interaction with other vertebrate species.

Careers in BIOLOGY

Veterinarian

Job Description Veterinarians are doctors who are trained to protect the health of animals, such as pets and livestock. Some veterinarians also protect animals and people from the diseases that they each carry.

Focus on a Veterinarian

Dr. Jack Walther grew up on a ranch, so animals were always part of his life. Today, Dr. Walther practices two days each week in a veterinary practice with another veterinarian. When he arrives each day, animal patients who have experienced overnight injuries or illnesses are already waiting. Dr. Walther handles these emergencies. He also sees patients with appointments for vaccinations,

surgical follow-ups, exams, and other routine needs. Dr. Walther emphasizes that a veterinarian's role extends beyond just treating the family pet. "Vets also work to conserve animal resources; for example, they help study and preserve endangered species in Africa."

Veterinarians are also at the forefront in protecting animals and people against diseases, such as Lyme disease and West Nile virus. "As a vet, you help not only the animal world, but also humanity."

Education and Skills

- **High school**—three years of science courses and four years of math courses
- **College**—bachelor of science including course work in biology, mathematics,



chemistry, and physics; four years of additional schooling to earn doctor of veterinary medicine (D.V.M.) degree

- **Skills**—self-motivation, curiosity, patience, ability to work independently, ability to work with animals



For more about careers, visit go.hrw.com and type in the keyword **HM6 Careers**.



(a) NO SYMMETRY



(b) RADIAL SYMMETRY



(c) BILATERAL SYMMETRY

BODY STRUCTURE

Animal bodies range from those that lack true tissues and an organized body shape to those that have very organized tissues and a consistent body shape.

Patterns of Symmetry

A *body plan* describes an animal's shape, symmetry, and internal organization. **Symmetry** is a body arrangement in which parts that lie on opposite sides of an axis are identical. An animal's body plan results from the animal's pattern of development. Sponges have the simplest body plan of all animals. Sponges, as shown in Figure 32-5a, are *asymmetrical*—they do not display symmetry. Animals that have a top and bottom side, but no front, back, right, or left end, display **radial symmetry**—a body plan in which the parts are organized in a circle around an axis. Cnidarians, such as the sea anemone in Figure 32-5b, are radially symmetrical.

Most animals have a **dorsal** (back) and **ventral** (abdomen) side, an **anterior** (toward the head) and **posterior** (toward the tail) end, and a right and left side, as shown by the squirrel in Figure 32-5c. Such animals have two similar halves on either side of a central plane and are said to display **bilateral symmetry**. Bilaterally symmetrical animals tend to exhibit **cephalization** (SEF-uh-li-ZAY-shuhn)—the concentration of sensory and brain structures in the anterior end of the animal. As a cephalized animal moves through its environment, the anterior end precedes the rest of the body, sensing the environment.

Germ Layers

Germ layers are tissue layers in the embryos of all animals except sponges, which have no true tissues. The embryos of cnidarians and ctenophores have two germ layers. All other animals have three germ layers. Every organ and tissue arises from a germ layer.

FIGURE 32-5

(a) The sponge lacks a consistent pattern of structure. (b) The sea anemone, an aquatic animal, displays radial symmetry. (c) The squirrel displays bilateral symmetry and cephalization.

Word Roots and Origins

cephalization

from the Greek word *kephale*, meaning "head"

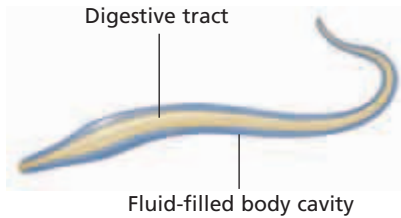


FIGURE 32-6

The body of a roundworm is held erect by its fluid-filled body cavity, which is firm but flexible, like a balloon filled with water.

Body Cavities

Most animals have some type of body cavity, a fluid-filled space that forms between the digestive tract and the outer wall of the body during development. Some animals, such as flatworms, have three germ layers but have a solid body. These animals lack a body cavity.

In the roundworm shown in Figure 32-6, the body cavity aids in movement by providing a firm, fluid-filled structure against which muscles can contract. The body cavity also allows some degree of movement of the exterior of the body with respect to the internal organs, resulting in more freedom of movement for the animal. Finally, the fluid in the body cavity acts as a reservoir and medium of transport for nutrients and wastes, which diffuse into and out of the animal's body cells.

Body Structure and Relatedness

Biologists use similarities in body plans and patterns of development to help them classify animals and hypothesize about the evolutionary history of animals. Biologists use information from extant (living) species and extinct species to develop phylogenetic diagrams, such as the one shown in Figure 32-3. Animal phyla shown on the same branch of the phylogenetic diagram, such as flatworms and rotifers, are thought to be related to each other more closely than they are to other animals and are characterized by similarities in morphology and rRNA sequences. Conversely, animals shown in different parts of the diagram are thought to be more distantly related.

- Multicellularity and a limited degree of cell specialization characterize the sponges. Sponges have no organized body shape and no true tissues.
- True tissues in two layers are found in the cnidarians and the ctenophores.
- True tissues in three layers and bilateral symmetry characterize all of the other animal phyla.

SECTION 1 REVIEW

1. What are the four characteristics common to most animals?
2. Identify how nervous tissue and muscle tissue are interrelated.
3. Summarize how unicellular organisms may have given rise to the first animals.
4. What are the features that all vertebrates share at some point in their development?
5. Identify the body-symmetry type that includes both an anterior end and a posterior end.
6. Name three body features that taxonomists use to help develop phylogenetic trees.

CRITICAL THINKING

7. **Recognizing Relationships** Animals such as sponges lack nervous tissue and muscle tissue. What does this tell you about sponges?
8. **Relating Concepts** In 1994, Western scientists first observed the Vietnamese saola, a hoofed mammal. The saola was shown to be related to wild cattle and buffalo. How do you think scientists identified the saola's closest relatives?
9. **Evaluating Information** How is cephalization advantageous to an animal in finding food?

INVERTEBRATES AND VERTEBRATES

Comparative anatomy, the study of the structure of animal bodies, is one of the oldest disciplines in biology. Some modern scientists work to establish the relationships between different animals, while others try to establish the relationships between the form and function of morphological features of animals and the role of these features in animal ecology.

INVERTEBRATE CHARACTERISTICS

Although it may be difficult for us to see many similarities between a clam and an octopus, they are classified in the same phylum. Adult invertebrates show a tremendous amount of morphological diversity.

Symmetry

Most invertebrates display radial or bilateral symmetry. The radial symmetry of a jellyfish, which drifts rather than swims, allows the animal to receive stimuli from all directions. Most invertebrates have bilateral symmetry, which is an adaptation to a more motile lifestyle. Bilateral symmetry allows for cephalization, which is present in varying degrees in different animals. Some bilaterally symmetric invertebrates, such as the sea hare shown in Figure 32-7, are not highly cephalized. Members of *Aplysia* do not have a true brain and are capable of only basic responses to the environment. Other invertebrates, such as squids and octopuses, are highly cephalized and have a distinct head and a nervous system dominated by a well-organized brain.

Segmentation

Segmentation in animals refers to a body composed of a series of repeating similar units. Segmentation is seen in its simplest form in the earthworm, an annelid in which each unit of the body is very similar to the next one. Within the phylum Arthropoda, however, segments may look different and have different functions. In the arthropod shown in Figure 32-8 on the next page, fusion of the anterior segments has resulted in a large structure that includes the animal's head and chest regions.

OBJECTIVES

- **Compare** symmetry, segmentation, and body support in invertebrates and vertebrates.
- **Describe** the differences in the respiratory and circulatory systems of invertebrates and vertebrates.
- **Compare** the digestive, excretory, and nervous systems of invertebrates and vertebrates.
- **Contrast** reproduction and development in invertebrates and vertebrates.

VOCABULARY

segmentation
exoskeleton
gill
open circulatory system
closed circulatory system
hermaphrodite
larva
endoskeleton
vertebra
integument
lung
kidney

FIGURE 32-7

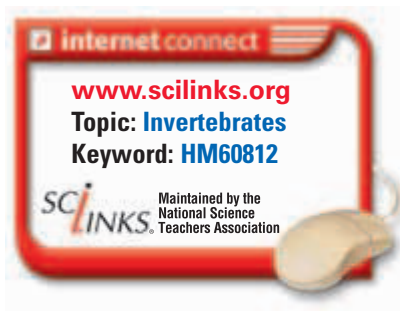
The California sea hare, *Aplysia californica*, is a shell-less mollusk that has a simple nervous system.





FIGURE 32-8

In animals such as this crayfish, *Procambarus* sp., segments are fused, producing larger structures. The head and chest structure in this crayfish results from the fusion of several segments. Segments may also give rise to other structures, such as limbs. The crayfish's exoskeleton is also clearly visible.



Support of the Body

Invertebrate bodies have diverse means of support. Sponges have a simple skeleton that supports their soft tissue; the dried, brown, irregularly shaped “natural sponge” found in stores is this skeleton. The bodies of some other invertebrates, such as roundworms, are supported by the pressure of their fluid-filled body cavity.

An **exoskeleton** is a rigid outer covering that protects the soft tissues of many animals, including arthropods, such as crustaceans, which include crayfish, shown in Figure 32-8. An exoskeleton limits the size and may impede the movement of the organism. Also, an exoskeleton does not grow and must be shed and replaced as the animal grows.

Respiratory and Circulatory Systems

Animals produce carbon dioxide, CO_2 , as a byproduct of metabolism. Therefore, carbon dioxide in the blood must be exchanged with oxygen, O_2 , from the environment. This process, called *gas exchange*, occurs most efficiently across a moist membrane. In the simplest aquatic invertebrates, gas exchange occurs directly across the body covering. Aquatic arthropods and mollusks, however, have **gills**, organs that consist of blood vessels surrounded by a membrane and are specialized for gas exchange in water.

In most animals, the circulatory system moves blood or a similar fluid through the body to transport oxygen and nutrients to cells. At the same time, carbon dioxide and wastes are transported away from the cells. Sponges and cnidarians have no circulatory system, so nutrients and gases are exchanged directly with the environment by diffusion across cell membranes. Arthropods and some mollusks have an **open circulatory system**, in which circulatory fluid is pumped by the heart through vessels and into the body cavity and is then returned to the vessels. Annelids and other mollusks have a closed circulatory system. In a **closed circulatory system**, blood is pumped by a heart and circulates through the body in vessels that form a closed loop. The exchange of gases, nutrients, and wastes occurs between body cells and very small blood vessels that lie near each cell.

Digestive and Excretory Systems

In sponges, digestion occurs within individual cells. In cnidarians, a central chamber with one opening serves as the digestive system. Most other invertebrates, however, have a digestive tract, or *gut*, running through their body. In these animals, food is broken down in the gut, and the nutrients are absorbed by specialized cells that line the gut.

In simple aquatic invertebrates, wastes are excreted as dissolved ammonia, NH_3 . In terrestrial invertebrates, specialized excretory structures filter ammonia and other wastes from the body cavity. The ammonia is then converted to less toxic substances, and water is reabsorbed by the animal before the waste is excreted.

Nervous System

The extraordinary degree of diversity among invertebrates is reflected in their nervous systems. Sponges have no neurons, although individual cells can react to environmental stimuli in much the same way that protozoa can. Neurons evolved in cnidarians, which have a very simple, loosely connected nervous system. Within a single invertebrate phylum, Mollusca, we can trace a step-wise progression of cephalization and the evolution of the brain.

The mollusks have very diverse nervous systems. Recall the sea hare, shown in Figure 32-7. Although its head is not well defined and its nervous system can perform only simple information processing, the sea hare can learn to contract a part of its body in response to certain stimuli. Contrast this simple behavior with that of a highly cephalized mollusk, such as the octopus. The octopus shows very complex decision-making behavior, and it can build a shelter from debris it finds on the ocean floor.

Reproduction and Development

Invertebrates are capable of some form of sexual reproduction, and many can also reproduce asexually. Some invertebrates, such as earthworms, are hermaphrodites. A **hermaphrodite** (huhr-MAF-roh-DIET) is an organism that produces both male and female gametes, allowing a single individual to function as both a male and a female.

Invertebrates may undergo indirect or direct development. Animals that undergo *indirect development* have an intermediate larval stage, as is shown in Figure 32-9. A **larva** (plural, *larvae*) is a free-living, immature form of an organism that is morphologically different from the adult. Larvae often exploit different habitats and food sources than adult organisms do. As a result, organisms in each stage are more likely to survive. Many insects, which constitute a class of arthropods, have indirect development.

In contrast, in *direct development*, the young animal is born or hatched with the same appearance and way of life it will have as an adult; no larval stage occurs. Most invertebrates undergo indirect development. A few, such as grasshoppers, undergo direct development.

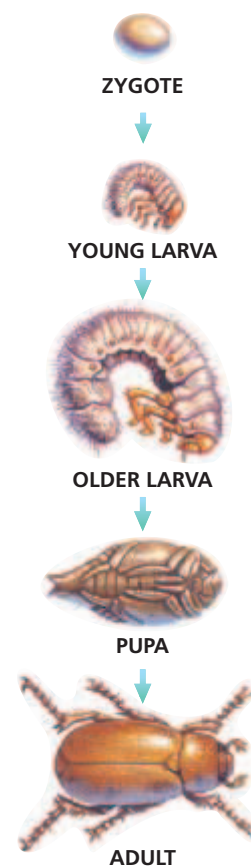


FIGURE 32-9

Animals with indirect development, such as this beetle, have an intermediate, larval stage. A larva is an immature form that exhibits physical traits that are different from those of the adult form.

VERTEBRATE CHARACTERISTICS

Vertebrates are chordates that have a backbone. Classes of vertebrates include fishes, amphibians, reptiles, birds, and mammals. All vertebrate classes except fishes spend part or all of their life on land. Many characteristics of terrestrial vertebrates are adaptations to life on land and fall into two broad categories: support of the body and conservation of water.

Segmentation and Support of the Body

Although it is not immediately apparent, vertebrates are segmented animals. Segmentation is evident in the ribs and the **vertebrae** (vuhr-tuh-bree), the repeating bony units of the backbone. As terrestrial vertebrates evolved from aquatic vertebrates, their limbs and associated muscles evolved to give the animals better support and greater mobility. For example, the legs of amphibians, the first land vertebrates to evolve, are positioned to the side of the body, as shown in Figure 32-10a. However, the legs of mammals, such as the deer shown in Figure 32-10b, are positioned directly beneath the body, allowing the animal to move faster and with a longer stride. Humans show an extreme version of this trait: we are bipedal, and our head is positioned directly over our body.

Vertebrates have an **endoskeleton**, an internal skeleton made of bone and cartilage, which includes the backbone. The endoskeleton grows as the animal grows.

Body Coverings

The outer covering of an animal is called the **integument** (in-TEG-yoo-muhnt). Although the integuments of fishes and most amphibians are adapted only to moist environments, the integuments of most terrestrial vertebrates are adapted to the dry conditions of a terrestrial environment. All animal bodies are composed of water-filled cells, and if the water content of the cells is reduced appreciably, the animal will die. Thus, the outer covering of terrestrial vertebrates, such as reptiles, birds, and mammals, is largely watertight. Integuments also serve other purposes. The moist skin of an amphibian functions as a respiratory organ for the exchange of gases. The scales of a reptile help protect it from predators. The feathers of birds and the fur of mammals efficiently insulate the body.

Respiratory and Circulatory Systems

Gas exchange occurs in the gills of aquatic vertebrates, including fishes and larval amphibians, but these gills do not function out of water. **Lungs** are organs for gas exchange composed of moist, membranous surfaces deep inside the animal's body. Lungs evolved in terrestrial vertebrates.

Vertebrates have a closed circulatory system with a multichambered heart. In some vertebrates, the multichambered heart separates oxygenated and deoxygenated blood, improving the efficiency of the circulatory system over that found in other vertebrates and many invertebrates.

Digestive and Excretory Systems

Digestion occurs in the gut, which runs from the mouth, at the anterior end, to the anus, at the posterior end. In many vertebrates, the gut is very long and folded, which helps increase the surface area over which nutrients can be absorbed. The human digestive tract is about 7 m (23 ft) long.



(a)



(b)

FIGURE 32-10

(a) The legs of amphibians, such as this tree frog, *Agalychnis saltator*, are sharply bent and positioned away from the body. (b) The legs of terrestrial mammals, such as this deer, *Odocoileus virginianus*, are straighter than those of amphibians, providing greater mobility and speed.

Both vertebrates and invertebrates must deal with the very toxic ammonia their bodies produce. Most vertebrates must expel wastes while conserving water. Like invertebrates, most vertebrates convert ammonia to less toxic substances. In most vertebrates, organs called **kidneys** filter wastes from the blood while regulating water levels in the body.

Nervous System

Vertebrates have highly organized brains, and the control of specific functions occurs in specific centers in the brain. The structure and function of the nervous system vary among vertebrate classes. For example, much of a fish's brain processes sensory information. Fishes have limited neural circuitry devoted to decision making. A fish's responses to stimuli in its environment are rigid, that is, they vary little from situation to situation and from fish to fish.

Other animals, such as dogs, display complex and flexible behavior. Much of the tissue in the dog's brain is given over to decision making, and its brain is large with respect to body size.

Reproduction and Development

In most fish and amphibian species, eggs and sperm are released directly into the water, where fertilization takes place. In reptiles, birds, and mammals, the egg and sperm unite within the body of the female, increasing the likelihood that an egg will be fertilized.

The fertilized eggs of many fishes, amphibians, reptiles, and birds develop outside the body. A developing embryo is nourished by the egg yolk and protected by jellylike layers or a shell. The zygotes of some fishes, amphibians, and reptiles remain inside the body of the female, nourished by the yolk until they hatch. In contrast, most mammals give birth to live offspring. Embryos of placental mammals develop in the female's body, nourished by the mother's blood supply until the young are born. With the exception of amphibians and some fishes, vertebrates undergo direct development. So, the young and the adults can share the same resources—an advantage if those resources are plentiful.



Quick Lab

Identifying Animal Characteristics

Materials 3 × 5 in. note cards (20), 5 pictures of vertebrates, 5 pictures of invertebrates

Procedure

1. Working in pairs, one partner will write one different vertebrate characteristic on each of 10 note cards. The other partner will write one different invertebrate characteristic on each of 10 note cards.
2. Place the animal pictures upside down in a stack. One partner (the dealer) will shuffle and deal all the cards and turn over one animal picture.
3. The nondealer plays first by laying down as many cards as possible that describe characteristics of the pictured animal. If no card matches, the play is passed to the other player. When neither partner can play, another picture is turned up and play continues.
4. Play ceases when neither student can play or when no pictures are left.

Analysis What are the disadvantages of using only morphological characteristics to identify an organism?

SECTION 2 REVIEW

1. Identify the primary function for the body covering of terrestrial animals.
2. Compare the structure of exoskeletons and endoskeletons.
3. Compare a closed circulatory system to an open circulatory system.
4. Compare the nervous systems of vertebrates and invertebrates.
5. Describe briefly invertebrate and vertebrate development.

CRITICAL THINKING

6. **Inferring Relationships** How might the segmented bodies of arthropods help them survive?
7. **Recognizing Relationships** How is the structure of the nervous system related to an animal's behavior?
8. **Comparing Concepts** Compare the advantages and disadvantages of the two types of development (direct and indirect development).

MILESTONES

IN

Developmental Biology

Timeline

1817 Christian Pander identifies germ layers.



1828 Karl Ernst von Baer discovers mammalian eggs and the fate of neural folds and opposes preformation.

1875 Oscar Hertwig observes fertilization.

1883 August Weismann publishes germ plasm theory.

1924 Spemann and Mangold discover the neural organizer.



1950s to present
Stem cell research begins and continues.



1969 Lewis Wolpert studies pattern formation and positional information in embryos.

1983 Researchers discover homeotic genes.



1996 Scottish researchers clone Dolly the sheep from a mammary cell.

Present Researchers find stem cells in many types of adult tissues.

People have known only since the 1800s that eggs and sperm unite at fertilization. In a little more than a century, researchers have come to understand a great deal about how one cell divides repeatedly to form a mass of cells and how a complex multicellular organism develops from that mass of dividing cells. As in many areas of biology, the speed and sophistication of research in developmental biology has been remarkable.

Until the 19th century, biologists believed that embryos grew from very small versions of complete organisms. In 1817, a Russian naturalist named Christian Pander observed developing chicks and described the three embryonic layers now called *ectoderm*, *mesoderm*, and *endoderm*. In 1828, Karl Ernst von Baer published a paper in which he concluded that the neural fold of an animal embryo gives rise to the animal's nervous system and notochord.

In 1875, German embryologist Oscar Hertwig first observed the union of the nuclei of male and female gametes at fertilization. Soon after, in 1883, another German biologist, August Weismann, laid out his germ plasm theory. This theory states that the body has germ cells, which pass along hereditary traits, and somatic cells, which do not pass along traits to new generations.

In 1924, German embryologists Hans Spemann and Hilde Mangold studied how cells “know” when to divide and what to do in a growing embryo. They hypothesized that one group of cells might “organize” the rest of the cells. Spemann and Mangold discovered that cells that were transplanted from the blastopore region of a blastula to another region of the blastula caused a new nervous system to form. Today, scientists are still searching for the signal these “organizer cells” give. Part of that search has been the study of pattern formation, or how cells respond to signals and form cells and tissues that have particular functions.

In 1969, Lewis Wolpert proposed that morphogens, or pattern-directing substances, diffuse through a body region—a limb bud, for example—and cause fingers or toes to form in a certain order.

In 1983, researchers discovered genes called *homeotic genes* that help control where limbs, such as the legs or antennae, grow on a fruit fly embryo.

Another recent line of research centers on stem cells. Stem cells are undifferentiated cells that are found in embryos and adult tissues and that can give rise to new cells. Researchers have recently discovered that adult stem cells exist in many more types of tissues than were once thought possible. A related line of research led to the birth of Dolly in 1996, the first mammal to be cloned from an adult somatic cell.

Review

1. What does the germ plasm theory state?
2. **Critical Thinking** Why are “organizer cells” important to the growing embryo?
3. **Critical Thinking** How does the production of genetic clones affect the germ plasm theory?

Internet connect

www.scilinks.org
Topic: Development of Mammalian Embryo
Keyword: HM60398

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FERTILIZATION AND DEVELOPMENT

Development of a multicellular animal from an egg cell is a truly remarkable process. Each cell in an animal has the same set of genes that are used to build the animal, yet animals have many different kinds of cells. From the fertilized egg come large numbers of cells—trillions in humans—that consistently give rise to structural features of the animal body.

FERTILIZATION AND EARLY DEVELOPMENT

In animals, **fertilization** is the union of female and male gametes to form a zygote. Fertilization results in the combination of the haploid sets of chromosomes from two individuals into a single diploid zygote.

Gametes

In most animal species, the sperm cell, shown in Figure 32-11, is specialized for movement—it is very streamlined and small. The head of the sperm contains chromosomes, and the tail of the sperm is composed of a long flagellum.

The egg, also shown in Figure 32-11, is typically large because it has a large store of cytoplasm and yolk. The size of an egg produced by a given species seems to depend on how long the food supply in the yolk must last. For aquatic animals in which the embryo begins to feed itself early, eggs are small, and there is little yolk. In sharp contrast, the embryos of birds must live on the yolk until they hatch. In these eggs, the yolk volume is very large.

Fertilization

At the start of fertilization, the sperm's cell membrane fuses with the egg's cell membrane, and the nucleus of the sperm enters the cytoplasm of the egg. The fusion of the cell membranes of the egg and sperm causes an electrical change in the egg membrane that blocks entry to the egg by other sperm cells. The sperm nucleus merges with the egg nucleus to form the diploid nucleus of the zygote. Once a zygote is formed, replication of DNA begins, and the first cell division soon follows.

OBJECTIVES

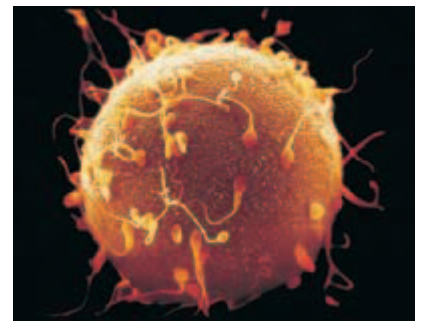
- List the steps of fertilization and development through gastrulation.
- List two body parts formed from each germ layer.
- Identify the three different body cavity structures of animals.
- Name the categories of animals that undergo spiral cleavage and radial cleavage.
- Contrast the two processes of coelom formation.

VOCABULARY

fertilization
cleavage
blastula
gastrulation
gastrula
archenteron
blastopore
ectoderm
endoderm
mesoderm
acoelomate
pseudocoelom
coelom
protostome
deuterostome
schizocoely
enterocoely

FIGURE 32-11

The small, flagellated sperm is adapted for motility and speed. It must seek out and fertilize the much larger, yolk-filled egg.



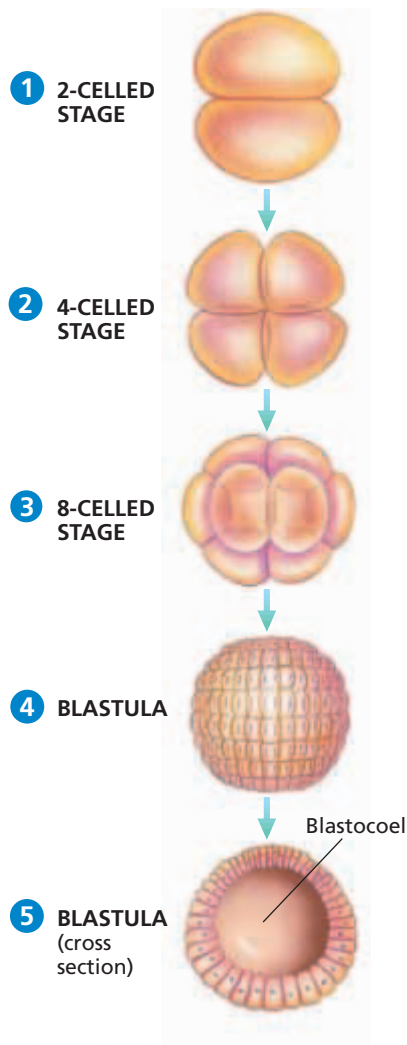


FIGURE 32-12

During cleavage, the zygote divides repeatedly without undergoing cell growth, producing a many-celled hollow blastula.

Cleavage and Blastula Formation

The series of cell divisions that occurs immediately following fertilization is termed **cleavage**. Figure 32-12, steps 1, 2, and 3 show that as cleavage progresses, the number of cells increases, from 2 to 4, then to 8, and so on. During cleavage, mitotic divisions rapidly increase the number of cells, but the cells do not grow in size. Thus, cleavage yields smaller and smaller individual cells. Cleavage increases the surface area-to-volume ratio of each cell, which enhances gas exchange and other environmental interactions.

In most species, cleavage produces a raspberry-shaped mass of 16 to 64 cells, as shown in step 4. As the number of dividing cells further increases, the mass becomes a hollow ball of cells called a **blastula**, shown in step 5. The central cavity of a blastula is called the *blastocoel* (BLAS-toe-SEEL).

Gastrulation and Organogenesis

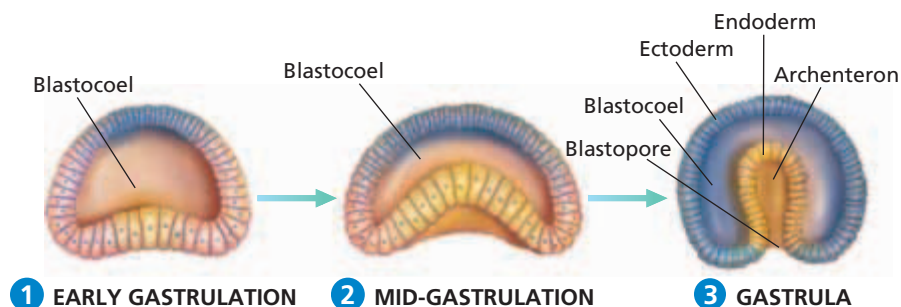
At the start of the next stage of development, shown in Figure 32-13, an area of the blastula begins to collapse inward. As shown in steps 1 and 2, reorganization of the cells of the hollow blastula begins with the inward movement of cells at one end of the blastula. This process, called **gastrulation**, transforms the blastula into a multilayered embryo, called the **gastrula**, shown in step 3. Gastrulation is marked by changes in the shape of cells and the way the cells interact with each other.

As the inward folding continues, the now cup-shaped embryo enlarges, and a deep cavity, called the **archenteron**, or primitive gut, develops. The open end of the archenteron is called the **blastopore**. Forming the outer layer of the gastrula is the outer germ layer, the **ectoderm**, shown in blue in step 3. The inner germ layer, the **endoderm**, is shown in yellow. In most phyla, the gastrula does not remain a two-layer structure. As development progresses, a third layer, the **mesoderm**, forms between the endoderm and the ectoderm.

Each of the germ layers formed during gastrulation develops into certain organs in a process called *organogenesis*. The endoderm forms the lining of the urinary system, the reproductive system, and most of the digestive tract; it also forms the pancreas, liver, lungs, and gills. The ectoderm forms the outer layer of skin, hair, nails, and the nervous system. The mesoderm forms many body parts, including the skeleton, muscles, the inner layer of skin, the circulatory system, and the lining of the body cavity.

FIGURE 32-13

Echinoderms, such as the sea urchin, undergo the gastrulation process shown here. The blastula reorganizes and forms the cup-shaped gastrula. Other phyla have somewhat different patterns of gastrulation.



PATTERNS OF DEVELOPMENT

The distinct patterns of cleavage and formation of body-plan features found in different animal phyla are clues to the phylogenetic history of the organisms.

Types of Body Cavities

Animals, such as flatworms, that do not have a body cavity are called **acoelomates** (uh-SEE-luh-MAYTS). The interior of the animal is solid, as shown in Figure 32-14a. The endodermic gut, shown in yellow, and the outer covering of the animal, shown in blue, are connected by the solid tissue of the mesoderm.

However, most animal phyla have body cavities that separate their digestive tract from the outer body wall. Within this group, there are differences in how the body cavity develops. In some phyla, including rotifers and roundworms, the mesoderm lines the interior of the coelom but does not surround the exterior of the endodermic gut. A cavity that is not completely lined by mesoderm is called a **pseudocoelom** (soo-doh-SEE-luhm), which means “false body cavity.” Roundworms have pseudocoeloms such as the one shown in Figure 32-14b and are thus called *pseudocoelomates*. In pseudocoelomates, mesoderm lines the fluid-filled body cavity, and the endodermic gut is suspended in this fluid.

A cavity completely lined by mesoderm is called a **coelom** (SEE-luhm), as shown in Figure 32-14c. Animals that have coeloms are called *coelomates* (SEE-luh-MAYTS). In coelomates, mesoderm lines the body cavity and surrounds and supports the endodermic gut. The mesoderm also forms the tissues of attachment for the organs located in the coelom, such as the liver and the lungs. Mollusks, annelids, arthropods, chordates, and echinoderms are coelomates.

Cleavage and Blastopore Fate

Recall from Figure 32-3 that echinoderms and chordates share a branch of the phylogenetic diagram of animals; and mollusks, annelids, and arthropods share another branch. There are two distinct patterns of development in animals with a coelom. In the embryos of mollusks, arthropods, and annelids, the blastopore develops into a mouth, and a second opening forms at the other end of the archenteron, forming an anus. These organisms are called **protostomes** (PROHT-oh-STOHMZ), which means “first mouth.”

Many protostomes undergo *spiral cleavage*, in which the cells divide in a spiral arrangement. In the embryos of echinoderms and chordates, the blastopore develops into an anus, and a second opening at the other end of the archenteron becomes the mouth. These organisms are called **deuterostomes** (DOOT-uh-oh-STOHMZ), which means “second mouth.” Most deuterostomes undergo *radial cleavage*, in which the cell divisions are parallel to or at right angles to the axis from one pole of the blastula to the other.

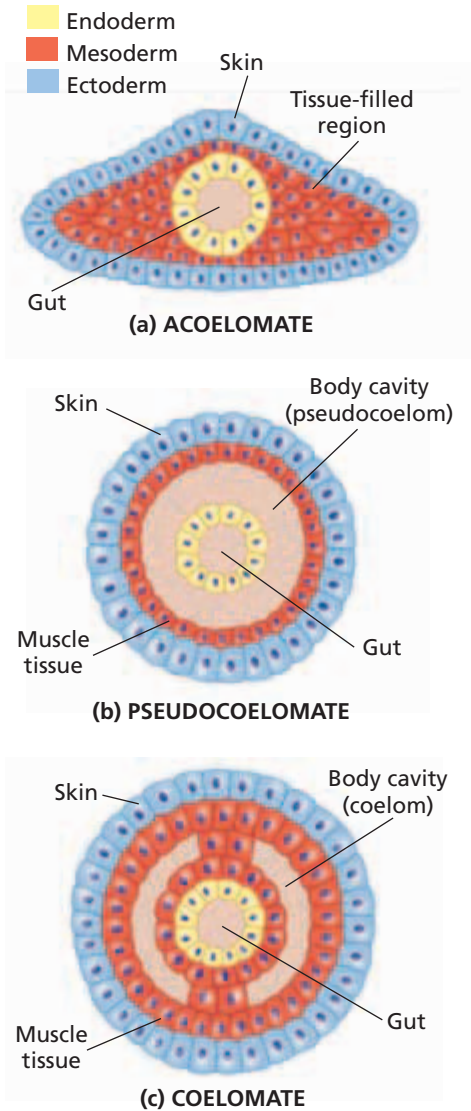


FIGURE 32-14

In three-layered acoelomates (a), the endodermic gut is surrounded by a solid layer of mesoderm. In pseudocoelomates (b), the endodermic gut is suspended in a fluid-filled cavity that is surrounded by mesoderm. In coelomates (c), the endodermic gut is surrounded by and suspended by mesoderm, which also surrounds the coelom, or body cavity.

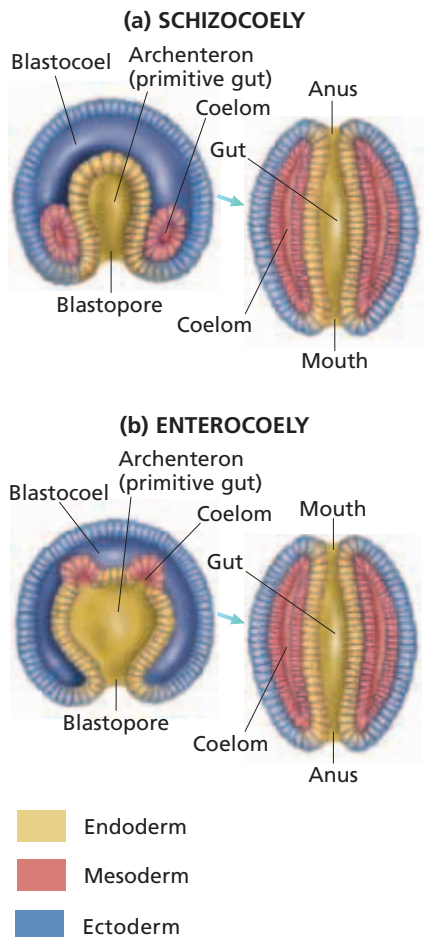


FIGURE 32-15

In protostomes, the coelom arises in a process called schizocoely (a), and the blastopore becomes the mouth. In deuterostomes, the coelom arises by enterocoely (b), and the blastopore becomes the anus.

Protostomes and deuterostomes also differ in how early the cells of the embryo specialize. If the cells of some protostome embryos are separated at the four-cell stage of development, each cell will develop into only one-fourth of a complete embryo, and the developing organism will die. Thus, the path of each cell is fixed early in the development of the protostome in a pattern called *determinate cleavage*.

In contrast, if the cells of most four-celled deuterostome embryos are separated, each cell will embark on its own path to become a separate organism. This type of development is called *indeterminate cleavage*. Indeterminate cleavage is responsible for the development of identical twins in humans.

Coelom Formation

The way in which the coelom forms in many protostomes differs from the way it forms in many deuterostomes. Figure 32-15a shows coelom formation in protostomes. Cells located at the junction of the endoderm and ectoderm (at the rim of the cup-shaped embryo) move toward the interior of the gastrula. Rapid division of these cells (shown in pink) in the blastocoel forms the mesoderm. The mesoderm then spreads and splits to form the coelom. This process of coelom formation is called **schizocoely** (SKIZ-oh-SEEL-ee), or “split body cavity.”

Figure 32-15b shows coelom formation in deuterostomes. The mesoderm forms when the cells lining the dorsal, or top, part of the archenteron begin dividing rapidly. These rapidly dividing cells (shown in pink) form pouches that become mesoderm. The coelom develops within the mesodermal pouches. This process of coelom formation is called **enterocoely** (EN-tuhr-oh-SEEL-ee), which means “gut body cavity.” During both enterocoely and schizocoely, mesodermal cells spread out to completely line the coelom, and the blastocoel disappears. Thus, in both protostomes and deuterostomes, mesoderm lines the interior of the outer body wall and surrounds the gut.

SECTION 3 REVIEW

1. Beginning with fertilization, list the steps of development through mesodermal formation.
2. Name the three germ layers and two body parts that arise from each layer.
3. Compare the development of protostomes and deuterostomes.
4. Identify how the mesoderm is formed in schizocoely, and differentiate it from the process in enterocoely.
5. Identify the type of cleavage that can give rise to identical human twins.

CRITICAL THINKING

6. **Inferring Relationships** Which adaptive advantage is associated with indeterminate cleavage?
7. **Recognizing Relationships** What is the relationship between endoderm formation and coelom formation in echinoderms?
8. **Predicting Patterns** While exploring a tide pool, you find a periwinkle, which is a marine snail. Which kind of development does a periwinkle have? Explain.

CHAPTER HIGHLIGHTS

SECTION 1

The Nature of Animals

- Animals are either invertebrates—lack a backbone—or vertebrates—have a backbone.
- Animals are multicellular organisms that lack cell walls and are heterotrophic. Most animals reproduce sexually and can move.
- Movement and response to the environment are governed by an animal's nervous and muscle tissues.
- The first animals may have evolved from colonial protists.
- At some stage of their lives, chordates have a notochord, dorsal nerve cord, postanal tail, and pharyngeal pouches.
- Most animals have tissues and radial or bilateral symmetry. Bilateral symmetry is associated with cephalization. Most animals have a body cavity that aids movement and the transport of nutrients and wastes.
- Animals are classified by degree of cell specialization, number of tissue layers, type of symmetry, and most recently, by sequencing of rRNA.

Vocabulary

animal (p. 651)
vertebrate (p. 651)
invertebrate (p. 651)
specialization (p. 651)
ingestion (p. 652)

zygote (p. 652)
differentiation (p. 652)
chordate (p. 654)
notochord (p. 654)
dorsal nerve cord (p. 654)

pharyngeal pouch (p. 654)
symmetry (p. 655)
radial symmetry (p. 655)
dorsal (p. 655)
ventral (p. 655)

anterior (p. 655)
posterior (p. 655)
bilateral symmetry (p. 655)
cephalization (p. 655)
germ layer (p. 655)

SECTION 2

Invertebrates and Vertebrates

- Symmetry, segmentation, and type of skeleton are related to the lifestyle of an animal.
- Invertebrates include morphologically diverse phyla, the members of which are mostly radially or bilaterally symmetrical.
- In an open circulatory system, circulatory fluid is pumped by the heart through vessels and into the body cavity and is then returned to the vessels.
- In a closed circulatory system, blood is pumped by a heart and circulates throughout the body in tubelike vessels that form a closed loop.
- Vertebrates have highly organized brains in which specific functions occur in specific centers of the brain. The larger the decision-making portion of the brain is, the more complex and flexible the behavior of the animal.
- Most animals reproduce sexually. Fertilization and development may be external or internal.

Vocabulary

segmentation (p. 657)
exoskeleton (p. 658)
gill (p. 658)

open circulatory system (p. 658)
closed circulatory system (p. 658)

hermaphrodite (p. 659)
larva (p. 659)
endoskeleton (p. 660)
vertebra (p. 660)

integument (p. 660)
lung (p. 660)
kidney (p. 661)

SECTION 3

Fertilization and Development

- Development includes cleavage, blastula formation, gastrulation, and organogenesis. The germ layers include ectoderm, endoderm, and mesoderm.
- Ectoderm forms, among other things, the outer layer of skin; endoderm forms many internal organs; and mesoderm forms the skeleton and muscles.
- Acoelomates are animals without a body cavity. Pseudocoelomates are animals whose body cavity is not completely lined by mesoderm. Coelomates have a body cavity completely lined by mesoderm.
- Protostomes have spiral cleavage and schizocoely. Deuterostomes have radial cleavage and enterocoely.

Vocabulary

fertilization (p. 663)
cleavage (p. 664)
blastula (p. 664)
gastrulation (p. 664)
gastrula (p. 664)

archenteron (p. 664)
blastopore (p. 664)
ectoderm (p. 664)
endoderm (p. 664)
mesoderm (p. 664)

acoelomate (p. 665)
pseudocoelom (p. 665)
coelom (p. 665)
protostome (p. 665)
deuterostome (p. 665)

schizocoely (p. 666)
enterocoely (p. 666)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
 - radial symmetry* and *bilateral symmetry*
 - open circulatory system* and *closed circulatory system*
 - vertebrate* and *invertebrate*
 - spiral cleavage* and *radial cleavage*
- Explain the relationship between cell specialization and differentiation.
- Choose the term that does not belong in the following group, and explain why it does not belong: *notochord*, *cephalization*, *dorsal nerve cord*, and *radial symmetry*.
- Word Roots and Origins** The word *blastopore* comes from the Greek *blastos*, which means “bud,” and *poros*, which means “passage.” Using this information, explain why the term *blastopore* is a good name for the structure it describes.

UNDERSTANDING KEY CONCEPTS

- Describe** each of the four characteristics that define animals.
- Explain** how neural tissue and muscle tissue work together in an animal's body to allow the animal to respond to its environment.
- Describe** the probable changes that early colonial flagellates underwent as they evolved into the first animals.
- List** the four features common to all chordates at some time in their life. What has happened to two of these features in an adult human?
- Summarize** what happened to the position of the body with respect to the legs as vertebrates adapted to life on land.
- Explain** how having a body cavity aids movement.
- Infer** the relationship between two phyla that are represented on the same branch of a phylogenetic tree. Which features are used to determine the relationship?
- Identify** the most probable type of movement that an organism with bilateral symmetry would exhibit.
- Contrast** segmentation in invertebrates with segmentation in vertebrates.
- Sequence** the development of the nervous system from cnidarians to mammals.
- Identify** the structure that the archenteron becomes in a developing animal.
- Name** two body parts formed by each of the following: endoderm, mesoderm, and ectoderm.
- Identify** how a closed circulatory system differs from an open circulatory system.
- Compare** schizocoely with enterocoely.
- CONCEPT MAPPING** Use the following terms to create a concept map that shows development from blastula to coelom formation in a deuterostome: *blastula*, *gastrula*, *archenteron*, *blastopore*, *anus*, *mouth*, *germ layers*, *ectoderm*, *endoderm*, *mesoderm*, and *coelom*.

CRITICAL THINKING

- Recognizing Relationships** Considering that an endoskeleton can support more weight than an exoskeleton, would a large-bodied animal with an exoskeleton be more likely to live in the water or on land? Explain.
- Recognizing Relationships** On mammals and birds, the head is positioned higher with respect to the body than it is on amphibians and reptiles. Why might it be helpful to have a head positioned over the body?
- Interpreting Graphics** Observe the animal pictured below, and answer the following questions.
 - Which kind of symmetry does the animal display?
 - Is this animal cephalized?
 - How many germ layers does this animal have?
 - How many openings does this animal's digestive system have?
 - Does this animal have neurons?



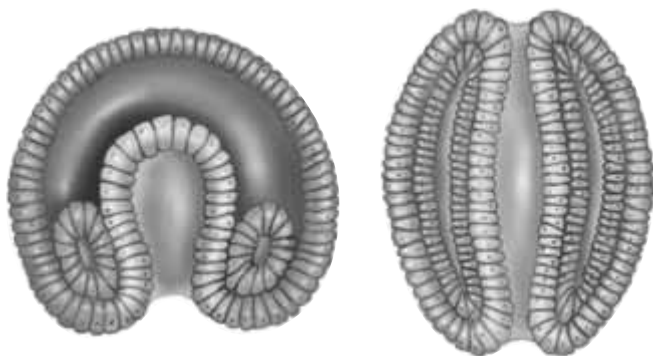


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- What is the name for the process that leads to cell specialization in multicellular organisms?
 - evolution
 - fertilization
 - differentiation
 - asexual reproduction
- What process takes place as a zygote begins to divide after fertilization?
 - meiosis
 - cleavage
 - gastrulation
 - organogenesis
- Which animals do not have true tissues?
 - sponges
 - chordates
 - cnidarians
 - ctenophores
- What are the basic tissue types in an embryo called?
 - coeloms
 - germ layers
 - notochords
 - pharyngeal pouches

INTERPRETING GRAPHICS: The diagrams below illustrate a certain type of body-cavity development. Study the diagrams to answer the questions that follow.



- The diagrams represent the development of which of the following animals?
 - fish
 - sponge
 - octopus
 - cnidarian
- With which of the following is the process illustrated above associated?
 - acoelomates
 - deuterostomes
 - spiral cleavage
 - indeterminate cleavage

DIRECTIONS: Complete the following analogy.

7. Ectoderm : skin :: mesoderm :
- lungs
 - nerves
 - vertebrae
 - intestines

INTERPRETING GRAPHICS: The diagrams below illustrate different organisms. Study the images to answer the question that follows.



Sponge



Beetle



Jellyfish

8. Which of the organisms has radial symmetry?
- beetle
 - sponge
 - jellyfish
 - both the beetle and the sponge

SHORT RESPONSE

Ctenophores and cnidarians are considered closely related to one another. Chordates and echinoderms are also considered closely related to one another.

Explain how it was determined that cnidarians are less closely related to chordates and echinoderms than they are to ctenophores.

EXTENDED RESPONSE

Three types of body-cavity organization exist in animals—acoelomate, pseudocoelomate, and coelomate.

Part A Explain how acoelomates and pseudocoelomates differ from coelomates.

Part B Explain how the lining of the body cavity develops in coelomates.

Test TIP

If you find particular questions difficult, put a light pencil mark beside them and keep working. (Do not write in this book.) As you answer later questions, you may find information that helps you answer the difficult questions.

Dissecting a Sheep's Heart

OBJECTIVES

- Describe the appearance of the external and internal structures of a sheep's heart.
- Name the structures and functions of a sheep's heart.

PROCESS SKILLS

- observing structures
- identifying
- demonstrating

MATERIALS

- sheep's heart
- dissecting tray
- blunt metal probe
- scissors
- scalpel
- tweezers

Background

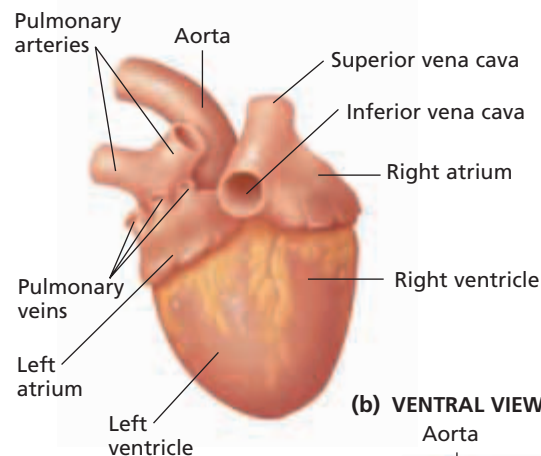
1. The heart has a left and a right side. It has two upper chambers, the left and right atria, and two lower chambers, the left and right ventricles. Why do multiple chambers result in a more efficient heart?
2. Blood enters the heart from the body through the superior or inferior vena cava. The blood then enters the right atrium and flows through valves into the right ventricle. Blood flows from the right ventricle through the pulmonary artery to the lungs. What process occurs in the lungs?
3. Oxygenated blood flows from the lungs through the pulmonary veins to the left atrium. Then, it flows through valves into the thick-walled left ventricle. Blood flows from the left ventricle through the large aorta to the rest of the body.

Procedure

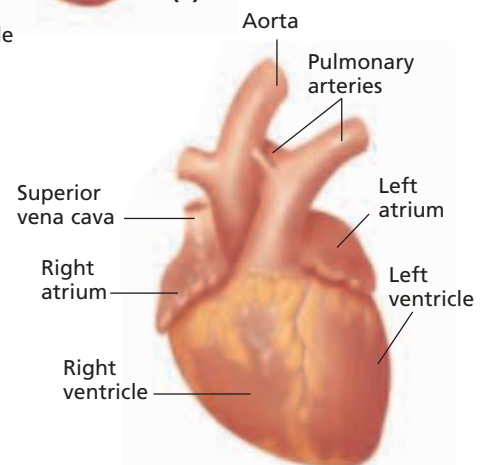
1. In this lab, you will observe the external structure of a sheep's four-chambered heart and dissect the heart to study its internal structure.
2.    Put on safety goggles, gloves, and a lab apron.

3. Place a sheep's heart in a dissecting tray. Turn the heart so that the ventral surface is facing you, as shown in the diagram below. Use the diagram of the ventral view to locate the left and right atria, the left and right ventricles, the aorta, the superior and inferior vena cava, and the pulmonary arteries. Turn the heart over. Use the diagram of the dorsal view to locate once again the structures just named, as well as the pulmonary veins.
4. Use a blunt metal probe to explore the blood vessels that lead into and out of the chambers of the heart.
5. Locate a diagonal deposit of fat along the lower two-thirds of the heart. This serves as a guideline to mark the wall between the two ventricles. Use this fatty deposit to guide your incision into the heart.

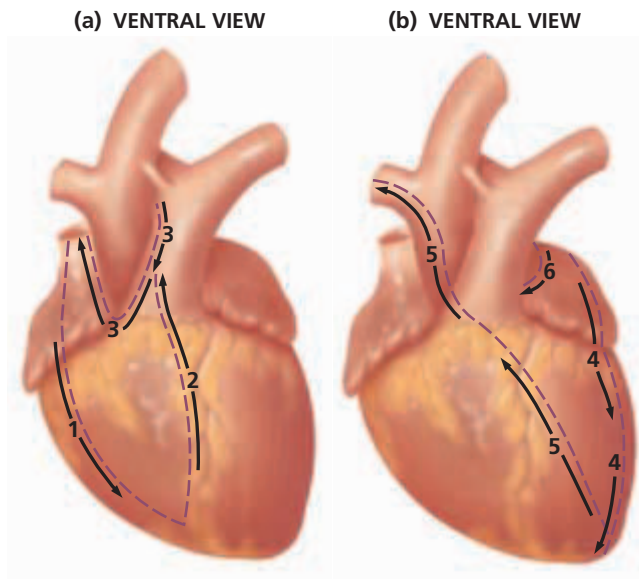
(a) DORSAL VIEW



(b) VENTRAL VIEW



6. Follow the cutting diagram below very carefully to study the anatomy of the right side of the heart.
7.  Again, turn the heart with the ventral surface facing you and the apex pointing downward. Use scissors to cut along line 1. **CAUTION** Always cut in a direction away from your face and body. Cut just deep enough to go through the atrial wall. Continue the cut into the right ventricle. With a probe, push open the heart at the cut, and examine the internal structure.



8. Cut along line 2, and extend the cut upward toward the pulmonary artery. Cut just deep enough to go through the ventricle wall. Complete the cut on line 3. Cut downward along the pulmonary artery, around through the wall of the right atrium, and upward along the right superior vena cava.
9. With tweezers, carefully lift the resulting flap to expose the structures underneath.
10. Follow the cutting diagram above very carefully to study the anatomy of the *left* side of the heart.
11. Start to cut on line 4 at the top of the left atrium, and continue into the left ventricle. Cut just deep enough to go through the ventricle wall.
12. Cut on line 5 across the middle of the left ventricle into the aorta. Leave a small margin between this cut

and the cut previously made for line 2. Begin to cut on line 6 on the left atrium where cut 4 began. Extend this cut around and through the pulmonary artery upward on the aorta to the right of cut 5.

13. With tweezers, carefully lift up the resulting flap to expose the structure underneath.
14. Observe the thick septum dividing the left and right ventricles. Also, note the greater thickness of the walls of the left ventricle.
15. Locate the tricuspid valve between the right atrium and ventricle. Locate the mitral valve between the left atrium and ventricles. Observe that the valves are connected by fibers to the inner surface of the ventricle. Use a probe to explore the openings in the valves.
16.  With a scalpel, cut across a section of the aorta and a section of the vena cava. Compare the thickness of their walls.
17.  Dispose of your materials according to the directions from your teacher.
18.  Clean up your work area, and wash your hands before leaving the lab.

Analysis and Conclusions

1. Trace the path of blood from the right atrium to the aorta.
2. Pulmonary circulation carries blood between the heart and the lungs. Systemic circulation carries blood to the rest of the body. In which chambers of the heart does pulmonary circulation begin and end? In which chambers does systemic circulation begin and end?
3. What is the function of the septum separating the left and right ventricles?
4. What is the function of the mitral and tricuspid valves?
5. Why are the walls of the left ventricle thicker than the walls of the right ventricle?

Further Inquiry

The heartbeat originates in a small bundle of tissue in the right atrium. This bundle is the sinoatrial, or S-A, node. Read about the S-A node. What does it do? Why is the S-A node known as the pacemaker?

SPONGES, CNIDARIANS, AND CTENOPHORES



Delicate polyps of Monet's tube coral, *Dendrophyllia gracilis*, extend from a hard skeleton.

SECTION 1 *Porifera*

SECTION 2 *Cnidaria and Ctenophora*

PORIFERA

Invertebrates are animals that do not have backbones.

Invertebrates include more than a million species. About 97 percent of all known animal species are invertebrates, among the simplest of which are sponges.

BODY PLAN OF SPONGES

Sponges are aquatic animals that make up the phylum Porifera (pohr-IF-uh-uh). These simple organisms clearly represent the transition from unicellular to multicellular life. Because sponges are heterotrophic, multicellular organisms that do not have cell walls, they are classified as animals. Sponges have no gastrula stage, exhibit less cell specialization than most other animals, and have no true tissues or organs.

But sponges do have a key property of all animal cells—cell recognition. A living sponge can be passed through a fine mesh, which separates the sponge into individual cells. These separated cells will then regroup to form a new sponge. There are about 5,000 named species of sponges. About 150 species live in fresh water, while the rest are marine.

Sponges are so unlike other animals that early biologists thought that sponges were plants. Most sponges do resemble plants in some ways. For example, adult sponges are **sessile** (SEHS-il), which means that they attach themselves firmly to a surface and do not move. Sponges grow in many shapes, sizes, and colors and often look like mossy mats, cactuses, or blobs of fungus. Sponges can be as small as 1 cm (0.4 in.) in length or as large as 2 m (6.6 ft) in diameter.

The basic body plan of a sponge consists of two layers of cells separated by a jellylike substance called **mesohyl** (MEHZ-oh-hil). In the simplest sponges, the body wall forms a hollow cylinder that is closed at the bottom and open at the top. The interior of the cylinder is lined with flagellated cells called **choanocytes** (koh-AN-oh-siets), or collar cells. By beating their flagella, choanocytes draw water into the sponge through numerous pores, called **ostia** (AHS-tee-uh) (singular, **ostium**), that penetrate the body wall. In fact, the name *Porifera* comes from a Latin word meaning “pore-bearer.” The water that is pumped into the interior of the sponge leaves through the **osculum** (AHS-kyoo-luhm), the opening at the top of the sponge that you can see in Figure 33-1 on the next page.

A sponge would collapse without some type of supporting structure. In some sponges, support is provided by a simple skeleton made of a network of tough, flexible protein fibers called **spongin** (SPUHN-jin). Other sponges have skeletons consisting of **spicules**.

OBJECTIVES

- **Describe** the basic body plan of a sponge.
- **Describe** the process of filter feeding in sponges.
- **Contrast** the processes of sexual and asexual reproduction in sponges.

VOCABULARY

sponge
sessile
choanocyte
ostium
osculum
spongin
spicule
filter feeding
amoebocyte
gemmule
regeneration
hermaphrodite



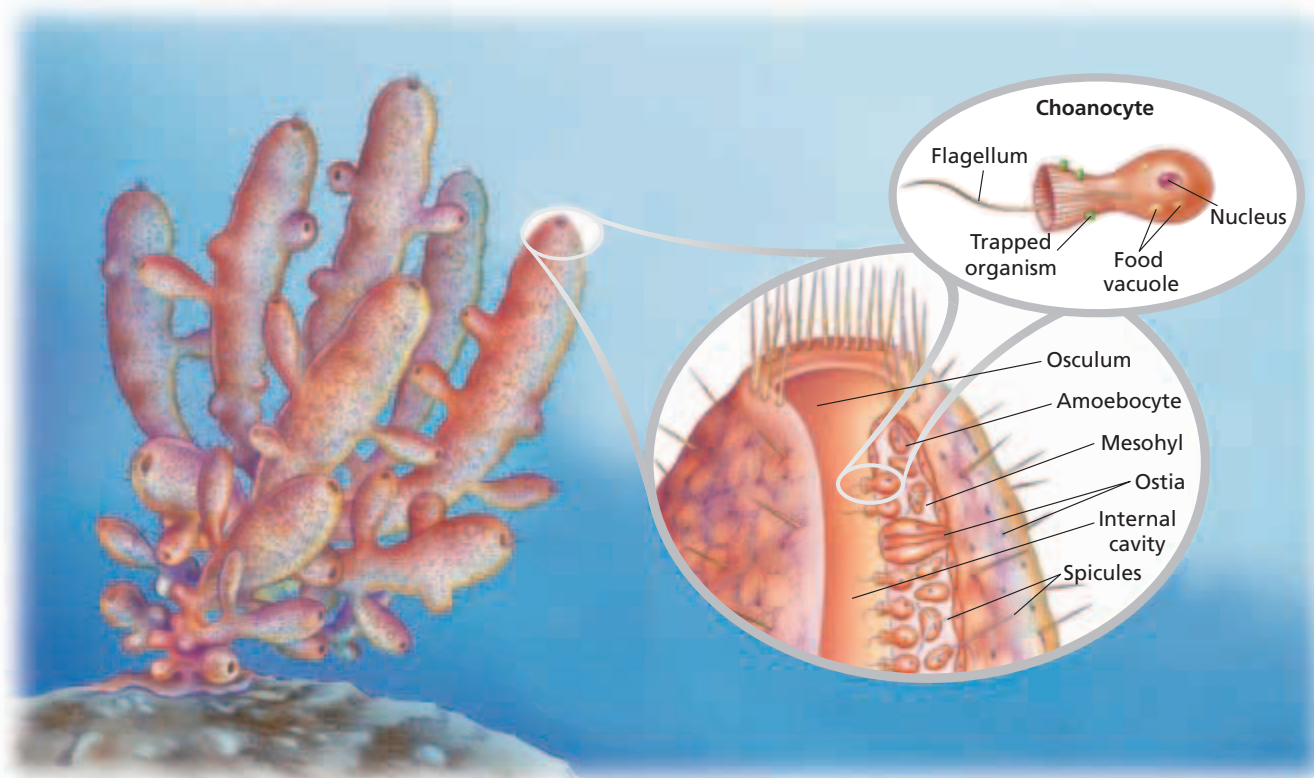


FIGURE 33-1

The body of a sponge is a hollow cylinder. Water is drawn into the cylinder through many small pores (ostia) in the sponge's body and exits through the osculum, the opening at the top of the sponge.

Spicules are tiny, hard particles of calcium carbonate or silicon dioxide that are often shaped like spikes. Calcium carbonate is one of the compounds that give bones and teeth their hardness, and silicon dioxide is the major component of glass and quartz. Still other sponges have a combination of spongin and spicules.

FEEDING AND DIGESTION IN SPONGES

Because they are sessile, sponges cannot pursue food. Instead, most sponges feed by sieving food out of the water. The flagella of the choanocytes beat, drawing water through the ostia. The choanocytes trap plankton and other tiny organisms in their small, hairlike projections. This feeding method is called **filter feeding**. Other food of filter-feeding sponges includes bits of organic matter. However, scientists have discovered a sponge species that uses movable filaments covered with hooked spicules to snare small shrimp. The shrimp are then absorbed into the sponge's body.

The food that a sponge collects is engulfed and digested by the choanocytes. Nutrients pass from the choanocytes to cells that crawl about within the body wall and deliver the nutrients to the rest of the body. Scientists call these crawling cells **amoebocytes** (uh-MEE-buh-siets) because the cells resemble amoebas. Locate the amoebocytes in Figure 33-1. Carbon dioxide and other wastes produced by the sponge's cells diffuse into the water that passes through the sponge. The water carries these wastes as it flows out through the osculum, thus removing the wastes from the sponge.

Word Roots and Origins

amoebocyte

from the Greek *amoibe*, meaning "change," and *kytos*, meaning "cavity"

REPRODUCTION IN SPONGES

Sponges can reproduce asexually by forming small buds that break off and live separately. The sponge shown in Figure 33-1 has many buds. During droughts or cold weather, some freshwater sponges produce internal buds called **gemmules** (JEM-yoolz). Each gemmule is a food-filled ball of amoebocytes surrounded by a protective coat made of organic material and spicules. Gemmules can survive harsh conditions that may kill adult sponges. When conditions improve, sponge cells emerge from the gemmules and grow into new sponges. Sponges also have remarkable powers of **regeneration**, the regrowth of missing cells, tissues, or organs. A very small piece of a sponge can regenerate a complete new sponge.

Sponges can also reproduce sexually. As you can see in Figure 33-2, sperm released into the water from one sponge enter the pores of a second sponge. Choanocytes in the second sponge engulf the sperm and transfer them to amoebocytes. Amoebocytes carry the sperm to an egg located in the mesohyl. After the egg is fertilized, it develops into a larva. A *larva* is an immature stage of an animal that is usually very different in form from the adult. Flagella on the larva's surface enable the larva to leave its parent sponge. Eventually, the larva settles and attaches to an object. Its cells then reorganize to form an adult sponge.

Some species of sponges have separate sexes, but in most species, each individual is a hermaphrodite (her-MAF-roh-diet). **Hermaphrodites** produce both eggs and sperm. Self-fertilization rarely happens in hermaphroditic species. Instead, the sperm of one individual usually fertilize the eggs of another individual. Because each hermaphrodite produces eggs, the chances of fertilization are greater than in species in which only females produce eggs. Hermaphroditism is common in invertebrates that are sessile, that move slowly, or that live in low-density populations.

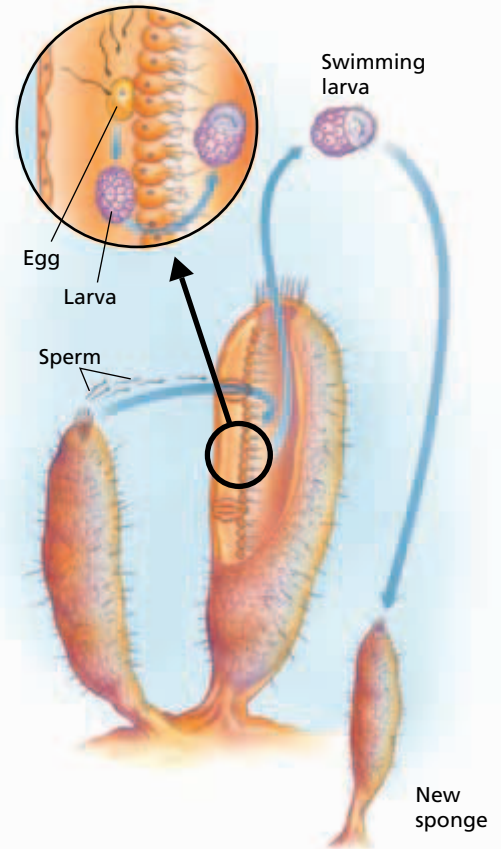


FIGURE 33-2

Sponges can reproduce asexually and sexually. In sexual reproduction, the union of an egg and sperm ultimately results in a swimming larva that escapes from the parent sponge and grows into a new individual.

SECTION 1 REVIEW

1. Compare the structure of spongin with the structure of spicules.
2. Describe two different ways in which sponges feed.
3. How do gemmules help some freshwater sponges survive unfavorable conditions?
4. What role do amoebocytes play in the sexual reproduction of sponges?

CRITICAL THINKING

5. **Analyzing Concepts** How is it possible for a population of sponges, which are sessile animals, to disperse?
6. **Applying Information** Consider the properties of a natural bath sponge that is made of spongin. How is a spongin skeleton adaptive?
7. **Relating Concepts** Why are sponges not considered to be predators?

SECTION 2

OBJECTIVES

- **Describe** the basic body plan of a cnidarian.
- **Summarize** how cnidarians feed.
- **Describe** the nervous system of cnidarians.
- **Identify** and give examples of the four classes of cnidarians.
- **Describe** the common characteristics of ctenophores.

VOCABULARY

medusa
polyp
epidermis
gastrodermis
mesoglea
gastrovascular cavity
tentacle
cnidocyte
nematocyst
nerve net
planula
coral reef
colloblast
apical organ
bioluminescence

CNIDARIA AND CTENOPHORA

Cnidaria (nie-DER-ee-uh) and Ctenophora (tee-NAHF-uh-uh) are two phyla of radially symmetrical invertebrates. The animals in these phyla are more complex than the sponges. Their cells are organized into tissues, and they have a few simple organs. All members of the phyla Cnidaria and Ctenophora are aquatic, and most live in the ocean.

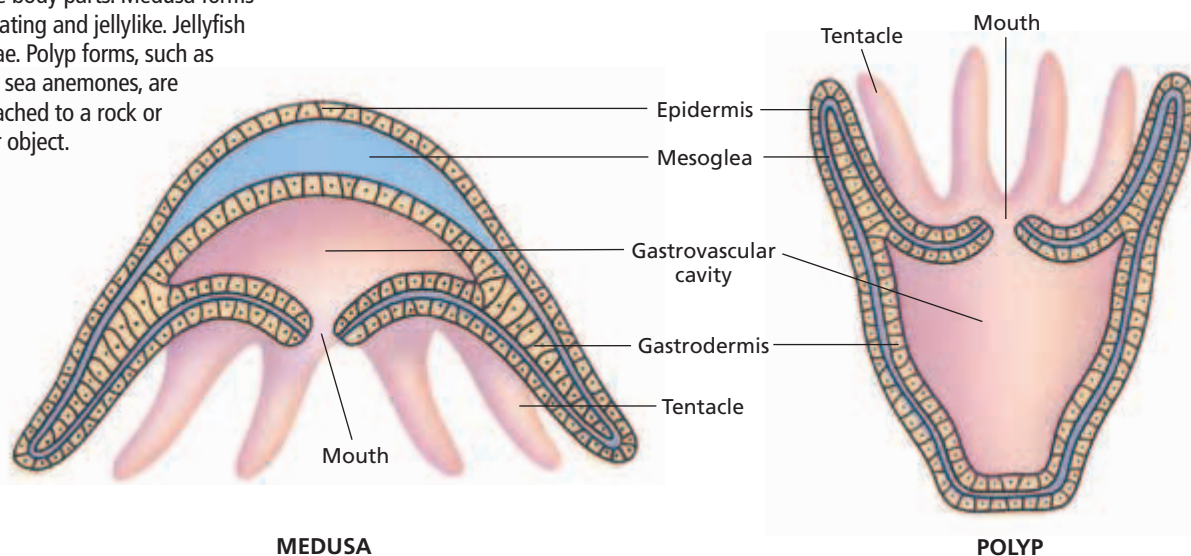
BODY PLAN OF CNIDARIANS

Tiny freshwater hydra, stinging jellyfish, and flowerlike coral all belong to the phylum Cnidaria. Animals in this phylum are called **cnidarians**. As you can see in Figure 33-3, the body of a cnidarian may be either bell-shaped or vase-shaped. The bell-shaped **medusa** (me-DOO-suh) is specialized for swimming. In contrast, the vase-shaped form, called a **polyp** (PAHL-ip), is specialized for a sessile existence.

Figure 33-3 also shows that all cnidarians have bodies constructed of two cell layers—an outer **epidermis** and an inner **gastrodermis**. Between these layers is a jellylike material known as **mesoglea** (mez-uh-GLÉE-uh). In the center of the body is a hollow gut called the **gastrovascular cavity**, which has a single opening, or mouth. Surrounding the mouth are numerous flexible extensions called **tentacles**.

FIGURE 33-3

The contrasting forms of medusae and polyps result from different arrangements of the same body parts. Medusa forms are free-floating and jellylike. Jellyfish are medusae. Polyp forms, such as hydras and sea anemones, are usually attached to a rock or some other object.



FEEDING AND DEFENSE IN CNIDARIANS

One of the distinguishing features of cnidarians is the presence of cnidocytes (NIE-duh-siets), which give the phylum its name. **Cnidocytes** are specialized cells used for defense and capturing prey. Figure 33-4 shows a type of cnidocyte organelle called a **nematocyst** (nuh-MAT-uh-sist), which has a long filament coiled up inside it. In some cnidarians, the cnidocytes are concentrated in the epidermis, especially on the tentacles. When an object brushes against the “trigger” on a cnidocyte, the nematocyst inside it suddenly pushes the filament out of the cell with great force. Some nematocysts have filaments with sharp tips and spines that puncture the object and inject poison. Others have filaments that adhere to the object by wrapping around it.

The relationship between structure and function is clearly seen in the way cnidarians feed. The tentacles capture small animals with their nematocysts and paralyze them with the poison they inject. The tentacles then push the prey into the gastrovascular cavity through the mouth. After enzymes inside the gastrovascular cavity break up the prey, cells lining the cavity absorb the nutrients. Undigested food and waste are expelled through the mouth.

NERVOUS SYSTEM IN CNIDARIANS

Nervous responses in cnidarians are controlled by a diffuse web of interconnected nerve cells called a **nerve net**. In many cnidarians, like the polyp shown in Figure 33-5, the nerve net is distributed uniformly throughout the entire body. There is no brain or similar structure that controls the rest of the nerve net. But in the medusa form of some cnidarians, such as jellyfish, some of the nerve cells are clustered in rings around the edge of the bell-shaped body.

The nerve net enables cnidarians to respond to specific stimuli in their environment. For example, when cells in the epidermis are touched, they relay a signal to nerve cells. The nerve cells, in turn, transmit a signal via the nerve net to contractile cells, which can cause the animal to withdraw from the stimulus. In cnidarians with the simplest nerve nets, a stimulus anywhere on the body causes signals to be sent through the nerve net in all directions. These signals bring about a contraction of the entire body.

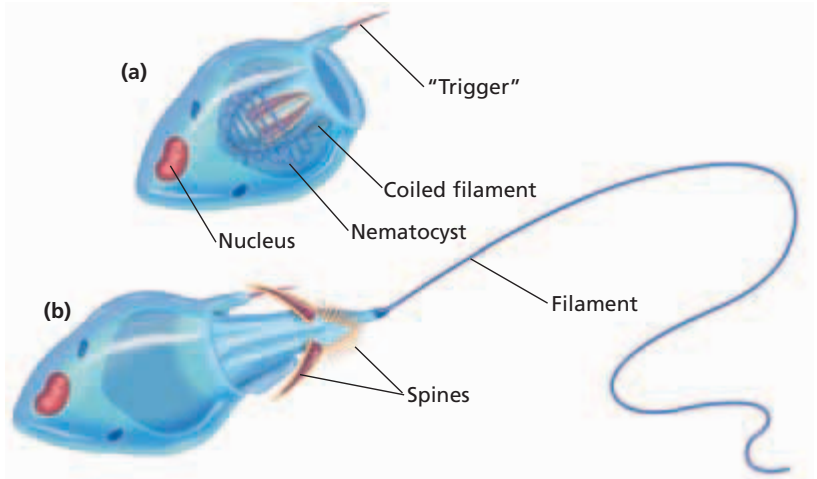
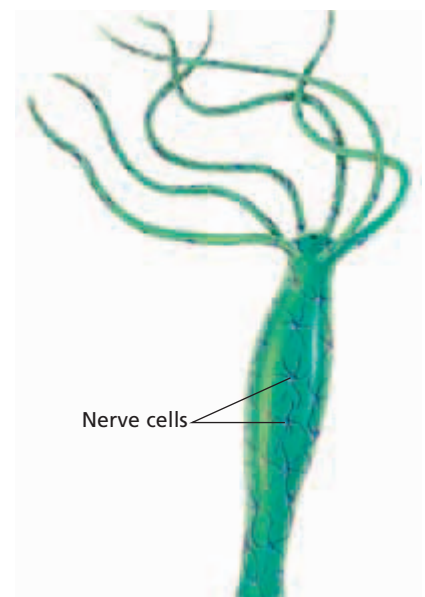


FIGURE 33-4

(a) The nematocyst inside this cnidocyte contains a coiled filament. (b) When something touches the “trigger,” the nematocyst suddenly ejects the filament.

FIGURE 33-5

The interconnected nerve cells in the nerve net of this cnidarian (a hydra), coordinate the animal’s responses to its environment.





Quick Lab

Identifying Poriferans, Ctenophorans, and Cnidarians

Materials pencil, paper, a picture of either a poriferan, a ctenophoran, or a cnidarian

Procedure

1. Prepare a dichotomous key to differentiate among poriferans, ctenophorans, and cnidarians by writing paired statements for the three phyla and for classes Hydrozoa, Scyphozoa, and Anthozoa.
2. Statement (1a) describes cnidarians and leads to paired statements (3). Statement (1b) leads to paired statements (2).
3. Statement (2a) describes Porifera and (2b) describes Ctenophora.
4. Statement (3a) describes Hydrozoa and (3b) leads to paired statements (4).
5. Statement (4a) describes Scyphozoa and (4b) describes Anthozoa.
6. Get a picture from your teacher. Identify the animal using your dichotomous key. Exchange keys with another group, and use the new key to identify your animal.

Analysis What do sponges and hydrozoans have in common? How do comb jellies differ from cnidarians? What is the dominant body form in each cnidarian class?

The nerve net also coordinates the complex activities of the body that are necessary for feeding and traveling through the environment. The movements by which the tentacles bring prey to the mouth and push it into the gastrovascular cavity are controlled by the nerve net, as are the rhythmic contractions of the body that propel swimming medusae through the water.

CLASSIFICATION OF CNIDARIANS

Scientists recognize four classes of cnidarians—Hydrozoa, Cubozoa, Scyphozoa, and Anthozoa. The members of these classes are known as hydrozoans, cubozoans, scyphozoans, and anthozoans, respectively. Some species of hydrozoans live only as polyps, some live only as medusae, some alternate between these two forms, and some live as mixed colonies of polyps and medusae. Cubozoans and scyphozoans spend most of their lives as medusae, while anthozoans live only as polyps.

Class Hydrozoa

The class Hydrozoa includes about 3,700 species, most of which live as colonial organisms in the oceans. Examples of colonial hydrozoans are species of the genus *Obelia*. As Figure 33-6a illustrates, one such species has many polyps attached to branched stalks. Some of the polyps function in gathering food, while others are responsible for reproduction.

The hydrozoan shown in Figure 33-6b, the Portuguese man-of-war (genus *Physalia*), exists as a colony of medusae and polyps. Its gas-filled float, which can measure as much as 30 cm (1 ft) across, keeps the colony at the surface of the ocean. The polyps in the colony are specialized for feeding, digestion, or sexual reproduction. Tentacles up to 20 m (65 ft) long dangle from the feeding polyps and carry large numbers of cnidocytes. The Portuguese man-of-war preys mostly on small fish, but its cnidocytes contain a neurotoxin (nerve poison) that can be painful and even fatal to humans.

FIGURE 33-6

These organisms are two examples of colonial hydrozoans. (a) This colony of the genus *Obelia* is made up of polyps, although organisms of the genus *Obelia* can produce medusae that reproduce sexually. (b) The tentacles of this Portuguese man-of-war, genus *Physalia*, have both polyps and medusae that have specialized functions.



(a)



(b)

One hydrozoan that has been extensively studied is the hydra. Hydras are not typical hydrozoans because they exist only as polyps, they are not colonial, and they live in fresh water. Hydras range from 1 to 4 cm (0.4 to 1.6 in.) in length. Most hydras are white or brown, but some, like the one shown in Figure 33-7, appear green because of the algae that live symbiotically inside cells of their gastrodermis. Hydras can be found in quiet ponds, lakes, and streams. They attach themselves to rocks or water plants by means of a sticky secretion produced by cells at the hydra's base.

A hydra can leave one place of attachment and move to another. This can happen when the base secretes bubbles of gas, which cause the hydra to float upside down on the surface of the water. Hydras can also move by tumbling. This movement occurs when the tentacles and the mouth end bend over and touch the surface to which the hydra was attached while the base pulls free.

During warm weather, hydras generally reproduce asexually. Small buds, such as the one you can see in Figure 33-8, develop on the outside of the hydra's body. These buds grow their own tentacles and then separate from the body and begin living independently.

Sexual reproduction usually occurs in the fall, when low temperatures trigger the development of eggs and sperm. The eggs are produced by meiosis along the body wall in swellings called *ovaries*. Motile sperm are formed by meiosis in similar swellings called *testes*. In some species, eggs and sperm are produced in the same hermaphroditic individual, as indicated in Figure 33-8. In other species, the individuals are either male or female. In either case, sperm are released into the water, and those that reach ovaries can fertilize egg cells. Each fertilized egg then divides and grows into an embryo. A hard covering protects the embryo through the winter, and in the spring the embryo hatches and develops into a new hydra.



FIGURE 33-7

The green color of this hydra, *Chlorohydra viridissima*, comes from the algae that live inside the hydra's cells. (LM 30×)

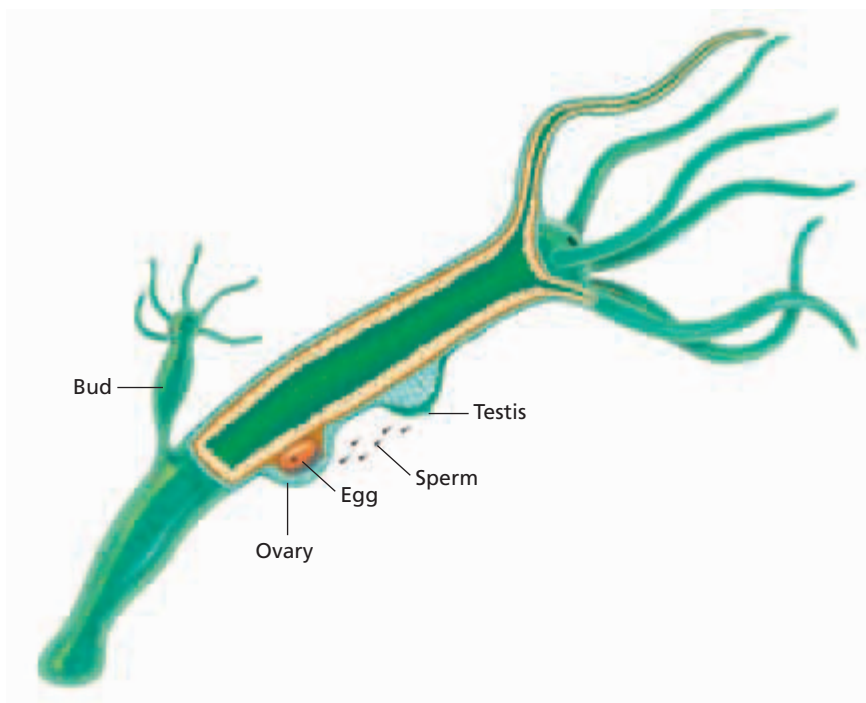


FIGURE 33-8

Hydras can reproduce either asexually, by forming buds, or sexually, by producing sperm that fertilize eggs.

Class Cubozoa

The cubozoans, or box jellies, were once classified in the class Schyphozoa. As their name implies, cubozoans have cube-shaped medusae. Their polyp stage is inconspicuous and has never been observed in some species. Most box jellies are only a few centimeters in height, although some reach 25 cm (10 in.) tall. A tentacle or group of tentacles is found at each corner of the “box.” The cnidocytes of some species, such as the sea wasp, can inflict severe pain and even death among humans. The sea wasp lives in the ocean along the tropical northern coast of Australia.

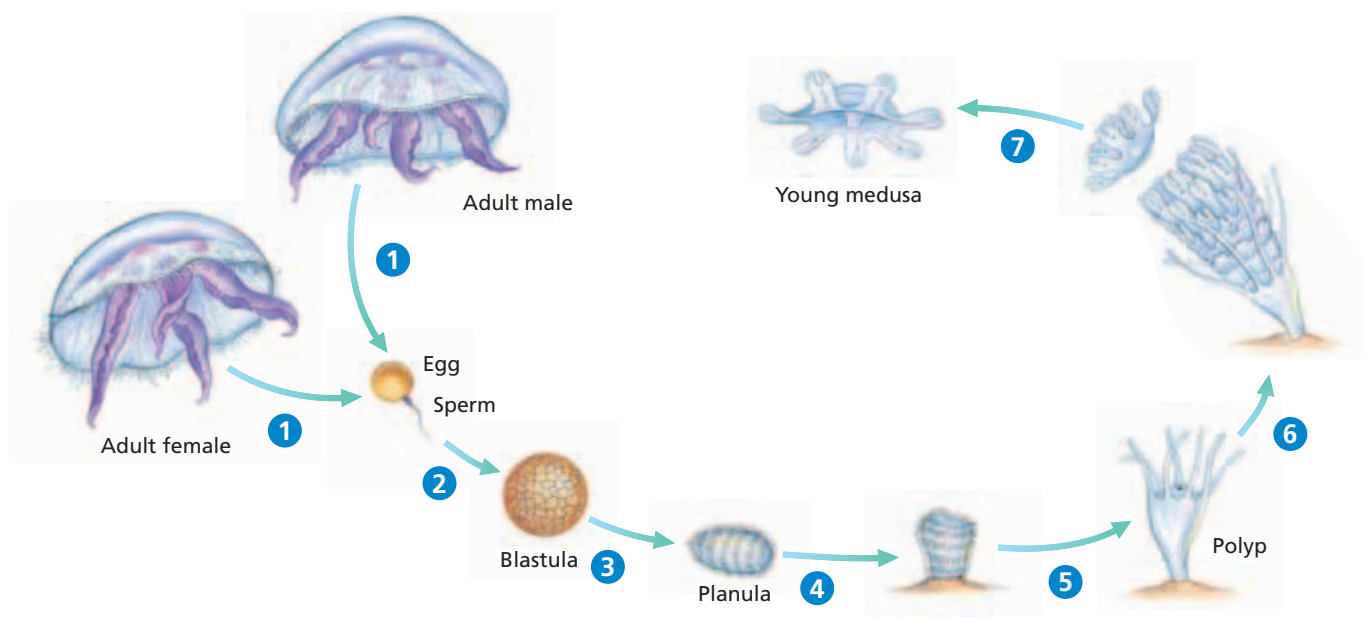
Class Scyphozoa

The name *Scyphozoa* (sie-foh-ZOH-uh) means “cup animals,” which describes the medusa, the dominant form of the life cycle of this class. There are more than 200 species of scyphozoans, known commonly as jellyfish. The cups of the medusae range from 2 cm (0.8 in.) to 4 m (13 ft) across, and some species have tentacles that are several meters long. Pulsating motions of the cup propel the jellyfish through the water. Like the Portuguese man-of-war, some jellyfish carry poisonous nematocysts that can cause severe pain and even death in humans.

The common jellyfish, genus *Aurelia*, is a scyphozoan whose life cycle includes both medusa and polyp forms. As you can see in Figure 33-9, step ①, adult medusae release sperm and eggs into the water, where fertilization occurs. The resulting zygote divides many times to form a blastula, as shown in step ②. In step ③, the blastula then develops into a ciliated larva called a **planula** (PLAN-yuh-luh). In step ④, the planula attaches to the ocean bottom. The planula becomes a polyp by developing a mouth and tentacles at the unattached end as shown in step ⑤. As the polyp grows, shown in step ⑥, it forms a stack of medusae. Finally, as shown in step ⑦, the medusae detach and develop into free-swimming jellyfish.

FIGURE 33-9

The common jellyfish, genus *Aurelia*, reproduces when sperm from an adult male medusa fertilize eggs from an adult female medusa. Each fertilized egg produces a blastula, which develops into larva known as a planula. The planula forms a polyp, which produces more medusae.



Class Anthozoa

The name *Anthozoa* means “flower animals,” which is a fitting description for the approximately 6,100 marine species in this class. Two examples of anthozoans are sea anemones and corals, which are shown in Figure 33-10.

Sea anemones are polyps commonly found in coastal areas, where they attach themselves to rocks and other submerged objects. Anemones feed on fishes and other animals that swim within reach of their tentacles. However, some anemones in the Pacific Ocean have a symbiotic relationship with the clownfish, as Figure 33-11 demonstrates. The two animals share food and protect each other from predators. The movements of the clownfish also help prevent sediments from burying the anemone. The clownfish produces a slimy mucus that prevents the anemone from firing its nematocysts when the clownfish touches the anemone’s tentacles.

Corals are small polyps that usually live in colonies. Each polyp cements its calcium carbonate skeleton to the skeletons of adjoining polyps in the colony. When the polyps die, their hardened skeletons remain, serving as the foundation for new polyps. Over thousands of years, these polyps build up large, rocklike formations known as **coral reefs**. Only the top layer of the reef contains the living polyps. Coral reefs provide food and shelter for an enormous and colorful variety of fishes and invertebrates.

Nearly all coral reefs are restricted to a band of ocean within 30 degrees north or south of the equator. Most form at shallow depths in warm, clear waters. These conditions are necessary in order for photosynthesis to be carried out by the algae that live symbiotically inside coral cells. These corals depend on the algae to provide oxygen and to speed up the accumulation of calcium from the sea water. The algae in turn depend on the corals to supply vital nutrients.



(a)



(b)

FIGURE 33-10

Anthozoans, including this crimson anemone, *Cribinopsis fernaldi* (a), and this golden cup coral, genus *Tubastraea* (b), live as polyps along ocean coasts.

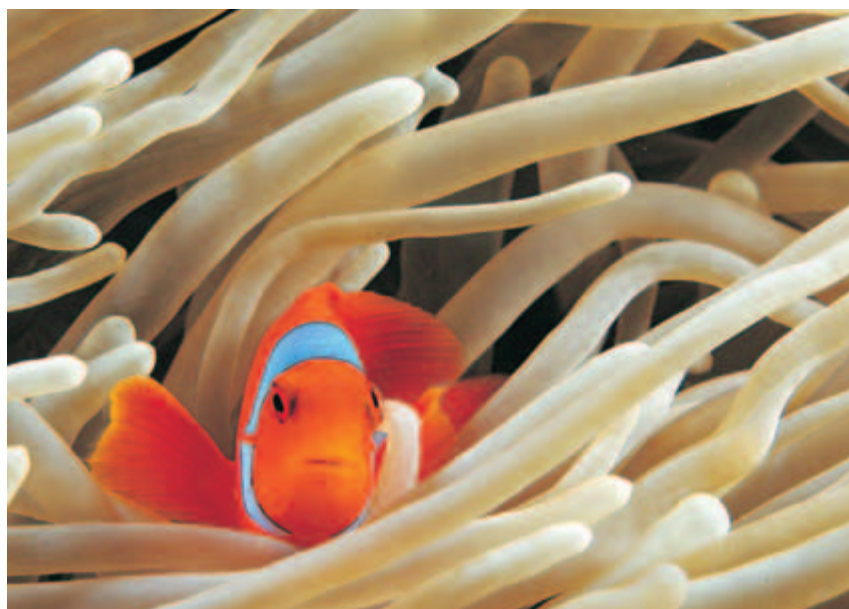


FIGURE 33-11

The clownfish, *Amphiprion ocellaris*, lives symbiotically among the tentacles of sea anemones. The anemone’s stinging tentacles protect the clownfish from predators. The clownfish, in turn, drives away other fish that try to feed on the anemone.

Word Roots and Origins

colloblast

from the Greek *kolla*, meaning "glue," and *blastos*, meaning "bud"

PHYLUM CTENOPHORA

The phylum Ctenophora includes about 100 species of marine animals known as *ctenophores*. A typical ctenophore is shown in Figure 33-12. *Ctenophora* means "comb holder" and refers to the eight comblike rows of cilia that run along the outside of the animal. Ctenophores resemble jellyfish and are often called *comb jellies*.

Ctenophores differ from jellyfish and other cnidarians in several ways. Rather than pulsating like jellyfish, they move through the water by beating their cilia. They are the largest organisms to move in this fashion. Also, ctenophores do not have cnidocytes. Instead, many have cells called **colloblasts**, which secrete a sticky substance that binds to their prey. Colloblasts are usually located on two tentacles. Ctenophores also have a sensory structure called an **apical organ** at one end of their body. This organ enables ctenophores to sense their orientation in the water. Nerves running from the apical organ coordinate the beating of the cilia. Most ctenophores are hermaphroditic.

One of the most striking features of ctenophores is their **bioluminescence** (BIE-oh-loo-muh-NES-ens), or production of light by means of a chemical reaction. Bioluminescent ctenophores often occur in large swarms near the surface of the ocean, which creates a spectacular display at night.

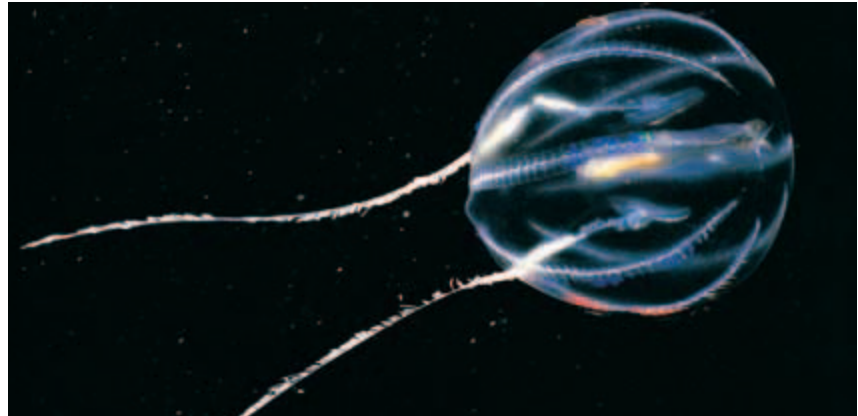


FIGURE 33-12

A ctenophore, *Pleurobrachia pileus*, moves through the water by beating its cilia, trailing its two long tentacles behind it.

SECTION 2 REVIEW

1. What are three characteristics that all cnidarians have in common?
2. Describe how cnidarians feed.
3. What is the name given to the nervous system of a cnidarian?
4. Describe one organism from each of the four cnidarian classes.
5. Describe three characteristics of ctenophores.

CRITICAL THINKING

6. **Analyzing Concepts** What might be the advantage of having a life cycle consisting of both medusae and polyps instead of only polyps?
7. **Inferring Relationships** Explain how having nerve cells clustered around the edge of its bell is advantageous to a medusa.
8. **Making Comparisons** Compare the movement of cnidarians with the movement of ctenophores.

CHAPTER HIGHLIGHTS

SECTION 1

Porifera

- The phylum Porifera is made up of sponges, sessile invertebrates that have no true tissues or organs. The simplest types of sponges are shaped like hollow cylinders.
- The body wall of a sponge is composed of two layers of cells that are separated by a jellylike substance called *mesohyl*. The body is supported by a skeleton made of spongin, spicules, or both.
- Choanocytes which line the inside of a sponge beat their flagella, thus drawing a current of water into the sponge through pores (ostia) in the body wall. Water leaves through the osculum, an opening at the top of the sponge.
- Sponges feed by filtering small organisms and organic matter out of the water that passes through their body. Nutrients are distributed through the body by amoebocytes, cells which crawl about within the body wall.
- Sponges can reproduce asexually, through budding or regeneration, and sexually, through the joining of egg and sperm. Most sponges are hermaphroditic, meaning that a single animal can produce both eggs and sperm.

Vocabulary

sponge (p. 673)
sessile (p. 673)
choanocyte (p. 673)

ostium (p. 673)
osculum (p. 673)
spongin (p. 673)

spicule (p. 673)
filter feeding (p. 674)
amoebocyte (p. 674)

gemmule (p. 675)
regeneration (p. 675)
hermaphrodite (p. 675)

SECTION 2

Cnidaria and Ctenophora

- Animals in the phylum Cnidaria can be either sessile polyps or swimming medusae. Some cnidarians alternate between polyp and medusa stages during their life cycles.
- The body of a cnidarian consists of two cell layers—an outer epidermis and an inner gastrodermis—separated by a jellylike mesoglea.
- Cnidarians have cells called *cnidocytes*, which contain organelles known as *nematocysts*. When a cnidocyte is stimulated, its nematocyst ejects a filament that can paralyze or ensnare prey.
- The cnidarian nervous system is a diffuse web of interconnected nerve cells called a *nerve net*.
- The four classes of cnidarians are Hydrozoa (which includes animals such as the hydra), Cubozoa (box jellies), Scyphozoa (jellyfish), and Anthozoa (which includes sea anemones and corals).
- Hydrozoans may live as polyps, medusae, or mixed colonies of polyps and medusae. Cubozoans and scyphozoans spend most of their lives as medusae. Anthozoans live only as polyps.
- Animals in the phylum Ctenophora move through the water by beating the cilia that occur in eight rows on the outside of a ctenophore's body.
- Ctenophores capture prey with a sticky substance secreted by cells called *colloblasts*, which are usually located on a pair of tentacles.
- An apical organ at one end of the body enables ctenophores to sense their orientation in the water. Most ctenophores are hermaphroditic, and many are bioluminescent.

Vocabulary

medusa (p. 676)
polyp (p. 676)
epidermis (p. 676)
gastrodermis (p. 676)

mesoglea (p. 676)
gastrovascular cavity (p. 676)
tentacle (p. 676)
cnidocyte (p. 677)

nematocyst (p. 677)
nerve net (p. 677)
planula (p. 680)
coral reef (p. 681)

colloblast (p. 682)
apical organ (p. 682)
bioluminescence (p. 682)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
 - spongin* and *spicule*
 - medusa* and *polyp*
 - epidermis* and *gastrodermis*
 - cnidocyte* and *nematocyst*
 - gemmule* and *planula*
- Explain the relationship between choanocytes and filter feeding.
- Use each of the following terms in a separate sentence: *colloblast*, *apical organ*, and *bioluminescence*.
- Word Roots and Origins** The name *Porifera* is derived from the Latin *porus*, meaning “channel,” and *ferre*, meaning “to bear.” Explain why *Porifera* is a good name for the sponge phylum.

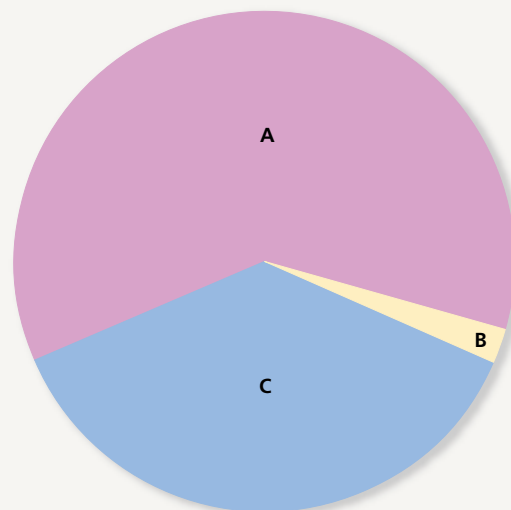
UNDERSTANDING KEY CONCEPTS

- Identify** three ways that sponges represent the transition from unicellular to multicellular life.
- Describe** why sponges are classified as animals.
- Outline** the path of water through a sponge, and explain what causes water to flow through a sponge.
- Explain** how a sponge’s usual method of feeding is suited to its sessile lifestyle.
- State** the role of gemmules in the life cycles of sponges.
- Summarize** the process of sexual reproduction in sponges.
- Compare** the two types of body structures that cnidarians have.
- Describe** how a cnidarian captures and ingests its prey.
- Identify** the specialization that is found among the individuals that make up a Portuguese man-of-war.
- Name** the benefits a sea anemone obtains from its symbiotic relationship with a clownfish. What benefit does the clownfish obtain from this relationship?
- Explain** why coral reefs are limited to warm, shallow environments.
- Compare** ctenophores and cnidarians. Include at least two differences.

-  **CONCEPT MAPPING** Use the following terms to create a concept map that sequences sexual reproduction in a typical scyphozoan: *adult male*, *medusa*, *blastula*, *adult female*, *sperm*, *polyp*, *planula*, and *egg*.

CRITICAL THINKING

- Applying Information** A single species of sponges may assume various appearances, depending on substrate, availability of space, and the velocity and temperature of water currents. How might these factors make the classification of sponges confusing? What features besides outward appearance can biologists use to classify sponges and eliminate some of this confusion?
- Analyzing Information** Sponge larvae have flagella on the outside of their bodies, while adult sponges have flagella lining their internal cavity. How is this structural difference related to functional differences between the larval and adult stages of sponges?
- Analyzing Patterns** Hydras generally reproduce asexually during warm weather and sexually in cooler weather. Based on what you have learned about the hydra embryo, what is the advantage of reproducing sexually when the weather turns cool?
- Predicting Results** What would happen to a coral reef if pollution or sediment caused the water around the reef to become less clear? Explain your answer.
- Interpreting Graphics** The pie chart below shows the relative numbers of hydrozoans, scyphozoans, and anthozoans. Which segment of the chart represents scyphozoans? How do you know?



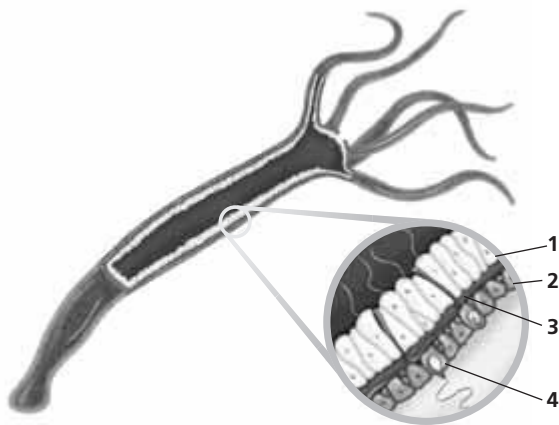


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- Why are spongin and spicules important to a sponge?
 - They digest food.
 - They remove wastes.
 - They provide support.
 - They produce offspring.
- Which of the following structures are involved in both feeding and sexual reproduction in sponges?
 - spicules and gemmules
 - amoebocytes and spongin
 - gemmules and choanocytes
 - choanocytes and amoebocytes
- Which of the following is *not* a characteristic of cnidarians?
 - tentacles
 - choanocytes
 - nematocysts
 - gastrovascular cavity
- What do colloblasts do?
 - They produce light.
 - They secrete a sticky substance.
 - They draw water through sponges.
 - They form medusae that live in colonies.

INTERPRETING GRAPHICS: The diagram below illustrates a hydra. Study the diagram to answer the questions that follow.

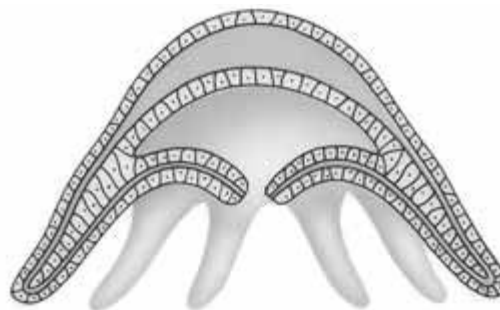


- Identify the substance found at point 3.
 - osculum
 - mesoglea
 - gastrodermis
 - gastrovascular cavity
- Which structure is involved in defense?
 - 1
 - 2
 - 3
 - 4

DIRECTIONS: Complete the following analogy.

7. Sponge : osculum :: hydra :
- mouth
 - tentacle
 - nerve net
 - nematocyst

INTERPRETING GRAPHICS: The diagram below illustrates a medusa. Study the diagram to answer the question that follows.



8. In which class is this body form dominant?
 - Cnidaria
 - Hydrozoa
 - Anthozoa
 - Scyphozoa

SHORT RESPONSE

Cnidarians have two tissue layers, which is an important evolutionary advancement.

Describe the tissue layers and general body structure of a typical cnidarian.

EXTENDED RESPONSE

Sponges reproduce asexually in a number of ways.

Part A Describe three forms of asexual reproduction in sponges.

Part B Explain how reproducing asexually is advantageous to sponges.

Test TIP

You can sometimes figure out an answer to a question before you look at the answer choices. After you answer the question in your mind, compare your answer with the answer choices. Choose the answer that most closely matches your own answer.

Observing Hydra Behavior

OBJECTIVES

- Observe live specimens of hydra.
- Determine how hydras respond to different stimuli.
- Determine how hydras capture and feed on prey.

PROCESS SKILLS

- observing
- relating structure to function

MATERIALS

- silicone culture gum
- microscope slide
- hydra culture
- 2 medicine droppers
- compound microscope
- methylene blue solution
- vinegar
- stereomicroscope
- filter paper cut into pennant shapes
- forceps
- concentrated beef broth
- culture of *Daphnia pulex* or *Daphnia magna*

Background

1. How do animals respond to stimuli in their environment?
2. What is a hydra?



3. What characteristics do hydras share with other cnidarians?
4. How does a sessile animal, such as a hydra, obtain food?
5. What is a nematocyst?

PART A Close-up Examination of a Hydra

1.  **CAUTION** Slides break easily. Use care when handling them. Using a long piece of silicone culture gum, make a circular "well" on a microscope slide, as shown in the illustration below.



2.  **CAUTION** You will be working with a live animal. Be sure to treat it gently and to follow directions carefully. With a medicine dropper, gently transfer a hydra from the culture dish to the well on the slide, making sure the hydra is in water. The hydra should be transferred quickly; otherwise, it may attach itself to the medicine dropper. Allow the hydra to settle. As you go through the following steps, add water to the slide periodically to replace water that has evaporated, and keep the hydra wet.
3. Examine the hydra under the low-power setting of a compound microscope. Add a drop of methylene blue solution to the well containing the hydra to make the tentacles more visible. Identify and draw the hydra's body stalk, mouth, and tentacles in your lab report.
4. In your lab report, make a data table like the one shown on the next page. As you complete the following steps, record your observations in your data table.
5. As you continue to observe the hydra at low power, add a drop of vinegar to the well. Record what happens to the bumps on the tentacles. These bumps are cnidocytes.

- Transfer the hydra to the culture dish labeled "Used hydras." Rinse the well on your microscope slide with water to remove all traces of methylene blue and vinegar.

PART B Feeding Behavior

- Hydras eat small invertebrates, such as daphnia. With a medicine dropper, gently transfer another hydra to the well on your slide. Then transfer live daphnia to the well in the same manner.
- Observe the hydra carefully with the high-power setting of the stereomicroscope. Watch for threadlike nematocysts shooting out from the hydra. Some nematocysts release a poison that paralyzes prey. If the hydra does not respond after a few minutes, obtain another hydra from the culture dish and repeat this procedure.
- Observe the way the hydra captures and ingests daphnia, and record your observations in your data table. Record how long it takes for a hydra to ingest a daphnia.
- Transfer the hydra to the culture dish labeled "Used hydras." Rinse the well on your microscope slide with water to remove the daphnia.

PART C Response to Stimuli

- Transfer another hydra to the well on your slide, and examine it with the high-power setting of the stereomicroscope. Using forceps, move the long tip of a pennant-shaped piece of filter paper near the hydra's tentacles. Be careful not to touch the hydra with the filter paper. Observe the hydra's response to the filter paper, and record your observations in your data table.
- Now observe how the hydra responds to a chemical stimulus. Dip the same piece of filter paper in beef broth, and repeat the procedure in step 11. Again, be

careful not to touch the hydra. Record the hydra's response to the beef broth in your data table.

- Finally, investigate how the hydra responds to touch. Using the long tip of a clean pennant-shaped piece of filter paper, touch the hydra's tentacles, mouth, and body stalk. **CAUTION Touch the hydra gently.** Record your observations in your data table.
-   Transfer the hydra to the culture dish labeled "Used hydras." Clean up your materials and wash your hands before leaving the lab.

Analysis and Conclusions

- Is vinegar a normal part of a hydra's diet? Why do you think vinegar is used in this step?
- Based on your observations, how do you think a hydra behaves when it is threatened in its natural habitat?
- Describe a hydra's feeding behavior.
- What happens to food that has not been digested by a hydra?
- What was the purpose of using the clean filter paper in step 11?
- Did the hydra show a feeding response or a defensive response to the beef broth? Explain.
- How is a hydra adapted to a sessile lifestyle?
- How is the feeding method of a hydra different from that of a sponge?
- Do you think the hydra's response is triggered by water vibrations, by chemicals that daphnia releases, or by daphnia touching the hydra?

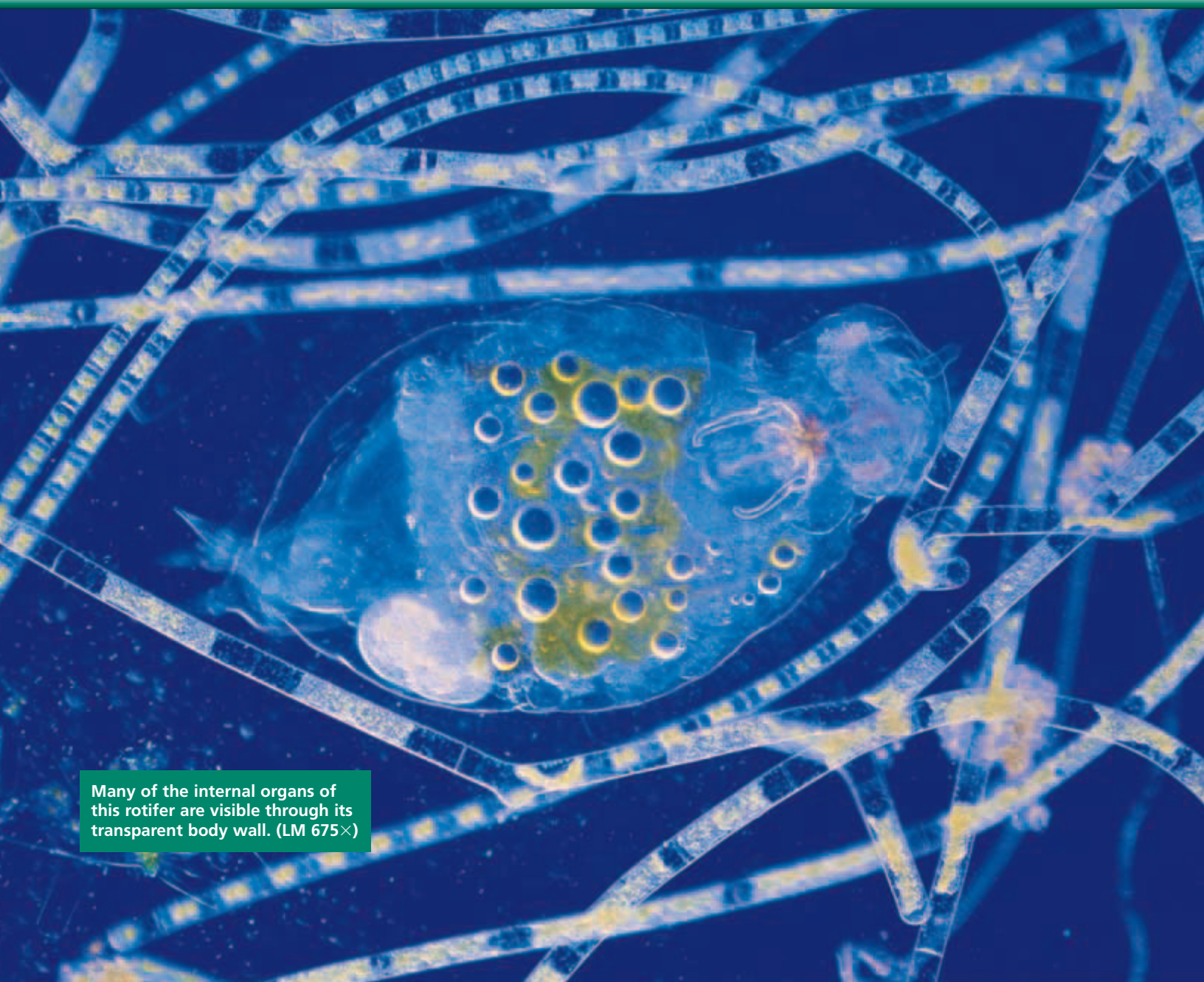
Further Inquiry

Design an experiment to determine how hydras respond to other stimuli, such as light.

OBSERVATIONS OF HYDRA BEHAVIOR

Behavior	Observations
Response to vinegar	
Feeding behavior	
Response to filter paper	
Response to beef broth on filter paper	
Response to touch with filter paper	

FLATWORMS, ROUNDWORMS, AND ROTIFERS



Many of the internal organs of this rotifer are visible through its transparent body wall. (LM 675 $\times$)

SECTION 1 *Platyhelminthes*

SECTION 2 *Nematoda and Rotifera*

PLATYHELMINTHES

The phylum Platyhelminthes (PLAT-ee-hel-MINTH-eez) includes organisms called flatworms. Their bodies develop from three germ layers and are more complex than those of sponges, cnidarians, and ctenophores. Flatworms have bilaterally symmetrical bodies, with dorsal and ventral surfaces, right and left sides, and anterior and posterior ends.

STRUCTURE AND FUNCTION OF FLATWORMS

Flatworms are the simplest animals with bilateral symmetry. The tissues in bilaterally symmetrical animals develop from three germ layers: ectoderm, mesoderm, and endoderm. In flatworms, the three germ layers are pressed against one another to form a solid body. Because flatworms do not have a hollow body cavity between the endoderm and the mesoderm, they are acoelomates.

The acoelomate body plan gives flatworms the thin, dorsoventrally flattened bodies for which they are named. This body shape ensures that no cell in a flatworm is very far from the animal's external environment. Thus, the cells can exchange oxygen and carbon dioxide directly with the environment through diffusion, allowing flatworms to survive without a circulatory system or respiratory system. Like cnidarians, most flatworms have a gastrovascular cavity, which is a gut with a single opening, and a mouth. Food is taken in and digested in the gastrovascular cavity, and any undigested material is eliminated through the same opening. Most of the sensory organs and nerve cells of flatworms, such as the marine species shown in Figure 34-1, are located at the anterior end of the body. This characteristic is known as *cephalization*.

The classification of Platyhelminthes has undergone many recent changes. Currently, the more than 20,000 species of flatworms are divided into four classes: Turbellaria, Trematoda, Monogenea, and Cestoda. Trematodes, monogeneans, and cestodes (SES-tohdz) live as parasites on or inside other animals. Almost all turbellarians are nonparasitic, free-living organisms found in marine and freshwater habitats and in moist terrestrial environments. Parasitic flatworms probably evolved from free-living organisms. As parasites evolved, some organs that were advantageous to free-living became modified for parasitism, while other organs were lost entirely.

OBJECTIVES

- **Summarize** the distinguishing characteristics of flatworms.
- **Describe** the anatomy of a planarian.
- **Compare** free-living and parasitic flatworms.
- **Diagram** the life cycle of a fluke.
- **Describe** the life cycle of a tapeworm.

VOCABULARY

pharynx
flame cell
cerebral ganglion
eyespot
fluke
tegument
primary host
intermediate host
schistosomiasis
scolex
proglottid
cyst

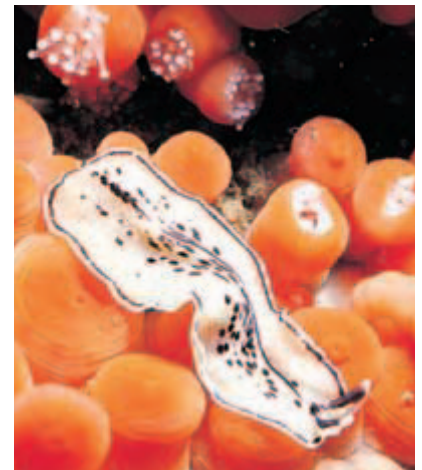


FIGURE 34-1

Many of the sensory organs in this marine flatworm, of the genus *Eurylepta*, are concentrated in the two tentacles at the anterior end of its body. This characteristic is an example of cephalization.

Word Roots and Origins

pharynx

from the Greek *pharynx*,
meaning "throat"

CLASS TURBELLARIA

The majority of the approximately 4,500 species in the class Turbellaria live in the ocean. However, the most familiar turbellarian is the freshwater planarian *Dugesia*, shown in Figure 34-2. Planarians have a spade-shaped anterior end and a tapered posterior end. They move through the water by swimming with a wave-like motion of their body. Over solid surfaces, planarians glide on a layer of mucus that they secrete, propelled by the cilia that cover their bodies.

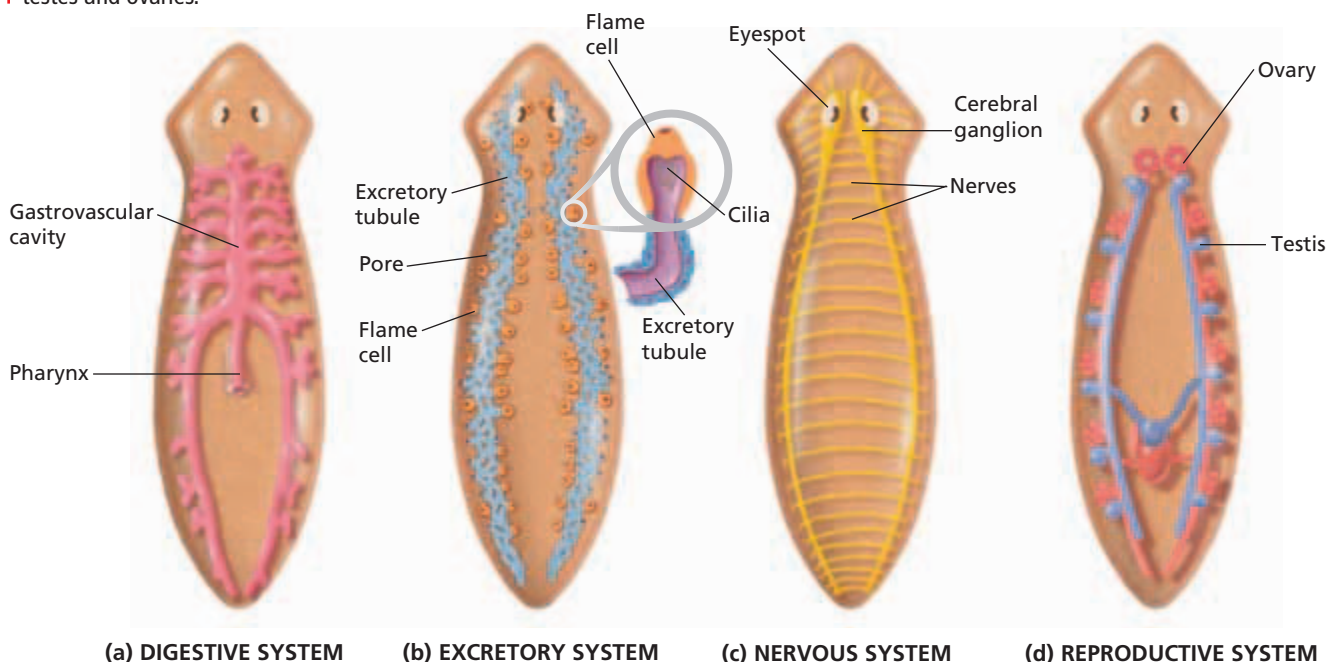
Digestion and Excretion in Planarians

Planarians feed by scavenging for bits of decaying plant or animal matter. They also prey on smaller organisms, such as protozoa. Food is ingested through a muscular tube called the **pharynx** (FAR-ee-nks), which the planarian extends from the middle of its body. As Figure 34-2a indicates, the pharynx leads to the highly branched gastrovascular cavity. Cells lining the cavity secrete digestive enzymes and absorb nutrients and small pieces of food. The nutrients then diffuse to other body cells.

Organisms that live in fresh water must deal with the water that constantly enters their bodies by osmosis. Planarians eliminate excess water through a network of excretory tubules that run the length of the body. Figure 34-2b shows that each tubule is connected to several **flame cells**, which are so named because they enclose tufts of beating cilia that resemble flickering candle flames. The beating of cilia in the flame cells draws in the excess water. The water is then transported through the tubules and excreted from numerous pores scattered over the body surface.

FIGURE 34-2

The organ systems of a planarian allow it to maintain its free-living existence. (a) The digestive system consists of the pharynx and gastrovascular cavity, which has many branches. (b) In the excretory system, flame cells collect excess water, which travels through excretory tubules to pores on the surface of the body. (c) The nervous system is a ladderlike arrangement of nerves with two cerebral ganglia at the anterior end. (d) Since planarians are hermaphrodites, their reproductive system includes both testes and ovaries.



Neural Control in Planarians

The planarian nervous system is more complex than the nerve net of cnidarians, as Figure 34-2c illustrates. Two clusters of nerve cells at the anterior end, the **cerebral ganglia** (suh-REE-bruhl GAN-gee-uh), serve as a simple brain. They receive information from sensory cells and transmit signals to the muscles along a ladderlike arrangement of nerves. A planarian's nervous system gives it the ability to learn. For example, a planarian normally moves away from light, but it can be trained to remain still when illuminated.

Planarians sense the intensity and direction of light with two cup-shaped **eyespots** located near the cerebral ganglia. You can see the eyespots in Figure 34-2c. Other sensory cells respond to touch, water currents, and chemicals in the environment. These cells are distributed over the body, but most are concentrated at the anterior end.

Reproduction in Planarians

Because planarians are free-living and motile, they can encounter and mate with other individuals of the same species. As shown in Figure 34-2d, planarians are hermaphrodites—they have both male sex organs (testes) and female sex organs (ovaries). When two planarians reproduce sexually, they simultaneously fertilize each other. Their eggs are laid in protective capsules that stick to rocks or debris and hatch in two to three weeks.

Planarians also reproduce asexually, generally during the summer. During asexual reproduction, the body constricts just behind the pharynx. While the posterior part of the worm is attached to a solid surface, the anterior part moves forward until the worm splits in two. This type of asexual reproduction is known as *fission*. The two halves then regenerate their missing parts to produce two complete planaria. During regeneration, each part of the planarian retains information about its original orientation in the body. If a piece is cut from the middle of a planarian, the anterior end of the piece will always regenerate a head and the posterior end of the piece will regenerate a tail.

CLASSES TREMATODA AND MONOGENEA

The classes Trematoda and Monogenea consist of parasitic **flukes**, leaf-shaped flatworms that parasitize many kinds of animals, including humans. Some flukes are endoparasites that live in the blood, intestines, lungs, liver, or other organs. Others are ectoparasites that live on the external surface of aquatic hosts, such as fish and frogs. Trematodes tend to parasitize a wide range of hosts, whereas monogeneans are mostly ectoparasites of fish and other aquatic animals.





FIGURE 34-3

Suckers on this blood fluke, *Schistosoma mansoni*, attach the fluke to the blood vessels of its host. (SEM 550×)

FIGURE 34-4

In the life cycle of schistosomes, fertilized eggs are released into the host's blood vessels. The eggs pass out of the primary host in feces or urine. In water, the eggs develop into ciliated larvae. The larvae burrow into certain species of snails, which serve as intermediate hosts. The larvae develop tails, escape from the snail, and swim about. The tailed larvae bore through the exposed skin of a person and settle in his or her blood vessels. There, the larvae develop into adults, and the cycle repeats.

Structure of Flukes

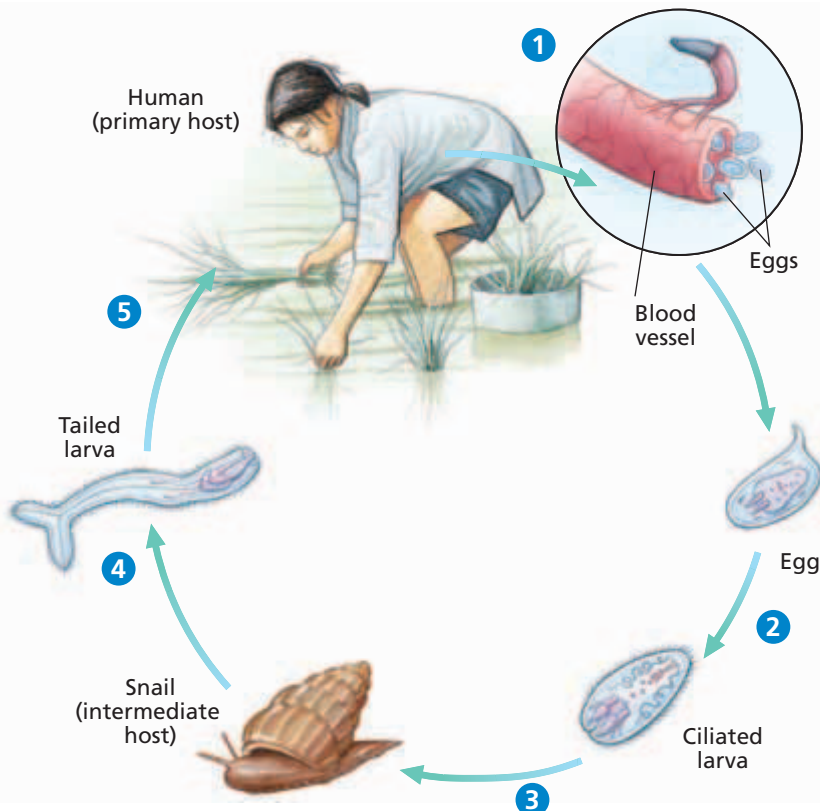
A fluke clings to the tissues of its host by an anterior sucker and a ventral sucker, which are shown in Figure 34-3. The anterior sucker surrounds the fluke's mouth, which draws the host's body fluids into the gastrovascular cavity. A fluke's nervous system is similar to a planarian's, but flukes have no eyespots and their other sensory structures are very simple. The external surface of a fluke is covered by a layer called the **tegument**. The outer zone of the tegument consists of a layer of proteins and carbohydrates that makes the fluke resistant to the defenses of the host's immune system. The tegument also protects the fluke against the enzymes secreted by the host's digestive tract.

Reproduction and Life Cycle of Flukes

Most flukes have highly developed reproductive systems and are hermaphroditic. Fertilized eggs are stored in a fluke's uterus, which is a long, coiled tube, until they are ready to be released. Each fluke may release tens of thousands of eggs at a time.

Flukes have complicated life cycles that involve more than one host species. A good example is provided by the trematode blood flukes of the genus *Schistosoma*, as shown in Figure 34-4. Adult schistosomes live inside human blood vessels. Therefore, a human is the schistosome's **primary host**, the host from which the adult parasite gets its nourishment and in which sexual reproduction occurs. Unlike most flukes, schistosomes have separate sexes. Eggs produced by the female are fertilized by the male. In step ①, some of the fertilized eggs make their way to the host's intestine or bladder and are excreted with the feces or urine. Human feces and

urine often pollute freshwater supplies in regions with poor sewage control. In step ②, the eggs that enter fresh water develop into ciliated larvae that swim. In step ③, if the larvae encounter a snail of a particular species, such as one of genus *Oncomelania*, within a few hours, they burrow into the snail's tissues and begin to reproduce asexually. The snail serves as the schistosome's **intermediate host**, the host from which the larvae derive their nourishment. ④ Eventually, the larvae develop tails and escape from the snail. ⑤ These tailed larvae swim through the water. If they find the bare skin of a human, they penetrate the skin, enter a blood vessel, and develop into adults. The cycle then begins again. Exposure to the larvae can happen when humans swim, bathe, wash clothes, or work in fresh water that contains the larvae.



Not all schistosome eggs leave the human body, however. Many are carried by the blood to the lungs, intestines, bladder, and liver, where they may block blood vessels and cause irritation, bleeding, and tissue decay. The eggs may also penetrate the walls of veins and the small intestine or urinary bladder, where they can cause much tissue damage and bleeding. The resulting disease, called **schistosomiasis** (SHIS-tuh-soh-MIE-uh-suhs), can be fatal. It affects about 200 million people worldwide, mostly in Asia, Africa, and South America.

Other kinds of flukes cause less-serious diseases in humans. For example, a small brown fluke common in freshwater lakes of North America is responsible for swimmer's itch, a condition characterized by minor skin irritation and swelling.

Word Roots and Origins

schistosomiasis

from the Greek *schizein*, meaning "to cleave or split," and *soma*, meaning "body," and *iasis*, meaning "process or condition"

CLASS CESTODA

About 5,000 species of tapeworms make up the class Cestoda. Tapeworms can live in the intestines of almost all vertebrates. Humans may harbor any of seven different species. Tapeworms enter their host when the host eats raw or undercooked food containing eggs or larvae. A tapeworm infection may cause digestive problems, weight loss, lack of energy, and anemia, which is a decrease in the number of red cells in the blood.

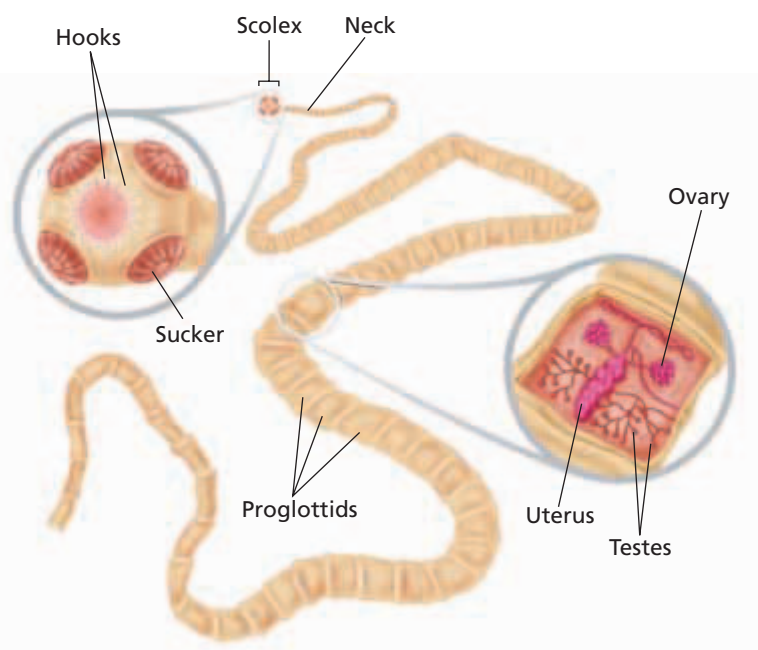
Structure of Tapeworms

Like flukes, tapeworms are surrounded by a tegument that protects them from their host's defenses. As Figure 34-5 shows, at the anterior end of a tapeworm is a knob-shaped organ called the **scolex** (SKOH-leks), which has hooks and suckers that enable the worm to attach to its host. A short neck connects the scolex with a long series of body sections called **proglottids** (proh-GLAHT-idz). As a tapeworm grows, it adds proglottids just behind the neck, pushing the older proglottids toward the rear. A single tapeworm may have 2,000 proglottids and exceed 10 m (33 ft) in length.

The excretory system and nervous system of a tapeworm are similar to those of other flatworms. However, tapeworms lack eyespots and other light-sensitive structures, and they have no mouth, gastrovascular cavity, or other digestive organs. They absorb nutrients directly from the host's digestive tract through their tegument. The tegument is highly folded, which increases the surface area available for absorption.

FIGURE 34-5

A tapeworm grows by adding proglottids behind its scolex. Each proglottid contains both male and female reproductive organs.



Reproduction and Life Cycle of Tapeworms

FIGURE 34-6

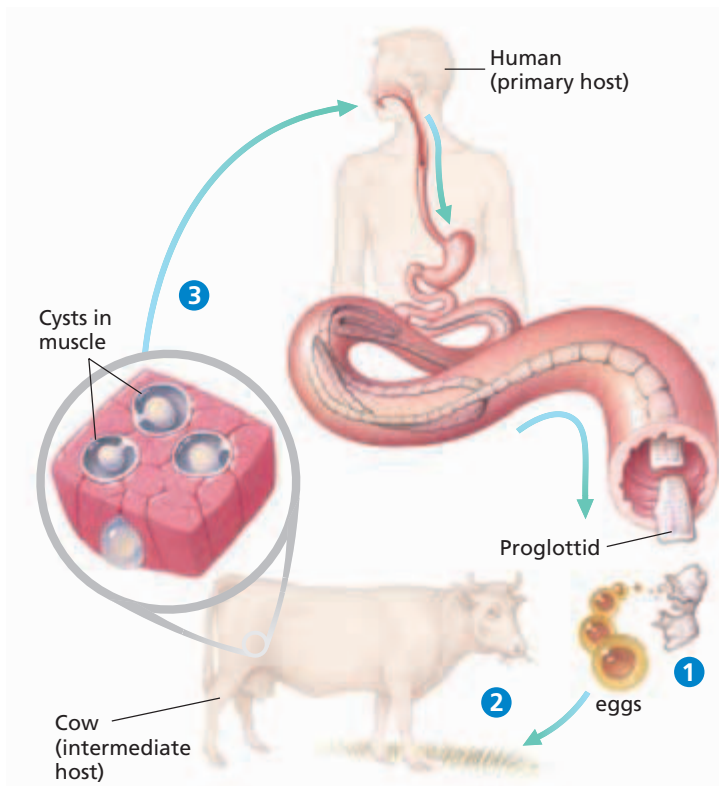
Adult beef tapeworms live in the intestines of humans, their primary hosts. ❶ Proglottids pass out of the primary host in feces, crawl onto vegetation, and release the tapeworm eggs. ❷ A cow, the intermediate host, ingests the eggs when it eats the vegetation. The eggs hatch into larvae that form cysts in the cow's muscles. ❸ When a person eats undercooked beef, the larvae in the beef develop into adult tapeworms in the person's intestine, and the cycle repeats.

Nearly all tapeworms are hermaphrodites. You can see in Figure 34-5 that each proglottid contains both male and female reproductive organs, but little else. As the proglottids move to the rear of the tapeworm, they grow, mature, and begin producing eggs. The oldest proglottids are almost completely filled with 100,000 or more eggs. Eggs in one proglottid are usually fertilized by sperm from a different proglottid, either in the same individual or a different individual if the host has more than one tapeworm.

The life cycle of the beef tapeworm, *Taenia saginatus*, is illustrated in Figure 34-6. Like the blood fluke, the beef tapeworm has two hosts. The primary host is a human. In the human intestine, mature proglottids break off from the adult and are eliminated with

the host's feces. If the feces are deposited on the ground, the proglottids crawl out of the feces and onto nearby vegetation. The eggs they release may remain alive for several months before the vegetation is eaten by a cow, the intermediate host. Inside the cow, the eggs develop into larvae that burrow through the cow's intestine and enter the bloodstream. The larvae then make their way to muscle tissue and form **cysts**, or dormant larvae surrounded by protective coverings. Humans become infected when they eat beef that has not been cooked well enough to kill the worms inside the cysts. Once a cyst enters the human intestine, the cyst wall dissolves and releases the worm. The worm then attaches to the intestinal wall and develops into an adult, beginning the cycle again.

Another tapeworm that infects humans is the pork tapeworm, *Taenia solium*. Its life cycle is similar to that of the beef tapeworm, except that a pig serves as the intermediate host.



SECTION 1 REVIEW

1. Why are flatworms called acoelomates?
2. What is a flame cell?
3. Describe two ways in which a planarian shows cephalization.
4. Summarize how the schistosome blood fluke affects the human body.
5. Explain why tapeworms can survive without a digestive system.

CRITICAL THINKING

6. **Recognizing Relationships** Why is it an adaptive advantage that a parasite not kill its host?
7. **Analyzing Concepts** How is asexual reproduction advantageous to a free-living flatworm?
8. **Applying Concepts** Considering the life cycle of the schistosome blood fluke, recommend an effective way of controlling the spread of schistosomiasis.

NEMATODA AND ROTIFERA

Members of the phyla Nematoda (nee-muh-TOHD-uh) and Rotifera (roh-TIF-uh-uh) have bilaterally symmetrical bodies that contain a fluid-filled space. This space holds the internal organs and serves as a storage area for eggs and sperm. It also supports the body and provides a structure against which the muscles can contract.

PHYLUM NEMATODA

The phylum Nematoda is made up of **roundworms**, worms with long, slender bodies that taper at both ends. Roundworms are among several phyla of animals known as *pseudocoelomates*. Pseudocoelomates are so named because they have a pseudocoelom, which is a hollow, fluid-filled cavity that is lined by mesoderm on the outside and endoderm on the inside.

Roundworms range in length from less than 1 mm to 120 cm (4 ft). In contrast to cnidarians, ctenophores, and flatworms, which have a gastrovascular cavity with a single opening, roundworms have a digestive tract with two openings. Food enters the digestive tract through the mouth at the anterior end, and undigested material is eliminated from the anus (AY-nuhs) at the posterior end. A digestive tract represents a significant advancement over a gastrovascular cavity because food moves through the tract in only one direction. This allows different parts of the tract to be specialized for carrying out different functions, such as enzymatic digestion and absorption of nutrients. Most roundworms have separate sexes and are covered by a protective, noncellular layer called the **cuticle** (KYOO-ti-kuhl).

About 15,000 species of roundworms are known, but biologists estimate that there may be 500,000 or more species. The vast majority of roundworm species are free-living on land, in salt water, and in fresh water. One free-living roundworm, *Caenorhabditis elegans*, is a favorite organism of scientists studying developmental biology. However, more than 150 species of roundworms are parasites of plants and animals. These parasitic species damage plant crops and can harm livestock, pets, and humans. Humans are host to about 50 roundworm species. As you read about these roundworms, notice their adaptations for parasitism.

OBJECTIVES

- **Describe** the body plan of a nematode.
- **Outline** the relationships between humans and parasitic roundworms.
- **Describe** the anatomy of a rotifer.

VOCABULARY

roundworm
cuticle
hookworm
trichinosis
pinworm
filarial worm
elephantiasis
rotifer
mastax
cloaca
parthenogenesis

Eco Connection

Roundworms for Your Garden—Just Add Water

Some garden supply companies now sell kits containing millions of microscopic roundworms. When released, these roundworms seek out and kill hundreds of varieties of insect pests, fleas, and ticks. Other types of soil-dwelling roundworms consume bacteria and fungi that attack plants. These roundworms are also known as *beneficial nematodes*.

However, not all roundworms are good for plants. Some species parasitize the roots of plants. Effective pest control with roundworms requires a knowledge of which species are harmful and which are beneficial.



FIGURE 34-7

This pig intestine is completely blocked by roundworms of the genus *Ascaris*.

Ascaris

Ascaris (AS-kuh-ris) is a genus of roundworm parasites that live in the intestines of pigs, horses, and humans. Ascarids feed on the food that passes through the intestines of their host. As Figure 34-7 shows, they can become so numerous that they completely block the host's intestines if left untreated. The adult female can reach lengths of up to 30 cm (1 ft). The smaller male has a hooked posterior end that holds the female during mating.

One ascarid female can produce up to 200,000 eggs every day. The fertilized eggs leave the host's body in feces. If they are not exposed to sunlight or high temperatures, they can remain alive in the soil for years. Ascarid eggs enter the body of another host when the host ingests contaminated food or water. The eggs develop into larvae in the intestines. The larvae bore their way into the bloodstream and are carried to the lungs and throat, where they develop further. They are coughed up, swallowed, and returned to the intestines, where they mature and mate, completing the life cycle. If the infection is severe, larvae in the lungs can block air passages and cause bleeding from small blood vessels.

Hookworms

Hookworms are another group of intestinal parasites. As you can see in Figure 34-8, a hookworm's mouth has cutting plates that clamp onto the intestinal wall. Hookworms feed on their host's blood. Because they remove much more blood than they need for food, a heavy hookworm infection can cause anemia. Hookworm infections in children can result in slowed mental and physical development.

Like ascarids, hookworms release their eggs in the host's feces. The eggs produce larvae in warm, damp soil, and the larvae enter new hosts by boring through the host's feet. They then travel through the blood to the lungs and throat. Swallowing takes them to the intestines, where they develop into adults. Hookworms infect about one billion people worldwide. Most infections occur in tropical and subtropical regions.

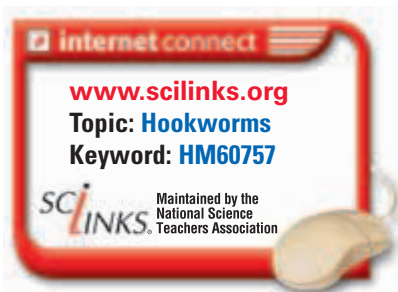
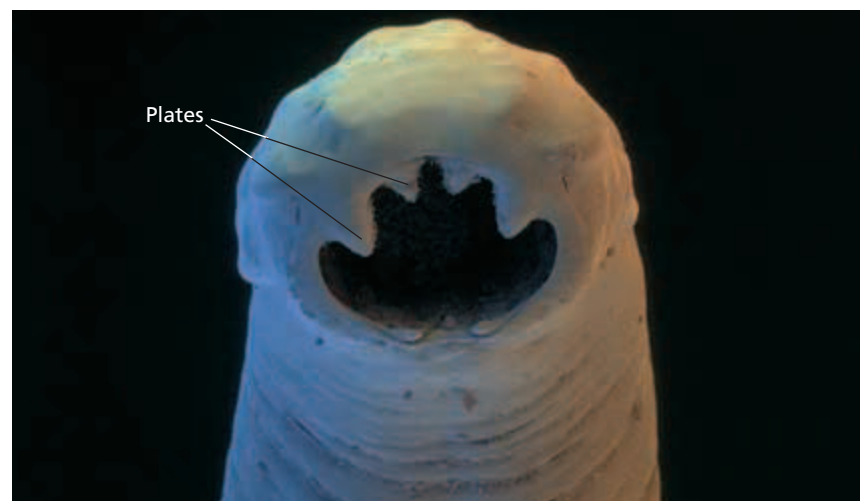


FIGURE 34-8

This SEM shows the hookworm *Ancylostoma duodenale*, which uses its plates to cut into the host's intestine, releasing blood on which the hookworm feeds.



Trichinella

Roundworms of the genus *Trichinella* infect humans and a variety of other mammals, including pigs. Adult trichina worms live embedded in the walls of the host's intestine. They produce larvae that travel through the bloodstream to the muscles, where they form cysts. Figure 34-9 shows several such cysts. People become infected when they eat undercooked meat—usually pork—that is contaminated with cysts. After they are eaten, the cysts release the larvae, which burrow into the intestinal wall and mature into adults. Trichina infections are responsible for the disease **trichinosis** (TRIK-i-NOH-sis), which causes muscle pain and stiffness. It can even cause death if large numbers of cysts form in the heart muscle. However, trichinosis is now rare in the United States. Farmers no longer feed raw meat to hogs, and meatpackers generally freeze pork, which kills the worms. Thoroughly cooking pork also prevents trichinosis.

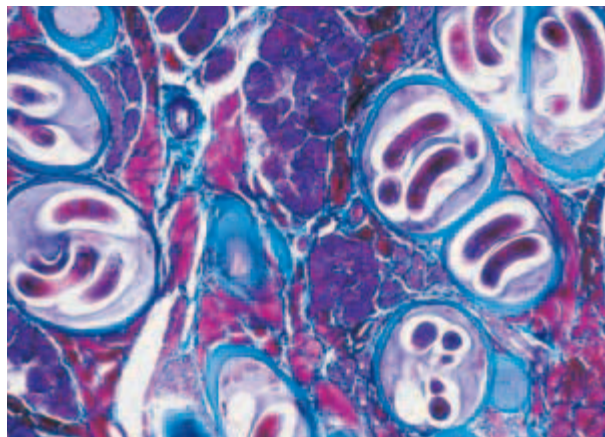


FIGURE 34-9

Larvae of the trichina roundworm coil up inside cysts in their hosts' muscle tissue. The cysts are stained blue in this light micrograph (100 $\times$).

Other Parasitic Roundworms

The most common roundworm parasite of humans in the United States is the **pinworm**, *Enterobius*. School-age children have the highest infection rate, which in some areas is as many as 50 percent of children. Despite their high rate of infection, pinworms do not cause any serious disease. Adult pinworms are 5–10 mm (0.2–0.4 in.) in length and resemble white threads. They live and mate in the lower portion of the intestine. At night, the females migrate out of the intestine and lay eggs on the skin around the anus. When an infected person scratches during sleep, the eggs are picked up by the person's hands and spread to anything the person touches. Eggs that are ingested hatch in the intestine, where the worms develop into adults.

Filarial (fuh-LAR-ee-uhl) **worms** are disease-causing roundworms that infect over 250 million people in tropical countries. The most dangerous filarial worms live in the lymphatic system, a part of the circulatory system that collects excess fluid around cells and returns it to the blood. The adult worms can be as long as 100 mm (4 in.). The larvae they produce enter the blood and are picked up by mosquitoes that draw blood from an infected person. The larvae develop into an infective stage inside the mosquitoes and are injected into the blood of another person when the mosquitoes feed again. Inside the new host, the larvae complete their development and settle in the lymphatic system. When they are present in large numbers, filarial worms can block the lymphatic vessels, causing fluid to accumulate in the limbs. In severe cases, the limbs become extremely swollen and skin hardens and thickens, a condition known as **elephantiasis** (EL-uh-fuhn-TIE-uh-sis). Another type of filarial worm infects dogs and cats. Large numbers of *Toxocara canis* or *T. cati* live in the heart and large arteries of the lungs and are responsible for heartworm disease.



Quick Lab

Comparing Flatworms and Roundworms

Materials living planarian, preserved specimens of tapeworms, male and female ascarids, hand lens or stereomicroscope

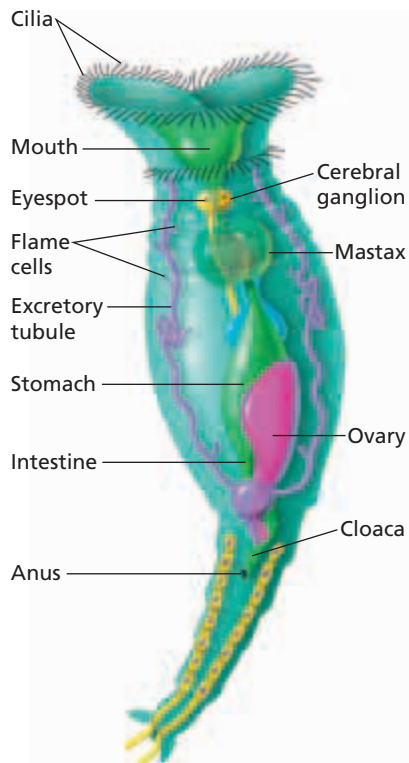
Procedure Examine specimens of tapeworms, ascarids, and the living planarian. Try to locate the following structures on each worm: anterior end, posterior end, mouth, eyespot, hooks, suckers, and anus. Draw each worm and label each of the features you located.

Analysis

1. List the features found on the posterior and anterior end of each worm.
2. Which worm has a separate mouth and an anus?
3. Which worm "absorbs" digested nutrients?
4. Which worm has a digestive tract?

FIGURE 34-10

Cilia surrounding the mouth of a rotifer sweep food into the animal's digestive tract.



PHYLUM ROTIFERA

Another group of pseudocoelomates are the approximately 1,750 species in the phylum Rotifera. Members of this phylum are called **rotifers**. Most rotifers are transparent, free-living animals that live in fresh water, although some live in salt water and damp soil. They typically range in length from 100 to 500 μm , with males being much smaller than females. Many rotifers can survive without water for long periods, during which they dry up and look like grains of sand. When wet conditions return, they absorb water and resume their activities. Although rotifers are tiny, they are truly multicellular and have specialized organ systems.

As Figure 34-10 illustrates, rotifers have a crown of cilia surrounding their mouth. Under a microscope, the crown with its beating cilia looks like a pair of rotating wheels. Rotifers use their cilia to sweep food—algae, bacteria, and protozoa—into their digestive tract. As shown in Figure 34-10, food moves from the mouth to the **mastax**, a muscular organ that breaks the food into smaller particles. The food is further digested in the stomach, and the nutrients are absorbed in the intestine. Indigestible material passes from the intestine to the **cloaca** (kloh-AY-kuh), a common chamber into which the digestive, reproductive, and excretory systems empty. Like planarians, rotifers use flame cells and excretory tubules to collect excess water in the body. The excess water, along with wastes from the intestine and eggs from the ovaries of females, leaves the cloaca through the anus. Rotifers exhibit cephalization, with a pair of cerebral ganglia and, in some species, two eyespots at the anterior end of the body.

Rotifer reproduction is unusual because some species of rotifers reproduce by parthenogenesis. In the process of **parthenogenesis** (PAHR-thuh-NOH-JEN-uh-sis), unfertilized eggs develop into adult females. Other species produce two types of eggs. Eggs of one type develop into small males that produce sperm. The sperm fertilizes eggs of the other type, which then form tough-coated zygotes that can survive poor conditions.

SECTION 2 REVIEW

1. Identify two features of roundworms and rotifers that flatworms do not have.
2. How are the roundworms of the genus *Ascaris* transmitted from one host to another?
3. Describe one parasitic adaptation of a roundworm.
4. What is the function of the cilia found on the anterior end of a rotifer?

CRITICAL THINKING

5. **Inferring Relationships** How are parasites able to recognize their hosts?
6. **Applying Concepts** Based on what you have learned about the life cycle of roundworms, recommend a method of preventing the transmission of parasitic roundworms.
7. **Analyzing Graphics** Study Figure 34-10. Suggest a function of the projections at the posterior end of the rotifer.

CHAPTER HIGHLIGHTS

SECTION 1

Platyhelminthes

- The phylum Platyhelminthes is made up of flatworms, the simplest animals that have bilateral symmetry. Flatworms have three germ layers and are acoelomates.
- Most flatworms have a gut with a single opening, called a gastrovascular cavity. These cephalized animals also have excretory, nervous, and reproductive systems.
- The class Turbellaria consists mostly of nonparasitic flatworms, including the freshwater planarian. Planarians are sexually reproducing hermaphrodites that can also reproduce asexually by splitting in two and regenerating the missing parts.
- The classes Trematoda and Monogenea consist of parasitic flukes. Fluke life cycles include two types of hosts: a primary host, from which the adults derive their nourishment, and an intermediate host, from which the larvae derive their nourishment.
- The class Cestoda consists of parasitic tapeworms. Tapeworms do not have a digestive system, so they absorb nutrients through their body surface. Tapeworms have a series of body sections called proglottids, each of which contains reproductive structures. Tapeworm life cycles involve primary and intermediate hosts.

Vocabulary

pharynx (p. 690)
flame cell (p. 690)
cerebral ganglion (p. 691)

eyespot (p. 691)
fluke (p. 691)
tegument (p. 692)

primary host (p. 692)
intermediate host (p. 692)
schistosomiasis (p. 693)

scolex (p. 693)
proglottid (p. 693)
cyst (p. 694)

SECTION 2

Nematoda and Rotifera

- The phylum Nematoda consists of roundworms. Most roundworms are free-living, but some are parasites of plants and animals.
- Nematodes are pseudocoelomates. They have a hollow, fluid-filled cavity called a pseudocoelom between the mesoderm and the endoderm. They also have a digestive tract with an anterior mouth and posterior anus.
- Ascarid worms infect people who consume food or water containing eggs of the genus *Ascaris*. The eggs develop into larvae that migrate through the body and mature in the intestines.
- Hookworm larvae in the soil burrow through a person's feet. The larvae migrate through the body and mature in the intestines.
- Trichina worms infect people who consume undercooked meat containing cysts of the genus *Trichinella*. The cysts release larvae that burrow into the intestinal wall and develop into adults.
- Pinworms live in the lower intestine and lay eggs on the skin around the anus. After being transmitted by the hands to other objects, the eggs may be ingested. They then hatch in the intestine, where the worms mature.
- Filarial worms include species that live in the human lymphatic system. Their larvae are transmitted between hosts by mosquitoes.
- Rotifers are pseudocoelomates, and most rotifers are free-living in fresh water. The cilia surrounding a rotifer's mouth sweeps food into its digestive tract.

Vocabulary

roundworm (p. 695)
cuticle (p. 695)
hookworm (p. 696)

trichinosis (p. 697)
pinworm (p. 697)
filarial worm (p. 697)

elephantiasis (p. 697)
rotifer (p. 698)
mastax (p. 698)

cloaca (p. 698)
parthenogenesis (p. 698)

CHAPTER REVIEW

USING VOCABULARY

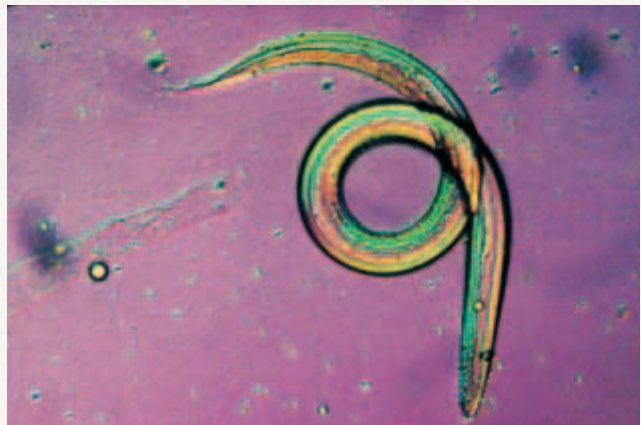
- For each pair of terms, explain how the meanings of the terms differ.
 - tegument* and *cuticle*
 - primary host* and *intermediate host*
 - schistosomiasis* and *trichinosis*
 - anus* and *cloaca*
 - scolex* and *mastax*
- Choose the term that does not belong in the following group, and explain why it does not belong: *eyespot*, *flame cell*, *gastrovascular cavity*, and *proglottid*.
- Explain the relationship between *proglottid* and *cyst*.
- Word Roots and Origins** The word part *roti-* means “wheel,” and the suffix *-fera* means “bearer.” Using this information, explain why *Rotifera* is a good name for the phylum of organisms it describes.

UNDERSTANDING KEY CONCEPTS

- Explain** how a flatworm can survive without a circulatory system or a respiratory system.
- Describe** the nervous system of a planarian.
- Summarize** reproduction in planarians.
- Sequence** the following steps in the schistosome life cycle in the correct order, beginning with the first step after sexual reproduction.
 - Tailed larvae swim through the water.
 - Ciliated larvae swim through the water.
 - Larvae penetrate the skin of a human.
 - Larvae reproduce asexually in the intermediate host.
 - Larvae burrow into a snail.
 - Larvae enter a blood vessel and develop into adults.
 - Fertilized eggs leave the primary host in the feces or urine.
- Compare** a primary host to an intermediate host.
- Identify** some adaptations of flukes to a parasitic way of life.
- Explain** how tapeworms are specialized for a parasitic way of life.
- Relate** how a person can avoid becoming infected by a beef tapeworm.
- Compare** the body plan of a nematode to that of a flatworm.
- Name** some features shared by roundworms and rotifers.
- Describe** feeding and digestion in a rotifer.
- CONCEPT MAPPING** Use the following terms to create a concept map that shows the major characteristics of the phylum Nematoda: *roundworms*, *pseudocoelomates*, *digestive tract*, *cuticle*, *free-living*, *water*, *parasites*, *separate sexes*, and *eggs*.

CRITICAL THINKING

- Applying Information** The Aswan High Dam across the Nile River in Egypt was completed in 1970. The dam was built to increase the supply of irrigation water, control major flooding, and provide a source of hydroelectric power. Since the dam was built, however, there has been an increase in the incidence of schistosomiasis in the region. Why do you think this has happened?
- Distinguishing Relevant Information** A person infected with a tapeworm may show symptoms such as tiredness, loss of weight, and anemia, which could indicate any number of diseases. How might a doctor be certain that the symptoms are caused by a tapeworm?
- Recognizing Relationships** Hookworm infections are extremely common in China, where rice is grown in paddies that are periodically flooded. Considering what you know about how hookworms invade the human body, why do you think hookworm infections are so common in this part of the world?
- Analyzing Patterns** Some rotifers can survive being dried out for as long as four years. When they are placed in water again, they revive. For what kind of environment might this characteristic be adaptive?
- Interpreting Graphics** Identify the animal in the photograph below. Give reasons for your choice.



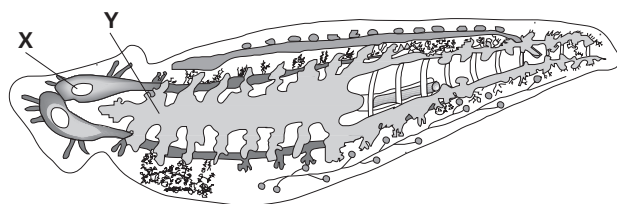


Standardized Test Preparation

DIRECTIONS: Choose the letter that best answers the question.

- What does a planarian use its pharynx for?
 - feeding
 - movement
 - reproduction
 - to respond to light
- Where do blood flukes of the genus *Schistosoma* reproduce asexually?
 - in water
 - inside a snail
 - inside a cow's intestine
 - inside a human's blood vessels
- What does a tapeworm use its scolex for?
 - to reproduce
 - to attach itself to its host
 - to eliminate excess water
 - to force food into its mouth
- Which of the following is true of most rotifers?
 - They are parasitic.
 - They live in the soil.
 - They feed with the help of cilia.
 - They have a gastrovascular cavity.

INTERPRETING GRAPHICS: The figure below shows the internal structure of a planarian of the genus *Dugesia*. Use the figure to answer the questions that follow.

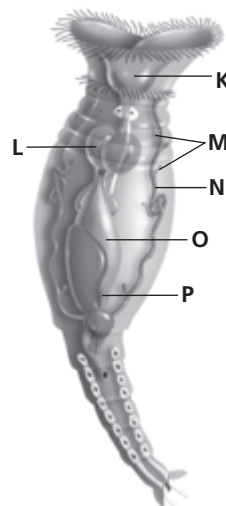


- What type of animal is shown in the figure?
 - flatworm
 - tapeworm
 - roundworm
 - rotifer
- What is the structure labeled X?
 - the brain
 - the mouth
 - an eyespot
 - a nerve cord
- What is the structure labeled Y?
 - the mouth
 - a flame cell
 - an eyespot
 - the gastrovascular cavity

DIRECTIONS: Complete the following analogy.

- fluke : schistosomiasis :: filarial worm :
 - trichinosis
 - elephantiasis
 - encysted meat
 - swimmer's itch

INTERPRETING GRAPHICS: The figure below shows the internal structure of a rotifer. Use the figure to answer the question that follows.



- Which structures are involved in excretion?
 - K and L
 - L and O
 - M and N
 - M and P

SHORT RESPONSE

Planarians and rotifers eliminate water through a network of excretory structures that run the length of the body.

Explain why the excretory structures in planarians and rotifers are called flame cells.

EXTENDED RESPONSE

Parasitic flatworms have life cycles that include primary and intermediate hosts.

Part A Distinguish between primary and intermediate hosts in flatworms.

Part B Sequence the life cycle of a beef tapeworm. Identify the primary and intermediate hosts.

Test TIP

Take the time to read each question completely on a standardized test, including all the answer choices. Consider each possible choice before determining which answer is correct.

Observing Flatworm Responses to Stimuli

OBJECTIVES

- Observe the feeding behavior of flatworms.
- Study the response of flatworms to different stimuli.

PROCESS SKILLS

- observing
- predicting
- experimenting
- collecting data
- analyzing data

MATERIALS

- *Dugesia*
- culture dish
- stereomicroscope
- blunt probe
- test tube
- test-tube rack
- aluminum foil
- wax pencil
- medicine dropper
- cork stopper
- stopwatch or clock
- sheet of white paper
- raw liver
- small flashlight



Background

1. What are some characteristics of the phylum Platyhelminthes?
2. *Dugesia*, a planarian, is a nonparasitic flatworm belonging to the class Turbellaria. What are some characteristics of this class?
3. What is meant by cephalization, and why is it an important evolutionary advance?
4. What structures does a planarian have that enable it to sense light? Where are these structures located?

PART A Observing a Flatworm

1.  **CAUTION** You will be working with a live animal. Be sure to treat it gently and to follow directions carefully. Use a medicine dropper to transfer one flatworm to a small culture dish. Gently cover the flatworm with water from the culture jar. Why should you use water from the culture jar instead of tap water?
2. Examine the flatworm under the low-power setting of the stereomicroscope. Notice the shape of its body. What kind of symmetry does it have? What structural features demonstrate this symmetry? Does the flatworm appear to have a distinct head and tail? How can you tell?
3. Observe how the flatworm moves. Look at the surface of the flatworm under high power. Can you see any structures that could account for its movement?

PART B Response to Touch

4. In your lab report, make a data table like the one shown. Predict the behavior you think will occur to each type of stimulus. As you complete the following steps, record your observations of the flatworm in your data table.
5. **CAUTION** Touch the flatworm gently. Using a blunt probe, gently touch the posterior end of the flatworm. Notice its response. Now gently touch its anterior end. How does the response compare? What can you conclude from your observations?

PART C Feeding Behavior

6. Place a tiny piece of raw liver in the culture dish. Observe the flatworm for several minutes and describe its feeding response in your data table. Why is it important to use flatworms that have not been recently fed?
7. After several minutes, use the probe to gently turn the flatworm over. What do you observe? If you have time, watch as the flatworm eats the liver. Where do you think the undigested waste will come out?

PART D Response to Gravity

8. With a wax pencil, draw a line around the middle of a test tube.
9. Fill the test tube almost full with water from the culture jar. Then use a medicine dropper to transfer one flatworm to the test tube. Seal the test tube with a stopper.
10. Hold the test tube horizontally and move it slowly back and forth until the flatworm is centered on the line you drew.
11. To test whether the flatworm can sense gravity, place the test tube vertically in the test-tube rack. Which way should the flatworm move to show a positive response to gravity? Observe the flatworm for several minutes. Use a stopwatch to measure the amount of time the flatworm spends above the line and below the line. Record the times in your data table.

PART E Response to Light

12. Check the lighting in the room. The light must be low and even during this part of the investigation.
13. Using a piece of aluminum foil, make a cover that is big enough to fit over the bottom half of the test tube

you used in Part D. Set the cover aside. Place the test tube horizontally on a white sheet of paper. Make sure the test tube is level. Why is it important for the test tube to be level in this experiment?

14. Position the flashlight so that it will shine directly on the test tube, but do not turn it on.
15. Wait for the flatworm to move to the center line. Then gently place the foil cover over the bottom half of the test tube, and turn the flashlight on. Which way should the flatworm move to show a positive response to light? Observe the movements of the flatworm. Use the stopwatch to measure the amount of time the flatworm spends in each half of the test tube. Record the times in your data table.
16.   Return the flatworm to the culture jar. Then clean up your materials, and wash your hands before leaving the lab.

Analysis and Conclusions

1. What evidence does the flatworm show of cephalization?
2. The flatworm has an incomplete digestive system and no circulatory system. How do you think a flatworm's food gets to the cells after it is digested?
3. How does the flatworm respond to gravity? Is the response positive or negative?
4. How does the flatworm respond to light? Is the response positive or negative?
5. Are the flatworm's anterior and posterior ends equally sensitive to light?

Further Inquiry

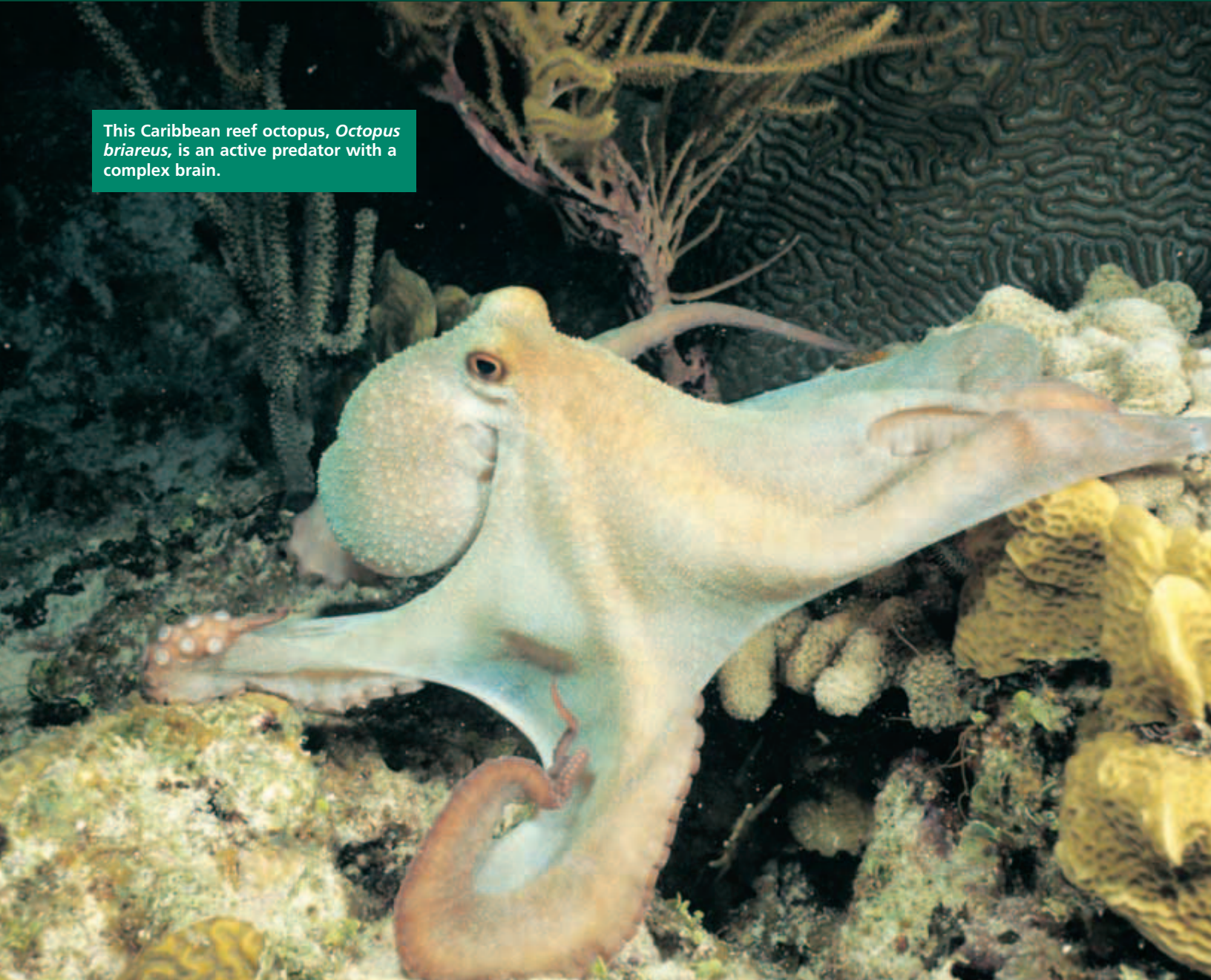
Design an experiment to study the responses of flatworms to other stimuli, such as vibrations or sound.

OBSERVATIONS OF FLATWORM BEHAVIOR

Behavior	Predictions	Observations
Response to touch with blunt probe		
Feeding behavior		
Response to gravity		
Response to light		

MOLLUSKS AND ANNELIDS

This Caribbean reef octopus, *Octopus briareus*, is an active predator with a complex brain.



SECTION 1 *Mollusca*

SECTION 2 *Annelida*

MOLLUSCA

Despite their very different appearances, invertebrates such as clams, snails, slugs, and octopuses belong to the same phylum, Mollusca (muh-LUHS-kuh). Members of this phylum are called mollusks, a name that comes from the Latin molluscus, which means “soft.” Although some mollusks have soft bodies, most have a hard shell that protects them.

CHARACTERISTICS OF MOLLUSKS

The phylum Mollusca is a diverse group of more than 112,000 species. Among animals, only the phylum Arthropoda has more species. Some mollusks are sedentary filter feeders, while others are fast-moving predators with complex nervous systems.

Mollusks are among several phyla of animals known as *coelomates*. Coelomates are so named because they have a true coelom, a hollow, fluid-filled cavity that is completely surrounded by mesoderm. Coelomates differ from pseudocoelomates, such as roundworms, which have a body cavity lined by mesoderm on the outside and endoderm on the inside.

A coelom has several advantages over a pseudocoelom. With a coelom, the muscles of the body wall are separated from those of the gut. Therefore, the body wall muscles can contract without hindering the movement of food through the gut. A coelom also provides a space where the circulatory system can transport blood without interference from other internal organs. The coelomate body plan is shared by annelids, which are discussed in the second half of this chapter, and by three other major phyla of animals: arthropods, echinoderms, and chordates, which include humans.

Another feature that is shared by most aquatic mollusks and annelids is a larval stage of development called a **trochophore** (TRAHK-oh-FAWR), illustrated in Figure 35-1. In some species, the trochophore hatches from the egg case and exists as a free-swimming larva. Cilia on the surface of a free-swimming trochophore propel the larva through the water and draw food into its mouth. As free-swimming trochophores are carried by ocean currents and tides, they contribute to the dispersal of their species. The presence of a trochophore in mollusks and annelids suggests that these two groups of animals may have evolved from a common ancestor.

OBJECTIVES

- Describe the key characteristics of mollusks.
- Describe the body plan of mollusks.
- Name the characteristics of three major classes of mollusks.
- Compare the body plans of gastropods, bivalves, and cephalopods.

VOCABULARY

trochophore
visceral mass
mantle
mantle cavity
ganglion
radula
gastropod
hemolymph
hemocoel
bivalve
incurrent siphon
excurrent siphon
cephalopod

FIGURE 35-1

A trochophore is a larva that develops from the fertilized egg of most mollusks and annelids. Cilia at both ends and in the middle propel free-swimming trochophores through the water.

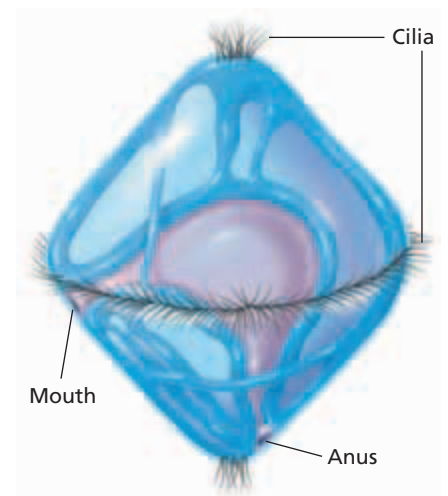
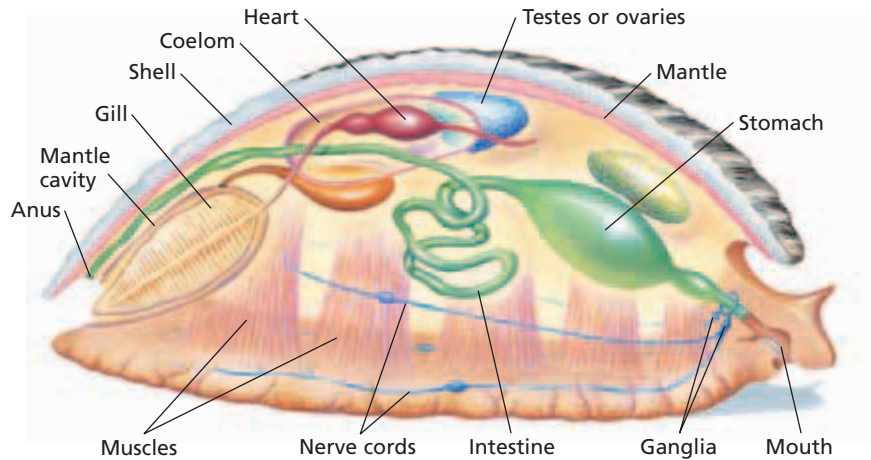


FIGURE 35-2

In the basic body plan of a mollusk, the body is divided into the head-foot and the visceral mass, which contains the internal organs. Covering the visceral mass is the mantle, which secretes the shell.

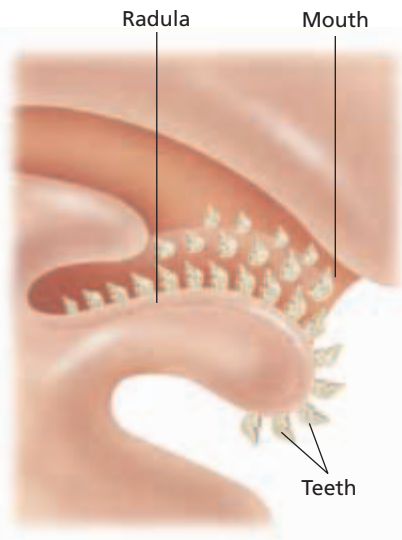
Visceral mass
Head-foot



BODY PLAN OF MOLLUSKS

FIGURE 35-3

Inside the mouth (a), many mollusks have a radula, a band of tissue covered with teeth that can scrape food from other surfaces. The SEM in (b) shows the sharp edges of these teeth (600 $\times$).



(a)



(b)

Figure 35-2 shows that the body of a mollusk is generally divided into two main regions: the head-foot and the visceral mass. As its name suggests, the head-foot consists of the head, which contains the mouth and a variety of sensory structures, and the foot, a large, muscular organ usually used for locomotion. Above the head-foot is the **visceral (VIS-uh-uhl) mass**, which contains the heart and the organs of digestion, excretion, and reproduction. As shown in Figure 35-2, the coelom is limited to a space around the heart. Covering the visceral mass is a layer of epidermis called the **mantle**.

In most mollusks, the mantle secretes one or more hard shells containing calcium carbonate. Although shells protect the soft bodies of mollusks from predators, they also reduce the surface area available for gas exchange. This disadvantage is offset by another structural adaptation: gills. Gills provide a large surface area that is in contact with a rich supply of blood. In this way, gills are specialized for the exchange of gases. Figure 35-2 also shows that the delicate gills of mollusks are protected within the **mantle cavity**, a space between the mantle and the visceral mass.

Unlike many coelomates, mollusks do not have segmented bodies. Like flatworms and roundworms, most mollusks are bilaterally symmetrical. This symmetry is apparent in the nervous system, which consists of paired clusters of nerve cells called **ganglia**. The ganglia are situated in the head-foot and visceral mass and are connected by two pairs of long nerve cords. Nerve cells in the ganglia control the muscles involved in locomotion and feeding and process sensory information from specialized cells that respond to light, touch, and chemicals in the environment.

The main feeding adaptation of many mollusks is the **radula (RAJ-u-luh)**. As Figure 35-3 shows, in most species the radula is a flexible, tonguelike strip of tissue covered with tough, abrasive teeth that point backward. Through evolution, the radula has become adapted for a variety of functions in different mollusks.

TABLE 35-1 Features of Three Classes of Mollusks

Class	External shell	Head	Radula	Locomotion
Gastropoda	one (most species)	yes	yes	crawling (most)
Bivalvia	two	no	no	sessile (most)
Cephalopoda	none (most species)	yes	yes	rapid swimming

Terrestrial snails use the radula to cut through the leaves of garden plants, while aquatic snails use it to scrape up algae or to drill holes in the shells of other mollusks. The cone shell has a harpoon-shaped radula with which it captures fish and injects venom.

Most biologists use structural differences to divide mollusks into seven classes. Three of these classes are discussed below: class Gastropoda (gas-TRAHP-uh-duh), class Bivalvia (bie-VALV-ee-uh), and class Cephalopoda (SEF-uh-LAHP-uh-duh). Table 35-1 summarizes the major features of these three classes.

Word Roots and Origins

gastropod

from the Greek *gaster*, meaning "stomach," and *pous*, meaning "foot"

CLASS GASTROPODA

The largest and most diverse class of mollusks is Gastropoda, whose members are called **gastropods** (GAS-troh-PAHDZ). Most of the 40,000 species of gastropods, including snails, abalones, and conchs, have a single shell. Others, such as slugs and nudibranchs, have no shell at all.

Gastropods undergo a process called *torsion* during larval development. During torsion, the visceral mass twists around 180 degrees in relation to the head. This twisting brings the mantle cavity, gills, and anus to the front of the animal, as shown in Figure 35-4. Because of torsion, a gastropod can withdraw its head into its mantle cavity when threatened. Coiling of the shell is unrelated to torsion.

Wavelike muscular contractions of the foot move gastropods smoothly over surfaces. You can see these contractions if you look closely at the underside of a snail or slug as it crawls across a windowpane or the side of an aquarium.

Gastropods have an open circulatory system, meaning that the circulatory fluid, called **hemolymph**, does not remain entirely within vessels. Instead, it is collected from the gills or lungs, pumped through the heart, and released directly into spaces in the tissues. These fluid-filled spaces compose what is known as a **hemocoel** (HEE-moh-SEEL), or blood cavity. From the hemocoel, the hemolymph returns via the gills or lungs to the heart.

FIGURE 35-4

In a gastropod, such as this snail, the mantle cavity, anus, and gills are near the head as a result of torsion during development.

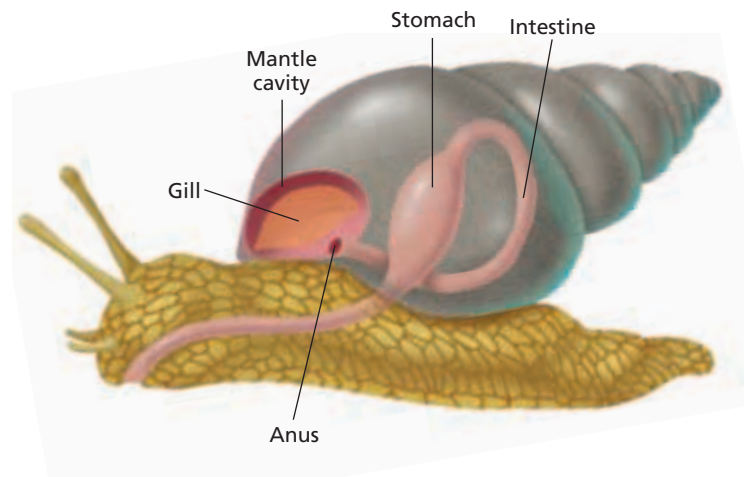




FIGURE 35-5

The extensions on the back of this horned nudibranch, *Hemissenda crassicornia*, provide a large surface area for gas exchange.

Snails

Snails are gastropods that live on land, in fresh water, and in the ocean. Two eyes at the end of delicate tentacles on the head help most snails locate food. If danger arises, the tentacles retract into the head. Aquatic snails respire through gills in the mantle cavity. In land snails, the mantle cavity acts as a modified lung that exchanges oxygen and carbon dioxide with the air. The thin membrane lining the mantle cavity must be kept moist to allow gases to diffuse through it. For this reason, land snails are most active when the air has a high moisture content. Snails survive dry periods by becoming inactive and retreating into their shells. They seal the opening to their shell with a mucous plug, which keeps them from drying out.

Other Gastropods

Slugs are terrestrial gastropods that look like snails without shells. Like land snails, slugs respire through the lining of their mantle cavity. They avoid drying out by hiding in moist, shady places by day and feeding at night.

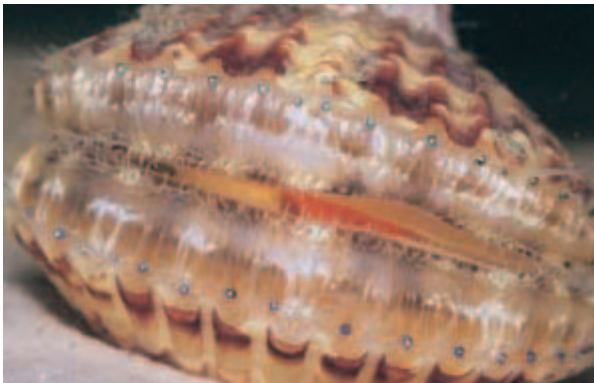
Nudibranchs (NOO-di-BRANGKS), such as the one in Figure 35-5, are marine gastropods that lack shells. *Nudibranch* means “naked gill,” which refers to the fact that gas exchange occurs across the entire body surface of these animals. The surface of most nudibranchs is covered with numerous ruffles or delicate, fingerlike extensions that increase the total area available for gas exchange.

Some gastropods show unusual adaptations of the foot. In pteropods, or “sea butterflies,” for example, the foot is modified into a winglike flap that is used for swimming rather than crawling.

Gastropods show great diversity in their reproduction. Many species have separate sexes, but hermaphrodites are common among aquatic and terrestrial gastropods.

FIGURE 35-6

The two valves that make up the hinged shell of a bivalve can separate, allowing water to circulate through the animal. Some bivalves, such as this scallop, of the genus *Argopecten*, have a row of eyes near the outer margin of each valve.



CLASS BIVALVIA

Members of the class Bivalvia include aquatic mollusks, such as clams, oysters, and scallops. These mollusks are called **bivalves** because, as Figure 35-6 shows, their shell is divided into two halves, or valves, connected by a hinge. A bivalve can close its shell by contracting the powerful adductor muscles that are attached to the inside surface of each valve. When the adductor muscles relax, the valves open.

Each valve consists of three layers that are secreted by the mantle. The thin outer layer protects the shell against acidic conditions in the water. The thick middle layer of calcium carbonate strengthens the shell. The smooth, shiny inner layer protects the animal’s soft body.

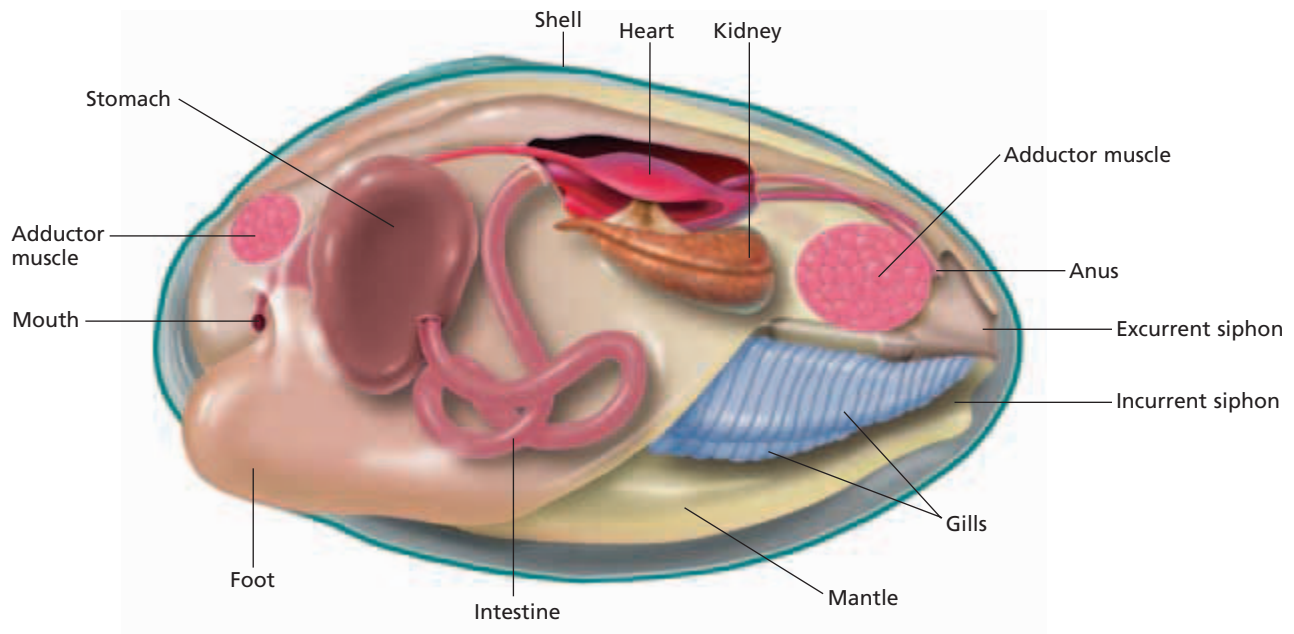


FIGURE 35-7

In this illustration, one valve has been omitted to show a clam's anatomy. The internal structure of a clam is typical of most bivalves.

In contrast with gastropods, which move about in search of food, most bivalves are sessile. Some species extend their muscular foot into the sand and fill the foot with hemolymph to form a hatchet-shaped anchor. The muscles of the foot then contract, pulling the animal down into the sand. As an adaptation for a sessile existence, bivalves usually are filter feeders. They are the only mollusks without a radula.

Bivalves lack a distinct head. Their nervous system consists of three pairs of ganglia: one pair near the mouth, another pair in the digestive system, and the third pair in the foot. The ganglia are connected by nerve cords. Nerve cells in the ganglia receive information from sensory cells in the edge of the mantle that respond to touch or to chemicals in the water. Some bivalves also have a row of small eyes along each mantle edge. Stimuli detected by these sensory structures can trigger nerve impulses that cause the foot to withdraw and the shell to close.

Clams

Clams are bivalves that live buried in mud or sand. The mantle cavity of a clam is sealed except for a pair of hollow, fleshy tubes called *siphons*, which you can see in Figure 35-7. Cilia beating on the gills set up a current of water that enters through the **incurrent siphon** and leaves through the **excurrent siphon**. As the water circulates inside the clam, the gills filter small organisms and organic debris from the water. The filtered material becomes trapped on the gills in a sticky mucus that moves in a continuous stream toward the mouth. Water passing over the gills also exchanges oxygen and carbon dioxide with the hemolymph.

Most species of clams have separate sexes. Marine clams reproduce by shedding sperm and eggs into the water, and fertilization occurs externally. The fertilized egg becomes a trochophore that eventually settles to the bottom and develops into an adult.



Quick Lab

Describing a Mollusk

Materials 2–3 bivalve shells, colored pencils, paper

Procedure

1. Using colored pencils, draw a bivalve shell on a sheet of paper.
2. Use Figure 35-7 to help you locate and label the adductor muscle scars, the mantle cavity, and the hinge area on the bivalve.

Analysis Describe the shell of the bivalve, including its color, its ridges, the appearance and texture of the mantle, and the location of the hinge area.

In some clam species, adults may weigh 200 kg (440 lb) and be more than 1 m (3.3 ft) across. In most freshwater clams, eggs are fertilized internally by sperm that enter through the incurrent siphon. The larvae that develop are discharged into the water through the excurrent siphon. If they contact a passing fish, they may live as parasites on its gills or skin for several weeks before settling to the bottom.

Other Bivalves

Oysters are bivalves that become permanently attached to a hard surface early in their development. Some are grown commercially as food or as sources of cultured pearls. Scallops can move through the water by repeatedly opening their valves and snapping them shut. This motion expels bursts of water, creating a form of jet propulsion. The teredo, or shipworm, is one of the few bivalves that does not filter-feed. Instead, it bores into driftwood or ship timbers and ingests the particles that are produced by the drilling. The wood cellulose is broken down by symbiotic bacteria that live in the shipworm's intestine.

CLASS CEPHALOPODA

Members of the class Cephalopoda include octopuses, squids, cuttlefishes, and chambered nautilus. These marine mollusks are called **cephalopods** (SEF-uh-loh-PAHDZ), a term that means “head-foot.” Cephalopods are specialized for a free-swimming, predatory existence. Extending from the head is a circle of tentacles, as you can see in Figure 35-8. The tentacles’ powerful suction cups allow cephalopods to grasp objects and capture prey. Cephalopods kill and eat their prey with the help of a pair of jaws that resemble a parrot’s beak.

The nervous system is more advanced in cephalopods than in any other group of mollusks. The cephalopod brain, which is the largest of any invertebrate brain, is divided into several lobes and contains millions of nerve cells. Octopuses, for example, can learn to perform tasks and discriminate between objects on the basis of their shape or texture. The sensory systems of cephalopods are also well developed. Most cephalopods have complex eyes that form images. The tentacles contain numerous cells that sense chemicals in the water.

Cephalopods have a closed circulatory system. Closed circulatory systems transport fluid more rapidly than open circulatory systems do. Thus, nutrients, oxygen, and carbon dioxide are carried quickly through the body of these highly active animals. Cephalopods also have separate sexes. The male uses a specialized tentacle to transfer packets of sperm from his mantle cavity to the mantle cavity of the female, where fertilization occurs. The female lays a mass of fertilized eggs and guards the eggs until they hatch. Unlike other mollusks, cephalopods develop from an egg into a juvenile without becoming a trochophore.

FIGURE 35-8

Most of the body of cephalopods is made up of a large head attached to tentacles. The tentacles of cephalopods, such as this cuttlefish, *Sepia latimanus*, surround their mouth. The streamlined body of many cephalopods enables them to swim rapidly in pursuit of prey.



Many cephalopods can release a dark fluid into the water to temporarily distract predators. They also have pigment cells called *chromatophores* (kroh-MAT-uh-FAWHRZ), which are located in the outer layer of the mantle. Chromatophores can produce a sudden change in the color of a cephalopod, allowing the animal to blend in with its surroundings.

Squids

Squids are cephalopods with ten tentacles. The longest two tentacles are used for capturing prey, and the other eight tentacles force the prey into the squid's mouth. The muscular mantle propels the squid swiftly through the water by pumping jets of water through an excurrent siphon. Most squids grow to about 30 cm (1 ft) in length, but a few species can be much longer. The giant squid, *Architeuthis*, may reach a length of 18 m (about 60 ft) and a weight of more than 900 kg (about 1 ton). *Architeuthis* is the world's largest known invertebrate.

Octopuses

Octopuses have eight tentacles and share many characteristics with squids, including their methods of escaping from predators. Instead of using jet propulsion to chase prey, however, octopuses are more likely to crawl along the ocean bottom with their tentacles or lie in wait in caves and rock crevices. Octopuses average 1 m (3.3 ft) or less in length, although the giant Pacific octopus may grow to a length of 9 m (about 30 ft).

Chambered Nautilus

Squids and cuttlefish have small internal shells. The chambered nautilus, shown in Figure 35-9, is the only existing cephalopod that has retained its external shell. The nautilus shell is coiled and divided into a series of gas-filled chambers separated by partitions. The soft body of the nautilus is confined to the outermost chamber. As the nautilus grows, it moves forward in its shell, makes a new partition, and fills the chamber behind the partition with gas. The gas makes the nautilus buoyant.

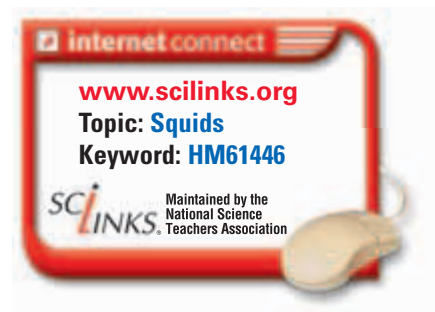


FIGURE 35-9

Although cephalopods evolved from shelled ancestors, the chambered nautilus is the only living cephalopod species with an external shell.

SECTION 1 REVIEW

1. What is one advantage of a true coelom over a pseudocoelom?
2. In which phyla of animals is the larval trochophore stage found?
3. For what purpose is a radula used?
4. Why are land snails more active when the air around them is moist?
5. Compare the circulatory systems of each of the three major classes of mollusks.

CRITICAL THINKING

6. **Analyzing Concepts** Suggest why an open circulatory system is sufficient to meet the needs of a gastropod.
7. **Making Comparisons** Compare the adaptations of clams and squids and relate them to lifestyle.
8. **Inferring Relationships** Which features of bivalves indicate that they are bilaterally symmetrical?

LEECHES: New Uses for an Old Remedy

Why are leeches called *bloodsuckers*? Do they really suck blood? Yes, they do, and their role as blood-suckers could help save your life.

For centuries, leeches were used in medical practice. In the second century C.E., the Greek physician Galen described the use of leeches in removing blood from patients in a procedure called *bloodletting*. An excess of blood in the body was believed to be responsible for a variety of illnesses, from headaches and fevers to heart disease. Physicians used the leeches to remove this “bad blood” from a patient’s body.

Bloodletting was common in Europe through the early 1800s. During the late 19th century, however, medical science discredited the idea that excess blood causes disease, and bloodletting fell out of favor.

Uses in Microsurgery

Leeches are making a comeback in medicine, although with new purposes. One of these purposes is to increase the success rate of surgical operations to reattach severed limbs, fingers, ears, or toes. Such operations involve microsurgery, a process in which surgeons reconnect tendons, blood vessels, and nerves by using tiny instruments and powerful microscopes. However, in some microsurgery, physicians cannot reconnect tiny, delicate blood vessels. As a result, circulation in the reattached limb, finger, or toe is impaired.

The tissues may become congested with blood. If congestion occurs, the tissues of the reattached part will not heal and will eventually die.

One solution to this congestion problem is to place leeches on the reattached body part. Once attached to the wound site, the leeches begin to suck out the accumulated blood, relieving congestion and allowing the tissues to remain healthy until the veins can grow back. At about \$10 each, leeches are a relatively inexpensive treatment for a serious problem.

Uses as Anticoagulants

Leeches have medical uses that go beyond their ability to remove blood. Scientists have known since the 1800s that leech saliva contains a powerful anticoagulant, a substance that inhibits blood clotting. The leech’s anticoagulant, called *hirudin*, can cause four hours or more of steady bleeding.



Leeches have been applied to this patient’s sutures across his upper back to reduce blood congestion. Each leech can remove up to 5 mL of blood.

The steady bleeding helps prevent blood from clotting so the leech can feed freely.

New Applications

Today, *hirudin* is made through genetic engineering, without the aid of leeches. It has proven useful in the treatment of some heart patients, particularly those who have had heart attacks, who suffer from angina, or who have undergone angioplasty, a procedure to open blocked arteries. One research study even indicated that *hirudin* may be effective against the spread of cancer.

The amazing uses that have been found for a substance in leech saliva are encouraging to medical researchers, who continue to explore how knowledge of invertebrate organisms can be beneficially combined with medical technology.

REVIEW

1. Outline how leeches help surgeons in microsurgery.
2. Identify three potential new applications of *hirudin*.
3. **Critical Thinking** Why is *hirudin* now made by genetic engineering instead of being taken directly from leeches?



ANNELIDA

Colorful feather-duster worms, common earthworms, and bloodsucking leeches are all members of the phylum *Annelida* (uh-NEL-i-duh). An animal in this phylum is called an annelid (AN-uh-LID), a term that means “little rings.” The name refers to the many body segments that make an annelid look as if it is composed of a series of rings.

CHARACTERISTICS OF ANNELIDS

The phylum *Annelida* consists of about 15,000 species of bilaterally symmetrical, segmented worms. Segmentation is the most distinctive feature of annelids. Like mollusks, annelids have a true coelom, but the coelom in annelids is divided into separate compartments by partitions. Division of the coelom represents an evolutionary advance over the earliest wormlike coelomates. In an undivided coelom, the force of muscle contraction in one part of the body is transmitted to other parts by the fluid in the coelom. A segmented coelom enables different parts of the body to contract or expand independently. In addition, duplication of some of the organ systems in each segment provides a form of insurance against injury. If one segment becomes disabled, the others can still function.

Most annelids have external bristles called **setae** (SEET-ee) (singular, *seta*), and some have fleshy protrusions called **parapodia** (PAR-uh-POH-dee-uh) (singular, *parapodium*). Both of these structures are visible in Figure 35-10. The number of setae and the presence or absence of parapodia provide the basis for dividing annelids into three classes: Oligochaeta (AHL-uh-goh-KEET-uh), Polychaeta (PAHL-i-KEE-tuh), and Hirudinea (HIR-yoo-DIN-ee-uh). All organ systems are well developed in most members of each class.

CLASS OLIGOCHAETA

Annelids of the class Oligochaeta generally live in the soil or in fresh water and have no parapodia. *Oligochaeta* means “few bristles,” and as the name suggests, these annelids have a few setae on each segment. The most familiar member of the class Oligochaeta is the earthworm. As you read about the earthworm, look for adaptations that enable this animal to lead a burrowing life.

OBJECTIVES

- **Identify** the structures that provide the basis for dividing annelids into three classes.
- **List** the advantages of body segmentation.
- **Describe** the structural adaptations of earthworms.
- **Compare** the three classes of annelids.

VOCABULARY

seta
parapodium
crop
gizzard
typhlosole
aortic arch
nephridium
clitellum
seminal receptacle
chitin

FIGURE 35-10

Numerous setae help this bearded fireworm, *Hermodice carunculata*, move through its environment. The setae extend from fleshy flaps called *parapodia*. Also known as bristle worms, bearded fireworms are members of the class Polychaeta.



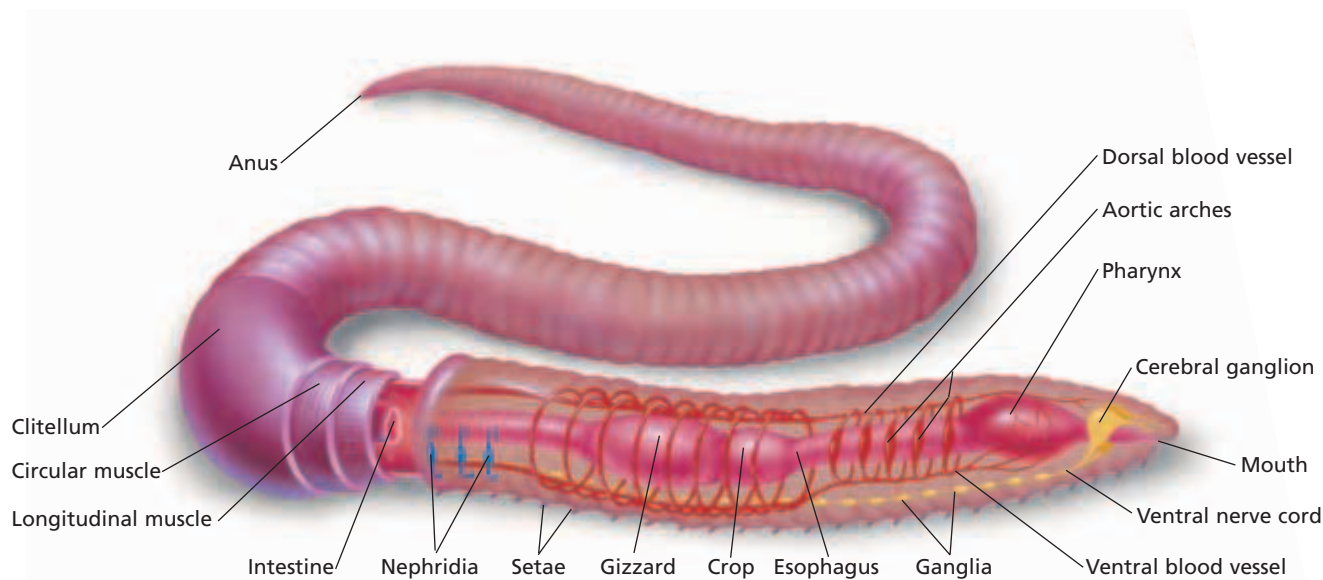


FIGURE 35-11

The segmentation of annelids, such as this earthworm, is visible both externally and internally. Some of the internal structures, such as ganglia and nephridia, are repeated in each segment.

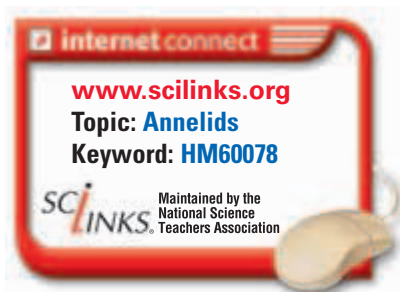
Structure and Movement

An earthworm's body is divided into more than 100 segments, most of which are virtually identical. Figure 35-11 shows that circular and longitudinal muscles line the interior body wall of an earthworm. To move, the worm anchors some of the middle segments by their setae and contracts the circular muscles in front of those segments. Contraction of the circular muscles increases the pressure of the coelomic fluid in those segments. This increased pressure elongates the animal and pushes the anterior end forward. Setae in the anterior segments then grip the ground as the longitudinal muscles contract, pulling the posterior segments forward. This method of locomotion in earthworms is an example of the kind of movement made possible by segmentation.

Feeding and Digestion

Earthworms ingest soil as they burrow through it. Soil is sucked into the mouth by the muscular pharynx. The soil then passes through a tube called the *esophagus* (ee-SAHF-uh-guhs) to a temporary storage area known as the **crop**. From the crop, the soil moves to a thick, muscular part of the gut called the **gizzard**. Find these parts of the digestive tract in Figure 35-11. The gizzard grinds the soil, releasing and breaking up organic matter. As the soil passes through the long intestine, digested organic compounds and nutrients in the soil are absorbed by the blood. An infolding of the intestinal wall called the **typhlosole** (TIF-luh-SOHL) increases the surface area available for digestion and absorption. Undigested material is eliminated from the earthworm's body through the anus.

Earthworms play an important role in maintaining the fertility of soil. By decomposing dead leaves and other organic materials, earthworms help release nutrients into the soil. The burrows made by earthworms allow air to penetrate into the soil, bringing oxygen to plant roots and soil microorganisms. Earthworms also loosen the soil, making it easier for roots to grow and for water to seep into the soil.



Circulation

A closed circulatory system transports oxygen, carbon dioxide, nutrients, and wastes through the body of an earthworm. The blood travels toward the posterior end through a ventral blood vessel and then returns to the anterior end through a dorsal blood vessel. As you can see in Figure 35-11, five pairs of muscular tubes, the **aortic** (ay-AWR-tik) **arches**, link the dorsal and ventral blood vessels near the anterior end of the worm. Contractions of the dorsal blood vessel and the aortic arches force blood through the circulatory system.

Respiration and Excretion

Oxygen and carbon dioxide diffuse directly through the skin, which contains many small blood vessels. This exchange of gases can take place only if the skin is moist. Therefore, earthworms avoid dry ground and extreme heat. Secretions of mucus and the presence of a thin cuticle also help keep an earthworm's skin moist.

Earthworms eliminate cellular wastes and excess water through excretory tubules called **nephridia** (nee-FRID-ee-uh), some of which are shown in Figure 35-11. Each segment, except the first three and the last one, contains a pair of nephridia. As coelomic fluid passes through the nephridia, some of the water is reabsorbed by blood vessels. The remaining fluid and the wastes dissolved in it are released from the body through pores on the ventral surface.

Neural Control

The nervous system of an earthworm consists of a chain of ganglia connected by a ventral nerve cord. Most body segments contain a single ganglion. Nerves branching from each ganglion carry impulses to the muscles and from the sensory cells in that segment. In the most anterior segments, several ganglia are fused to form the cerebral ganglia, or brain, as you can see in Figure 35-11. One of the main functions of the cerebral ganglia is to process information from sensory structures that respond to light, touch, chemicals, moisture, temperature, and vibrations. Although these sensory structures are found in all segments, they are concentrated at the anterior end.

Reproduction

Earthworms are hermaphrodites, but an individual worm cannot fertilize its own eggs. Mating occurs when two earthworms press their ventral surfaces together with their anterior ends pointing in opposite directions. The worms are held together by their setae and by a film of mucus secreted by each worm's **clitellum** (klie-TEL-uhm). The clitellum, also shown in Figure 35-11, is a thickened section of the body. Each earthworm injects sperm into the mucus. The sperm from each worm move through the mucus to the **seminal receptacle** of the other, where the sperm are stored. The worms then separate. After several days, the clitellum of each worm secretes a tube of mucus and a tough carbohydrate known as **chitin** (KIE-tin).

Word Roots and Origins

nephridium

from the Greek *nephros*, meaning "kidney," and *idion*, meaning "small"

As this tube slides forward, it picks up the worm's eggs and the stored sperm from the other worm. Fertilization occurs inside the tube, which closes up to form a protective case. The young worms develop inside the case for 2–3 weeks before hatching.

CLASSES POLYCHAETA AND HIRUDINEA

About two-thirds of all annelids are members of the class Polychaeta. *Polychaeta* means “many bristles,” which refers to the numerous setae that help polychaetes move. The setae project from parapodia, some of which function in gas exchange. Polychaetes differ from other annelids in that they have antennae and specialized mouthparts. They are also the only annelids that have a trochophore stage. Most polychaetes live in marine habitats. Some are free-swimming predators that use their strong jaws to feed on small animals. Others feed on sediment as they burrow through it or use their tentacles to scour the ocean bottom for food.

Hirudinea is the smallest class of annelids, consisting of about 500 species of leeches. Most leeches live in calm bodies of fresh water, but some species live among moist vegetation on land. Leeches have no setae or parapodia. At each end of a leech's body is a sucker that can attach to surfaces. By attaching the anterior sucker and then pulling the rest of the body forward, leeches can crawl along solid objects. Aquatic leeches can also swim with an undulating movement of their body. Many leeches are carnivores that prey on small invertebrates, but some species, including the one shown in Figure 35-12, are parasites that suck blood from other animals. After attaching themselves to the skin of their host, parasitic leeches secrete an anaesthetic that prevents the host from feeling their presence. They also secrete a substance that prevents blood from clotting. If undisturbed, a leech can ingest 10 times its own weight in blood.

FIGURE 35-12

The leech *Haemadipsa* sp. is a parasite that sucks blood from animals, including humans. Other leeches are free-living carnivores.



SECTION 2 REVIEW

1. What are the advantages of a segmented body?
2. How are an earthworm's circular and longitudinal muscles used in locomotion?
3. How does an earthworm exchange oxygen and carbon dioxide with its environment?
4. How do polychaetes differ from earthworms?
5. Describe how some leeches are adapted to a parasitic lifestyle.

CRITICAL THINKING

6. **Analyzing Patterns** In earthworms, blood flows in opposite directions in the dorsal and ventral vessels. How is this helpful to the animal?
7. **Applying Information** Why do you see so many earthworms after a long rainy period?
8. **Recognizing Differences** How is the form of parasitism shown in some leeches different from that of a tapeworm or a liver fluke?

CHAPTER HIGHLIGHTS

SECTION 1

Mollusca

- Mollusks and annelids have true coeloms. Most aquatic mollusks and annelids develop from a trochophore.
- A mollusk's body is divided into the head-foot and the visceral mass, which contains the internal organs. Most mollusks have at least one shell, which is secreted by a layer of epidermis called the mantle. Aquatic mollusks have gills through which gas exchange takes place.
- The main feeding adaptation of most mollusks is the radula, a tonguelike structure that is modified in different species for scraping, drilling, or harpooning.
- Gastropods undergo torsion—the visceral mass twists during larval development. Snails and most other gastropods have a single shell, while some gastropods, such as slugs and nudibranchs, lack shells. Gastropods move by means of wavelike, muscular contractions of the foot and have an open circulatory system.
- Bivalves have a shell that is divided into two valves, which can be pulled together by contracting powerful adductor muscles. Bivalves lack a distinct head and have no radula. Most are sessile and filter food from the water. In clams, water enters through an incurrent siphon and exits through an excurrent siphon. Food is strained from the water as it passes through the gills.
- Cephalopods, including octopuses and squids, are free-swimming, predatory mollusks with numerous tentacles. They have an advanced nervous system with a large brain and well-developed sensory organs. Cephalopods have a closed circulatory system and do not pass through a trochophore stage during development.

Vocabulary

trochophore (p. 705)
visceral mass (p. 706)
mantle (p. 706)
mantle cavity (p. 706)

ganglion (p. 706)
radula (p. 706)
gastropod (p. 707)
hemolymph (p. 707)

hemocoel (p. 707)
bivalve (p. 708)
incurrent siphon (p. 709)
excurrent siphon (p. 709)

cephalopod (p. 710)

SECTION 2

Annelida

- Annelids have a true coelom and a body that is divided into many segments. Most annelids have external bristles called *setae*, and some have fleshy protrusions called *parapodia*.
- The number of setae and the presence or absence of parapodia provide the basis for dividing annelids into three classes: Oligochaeta, Polychaeta, and Hirudinea.
- Segmentation enables the different parts of the body to carry out various functions independently. It also protects the organism in case of injury because uninjured segments can continue to function.
- Members of the class Oligochaeta generally live in the soil or in fresh water. They have no parapodia and relatively few setae.
- The most familiar member of the class Oligochaeta is the earthworm, which feeds on organic matter as it burrows through the soil. Earthworms have a closed circulatory system. They exchange gases through their skin and eliminate cellular wastes and excess water through excretory tubules called *nephridia*.
- Polychaetes have numerous setae that project from parapodia. They also have antennae and specialized mouthparts, and they pass through a trochophore stage during their development. Most polychaetes live in the ocean.
- Members of the class Hirudinea, leeches live in fresh water or on land. They have no setae or parapodia. Many leeches are carnivores that prey on small invertebrates, but some are bloodsucking parasites.

Vocabulary

seta (p. 713)
parapodium (p. 713)
crop (p. 714)

gizzard (p. 714)
typhlosole (p. 714)
aortic arch (p. 715)

nephridium (p. 715)
clitellum (p. 715)
seminal receptacle (p. 715)

chitin (p. 715)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
 - crop* and *gizzard*
 - gastropod* and *cephalopod*
 - setae* and *parapodia*
 - clitellum* and *seminal receptacle*
 - incurrent siphon* and *excurrent siphon*
- Explain the relationship between *hemolymph* and *hemocoel*.
- Choose the term that does not belong in the following group, and explain why it does not belong: *incurrent siphon*, *excurrent siphon*, *trochophore*, and *crop*.
- Word Roots and Origins** The word *chitin* is derived from the Greek *chiton*, meaning “tunic.” Using this information, explain why the term *chitin* is a good name to describe this compound.

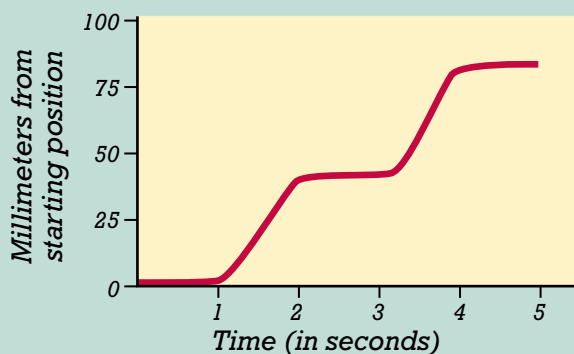
UNDERSTANDING KEY CONCEPTS

- Identify** the functions performed by the cilia on a free-swimming trochophore.
 - List** the main parts in the basic body plan of a mollusk.
 - Describe** the main feeding adaptation of mollusks.
 - Identify** whether or not each of the following mollusks has a shell: nudibranch, clam, snail, and slug.
 - Compare** the body structure of a cephalopod to that of a gastropod.
 - Distinguish** between polychaetes and other annelids. Which characteristics set polychaetes apart?
 - Explain** how segmentation aids movement in earthworms.
 - Summarize** why earthworms require a moist environment to survive.
 - Describe** the organization of an earthworm’s nervous system.
 - Identify** a typical environment in which one member of each of the three classes of annelids may be found.
 - Summarize** the characteristics shared by mollusks and annelids.
- CONCEPT MAPPING** Use the following terms to create a concept map that shows the major characteristics of the phylum Annelida: *annelids*, *segmented worms*, *coelomates*, *setae*, *parapodia*, *Oligochaeta*, *Polychaeta*, *Hirudinea*, *earthworm*, *bearded fireworm*, and *leech*.

CRITICAL THINKING

- Applying Information** If a grain of sand or other irritant gets inside the shell of a bivalve, the mantle coats the irritant with the material that lines the inner layer of the shell. Many layers of this material form a pearl. However, making a pearl consumes resources that an oyster could use for other purposes, such as strengthening its shell. Of what advantage is it to an oyster to make a pearl?
- Inferring Relationships** Clams reproduce by releasing sperm and eggs into the water. How might this process affect the reproductive success of these mollusks? Would you expect clams to release many sperm and eggs or only a few?
- Making Comparisons** Clams are aquatic, and earthworms are terrestrial. Nevertheless, the feeding methods of clams and earthworms are somewhat similar. Explain how they are similar.
- Interpreting Graphics** The graph below plots the movement of the anterior end of an earthworm over time. Was the anterior end moving or stationary during the periods represented by the horizontal sections of the graph? Which of the earthworm’s sets of muscles were contracting during the periods represented by the horizontal sections? Explain.

Earthworm Movement



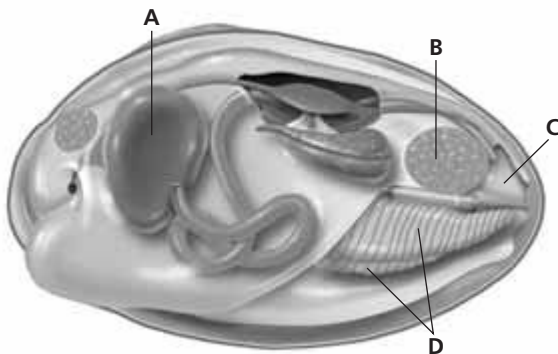


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- Why do terrestrial snails and slugs need an environment with a high moisture content?
 - to avoid drying out
 - to see and hear better
 - to swim more efficiently
 - to avoid being eaten by birds
- Which of the following are the only mollusks with a closed circulatory system?
 - snails
 - bivalves
 - gastropods
 - cephalopods
- Annelids are divided into three classes based partly on the number of which of the following?
 - setae
 - segments
 - nephridia
 - aortic arches
- Parapodia are a distinguishing characteristic of which class of annelids?
 - Bivalvia
 - Hirudinea
 - Polychaeta
 - Oligochaeta

INTERPRETING GRAPHICS: The diagram below shows the internal structure of a bivalve. Use the diagram to answer the questions that follow.

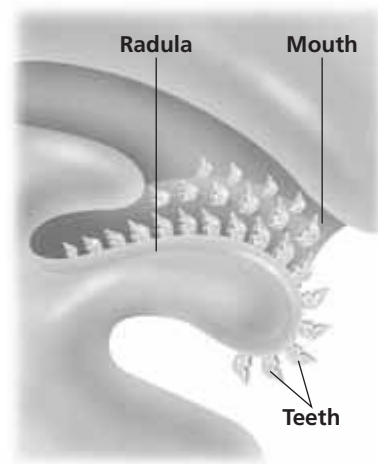


- Which of the structures is involved in respiration and feeding?
 - A
 - B
 - C
 - D
- What is structure B?
 - a gill
 - a siphon
 - the mantle
 - an adductor muscle

DIRECTIONS: Complete the following analogy.

7. Nephridium : excretion :: clitellum :
- digestion
 - circulation
 - respiration
 - reproduction

INTERPRETING GRAPHICS: The diagram below shows the mouth structure of a class of organisms. Use the diagram to answer the question that follows.



8. In which class of organisms is the mouth structure shown above most likely to be found?
- bivalves
 - hirudines
 - gastropods
 - polychaetes

SHORT RESPONSE

During larval development, gastropods undergo a process called *torsion*.

Describe the outcome of torsion.

EXTENDED RESPONSE

Earthworms have three main structural features that enable movement.

Part A Describe these features.

Part B Explain how they work together to enable movement.

Test TIP

In tests, analogies compare pairs of items, such as *kitten : cat :: puppy : ?* Before looking at the possible answers, identify how the first pair of terms is related. For example, since a kitten is a young cat, the relationship is baby : adult, so the correct answer would be *puppy : dog*.

Testing Earthworm Behavior

OBJECTIVES

- Observe a live earthworm.
- Test how an earthworm responds to light, moisture, and ammonia.
- Test the effect of temperature on heart rate.

PROCESS SKILLS

- observing
- hypothesizing
- experimenting
- collecting data
- analyzing data

MATERIALS

- safety goggles
- live earthworm
- shallow pan
- paper towels
- medicine dropper
- hand lens
- black paper or piece of cardboard
- fluorescent lamp
- 2 cotton swabs
- 3% aqueous ammonia solution
- 15 cm Petri dish
- thermometer
- stopwatch or clock with second hand
- 2 plastic tubs for water baths
- warm tap water
- ice cubes

Background

1. How does an earthworm benefit from cephalization?
2. Describe how gases enter and exit an earthworm's body.

PART A Observing an Earthworm

1.  **CAUTION** You will be working with a live animal. Be sure to treat it gently and to follow directions carefully. Place a moist paper towel in a pan, and place an earthworm on the paper towel. **CAUTION** Rinse the earthworm frequently with

water from a medicine dropper to prevent the worm from drying out.

2. Observe the behavior of the earthworm for a few minutes. Identify the earthworm's anterior and posterior ends by watching it move in the pan. As the worm crawls around in the pan, it will lead with its anterior end.
3. Locate the earthworm's clitellum. Is the clitellum closer to the anterior end or the posterior end? What is the function of the clitellum?
4. Identify the earthworm's dorsal and ventral surfaces by gently rolling the worm over. The dorsal surface will be on top after the worm rights itself.
5. Pick up the earthworm, and feel its skin with your fingers. One surface of the earthworm should feel slightly rougher than the other. The roughness is due to the hairlike setae that project from the earthworm's skin. On which surface are the setae located? Use a hand lens to examine the setae up close.
6. Return the earthworm to the pan, and use the hand lens to find a thick purple line running along the dorsal surface of the worm. This line is the dorsal blood vessel. Does the earthworm have an open or a closed circulatory system?
7. Draw a picture of the earthworm, and label its anterior and posterior ends, dorsal and ventral surfaces, clitellum, setae, and dorsal blood vessel.

PART B Earthworm Responses to Stimuli

8. In this part of the laboratory investigation, you will test the earthworm's responses to three different stimuli. With your lab partners, develop three separate hypotheses that describe an earthworm's responses to light, moisture, and a base. In your lab report, make a data table like the one on the next page.
9. To test the earthworm's response to light, cover half of the pan with black paper or cardboard. Check the lighting in the room. The light must be low and even during this test. Position the fluorescent lamp over the uncovered portion of the pan. Place the earthworm in the center of the pan, and observe its movements. Record your observations in your data table.

OBSERVATIONS OF EARTHWORM BEHAVIOR

Behavior	Observations
Response to light	
Response to moisture	
Response to water on a swab	
Response to ammonia	

10. To test the earthworm's response to moisture, turn off the fluorescent lamp, move it away from the pan, and remove the paper covering half of the pan. Place a piece of dry paper towel on one side of the pan and a piece of wet paper towel on the other side of the pan. Lay the earthworm across the two paper towels. Observe the earthworm's response to the two environments, and record your observations in your data table.
11. To test the earthworm's response to ammonia, make sure the paper towels on both sides of the pan are wet. Moisten a cotton swab with water. Hold the cotton swab first near the earthworm's anterior end and then near its posterior end. Do not touch the earthworm with the swab. Record your observations in your data table.
12.   **CAUTION** Wear safety goggles at all times during the following procedure. If you get ammonia on your skin or clothing, wash it off at the sink while calling to your teacher. If you get ammonia in your eyes, immediately flush it out at the eyewash station while calling to your teacher. Moisten a different cotton swab with ammonia solution, and repeat step 11. Do not touch the earthworm with the swab or the ammonia solution. Record your observations in your data table.

PART C Effect of Temperature on Heart Rate

13.  In this part of the laboratory investigation, you will examine how an earthworm's heart rate changes as its body temperature changes. Add enough tap water to a Petri dish to barely cover the bottom of the dish. Place an earthworm in the dish.

14. Using a hand lens, look for rhythmic contractions of the dorsal blood vessel. Each contraction represents a single heartbeat. Calculate the worm's heart rate by counting the number of contractions that occur in exactly one minute. This is easiest to do if one person counts contractions while another person watches a stopwatch or clock.
15.  **CAUTION** Glassware is fragile. Notify the teacher of broken glass or cuts. Do not clean up broken glass or spills involving broken glass unless the teacher tells you to do so. Place a thermometer next to the worm in the Petri dish, and measure the temperature. Record the worm's heart rate and the temperature in a table on the chalkboard.
16. Float the Petri dish containing the worm on top of either a warm-water bath or a cold-water bath. Place the thermometer next to the worm in the Petri dish, and watch the temperature until it reaches either 30°C (for the warm-water bath) or 10°C (for the cold-water bath).
17. Remove the Petri dish from the water bath, and immediately begin counting heartbeats for exactly one minute. After one minute, measure the temperature in the dish again. Calculate the average temperature to the nearest degree. Record the worm's heart rate and the average temperature in the table on the chalkboard.
18. Using data from the whole class, graph heart rate as a function of temperature. Draw the best-fit curve through the points.
19.   Return the earthworm to the container from which you obtained it. Clean up your materials, and wash your hands before leaving the lab.

Analysis and Conclusions

1. State whether your hypotheses in Part B were supported by your observations. Explain.
2. What is the adaptive advantage of the earthworm's responses to light and moisture?
3. List variables that, if not controlled, might have affected the results in Part B.
4. Describe the relationship between the earthworm's heart rate and temperature as shown by your graph.

Further Inquiry

Design an experiment to determine which colors of light an earthworm is sensitive to or which areas on an earthworm are sensitive to light.

ARTHROPODS



The jointed appendages and hard exoskeleton of this red reef lobster, *Enoplometopus occidentalis*, are characteristic of arthropods.

SECTION 1 *Phylum Arthropoda*

SECTION 2 *Subphylum Crustacea*

SECTION 3 *Subphyla Chelicerata and Myriapoda*

PHYLUM ARTHROPODA

Two-thirds of all animal species belong to the phylum Arthropoda (ahr-THRAHP-uh-duh). This phylum contains a variety of bilaterally symmetrical coelomates, including lobsters, crabs, spiders, and insects. These animals have adapted to almost every environment on Earth.

CHARACTERISTICS OF ARTHROPODS

The members of the phylum Arthropoda are called **arthropods** (AHR-thruh-PAHDS). Like annelids, arthropods are segmented animals. In arthropods, however, the body segments bear jointed extensions called **appendages**, such as legs and antennae. In fact, *arthropod* means “jointed foot.”

Another distinguishing feature of arthropods is their exoskeleton, which provides protection and support. As Figure 36-1 shows, the arthropod exoskeleton is made up of three layers that are secreted by the epidermis, which lies just beneath the layers. The waxy outer layer is composed of a mixture of protein and lipid. It repels water and helps keep terrestrial arthropods from drying out. The middle layer, which provides the primary protection, is made mainly of protein and **chitin**, a tough polysaccharide. In some arthropods, the middle layer is hardened by the addition of calcium carbonate. The inner layer also contains protein and chitin, but its flexibility at the joints allows arthropods to move freely. Muscles that attach to the inner layer on either side of the joints move the body segments.

Arthropods show a high degree of cephalization. Recall that *cephalization* is the concentration of sensory and brain structures at the anterior end. A variety of segmented appendages around the mouth serve as sensors and food handlers. Most arthropods have segmented antennae that are specialized for detecting chemicals. Most arthropods also have **compound eyes**—eyes made of many individual light detectors, each with its own lens. In addition, many arthropods have simpler structures that sense light intensity. These sensory structures on the head send nerve impulses to the brain, which coordinates the animal’s actions. As in annelids, impulses travel from the brain along a ventral nerve cord, which links ganglia in the other segments of the body.

All arthropods have open circulatory systems. Recall that in an open circulatory system, the heart pumps circulatory fluid through vessels that empty into spaces in the body.

OBJECTIVES

- **Describe** the distinguishing characteristics of arthropods.
- **Explain** the process of molting in an arthropod.
- **List** the five major subphyla of the phylum Arthropoda.

VOCABULARY

arthropod
appendage
chitin
compound eye
molting
trilobite
tagma
mandible
chelicera

FIGURE 36-1

The arthropod exoskeleton consists of three layers that cover the epidermis. Wax in the outer layer is secreted by wax glands. Sensory hairs projecting from the exoskeleton allow arthropods to respond to vibrations and chemicals in their environment.

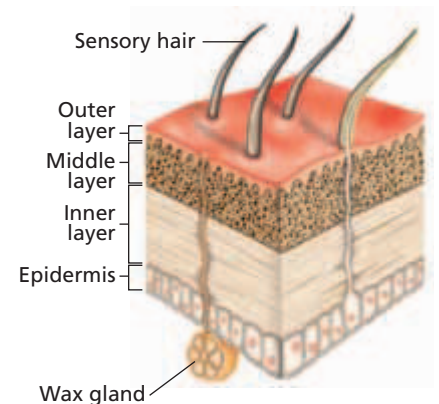




FIGURE 36-2

This green cicada, *Tibicen superbus*, is in the process of molting. The outer layer of its old exoskeleton appears brown.

Molting

A rigid exoskeleton limits the size to which an arthropod can grow. So, each arthropod periodically sheds its exoskeleton and makes a new one in the process of **molting**. An arthropod goes through many cycles of molting during its life. Figure 36-2 shows an insect in the process of molting.

A cycle of molting begins as the tissues of an arthropod gradually swell. When the pressure inside the exoskeleton is very strong, a hormone that triggers molting is produced. In response to this hormone, the cells of the epidermis secrete enzymes that digest the inner layer of the exoskeleton. At the same time, the epidermis begins to make a new exoskeleton by using the digested material. Eventually, the outer layer of the old exoskeleton loosens, breaks apart, and is shed. The new exoskeleton, which is flexible at first, stretches to fit the enlarged animal.

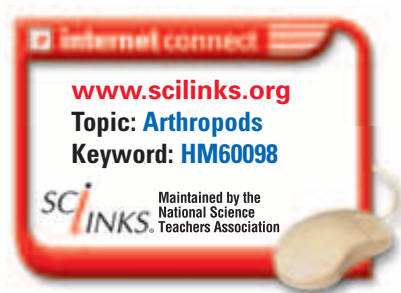
The new exoskeleton takes a few hours to a few days to become as hard as the one that it replaced. During this time, the animal is vulnerable to predators and, if the arthropod is terrestrial, to desiccation. For these reasons, arthropods usually stay in hiding from the time that they begin to molt until their new exoskeleton has hardened. The “soft-shelled crabs” sold in some restaurants are crabs that have been caught immediately after molting.

EVOLUTION AND CLASSIFICATION

Animals having arthropod characteristics first appeared about 545 million years ago. Because all arthropods have a true coelom, an exoskeleton, and jointed appendages, biologists infer that all arthropods evolved from a common ancestor. However, biologists are still uncertain about the order in which subgroups of arthropods evolved and the exact relationships between the subgroups. The phylogenetic diagram in Figure 36-3 shows possible evolutionary relationships between the highly diverse arthropod subgroups.

The similar characteristics of many modern subgroups of arthropods may be the result of convergent evolution. For example, ancient and extinct arthropods, such as **trilobites**, had many body segments and one pair of appendages on each segment. However, most living arthropod species have some segments that lack appendages and some segments that are fused into a larger structure called a **tagma** (plural, *tagmata*). The tagmata tend to be specialized for functions such as feeding, locomotion, and reproduction.

Arthropods are usually divided into five subphyla on the basis of differences in development and in the structure of appendages, such as mouthparts. The two major types of mouthparts are **mandibles**, which are jawlike, and **chelicerae** (singular, *chelicera*), which are pincerlike. The five main subphyla are Trilobita, Crustacea, Chelicerata, Myriapoda, and Hexapoda, as Figure 36-3 shows.



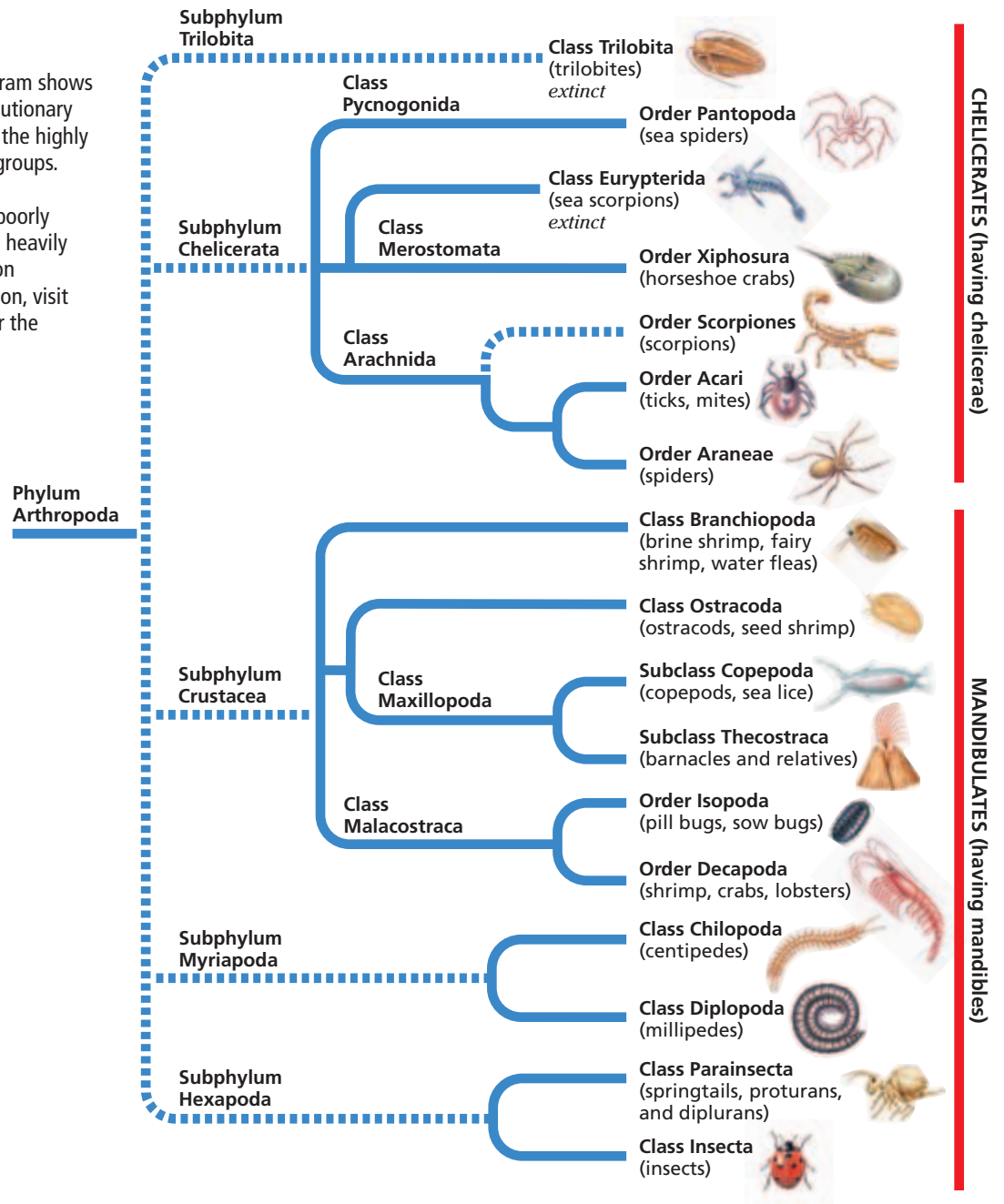
Word Roots and Origins

chelicera

from the Greek *chele*, meaning “claw,” and *keras*, meaning “horn”

FIGURE 36-3

This phylogenetic diagram shows hypotheses of the evolutionary relationships between the highly diverse arthropod subgroups. Dashed lines indicate relationships that are poorly understood or that are heavily debated. For updates on phylogenetic information, visit go.hrw.com and enter the keyword HM6 Phylo.



SECTION 1 REVIEW

1. What characteristics do all arthropods share?
2. What is the main function of each layer of an arthropod's exoskeleton?
3. What is a compound eye?
4. What is molting?
5. Identify the difference between chelicerae and mandibles.
6. Identify examples of organisms in each of the five main subphyla of arthropods.

CRITICAL THINKING

7. **Inferring Relationships** What is the adaptive advantage of having a layered exoskeleton?
8. **Analyzing Concepts** After its old exoskeleton has been shed but before the new one has hardened, an aquatic arthropod absorbs water and swells. How is this behavior adaptive?
9. **Interpreting Graphics** Study Figure 36-3. Are pill bugs more closely related to centipedes or to barnacles? Explain your answer.

SECTION 2

OBJECTIVES

- **Describe** the characteristics of crustaceans.
- **Compare** aquatic crustaceans with terrestrial crustaceans.
- **Explain** the functions of the appendages on a crayfish.
- **Summarize** digestion, respiration, circulation, excretion, and neural control in crayfish.

VOCABULARY

nauplius
cirrus
isopod
decapod
cephalothorax
thorax
carapace
abdomen
antenna
antennule
cheliped
swimmeret
telson
uropod
digestive gland
green gland

SUBPHYLUM CRUSTACEA

The subphylum Crustacea contains about 38,000 known species. Crustaceans are abundant in oceans, lakes, and rivers, and a few species live on land. Some crustaceans are sessile, while others move by walking on legs, swimming with paddle-like appendages, or drifting with the currents.

CHARACTERISTICS

Crustaceans are so diverse that their single defining characteristic is having two pairs of antennae. Also, most crustaceans have a pair of jawlike, chewing mouthparts called *mandibles*. In most crustaceans, each body segment has a pair of appendages, and at least some of those appendages are branched. Although some crustaceans have 60 or more body segments, most crustaceans have 16 to 20 segments, which are fused into several tagmata.

Some small crustaceans, such as pill bugs, respire through the thin areas of their exoskeleton. Larger crustaceans, such as crayfish and lobsters, use gills to respire. The exoskeletons of aquatic crustaceans, such as lobsters, often contain large amounts of calcium carbonate and thus are extremely hard.

During the development of many crustaceans, the embryo becomes a free-swimming larva called a **nauplius** (NAH-plee-uhs), which looks quite different from an adult of the species. As Figure 36-4 shows, a nauplius has three pairs of appendages and a single eye in the middle of its head. Through a series of molts, the nauplius eventually takes on the adult form.

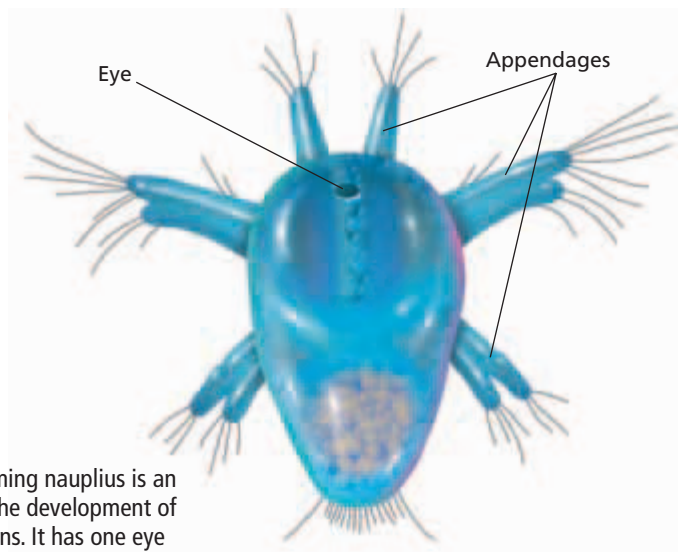


FIGURE 36-4

The free-swimming nauplius is an early stage in the development of most crustaceans. It has one eye and three pairs of appendages.

DIVERSITY OF CRUSTACEANS

Crustaceans exist in a range of sizes, but most are small. For example, copepods (KOH-puh-PAHDZ), as shown in Figure 36-5a, are no larger than the comma in this sentence. At the other end of the size spectrum is the Japanese spider crab, shown in Figure 36-5b. With a leg span of 4 m (13 ft), this crab is the largest living arthropod.



(a) Copepod, *Cyclops* sp.



(b) Japanese spider crab, *Macrocheira kaempferi*



(c) Goose barnacle, *Lepas anatifera*

Aquatic Crustaceans

Copepods are extremely abundant in marine environments and may be the most abundant animals in the world. Copepods are an important part of the ocean's *plankton*, the collection of small organisms that drift or swim weakly near the surface of the water.

In freshwater environments, on the other hand, much of the plankton is composed of crustaceans known as *water fleas*, which are about the size of copepods. A common type of water flea is a member of the genus *Daphnia*.

Barnacles, such as the one shown in Figure 36-5c, are marine crustaceans that are sessile as adults. Free-swimming barnacle larvae attach themselves to rocks, piers, boats, sea turtles, whales, and just about any other surface. They then develop a shell composed of calcium carbonate plates that completely encloses the body in most species. Their swimming appendages develop into six pairs of long legs called **cirri** (SIR-IE) (singular, *cirrus*). The cirri extend through openings in the shell, sweeping small organisms and food particles from the water and directing them to the mouth.

Terrestrial Crustaceans

Sow bugs and pill bugs are terrestrial members of a group of crustaceans called **isopods**. Terrestrial isopods lack adaptations for conserving water, such as a waxy cuticle, and can lose water quickly through their exoskeleton. Therefore, they live only in moist environments, such as under leaves and rocks, in crevices around garden beds, and in the spaces between house foundations and sidewalks. In addition, pill bugs are capable of rolling into a ball when disturbed or threatened with desiccation, as shown in Figure 36-6. Sow bugs and pill bugs generally feed on decaying vegetation, but they may also eat garden bulbs, vegetables, and fruits that lie on or in the soil.

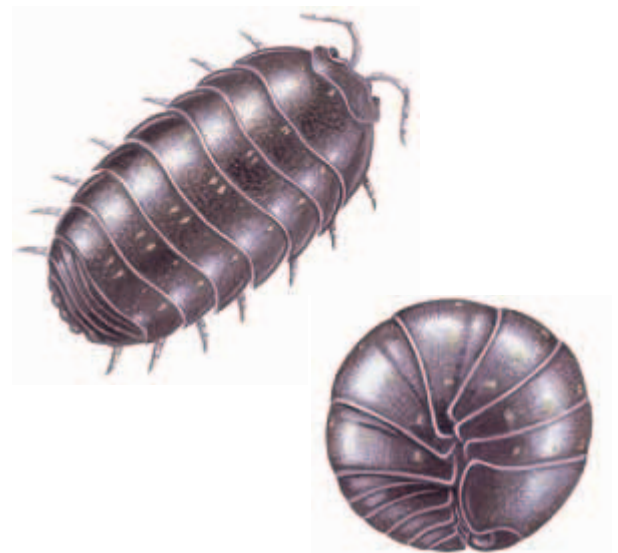


FIGURE 36-5

Crustaceans include tiny species, such as the copepod (a), as well as giant species, such as the Japanese spider crab (b). Some aquatic species, such as barnacles (c), attach themselves to submerged marine surfaces.

FIGURE 36-6

Pill bugs are isopod terrestrial crustaceans that are often found in moist environments with decaying vegetation. Pill bugs and sow bugs look similar, but only pill bugs roll up when disturbed.

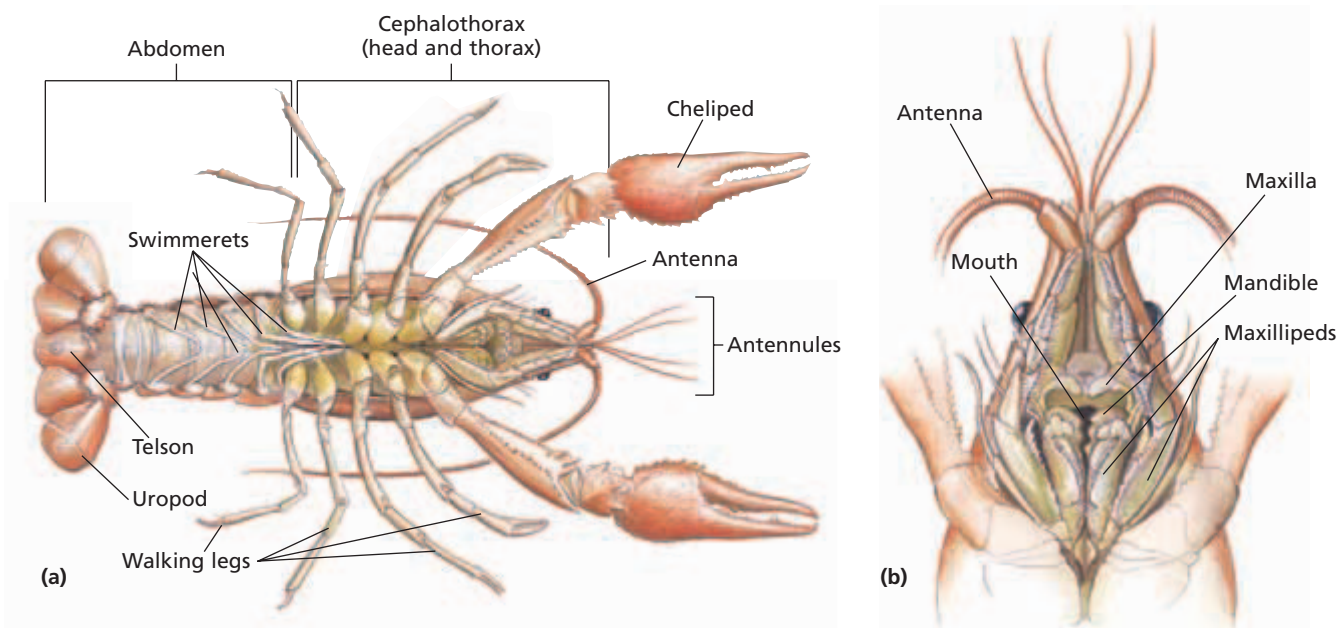


FIGURE 36-7

(a) Each of the 20 body segments of a crayfish has a pair of appendages, as seen in this ventral view. (b) Some of the appendages of the anterior cephalothorax are visible in this closer ventral view.

THE CRAYFISH

The crayfish is a freshwater crustacean that is well studied because of its size and abundance. Crayfish are structurally similar to lobsters, which are marine crustaceans. Crayfish, lobsters, crabs, and shrimp are **decapods** (DEK-uh-PAHDZ), or members of the order Decapoda. *Decapoda* means “10 feet,” a name used because these crustaceans have five pairs of legs that are used for locomotion.

External Structure

The crayfish’s external structure is shown in Figure 36-7. The body is divided into two major sections: the abdomen and the cephalothorax (SEF-uh-loh-THAWR-AKS). The **cephalothorax** consists of two tagmata: the head, which has five segments, and the **thorax**, which has eight segments and lies behind the head. The dorsal exoskeleton over the cephalothorax is a single, tough covering known as the **carapace** (KAR-uh-PAYS). The **abdomen**, the tagma behind the cephalothorax, is divided into six segments.

A pair of appendages is attached to each segment of the crayfish. Several pairs have specialized functions, as summarized in Table 36-1. The two pairs of **antennae** include the branched **antennules**, which serve as feelers sensitive to touch, taste, and balance. The long antennae are also feelers that respond to touch and taste. Crayfish use a pair of *mandibles* to chew food and use two pairs of *maxillae* (maks-IL-ee) and three pairs of *maxillipeds* (maks-IL-i-PEDS) to manipulate food. The posterior pair of maxillae also function in respiration, and the maxillipeds are sensitive to touch and taste. The most anterior pair of walking legs, the **chelipeds** (KEE-luh-PEDS), end in large pincers used for capturing food and for defense. The other four pairs of walking legs carry the crayfish over solid surfaces; two of these pairs end in small pincers that can grasp small objects. The **swimmerets**, which are attached to the five anterior abdominal segments, create water currents and function in reproduction.

TABLE 36-1
Crayfish Appendages

Appendage	Function
Antennule	touch, taste, and balance
Antenna	touch and taste
Mandible	chewing food
Maxilla	manipulating food and drawing water currents over gills
Maxilliped	touch, taste, and manipulating food
Cheliped	capturing food and defense
Walking leg	locomotion over solid surfaces
Swimmeret	creating water currents and transferring sperm (males)

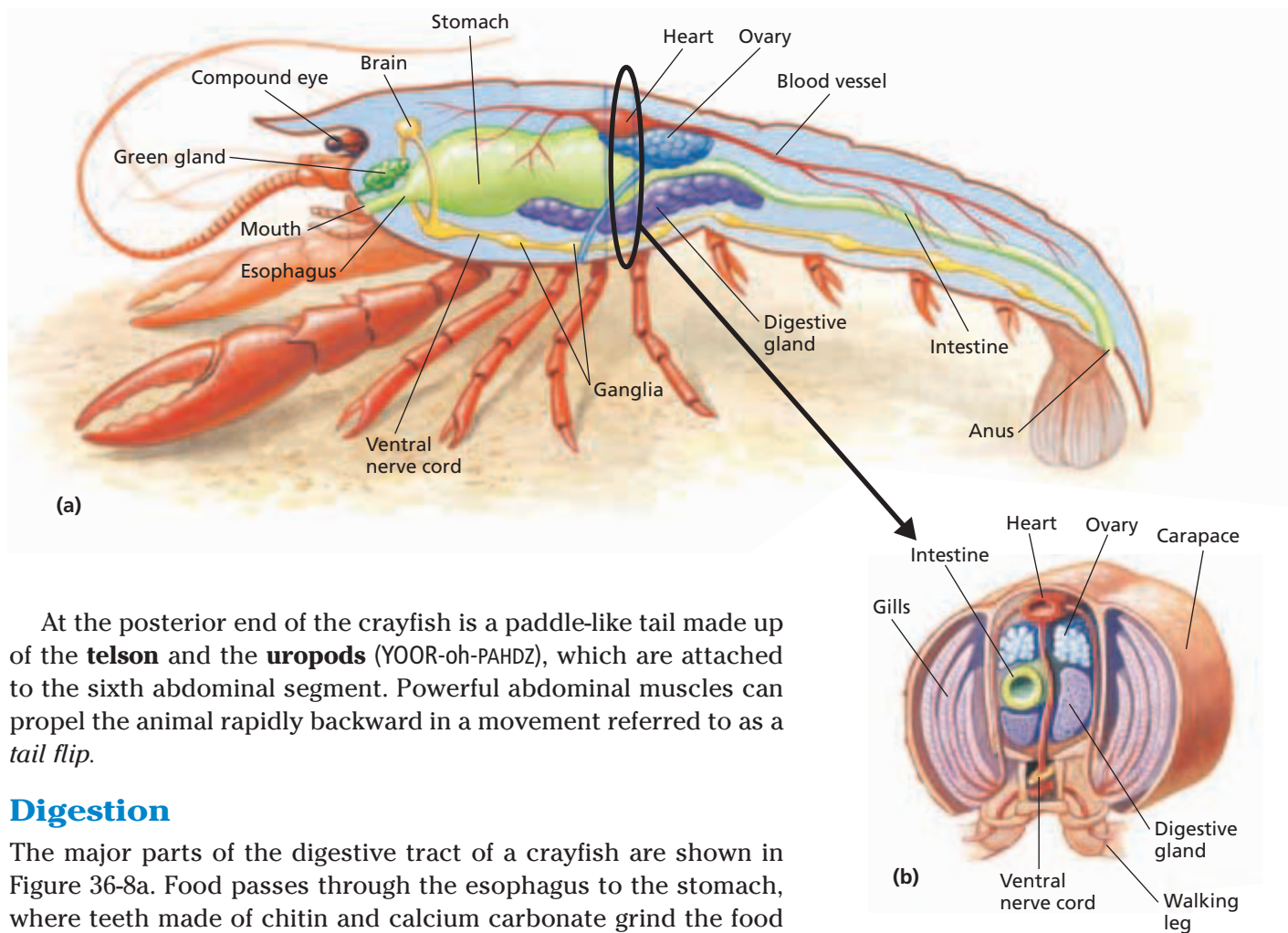


FIGURE 36-8

The major internal organs of a crayfish are seen in this cutaway side view (a) and cross section through the heart region (b).

At the posterior end of the crayfish is a paddle-like tail made up of the **telson** and the **uropods** (YOOR-oh-PAHDZ), which are attached to the sixth abdominal segment. Powerful abdominal muscles can propel the animal rapidly backward in a movement referred to as a *tail flip*.

Digestion

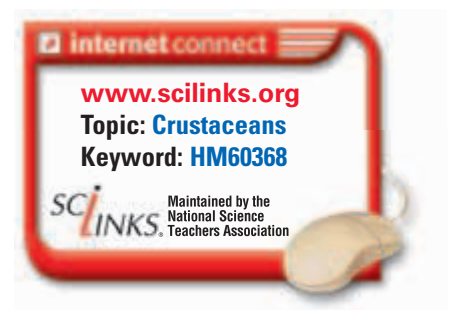
The major parts of the digestive tract of a crayfish are shown in Figure 36-8a. Food passes through the esophagus to the stomach, where teeth made of chitin and calcium carbonate grind the food into a fine paste. After the paste is mixed with enzymes secreted by a **digestive gland** near the stomach, it enters the intestine for further digestion and absorption. Waste leaves through the anus.

Respiration

Like most crustaceans, crayfish have featherlike gills for respiration. Figure 36-8b shows that the gills extend from the base of each walking leg into a chamber under the carapace. As a crayfish walks, its legs circulate water across its gills. Feathery branches on the posterior pair of maxillae also help direct water over the gills. Each gill is covered by an extension of the exoskeleton that is thin enough to permit gases to diffuse across the gill surface.

Circulation

The main components of the crayfish's open circulatory system are shown in Figures 36-8a and 36-8b. In this system, the dorsal heart pumps a circulatory fluid called *hemolymph* into several large vessels that carry the fluid to different regions of the body. The fluid leaves the vessels and enters spaces within the body, where it bathes the various tissues. It then passes through the gills, where it exchanges carbon dioxide with oxygen in the water. From the gills, the fluid returns to the dorsal part of the crayfish and enters the heart.





Quick Lab

Observing Crayfish Behavior

Materials crayfish in container with water, tapping instrument

Procedure



CAUTION Crayfish have powerful pincers that could injure you. Do not place your hand near the crayfish.

1. Gently tap the anterior (head) end of the crayfish, and record your observations.
2. Tap the posterior (tail) end of the crayfish, and record your observations.

Analysis Describe each behavioral response you observed when you tapped the crayfish. How does each behavior aid the animal in its survival?

Excretion

As freshwater organisms, crayfish live in a hypotonic environment. Remember that a *hypotonic environment* is one in which the concentration of solute molecules is lower than that in the organism's cells. Therefore, water constantly enters the tissues of a crayfish by osmosis. This excess water, along with wastes, is eliminated by excretory organs called **green glands**, which are visible in Figure 36-8a. The dilute fluid collected by the green glands leaves the body through a pore at the base of the antennae.

Neural Control

The nervous system of the crayfish is illustrated in Figure 36-8a. It is typical of arthropods and is similar to the nervous system of annelids. The crayfish brain consists of a pair of ganglia above the esophagus that receive nerve impulses from the eyes, antennules, and antennae.

Two bundles of nerve fibers extend from the brain and pass around either side of the esophagus to a ganglion that controls the mandibles, maxillae, and maxillipeds. The ventral nerve cord runs posteriorly from this ganglion, connecting a series of ganglia that control the appendages and muscles in the segments of the thorax and abdomen.

Sensory Organs

Crayfish sense vibrations and chemicals in the water with thousands of small sensory hairs that project from the exoskeleton. Sensory hairs are visible in Figure 36-1. These sensory hairs are distributed over the entire body, but they are especially concentrated on the antennules, antennae, mouthparts, chelipeds, and telson.

The compound eyes of a crayfish are set on two short, movable stalks. Each eye has thousands of light-sensitive units, each with its own lens. At the base of the antennules are organs that can detect the animal's orientation with respect to gravity.

SECTION 2 REVIEW

1. What characteristics are shared by most or all crustaceans?
2. What is the important role of copepods in marine ecosystems?
3. What are the functions of the mandibles and the chelipeds on a crayfish?
4. What structural adaptations of crayfish promote effective respiration in water?
5. Describe the type of circulation found in a crayfish.

CRITICAL THINKING

6. **Inferring Relationships** Why are the largest crustaceans found only in aquatic environments?
7. **Applying Information** Barnacles spend most of their adult life attached to a marine surface. What structural adaptations for this lifestyle do barnacles have?
8. **Inferring Relationships** What is the adaptive advantage of having two pairs of antennae?
9. **Predicting Results** What problem would a crayfish have if it were born without a green gland?

SUBPHYLA CHELICERATA AND MYRIAPODA

Unlike crustaceans, nearly all members of the subphyla Chelicerata, Myriapoda, and Hexapoda are terrestrial.

SUBPHYLUM CHELICERATA

The subphylum Chelicerata, made up of *chelicerates*, includes spiders, scorpions, mites, sea spiders, and horseshoe crabs. Chelicerates lack antennae and typically have six pairs of appendages. The first pair of appendages, the *chelicerae*, are modified into pincers or fangs. The major group in Chelicerata is class Arachnida (uh-RAK-ni-duh), which has more than 70,000 species.

Class Arachnida

Members of the class Arachnida, called **arachnids**, include spiders, scorpions, mites, and ticks. Like crayfish, arachnids have a body that is divided into a cephalothorax and an abdomen. The cephalothorax usually bears six pairs of jointed appendages: one pair of chelicerae; one pair of **pedipalps**, which aid in holding food and chewing; and four pairs of walking legs.

Anatomy of a Spider

Spiders range in length from less than 0.5 mm long to as long as 9 cm (3.5 in.) long in some tropical tarantula species. As Figure 36-9 shows, the body of a spider is very narrow between the cephalothorax and the abdomen.

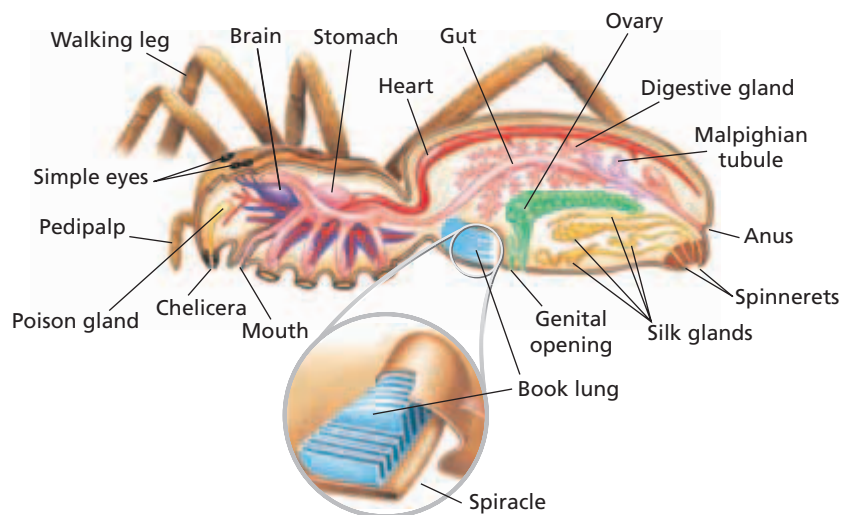


FIGURE 36-9

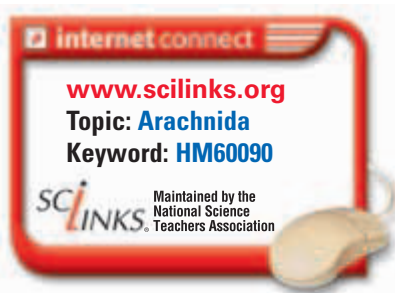
The major internal organs of a female spider are shown in this cutaway side view. The inset shows a closer view of a book lung, one of the spider's adaptations to life on land.

OBJECTIVES

- **List** the characteristics of arachnids, as represented by a spider.
- **Explain** the adaptations that spiders have for a predatory life on land.
- **Identify** the unique characteristics of scorpions, mites, and ticks.
- **Compare** the characteristics of millipedes and centipedes.

VOCABULARY

arachnid
pedipalp
spinneret
book lung
trachea
spiracle
Malpighian tubule



Word Roots and Origins

spiracle

from the Latin *spiraculum*,
meaning "air hole"

FIGURE 36-10

The black widow, *Latrodectus mactans* (a), and the brown recluse, *Loxosceles reclusa* (b), are the only two spiders in the United States whose venom is dangerous to humans. Note the red hourglass-shaped spot on the abdomen of the female black widow and the dark violin-shaped marking on the cephalothorax of the brown recluse.



(a)



(b)

The chelicerae of spiders are modified as fangs and are used to inject venom into prey. The venom is produced by poison glands in the cephalothorax and flows through ducts in the chelicerae to the tips of the fangs. Most spiders have eight simple eyes at the anterior end of the cephalothorax. Each simple eye has a single lens.

On the tip of the abdomen of many spiders are three pairs of organs called **spinnerets**, which are visible in Figure 36-9. Each spinneret is composed of hundreds of microscopic tubes that connect to silk glands in the abdomen. A protein-containing fluid produced in the silk glands hardens into threads as it is secreted from the spinnerets. Spiders use their silk threads to spin webs, build nests, and protect eggs. In some species, spider silk also aids in dispersing spiders from one habitat to another. Young spiders move to new habitats when the wind pulls them through the air by their threads.

The nervous, digestive, and circulatory systems of spiders are similar to those of crustaceans. Because spiders are terrestrial, however, their respiratory system is quite different. In some spiders, respiration occurs in **book lungs**, paired sacs in the abdomen with many parallel folds that resemble the pages of a book. The folds in a book lung provide a large surface area for gas exchange. Other spiders have a system of tubes called **tracheae** (TRAY-kee-ee) that carry air directly to the tissues from openings in the exoskeleton known as **spiracles**. Some spiders have both book lungs and tracheae.

The excretory system of spiders is also modified for life on land. The main excretory organs, called **Malpighian tubules** (mal-PIG-ee-uhn), are hollow projections of the digestive tract that collect body fluids and wastes and carry them to the intestine. After most of the water is reabsorbed, the wastes leave the body in a nearly solid form with the feces. Thus, the Malpighian tubules help spiders conserve water in terrestrial environments.

Life of a Spider

Spiders feed mainly on insects, although some can catch fish, frogs, and even birds. Different species of spiders are adapted to capture their prey in different ways. Some chase after prey, some hide beneath trapdoors waiting for prey to approach, and some snare prey in webs spun from silk. When an insect becomes trapped in the sticky web, the spider emerges from its hiding place near the edge of the web and paralyzes the insect with its venom. Many spiders also immobilize their prey by wrapping them in silk. They can then consume the body fluids of the prey at a later time.

Spider venom is usually harmless to humans, and most spiders bite only when threatened. There are, however, two kinds of spiders in the United States whose bites can be dangerous to humans. They are the black widow and the brown recluse, shown in Figure 36-10. The female black widow has a bright red or orange mark shaped like an hourglass on the ventral surface of its abdomen. The venom attacks the nervous system.

The brown recluse has a violin-shaped mark on the dorsal surface of its cephalothorax. Therefore, it is sometimes called the “violin spider.” The venom of the brown recluse kills and digests the tissues surrounding the bite.

A male spider is usually smaller than a female of the same species. When the male is mature, he transfers sperm to special sacs in the tips of his pedipalps. The sperm are then placed in the seminal receptacle of the female during mating. As soon as mating has occurred, the male darts away. If he is not quick enough, he may be eaten by the female. Eggs are fertilized as they pass out of the female into a silken case that she has spun. The female may carry the egg case with her or attach it to a web or plant. The young spiders hatch in about two weeks and undergo their first molt before leaving the case.

Scorpions

Scorpions, such as the one shown in Figure 36-11, differ from spiders in two ways. Scorpions have large, pincerlike pedipalps, which they hold in a forward position. They also have a segmented abdomen with a large stinger on the last segment, which can be curled over the body. Scorpions usually hide during the day and hunt at night, mostly for insects and spiders. They seize prey with their pedipalps and inject venom into the prey with their stinger. Only a few species of scorpions have venom that can be fatal to humans. Most scorpions live in tropical or semitropical areas, but some are found in dry temperate or desert regions.

Mites and Ticks

Mites and ticks are the most abundant and most specialized arachnids. About 30,000 species have been identified, but the actual number of species may be much larger than that. Unlike spiders and scorpions, mites and ticks have a completely fused cephalothorax and abdomen, with no sign of separation between them.

Most mites are less than a millimeter in length, and some are small enough to live on particles of dust. They can be found in freshwater, marine, and terrestrial habitats. Many mites are entirely free living, while others are parasites during at least part of their life cycle. Spider mites parasitize fruit trees and many other agricultural crops by sucking the fluid from their leaves. The larvae of harvest mites—also known as chiggers—attach themselves to the skin of vertebrates, including humans. They break the skin with their chelicerae and feed on blood, causing swelling and itching. Other species of mites live on the bodies of chickens, dogs, cattle, humans, and other animals, where they feed on sloughing skin, hair, and feathers.

Ticks, such as the one shown in Figure 36-12, are parasites that feed on their hosts by piercing the flesh and sucking on the blood. Many species of ticks carry bacteria and other microorganisms in their guts that may cause diseases in the host. While the tick feeds, the microorganisms are transmitted to the new host. Rocky Mountain spotted fever and Lyme disease are transmitted in this way. Ticks range in length from a few millimeters to 3 cm (about 1 in.).



FIGURE 36-11

Scorpions, such as *Paruroctonus mesaensis*, have pincerlike pedipalps and a venomous stinger at the end of their segmented abdomen.

FIGURE 36-12

The lone star tick, *Amblyomma americanum*, is an arachnid that parasitizes humans and other mammals.





(a) Millipede, *Narceus americanus*



(b) Giant centipede, *Scolopendra* sp.

FIGURE 36-13

(a) Millipedes have a rounded body and two pairs of legs on each body segment except the last two. (b) Centipedes have a flattened body and one pair of legs on each body segment except the first and the last two.

SUBPHYLUM MYRIAPODA

The subphylum Myriapoda (MIR-ee-AHP-uh-duh), includes the class Diplopoda (duh-PLAH-puh-duh), which consists of millipedes, and the class Chilopoda (KIE-LAHP-uh-duh), which consists of centipedes. *Myriapoda* means “many feet,” and is so named because myriapods have many body segments, most of which have one or two pairs of legs each. Unlike crustaceans, myriapods have one pair of unbranched antennae. They are terrestrial but lack a waxy exoskeleton. They avoid drying out by living in damp areas.

Class Diplopoda

Millipedes are members of the class Diplopoda. Most millipedes have two pairs of legs on each body segment except the last two. Millipedes may have up to 100 body segments and thus 200 pair of legs. The legs are well adapted for burrowing through humus and soil, but because the legs are short, millipedes move slowly. As Figure 36-13a shows, the bodies of millipedes are rounded.

Millipedes have short antennae and two groups of simple eyes on their head. They have poor vision but a good sense of smell. Millipedes use their maxillae and mandibles to chew on plants or decaying plant matter in the soil.

Class Chilopoda

Centipedes are members of the class Chilopoda. Centipedes may have as few as 15 or as many as 175 pairs of legs. In tropical regions centipedes reach lengths of 30 cm (12 in.). Their bodies are more flattened than those of millipedes, and their legs are longer relative to their body, as Figure 36-13b shows. Each body segment behind the head, except the first segment and the last two segments, has one pair of jointed legs. The appendages on the first segment are modified into a pair of poison claws. Long antennae and two clusters of simple eyes are located on the head. Centipedes can move quickly in search of earthworms, insects, and other prey. They kill the prey with their poison claws and use their mandibles and maxillae to tear it apart. Most centipedes are not harmful to humans.

SECTION 3 REVIEW

1. List the major characteristics of arachnids.
2. Describe the functions of pedipalps.
3. Describe three ways in which spiders are adapted for catching prey.
4. Name two ways in which scorpions differ from spiders.
5. How do mites and ticks differ?
6. How do millipedes and centipedes differ?

CRITICAL THINKING

7. **Making Comparisons** Compare the excretory system of a spider with that of a crayfish.
8. **Inferring Relationships** How might spiders benefit humans?
9. **Applying Information** A cubic meter of soil may contain thousands of millipedes. What role might millipedes serve in an ecosystem?

CHAPTER HIGHLIGHTS

SECTION 1

Phylum Arthropoda

- Arthropods are segmented animals that have jointed appendages, an exoskeleton, a high degree of cephalization, a ventral nerve cord, and an open circulatory system.
- To grow, an arthropod must shed its exoskeleton periodically in a process called molting.
- The five major subphyla of Arthropoda are Trilobita, Crustacea, Chelicerata, Myriapoda, and Hexapoda. The subphylum Trilobita consists of extinct animals called trilobites.

Vocabulary

arthropod (p. 723)
appendage (p. 723)
chitin (p. 723)

compound eye (p. 723)
molting (p. 724)

trilobite (p. 724)
tagma (p. 724)

mandible (p. 724)
chelicera (p. 724)

SECTION 2

Subphylum Crustacea

- Crustaceans, members of the subphylum Crustacea, have two pairs of antennae. Most crustaceans have a pair of chewing mouthparts called mandibles and one pair of branched appendages on each body segment. The exoskeletons of many crustaceans contain large amounts of calcium carbonate.
- Crustaceans include shrimps, lobsters, crabs, crayfish, barnacles, isopods, copepods, and water fleas. Most crustaceans are aquatic, use gills to respire, and have a larval stage called a nauplius.
- The body of a crayfish is divided into a cephalothorax and an abdomen. The appendages on each body segment are specialized for sensing, feeding, respiration, locomotion, or reproduction.
- Crayfish have a digestive gland that is near the stomach and that secretes enzymes for digestion. Walking circulates water across the gills for respiration. The circulatory system is open. Green glands assist in excretion of excess water that enters the body by osmosis.

Vocabulary

nauplius (p. 726)
cirrus (p. 727)
isopod (p. 727)
decapod (p. 728)

cephalothorax (p. 728)
thorax (p. 728)
carapace (p. 728)
abdomen (p. 728)

antenna (p. 728)
antennule (p. 728)
cheliped (p. 728)
swimmeret (p. 729)

telson (p. 729)
uropod (p. 729)
digestive gland (p. 729)
green gland (p. 730)

SECTION 3

Subphyla Chelicerata and Myriapoda

- Members of the subphylum Chelicerata lack antennae and have pincerlike mouthparts called chelicerae.
- Arachnids include spiders, scorpions, mites, and ticks. Their bodies are divided into a cephalothorax and an abdomen, and they usually have six pairs of jointed appendages: one pair of chelicerae, one pair of pedipalps, and four pairs of walking legs.
- Spiders have eight simple eyes and chelicerae modified as fangs. Spiders produce silk threads that are used for several functions. Spiders are terrestrial; they respire by means of book lungs, tracheae, or both. Malpighian tubules function to excrete wastes while conserving water.
- Scorpions have large, pincerlike pedipalps and a stinger on the last segment of the abdomen.
- Mites and ticks have a completely fused cephalothorax and abdomen. Many species are parasitic, and some spread diseases that affect humans.
- Members of the subphylum Myriapoda have antennae, mandibles, and unbranched appendages.
- Millipedes have rounded bodies and two pairs of jointed legs on each body segment except the last two segments. Centipedes have flattened bodies and one pair of jointed legs on each body segment except the first segment and the last two segments.

Vocabulary

arachnid (p. 731)
pedipalp (p. 731)

spinneret (p. 732)
book lung (p. 732)

trachea (p. 732)
spiracle (p. 732)

Malpighian tubule (p. 732)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
 - mandible* and *chelicera*
 - isopod* and *decapod*
 - cephalothorax* and *abdomen*
 - antennae* and *antennules*
 - cheliped* and *telson*
 - book lung* and *trachea*
- Explain the relationship between the cephalothorax and the thorax.
- Choose the term that does not belong in the following group, and explain why it does not belong: *book lung*, *trachea*, *spiracle*, and *green gland*.
- Word Roots and Origins** The word *tagma* is derived from the Greek *tassein*, which means “to arrange.” Using this information, explain why the term *tagma* is a good name for the structure that the term describes.

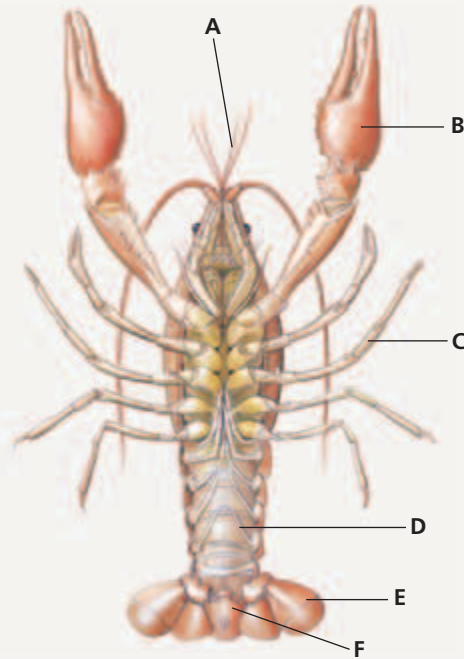
UNDERSTANDING KEY CONCEPTS

- Describe** the key characteristics of arthropods.
- Explain** how arthropods molt.
- List** the five subphyla of arthropods.
- Summarize** the major characteristics of crustaceans.
- Explain** how a barnacle feeds.
- Summarize** the function of the major appendages of a crayfish.
- Describe** excretion in a crayfish.
- Compare** respiration in crayfish and spiders.
- Identify** the main characteristics of the members of subphylum Chelicerata.
- Compare** the appendages on the cephalothorax of a spider with those of a scorpion.
- List** three predatory behaviors of spiders.
- Identify** a distinguishing characteristic of mites.
- Explain** how parasitic ticks spread diseases.
- Identify** the main characteristics of the members of subphylum Myriapoda.
- Compare** the characteristics of millipedes and centipedes.
- Summarize** how myriapods avoid desiccation in terrestrial environments.

-  **CONCEPT MAPPING** Use the following terms to create a concept map that sequences the process of molting in an arthropod: *exoskeleton*, *enzymes*, *inner layer*, *hormone*, *molting*, and *epidermis*.

CRITICAL THINKING

- Interpreting Graphics** Identify structures A through F in the diagram below.



- Applying Information** Arthropods first lived on land about 400 million years ago. They have survived several time periods in which other phyla became extinct. What characteristics might have enabled arthropods to survive and adapt?
- Relating Concepts** A water flea of the genus *Daphnia* eats algae. How might this organism acquire its food?
- Inferring Relationships** Barnacles are sessile crustaceans. What adaptation enables them to compete with motile organisms for food? What adaptation protects them from predators?
- Recognizing Relationships** The cephalothorax of a crayfish is covered by the carapace, a fused plate of exoskeleton. What are some advantages and disadvantages of this fused structure?
- Analyzing Patterns** Crayfish have a high concentration of sensory hairs on the telson. What might be the advantage of having so many sensory structures at the posterior end?

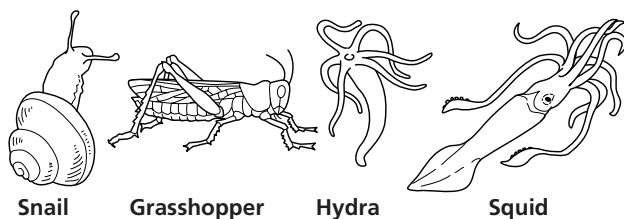


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- What do all arthropods have in common?
 - spiracles
 - antennae
 - a cephalothorax
 - jointed appendages
- What is the chitin-containing structure that protects and supports the body of an arthropod?
 - a tagma
 - a chelicera
 - an appendage
 - an exoskeleton
- Which of the following statements about compound eyes is true?
 - Compound eyes have a single lens.
 - Compound eyes are located on the abdomen of scorpions.
 - Compound eyes are found in all arthropods except crayfish.
 - Compound eyes are composed of many individual light detectors.
- What are the major respiratory organs of crayfish?
 - gills
 - lungs
 - tracheae
 - book lungs
- How do mites and ticks differ from spiders?
 - Mites and ticks have mandibles.
 - Mites and ticks have two pairs of antennae.
 - Mites and ticks have a unique respiratory system.
 - Mites and ticks have a fused cephalothorax and abdomen.

INTERPRETING GRAPHICS: The illustration below shows four animals. Use the illustration to answer the question that follows.



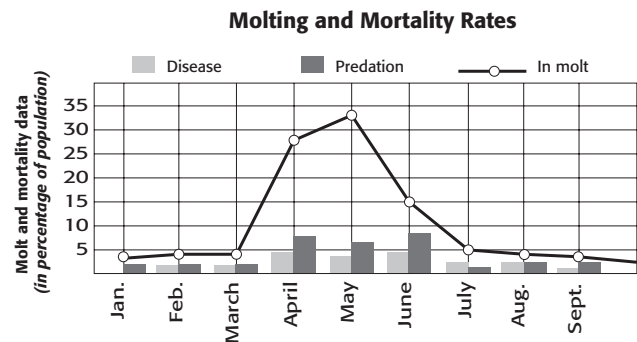
- Which animal has a chitinous exoskeleton?
 - snail
 - grasshopper
 - hydra
 - squid

DIRECTIONS: Complete the following analogy.

- mandible : crayfish :: chelicera :

- pill bug
- scorpion
- barnacle
- millipede

INTERPRETING GRAPHICS: The graph below shows data about molting and two causes of mortality in crabs. Use the graph to answer the questions that follow.



- During which months are you most likely to find crabs in molt?
 - January through March
 - April through June
 - July through September
 - September through January
- What is the relationship between molting and mortality?
 - Molting increases mortality.
 - Molting decreases mortality.
 - Molting has no effect on mortality.
 - Molting and mortality occur at different times of the year.

SHORT RESPONSE

Some arthropods are terrestrial and some are aquatic. Give examples and describe the distinguishing characteristics of each type of arthropod.

EXTENDED RESPONSE

All arthropods undergo many cycles of molting throughout their lifetime.

Part A Describe the process of molting in arthropods.

Part B Explain how the anatomical structure of an arthropod relates to the function of molting.

Test TIP

For questions involving graphs, be sure to study the graph's title, axes, and labels carefully before choosing a final answer.

Investigating Pill Bug Behavior

OBJECTIVES

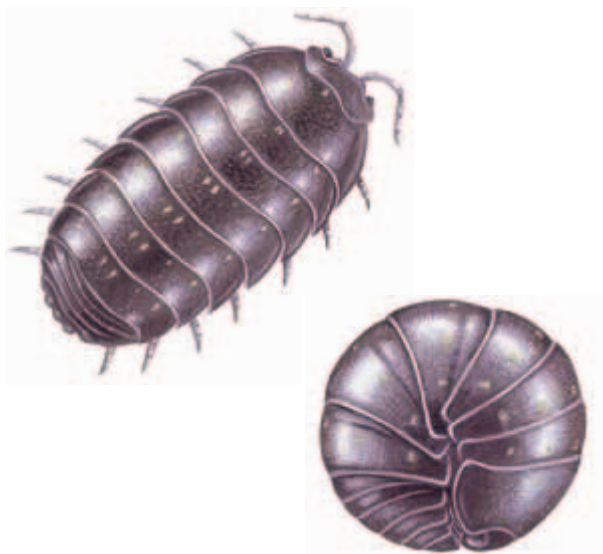
- Review characteristics of the phylum Arthropoda and the subphylum Crustacea.
- Observe the external anatomy of a living terrestrial isopod.
- Investigate the behavior of terrestrial isopods.

PROCESS SKILLS

- observing
- hypothesizing
- experimenting

MATERIALS

- 5 live pill bugs for each pair of students
- 1 plastic medicine dropper
- water
- potato
- 3 sheets of filter paper cut to fit a Petri dish
- Petri dish with cover
- aluminum foil
- bright lamp or flashlight
- 4 fabrics of different texture
- cellophane tape
- scissors



Background

1. What are the major characteristics of arthropods and crustaceans?
2. Where would you expect to find pill bugs?
3. How are pill bugs different from most other crustaceans?

PART A Response to Light

1. Put several drops of water on a piece of filter paper until the paper becomes slightly moist. Place the filter paper in the bottom of a Petri dish. Cover half the bottom of the Petri dish with aluminum foil.
2. Check the lighting in the room. The light must be low and even during this part of the investigation.
3.  **CAUTION** You will be working with live animals. Be sure to treat them gently and to follow directions carefully. Place five pill bugs in the center of the filter paper. Shine a lamp directly over the Petri dish so that half the filter paper is brightly illuminated and the other half is in darkness, shaded by the foil.
4. Based on your knowledge of the natural habitat of pill bugs, can you predict where they will go? Make a data table like the one shown, and record your prediction as well as the actual responses of the pill bugs.

PART B Response to Moisture

5.  **CAUTION** Sharp or pointed objects can cause injury. Handle scissors carefully. Cut a piece of filter paper in half. Moisten one of the halves with water and place it in the bottom of a Petri dish. Make sure that drops of water do not leak onto the bottom of the dish.
6. Place the dry half of the filter paper in the bottom of the Petri dish, leaving a 2 mm gap between it and the damp filter paper.
7. Place five pill bugs along the boundary between the wet and dry areas. Place the top on the dish.
8. To which side do you predict the pill bugs will move? Write your prediction in your data table. Observe the pill bugs for 3 to 5 minutes, and record your observations in your data table. Do your observations agree with your predictions?

PART C Response to Food

9. Again dampen a piece of filter paper and place it in the bottom of a Petri dish. Next place a thin slice of potato near the edge of the dish.
10. Place five pill bugs in the Petri dish opposite the potato slice, and place the lid on the dish.
11. Where do you predict the pill bugs will go? Write your prediction in your data table. Observe the pill bugs for 3 to 5 minutes, and record your observations in your data table. Do your observations agree with your predictions?

PART D Response to Surface Texture

12. Trace the outline of the bottom of a Petri dish on one of the fabrics. Cut the circle out of the fabric and fold it in half. Then cut along the fold to produce two half-circles.
13. Repeat step 12 using the other three fabrics. You should now have eight half-circles.
14. Tape together two half-circles, each of a different fabric. Place the full circle in the bottom of a Petri dish, tape side down, as shown in the figure below.
15. On a sheet of paper, draw the fabric circle and label the two types of fabric that make up the circle.



16. Place a pill bug in the center of the circle and observe its movements. One student should keep track of the amount of time the animal spends on each fabric. On the drawing that was made in step 15, the other student should draw the path the pill bug travels in the circle. After 5 minutes, stop your observations and record the amount of time the pill bug spent on each fabric.
17. Repeat steps 14–16 for two other pairs of fabrics.
18.   **Return the pill bugs to their container.** Clean up your materials and wash your hands before leaving the lab.

Analysis and Conclusions

1. In Part A, why was the entire filter paper moistened?
2. In Part B, why was there a slight separation between the wet and dry halves of the filter paper?
3. In Part C, why was the entire filter paper moistened?
4. In Part D, which fabric did the pill bugs prefer? Describe the texture of that fabric.
5. How do the responses of pill bugs to light, moisture, and food in these experiments reflect adaptations to their natural surroundings?
6. How is being able to detect surface texture a good adaptation for pill bugs in their natural habitat?

Further Inquiry

1. Design an experiment to investigate the response of pill bugs to temperature. Think carefully about how you will construct your apparatus. Seek approval from your teacher before you actually conduct this experiment. How do you think the pill bugs will respond?
2. Design an experiment to investigate whether pill bugs have preferences for certain types of food.

OBSERVATIONS OF PILL-BUG BEHAVIOR

Stimulus	Prediction	Observation
Light		
Moisture		
Food		

INSECTS

The leaf-footed bug, *Diactor bilineatus*, is a colorful member of the extremely diverse world of insects.

**SECTION 1** *The Insect World***SECTION 2** *Insect Behavior*

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THE INSECT WORLD

Insects have thrived for more than 400 million years, since long before the rise and fall of the dinosaurs. The story of insects is one of great biological success through evolution and adaptation. Today, insects account for more than half of all animal species on Earth.

CHARACTERISTICS AND CLASSIFICATION OF INSECTS

Many of the adaptations that have made insects successful are characteristics they share with other arthropods, such as a segmented body, jointed appendages, and an exoskeleton. Insects belong to the class Insecta in the subphylum Hexapoda (hek-SAP-uh-duh), which also includes three minor orders that are not considered true insects.

The body of an insect is divided into three tagmata: the head, thorax, and abdomen. Like myriapods, insects have mandibles and one pair of antennae on their head, and the antennae and other appendages are unbranched. The thorax has three pairs of jointed legs and, in many species, one or two pairs of wings. The abdomen is composed of 9 to 11 segments, and in adults it has neither wings nor legs.

Most insects are small. Among the smallest is a parasitic wasp, which is only 0.14 mm (0.005 in.) in length. Some insects are much larger. For example, the African Goliath beetle reaches 10 cm (4 in.) in length, and the atlas moth has a wingspan of more than 25 cm (10 in.). These two giants of the insect world are shown in Figure 37-1.

OBJECTIVES

- **Relate** the major characteristics of insects to insects' biological success.
- **List** both harmful and beneficial effects of insects on human society.
- **Describe** the external structure and organ systems of a grasshopper.
- **Compare** incomplete and complete metamorphosis in insects.
- **Describe** defensive adaptations in insects.

VOCABULARY

entomology
labrum
labium
tympanum
ovipositor
metamorphosis
incomplete metamorphosis
nymph
complete metamorphosis
pupa

FIGURE 37-1

The African goliath beetle (a) and Atlas moth (b) are among Earth's largest insects.



(a) African goliath beetle, *Goliathus goliatus*



(b) Atlas moth, *Attacus atlas*

Entomologist

Job Description Entomologists are scientists who study insects and other terrestrial arthropods. Studying insects can be exciting because insects make up more than half of all animal species. Some entomologists research the classification, distribution, or physiology of insects. Some work in medicine and agriculture to reduce insect pests that spread disease or to use beneficial insects to control harmful pests. Entomologists may work in universities, museums, private companies, or government agencies.

Focus on an Entomologist

“You look for beauty in the small,” says entomologist and professor Genaro López. López and his students are performing

field surveys of pseudoscorpions. “My students can be world experts if they continue to study them for another year or two!” notes López. López enjoys telling stories about scorpions, butterflies, dragonflies, and other insects, to capture his biology students’ attention. López specialized in entomology for several reasons. “I wanted to study an entire animal—and, I wanted to be outdoors,” says López. He also likes studying insects because they play roles that are important to humans.

Education and Skills

- **High school**—three years of science courses and four years of math courses.
- **College**—bachelor of science (B.S.) in biology, including course work in



ecology, botany, and zoology; for research, a master’s degree (M.S.) or doctoral degree (Ph.D.) is needed.

- **Skills**—attention to detail, self-motivation, curiosity, patience, ability to work independently, fitness, and survival skills.



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The study of insects and other terrestrial arthropods is called **entomology** (ENT-uh-MAHL-uh-jee), and the scientists who engage in it are known as *entomologists*. Entomologists classify insects into more than 25 orders based on characteristics such as the structure of mouthparts, number of wings, and type of development. Several of the more common insect orders are listed in Table 37-1.

The Success of Insects

Insects live almost everywhere in the world except in salt water. Entomologists have described and classified nearly 1 million insect species, or more species than exist in all other animal groups combined. In terms of their widespread distribution and great abundance, insects are very successful.

One of the most important factors responsible for the remarkable success of insects is their ability to fly, which enables them to escape from predators and disperse rapidly into new environments. Like other arthropods, insects also benefit from having a light but sturdy exoskeleton and jointed appendages that perform a variety of functions. In addition, most insects are small, so several species can inhabit different local environments within an area without competing with one another for food or other resources. Finally, insects generally have very short life spans and produce large numbers of eggs. Therefore, natural selection can occur more quickly in insects than in organisms that take longer to reach maturity.

TABLE 37-1 Common Insect Orders

Order (meaning of order name)	Approximate number of species	Examples	Type of metamorphosis	Characteristics	Significance to humans
Hemiptera ("half wing")	55,000	true bugs 	incomplete	two pairs of wings during part of life; piercing, sucking mouthparts	damage crops and garden plants; transmit disease
Homoptera ("like wing")	20,000	aphids mealy bugs cicadas 	incomplete	membranous wings held like roof over body (some species wingless); piercing, sucking mouthparts	damage crops, garden plants, and young trees
Isoptera ("equal wing")	2,000	termites 	incomplete	at times, two pairs of membranous wings; chewing mouthparts	decompose wood in buildings; recycle resources in forests
Odonata ("toothed")	5,000	dragonflies damselflies 	incomplete	two pairs of long, narrow, membranous wings; chewing mouthparts	destroy harmful insects; nymphs serve as food for freshwater fish
Orthoptera ("straight wing")	30,000	grasshoppers crickets katydids 	incomplete	two pairs of straight wings; chewing mouthparts	damage crops, garden plants, and stored foods
Coleoptera ("sheathed wing")	400,000	weevils ladybugs beetles 	complete	hard forewings, membranous hind wings; chewing mouthparts	destroy crops; damage trees; prey on other insects
Diptera ("two wing")	120,000	mosquitoes flies gnats 	complete	one pair of wings (hind pair reduced to knobs); sucking, piercing, or lapping mouthparts	carry diseases; destroy crops; pollinate flowers; act as decomposers
Hymenoptera ("membrane wing")	100,000	bees wasps ants 	complete	two pairs of membranous wings (some species wingless); biting, sucking or lapping mouthparts; some species social	pollinate flowers; make honey; destroy harmful insects; may sting
Lepidoptera ("scaled wing")	140,000	butterflies moths 	complete	large, scaled wings; chewing mouthparts in larvae, siphoning mouthparts in adults	pollinate flowers; larvae and pupae produce silk; larvae damage clothing and crops

FIGURE 37-2

Some insects are harmful to humans, but most are beneficial. (a) Termites, such as *Reticulitermes flavipes*, can destroy a building by feeding on the wood. (b) The blister beetle, *Lytta fulvipennis*, cross-fertilizes plants by spreading pollen from flower to flower as it searches for nectar. Insects serve important roles in many agricultural systems.



(a)



(b)

Insects and People

Because insects are so abundant, it is not surprising that they affect our lives in many ways. Some insects, such as grasshoppers, boll weevils, and corn earworms, compete with humans for food by eating crops. In fact, nearly every crop plant has some insect pest. Other insects spread diseases by biting humans or domesticated animals. For example, some fleas may transmit the bacteria that cause plague; some female mosquitoes may transmit the protists that cause malaria; and the tsetse fly may transmit the protists that cause African sleeping sickness. Termites, shown in Figure 37-2a, attack the wood in buildings, and some moths consume wool clothing and carpets.

Despite the problems some insects cause, it would be a serious mistake to think that the world would be better off without any insects. Insects play vital roles in almost all terrestrial and freshwater environments. They serve as food for numerous species of fish, birds, and other animals. Many kinds of insects, such as the beetle shown in Figure 37-2b, are essential for the cross-pollination of plants. It is estimated that insects pollinate 40 percent of the world's flowering plants, including many of those cultivated as food for humans and livestock. Insects also manufacture a number of commercially valuable products, including honey, wax, silk, and shellac. We tend to think of termites as destructive pests because of their effects on buildings, but by feeding on decaying wood, they also help recycle nutrients needed to maintain a healthy forest. Other insects recycle the nutrients contained in animal carcasses.

THE GRASSHOPPER

In this section, the grasshopper will be used to demonstrate some of the details of insect structure and function. As you read, remember that these details are not shared by all insects. The diversity of the insect world is so great that no typical insect exists.

External Structure

The major features of an adult grasshopper's external structure are illustrated in Figure 37-3. The body of a grasshopper clearly shows three tagmata. The most anterior tagma, the head, bears the mouthparts. It also has a pair of unbranched antennae as well as simple and compound eyes.

The middle tagma, the thorax, is divided into three parts: the prothorax, mesothorax, and metathorax. The *prothorax* attaches to the head and bears the first pair of walking legs. The *mesothorax* bears the forewings and the second pair of walking legs. The *metathorax* attaches to the abdomen and bears the hindwings and the large jumping legs. The muscles inside the jumping legs store energy when the legs are flexed. Release of this energy causes the legs to extend suddenly, launching the grasshopper into the air and away from danger. A flexible joint at the base of each leg provides the legs with great freedom of motion. Spines and hooks on the legs enable the grasshopper to cling to branches and blades of grass.

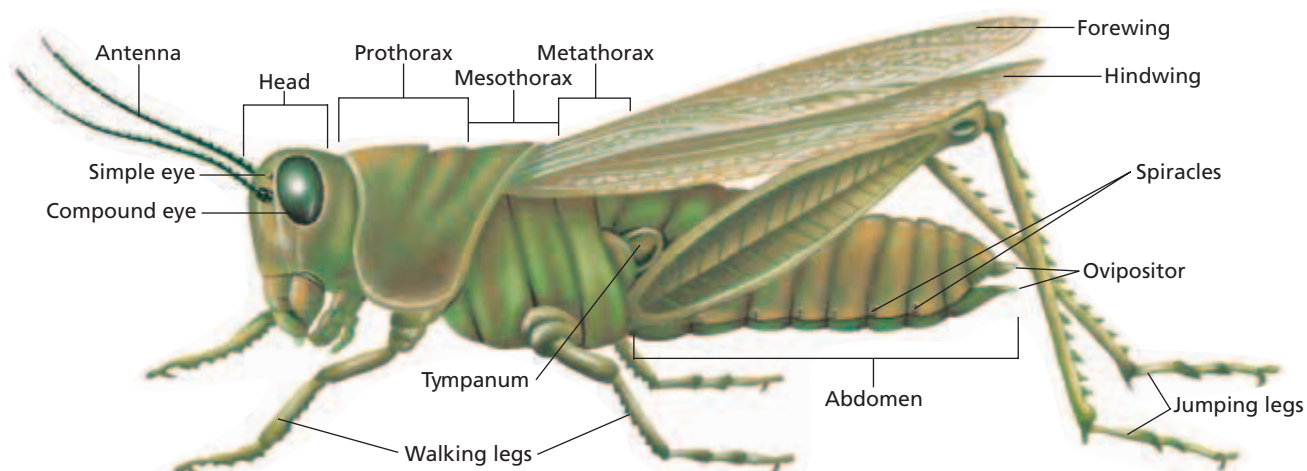
The leathery forewings cover and protect the membranous hindwings when the grasshopper isn't flying. Although the forewings help the grasshopper glide during flight, the hindwings actually propel it through the air. The wings are powered by muscles attached to the inside of the exoskeleton in the thorax. Note that insect wings develop as outgrowths of the thorax and are composed of exoskeleton material. Thus, they are not homologous to bird and bat wings, which develop from limb buds.

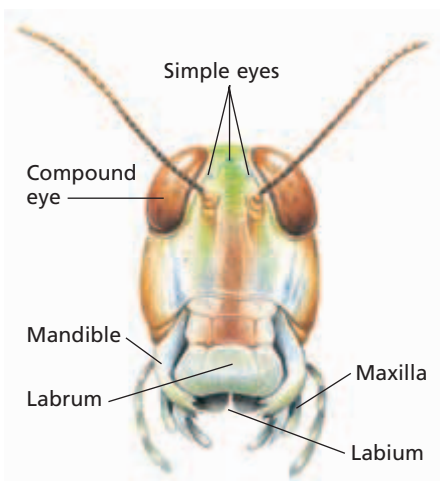
The segments in the most posterior tagma, the abdomen, are composed of upper and lower plates that are joined by a tough but flexible sheet of exoskeleton. The same flexible sheet also connects the segments to one another. The exoskeleton is covered by a waxy cuticle that is secreted by the cells of the epidermis. The rigid exoskeleton supports the grasshopper's body, and the cuticle retards the loss of body water. Both structures are adaptations for a terrestrial life.



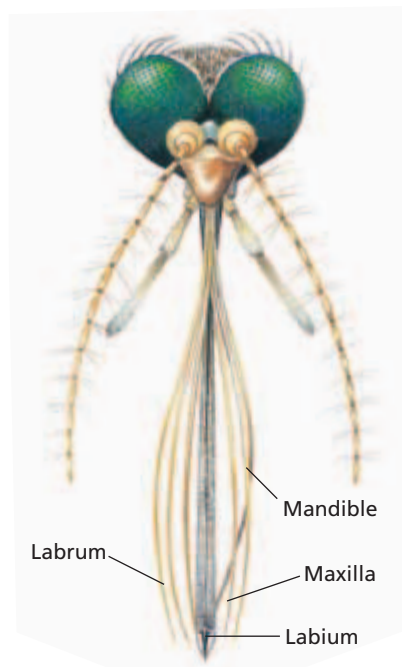
FIGURE 37-3

The external anatomy of a grasshopper shows features that are characteristic of most insects: a body consisting of a head, thorax, and abdomen; a pair of unbranched antennae; three pairs of jointed legs; and two pairs of wings.

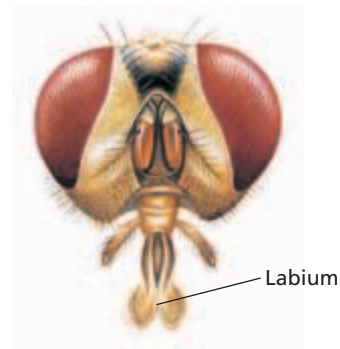




(a) GRASSHOPPER MOUTHPARTS



(b) MOSQUITO MOUTHPARTS



(c) HOUSEFLY MOUTHPARTS

FIGURE 37-4

Insect mouthparts are adapted for different functions in different species. Mouthparts are used for biting and chewing in grasshoppers (a), piercing and sucking in mosquitoes (b), and sponging and lapping in houseflies (c).

Feeding and Digestion

Grasshoppers feed on plants. The mouthparts of grasshoppers, shown in Figure 37-4a, are modified for cutting and chewing leaves and blades of grass. The **labrum** and **labium** are mouthparts that function like upper and lower lips, respectively. They hold the food in position so that the sharp-edged *mandibles* can tear off edible bits. Behind the mandibles are the *maxillae*, which also help hold and cut the food. Recall that all arthropods have mandibles and maxillae.

The mouthparts of other insects are specialized for the types of food they eat, as you can see in Figures 37-4b and 37-4c. For example, mosquitoes have long, thin mouthparts that fit together to form a needle-like tube, which the females use to pierce the skin of other animals and suck up blood. The mouthparts of many flies, in contrast, are soft, spongelike lobes that soak up fruit juices and other liquids.

The digestive tract of a grasshopper is shown in Figure 37-5. Food enters the mouth, is moistened by saliva from the *salivary* (SAL-uh-VER-ee) *glands* and then passes through the esophagus and into the *crop* for temporary storage. From the crop, food passes into the *gizzard*, where sharp, chitinous plates shred it. The shredded mass then enters a portion of the digestive tract called the *midgut*. There, the food is broken down by enzymes secreted by the *gastric ceca* (GAS-trik SEE-kuh), which are pockets that branch from the digestive tract. Nutrients are absorbed into the coelom through the *midgut*. Undigested matter enters the posterior section of the digestive tract, the *hindgut*, and leaves the body through the *anus*.

Circulation, Respiration, and Excretion

Nutrients and other materials are transported through the body of a grasshopper by an open circulatory system that is similar to that of the crayfish. Hemolymph flows through a large dorsal vessel called the *aorta* (ay-OHR-tuh), which is shown in Figure 37-5. The muscular heart, which is located in the abdomen and thorax, pumps the hemolymph forward through the aorta and into the part of the coelom nearest the head. The hemolymph then percolates through the coelom toward the abdomen and reenters the heart through small pores along its length.

Most animals transport oxygen and carbon dioxide through their circulatory system. However, insects exchange these gases with the environment through a complex network of air tubes called *trachea*. Trachea also serve this purpose in some spiders. In grasshoppers, air enters the tracheae through spiracles on the sides of the thorax and abdomen, as seen in Figures 37-3 and 37-5. The ends of the tracheae branch near the cells of the body and are filled with fluid. Oxygen diffuses into the cells from this fluid while carbon dioxide diffuses in the reverse direction. Air can be pumped in and out of the tracheae by the movements of the abdomen and wings.

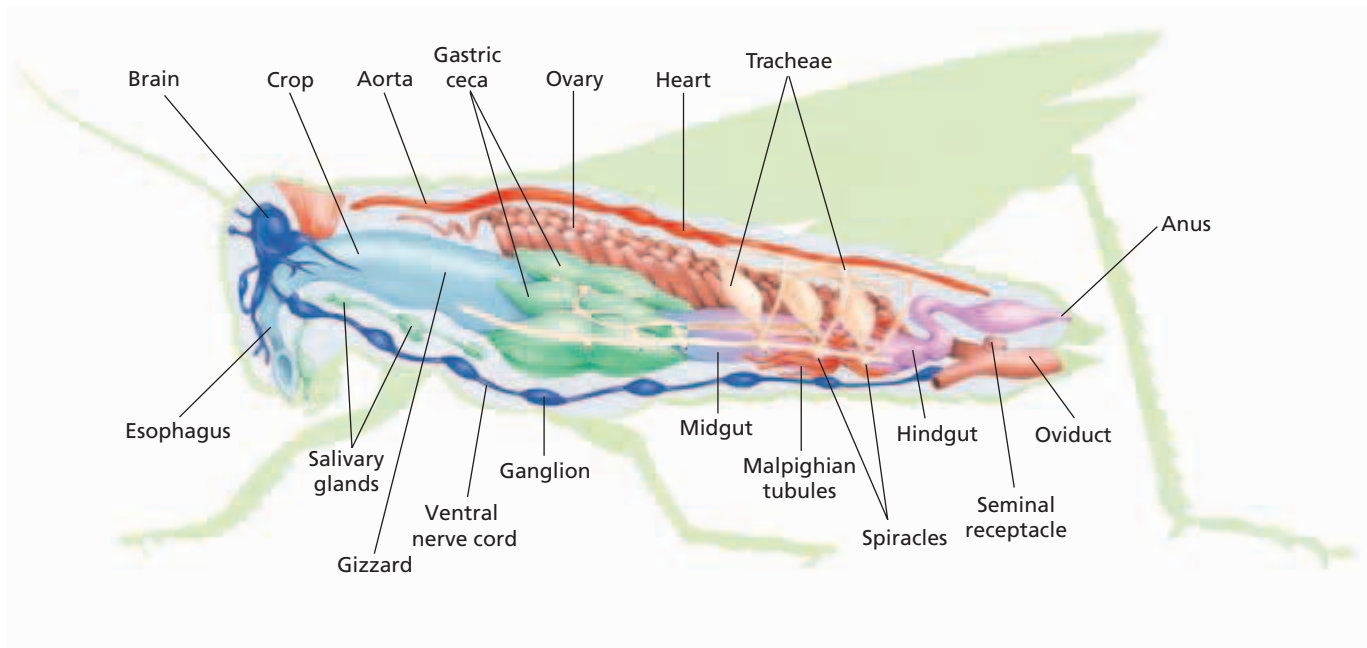


FIGURE 37-5

The major internal organs of a female grasshopper are seen in this cutaway side view.

Like spiders, insects have excretory organs called *Malpighian tubules* that collect water and cellular wastes from the hemolymph. As Figure 37-5 shows, the Malpighian tubules are attached to the digestive tract between the midgut and the hindgut. In insects that live in dry environments, the Malpighian tubules return most of the water to the hemolymph, producing a very concentrated mixture of wastes that is deposited in the hindgut and leaves the body with the feces. This is another method by which insects are adapted for life on land.

Neural Control

The grasshopper's central nervous system consists of a brain and a ventral nerve cord with ganglia located in each body segment. Nerves extend from the brain to the antennae, eyes, and other organs of the head. The antennae contain sensory structures that respond to touch and smell. The three simple eyes are arranged in a row just above the base of the antennae. The simple eyes function to sense the intensity of light. Two bulging compound eyes, which are composed of hundreds of individual light detectors and lenses, provide a wide field of view. In addition to sensing light intensity, the compound eyes can detect movement and form images.

Other nerves extend from the ganglia to the muscles and sensory structures in the thorax and abdomen. One such structure is a sound-sensing organ called the **tympanum** (TIM-puh-nuhm). The tympanum is a large, oval membrane that covers an air-filled cavity on each side of the first abdominal segment. Sounds cause the tympanum to vibrate, and the vibrations are detected by nerve cells that line the cavity. Tympana are also found in many other insects that use sound in communication, such as crickets and cicadas. In addition, sensory hairs that are similar to those of a crayfish are distributed over an insect's body. At the base of each hair is a nerve cell that is activated if the hair is touched or moved by vibration.

Word Roots and Origins

tympanum

from the Latin
tympanum, meaning
"drum"

Word Roots and Origins

ovipositor

from the Latin *ovum*, meaning "egg,"
and *positus*, meaning "to place"

Reproduction

Grasshoppers have separate sexes, as do all insects. During mating, the male deposits sperm into the female's seminal receptacle, where they are stored until the eggs are released by the ovaries. After release, the eggs are fertilized internally. The last segment of the female's abdomen forms a pointed organ called an **ovipositor** (OH-vuh-PAHZ-uht-uhr), which you can see in Figure 37-3. The female grasshopper uses her ovipositor to dig a hole in the soil, where she lays the fertilized eggs.

INSECT DEVELOPMENT

After hatching from the egg, a young insect must undergo several molts before it reaches its adult size and becomes sexually mature. Silverfish and a few other insects go through the molting process without changing body form. The majority of insects undergo some type of change in form as they develop into adults. This phenomenon of developmental change in form is called **metamorphosis** (MET-uh-MOHR-fuh-suhs). There are two main kinds of metamorphosis in insects: incomplete and complete.

Incomplete Metamorphosis

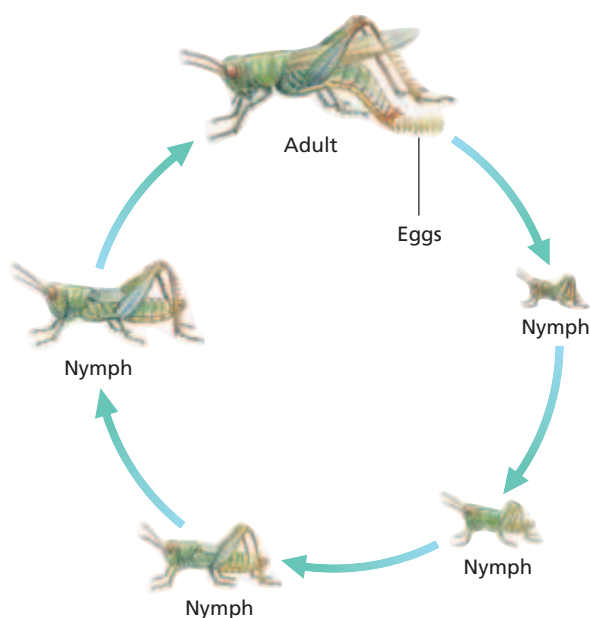
In **incomplete metamorphosis**, illustrated in Figure 37-6, a nymph hatches from an egg and gradually develops into an adult. A **nymph** is an immature form of an insect that looks somewhat like the adult, but it is smaller, and its wings and reproductive organs are undeveloped. The nymph molts several times. With each molt, the wings become larger and more fully formed. The final molt transforms the nymph into an adult that can reproduce and, in most species, fly. Insects that undergo incomplete metamorphosis include grasshoppers, mayflies, dragonflies, and termites. Several other examples are listed in Table 37-1.

Complete Metamorphosis

In **complete metamorphosis**, an insect undergoes two stages of development between the egg and the adult. In both of those stages, the insect looks substantially different from its adult form. Figure 37-7 illustrates complete metamorphosis in the monarch butterfly. A wormlike larva, commonly called a *caterpillar*, hatches from the egg. Insect larvae may or may not have legs on the thorax and may or may not have leglike appendages on the abdomen. The larva eats almost constantly, growing large on a diet composed mostly of leaves. Thus, it is the larval stage of most insects that causes the most damage to plants.

FIGURE 37-6

In incomplete metamorphosis, shown here in a grasshopper, a nymph hatches from an egg and molts several times before becoming an adult. Nymphs resemble adults but are not sexually mature and lack functional wings.



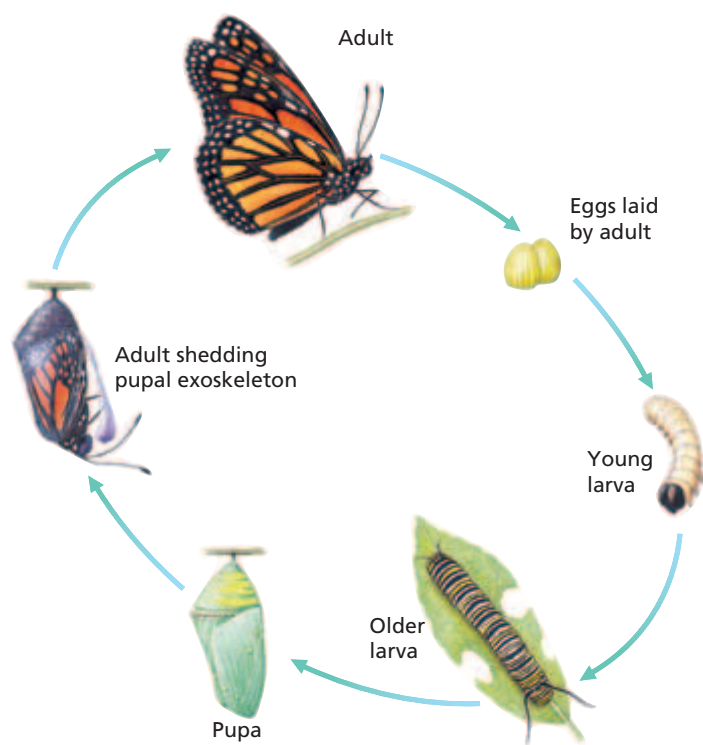
INCOMPLETE METAMORPHOSIS

The monarch larva molts several times as it grows. In the last larval stage, it develops bands of black, white, and yellow along its body. It continues to feed, but soon finds a sheltered spot and hangs upside down. Its body becomes shorter and thicker. Its exoskeleton then splits down the dorsal side and falls off, revealing a green pupa. A **pupa** (PYOO-puh), also called a *chrysalis* (KRIS-uh-luhs), is a stage of development in which an insect changes from a larva to an adult. The pupa of butterflies is enclosed in a protective case. Moth pupae are enclosed in a case called a *cocoon*. Inside the pupa, the larval tissues break down, and groups of cells called imaginal disks develop into the wings and other tissues of the adult. When metamorphosis is complete, the pupa molts into a sexually mature, winged butterfly. Most insects go through complete metamorphosis. Table 37-1 lists several examples besides butterflies and moths, such as beetles, mosquitoes, and bees.

Importance of Metamorphosis

In a life cycle based on complete metamorphosis, the larval and adult stages often fulfill different functions, live in different habitats, and eat different foods. Therefore, the larvae and adults do not compete for space and food. For example, mosquito larvae live in fresh water and feed by filtering small food particles out of the water. When they become adults, the mosquitoes leave the water and feed on plant sap or the blood of terrestrial animals.

Metamorphosis also enhances insect survival by helping insects survive harsh weather. For instance, most butterflies and moths spend the winter as pupae encased in chrysalises or cocoons, which are often buried in the soil.



COMPLETE METAMORPHOSIS

Eco Connection

Biological Control of Insects

Humans have been competing with insects for food since the development of agriculture. To limit the damage that insects do to our food crops, we have developed a variety of poisons that kill a broad range of insects. However, these poisons may kill beneficial as well as harmful insects; they persist in the environment, accumulating in animals at higher levels in the food web; and they select for strains of insects that are resistant to the poisons.

These drawbacks have led scientists to develop biological controls of insect pests. One type of biological control is the use of natural predators or parasites that attack specific kinds of insects. For example, the bacterium *Bacillus thuringiensis* is used to control cabbage worms, tomato worms, and other moth larvae.

Biological control also includes methods that interfere with the reproduction of insects. In the sterile-male approach, for instance, large numbers of male insects sterilized by radiation are introduced into an area. The females lay eggs that never develop, so the population size of the next generation is smaller. Used over several generations, this technique can nearly eliminate some pest species in selected areas.

FIGURE 37-7

In complete metamorphosis, shown here in the monarch butterfly, a larva hatches from an egg and goes through several molts before becoming a pupa, which then develops into an adult. Neither the larva nor the pupa resembles the adult.

INSECT DEFENSE

FIGURE 37-8

Batesian mimicry is shown by the harmless syrphid fly of the genus *Arctophila* (a), which looks very similar to the stinging bumblebee of the genus *Bombus* (b).



(a)



(b)

Insects have many defensive adaptations that increase their chances for survival. Some adaptations provide a passive defense. One form of passive defense that is frequently observed in insects is camouflage. Camouflage enhances survival by making it difficult for predators to recognize an insect. Insects often resemble parts of the plants on which they feed or hunt for food. For example, many varieties of stick insects and mantises look so much like twigs or leaves that they are easy to overlook unless they move.

Insects that are poisonous or taste bad as a defense often have bold, bright color patterns that make them clearly recognizable and warn predators away. This type of coloration is known as *warning coloration*. In some cases, several dangerous or poisonous species have similar warning coloration. For example, many species of stinging bees and wasps display a pattern of black and yellow stripes. This adaptation, in which one dangerous species mimics the warning coloration of another, is called *Müllerian* (myoo-LER-ee-uhn) *mimicry*. Figure 37-8 shows that the black and yellow stripes of bees and wasps are found in some species of flies, which lack stingers and are harmless. Mimicry of this type, in which a harmless species mimics the warning coloration of a dangerous species, is called *Batesian* (BAYTZ-ee-uhn) *mimicry*. Both Müllerian and Batesian mimicry discourage predators from preying on similarly marked species.

Other defensive adaptations of insects are more aggressive, such as the venomous stingers of female bees and wasps. One of the most elaborate adaptations is that of the bombardier beetle, which defends itself by spraying a stream of a noxious chemical. The beetle can even rotate an opening on its abdomen to aim the spray at an attacker.

SECTION 1 REVIEW

1. What are some major characteristics of the class Insecta that have contributed to the biological success of insects?
2. List two ways that insects are harmful to society and two ways that they are beneficial.
3. State the function of each of the following parts of an insect: labrum, tympanum, and ovipositor.
4. What are the differences between incomplete and complete metamorphosis?
5. Describe two types of mimicry in insects.

CRITICAL THINKING

6. **Inferring Relationships** What is the adaptive advantage of having a tracheal system rather than respiring through the skin?
7. **Analyzing Concepts** Metamorphosis is part of the process of growth and maturation of insects. What kind of internal mechanism likely controls metamorphosis?
8. **Recognizing Differences** The monarch butterfly and the viceroy butterfly have similar markings, and birds do not eat either type. Monarchs are distasteful to birds, but viceroys are not. What type of defensive adaptation is being described?

INSECT BEHAVIOR

One reason for the success of insects is their ability to engage in complex behaviors. This ability is made possible by insects' jointed appendages, elaborate sense organs, and relatively complex brains. Insects are capable of interpreting sensory information to escape from predators, find food and mates, and communicate with one another.

COMMUNICATION

Insects communicate in many different ways. Many insects use pheromones, sound, and light to communicate.

One of the most common forms of communication among insects is chemical communication involving pheromones. A **pheromone** (FER-uh-MOHN) is a chemical released by an animal that affects the behavior or development of other members of the same species through the sense of smell or taste. Pheromones play a major role in the behavior patterns of many insects. For example, you may have noticed ants, like those in Figure 37-9, marching along a tightly defined route on the ground. The ants are following a trail of pheromones left by the ants that preceded them. Such trails are often laid down by ants that have found a source of food as they make their way back to the nest. As other ants follow the trail, they too deposit pheromones, so the trail becomes stronger as more ants travel along it.

Pheromones are also used for other purposes. For example, honeybees use pheromones to identify their own hives and to recruit other members of the hive in attacking animals that threaten the hive. Some insects secrete pheromones to attract mates. The female silkworm moth, for example, can attract males from several kilometers away by secreting less than 0.01 μg of a pheromone. Sensory hairs on the large antennae of a male moth make it exquisitely sensitive to the pheromone.

Many insects communicate through sound. Male crickets produce chirping sounds by rubbing a scraper located on one forewing against a vein on the other forewing. They make these sounds to attract females and to warn other males away from their territories. Each cricket species produces several calls that differ from those of other cricket species. In fact, because many species look similar, entomologists often use a cricket's calls rather than its appearance to identify its species.

OBJECTIVES

- **Identify** three ways that insects communicate, and give an example of each.
- **Describe** the social organization of honeybees.
- **Explain** how honeybees communicate information about the location of food.

VOCABULARY

pheromone
social insect
innate behavior
worker bee
queen bee
drone
royal jelly
queen factor
altruistic behavior
kin selection

FIGURE 37-9

These leaf-cutter ants, *Atta colombica*, are following a pheromone trail as they carry sections of leaves to their underground nest.



Mosquitoes communicate through sound, too. Males that are ready to mate fly directly to the buzzing sounds produced by females. A male senses the buzzing by means of sensory hairs on his antennae that vibrate only at the frequency produced by females of the same species.

Insects may also communicate by generating flashes of light. Fireflies, for example, use light to find mates. Males emit flashes in flight, and females flash back in response. Each species of firefly has its own pattern of flashes, which helps males find females of the same species.

BEHAVIOR IN HONEYBEES

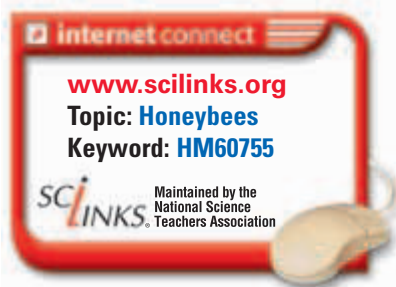
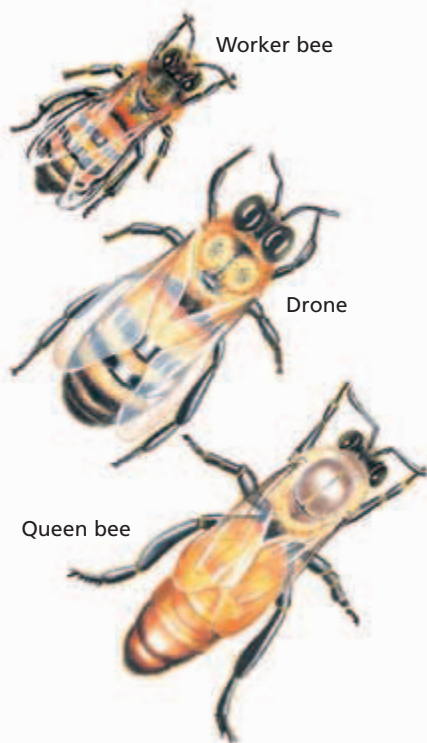


FIGURE 37-10

In a honeybee colony, worker bees perform the work of the hive, while drones and the queen are involved exclusively with reproduction.



Some insects, such as certain species of bees, wasps, ants, and termites, live in complex colonies. In these colonies, some individuals gather food, others protect the colony, and others reproduce. Insects that live in such colonies are called **social insects**. The division of labor among social insects creates great interdependence and a heightened need for communication. This section will look at the behavioral adaptations of one well-studied species of social insect, the honeybee. As you read about the complex behaviors of honeybees, keep in mind that these behaviors are neither taught nor learned. Instead, they are genetically determined. Genetically determined behavior is called **innate behavior**.

A honeybee colony consists of three distinct types of individuals, which are illustrated in Figure 37-10: worker bees, the queen bee, and drones. **Worker bees** are nonreproductive females that make up the vast majority of the hive population, which may reach more than 80,000. The workers perform all the duties of the hive except reproduction. The **queen bee** is the only reproductive female in the hive, and her only function is to reproduce. **Drones** are males that develop from unfertilized eggs. Their only function is to deliver sperm to the queen. Their mouthparts are too short to obtain nectar from flowers, so the workers must feed them. The number of drones in the hive may reach a few hundred during the summer, but when the honey supply begins to run low, the workers kill the drones and clear them from the hive.

Worker Bees

Worker bees perform many functions during their lifetime, which lasts about six weeks. After making the transition from pupa to adult, workers feed honey and pollen to the queen, drones, and larvae. During this stage, the workers are called **nurse bees**. They secrete **royal jelly**, a high-protein substance that they feed to the queen and youngest larvae. After about a week, worker bees stop secreting royal jelly and begin to secrete wax, which they use to build and repair the honeycomb. During this stage, they may also clean and guard the hive and fan their wings to circulate air through the hive.

The workers spend the last weeks of their life gathering nectar and pollen. A number of structural adaptations aid them in this work. Their mouthparts are specialized for lapping up nectar, and their legs have structures that serve to collect and transport pollen from flowers.

The workers do not use their ovipositors for egg laying. Instead, their ovipositors are modified into barbed stingers that the workers use to protect the hive. When a worker bee stings an animal, the stinger and attached venom sac are left behind in the victim as the bee flies away. The worker, having lost part of its body and much of its hemolymph, dies a day or two later. Wasps also have stingers that are modified ovipositors. Unlike honeybees, they can sting many times because their stinger is not barbed.

The Queen Bee

The queen bee develops from an egg identical to the eggs that develop into the workers. Queen bees develop only when selected larvae are fed a continuous diet of royal jelly throughout their larval development. As a queen matures, she secretes a pheromone called the **queen factor** that prevents other female larvae from developing into queens.

The queen's role is to reproduce. Within a few days after she completes metamorphosis and emerges as an adult, she flies out of the hive and mates in the air with one or more drones. During mating, millions of sperm are deposited in the queen's seminal receptacle, where they will remain for the five or more years of her life. Although the queen mates only once, she may lay thousands of eggs each year.

When the hive becomes overcrowded, the queen leaves the hive. As she leaves, she secretes a swarming pheromone that induces about half of the workers in the hive to follow her and form a swarm. Eventually the swarm finds another location for a new hive. Meanwhile, in the old hive, the remaining workers begin feeding royal jelly to other larvae. When a new queen emerges, she produces the queen factor, and in response, the workers destroy the other developing queens. The new queen departs on a mating flight, and the cycle begins again.

The Dances of the Bees

When honeybees leave the hive and find a source of pollen and nectar, how do they communicate the location of this food source to other workers in the hive? An Austrian biologist, Karl von Frisch (1886–1982), spent 25 years answering this question. His careful experimentation earned him a Nobel Prize in 1973.

To study bees, von Frisch built a glass-walled hive and placed feeding stations stocked with sweetened water near the hive. He noted that “scout bees” returning from the feeding stations would perform a series of dancelike movements in the hive. The scout bee would circle first to the right and then to the left, a behavior that von Frisch called the *round dance*.



Quick Lab

Interpreting Nonverbal Communication

Materials pencil, paper, wrapped candy pieces

Procedure

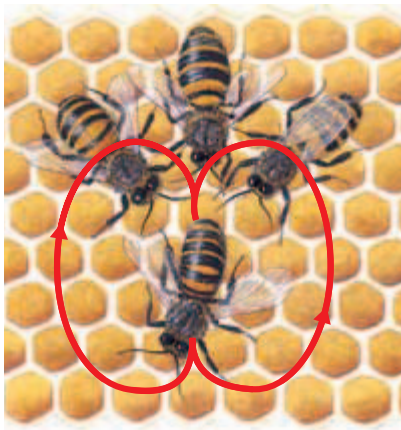
1. Choose one member of your group to play the part of the “scout” bee. The others in the group will be the “worker” bees.
2. Your teacher will secretly tell the scout bee where a “food source” (piece of candy) is located. The scout bee will develop a method of nonverbal communication to let the worker bees know where the food is hidden. The scout bee may not point to the food. Use Figure 37-11 for ideas on how to develop your method of communication.
3. When the food has been located, the scout will hide another piece of food and select a new scout. The new scout will develop a different way to tell the group where the food is located. Repeat the procedure until everyone in your group has been a scout.

Analysis

1. How effective were each of your scout's methods for showing the location of the food?
2. Did the worker bees improve their ability to find the food after several trials?
3. List and describe some types of nonverbal communication that humans use.



(a) ROUND DANCE



(b) WAGGLE DANCE

FIGURE 37-11

Honeybees use two types of dances to convey information about food sources. (a) The round dance indicates that a food source is nearby. (b) The waggle dance indicates the direction of food and the food's distance from the hive.

A round dance pattern is shown in Figure 37-11a. After many observations, von Frisch concluded that the round dance told other worker bees that a food source was near the hive, but it did not inform them of the exact location of the food.

Von Frisch also observed that when the food source was far from the hive, the scout bees would perform another type of dance on a vertical surface inside the hive. He called this dance the *waggle dance* because the scout bees wagged their abdomens from side to side. As you can see in Figure 37-11b, the pattern of the waggle dance is like a figure eight. The scout bee makes a circle in one direction, then a straight run while wagging her abdomen, and then another circle in the opposite direction from the first. The direction of food is indicated by the angle of the straight run on the vertical surface. Straight up, for example, indicates a direction toward the sun. The distance to the food source is indicated by the duration of the dance and the number of waggles on each run.

Altruistic Behavior

When worker bees sting an intruder to defend the colony, they cause their own deaths. This behavior is an example of **altruistic** (AL-troo-IS-tik) **behavior**, which is the aiding of other individuals at one's own risk or expense. The stinging of honeybees is an innate behavior. You might think that the genes directing this behavior would eventually be eliminated from the population, since dead bees can't reproduce. However, this does not happen.

Genetics explains why evolution has selected for altruistic behavior in honeybees. Worker bees are nonreproductive. Therefore, they cannot pass on their own genes by reproducing. However, they can pass on some of their genes by helping a closely related individual reproduce. By defending the colony, a worker bee increases the chances that the queen bee will survive. If the queen survives, she will produce more workers who will share many of the same genes. Thus, by behaving altruistically, a worker can cause more of her genes to be propagated in the population. This mechanism of propagating one's own genes by helping a closely related individual reproduce is called **kin selection**.

SECTION 2 REVIEW

1. Name three ways that insects communicate, and give an example of each way.
2. What determines whether a fertilized honeybee egg will develop into a worker or a queen?
3. How do honeybees behave when their hive is overcrowded?
4. How do honeybees convey information about the direction and distance of a food source that is far from the hive?

CRITICAL THINKING

5. **Evaluating Methods** How could you experiment to show that an insect is responding to pheromones and not visual cues?
6. **Recognizing Relationships** What is the adaptive advantage of innate behavior?
7. **Analyzing Current Research** Queen bees sometimes mate with drones from other hives. How might this behavior benefit the colony?

CHAPTER HIGHLIGHTS

SECTION 1

The Insect World

- The insect body is divided into three tagmata. The head has mandibles and one pair of unbranched antennae; the thorax has three pairs of jointed legs and, in many species, one or two pairs of wings; and the abdomen has 9 to 11 segments but neither wings nor legs in adults.
- Insects live in almost every terrestrial and freshwater environment. Factors responsible for their success include their ability to fly, exoskeleton, jointed appendages, small size, and short life span.
- Insects negatively affect humans by competing for food, transmitting diseases, and destroying buildings and other manufactured products. However, insects are also beneficial. They serve as food for other animals, pollinate flowers, make valuable products such as honey, and recycle nutrients.
- The mouthparts of insects are often specialized for tearing and cutting solid food or for sucking or soaking up liquid food.
- Insects have an open circulatory system that transports nutrients through the body. Gas exchange occurs by means of air-filled tracheae that reach deep into the body. Malpighian tubules remove cellular wastes from the hemolymph while conserving water.
- Insect sensory structures include simple and compound eyes, sound-sensing tympana in some species, and sensory hairs on the antennae and other body parts.
- Most insects go through metamorphosis. In incomplete metamorphosis, a nymph hatches from an egg and resembles the adult but has undeveloped reproductive organs and no wings. The nymph molts several times to become an adult.
- In complete metamorphosis, a wormlike larva hatches from an egg and molts several times before becoming a pupa. The pupa molts to produce the adult, which resembles neither the larva nor the pupa.
- Insects can defend themselves by stinging, using camouflage, or releasing noxious chemicals. Insects that are dangerous or taste bad often have warning coloration that makes them recognizable to predators. The warning coloration of a dangerous species may be mimicked by harmless species.

Vocabulary

entomology (p. 742)

labrum (p. 746)

labium (p. 746)

tympanum (p. 747)

ovipositor (p. 748)

metamorphosis (p. 748)

incomplete metamorphosis
(p. 748)

nymph (p. 748)

complete metamorphosis
(p. 748)

pupa (p. 749)

SECTION 2

Insect Behavior

- Insects communicate by releasing pheromones and by producing sounds and flashes of light.
- Honeybees live in complex colonies consisting mostly of nonreproductive female workers that perform all duties except reproduction. Reproduction in each colony is the exclusive function of one queen and a few hundred male drones.
- Honeybees communicate the direction and distance to food sources by performing dances inside the hive.
- In defending the colony, worker bees show altruistic behavior toward their close relatives in the colony. By doing so, they increase the propagation of their own genes.

Vocabulary

pheromone (p. 751)

social insect (p. 752)

innate behavior (p. 752)

worker bee (p. 752)

queen bee (p. 752)

drone (p. 752)

royal jelly (p. 752)

queen factor (p. 753)

altruistic behavior (p. 754)

kin selection (p. 754)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
 - mesothorax* and *metathorax*
 - labrum* and *labium*
 - nymph* and *pupa*
 - worker bee* and *queen bee*
 - royal jelly* and *queen factor*
- Explain the relationship between altruistic behavior and kin selection.
- Choose the term that does not belong in the following group, and explain why it does not belong: *egg*, *larva*, *pupa*, and *nymph*.
- Word Roots and Origins** The word *gastric* is derived from the Greek *gaster*, which means “stomach.” The word *cecum* is from the Latin *caecus*, which means “blind.” Using this information, explain why the term *gastric cecum* is a good name for the structure that the term describes.

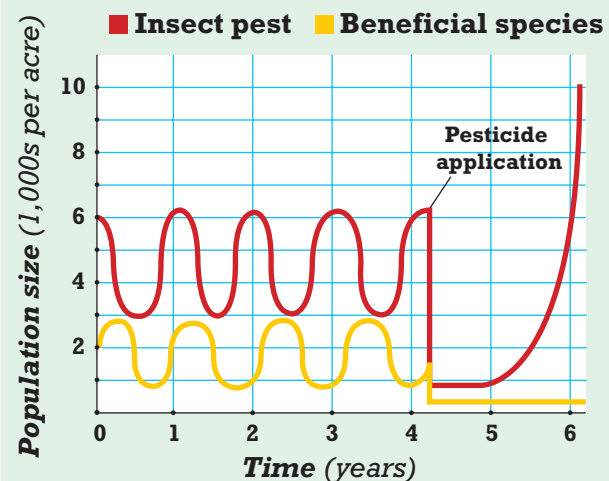
UNDERSTANDING KEY CONCEPTS

- Identify** three characteristics that entomologists use to divide insects into orders.
- State** how the small size of insects has contributed to the biological success of insects.
- Explain** why insect wings are not homologous to the wings of birds and bats.
- Describe** the function of spiracles in insects.
- Define** *tympanum*, and identify where it is located on a grasshopper.
- Explain** the function of warning coloration.
- Describe** how crickets produce sounds, and explain the functions of these sounds.
- Define** *innate behavior*, and give an example.
- List** the three types of individuals that make up a honeybee society.
- Compare** the round dance with the waggle dance of honeybees.
- Explain** how altruistic behavior may have been selected in honeybees.
- CONCEPT MAPPING** Use the following terms to create a concept map that sequences the stages of complete metamorphosis: *older larva*, *pupal exoskeleton*, *eggs*, *pupa*, *adult*, and *young larva*.

CRITICAL THINKING

- Relating Structure and Function** Insects and crustaceans both belong to the phylum Arthropoda and share many characteristics. Recall that the largest crustacean, the Japanese spider crab, has a leg span of 4 m. In contrast, the largest insects, such as the atlas moth, have a wingspan of only about 25 cm. Identify some possible reasons that the largest crustaceans are so much bigger than the largest insects.
- Applying Information** What characteristics may have helped insects survive the major climatic changes that led to the extinction of the dinosaurs and many other species about 65 million years ago?
- Interpreting Graphics** The graph below shows changes in size of two insect populations over a six-year period. One population is considered a pest, and the other is considered a beneficial species. At year 4, a pesticide is applied. Describe the relationship between the two species before year 4. How did the populations change after the application of the pesticide? Propose an explanation for the changes that occurred in the two populations after the use of pesticides.

Changes in Two Insect Populations



- Relating Concepts** Recall that squids and other cephalopods have a closed circulatory system, which supports their active lifestyle by circulating blood quickly through their bodies. Many insects, such as dragonflies and bees, are also very active, but all insects have an open circulatory system. How can insects maintain active lifestyles while having an open circulatory system?



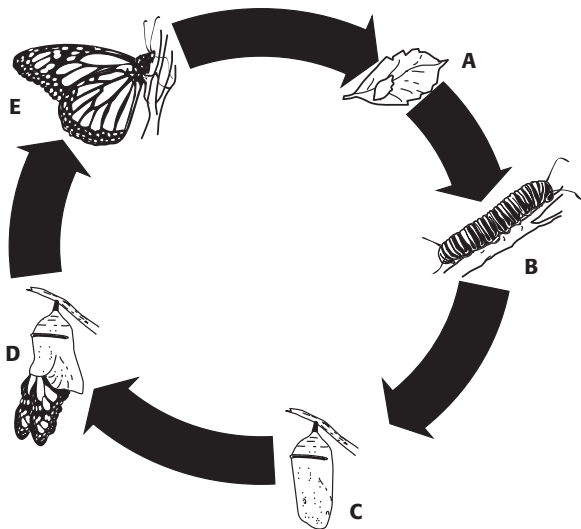
Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- What are an insect's legs and wings attached to?
 - head
 - thorax
 - labrum
 - abdomen
- What are the mouthparts of a grasshopper specialized for?
 - sucking fluids
 - lapping up liquids
 - cutting and tearing fibers
 - filtering food out of muddy water
- What is the term for the immature form of an insect that undergoes incomplete metamorphosis?
 - adult
 - pupa
 - infant
 - nymph

INTERPRETING GRAPHICS: The illustration below shows the life cycle of a butterfly. Use the illustration to answer the questions that follow.

Life Cycle of a Butterfly

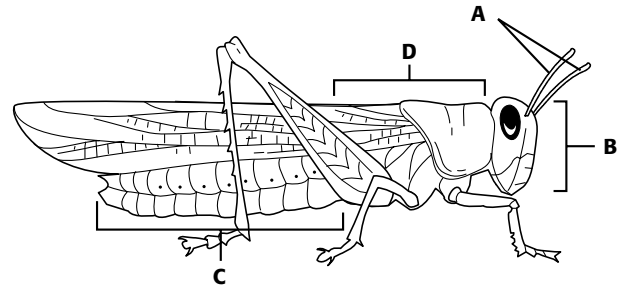


- What kind of life cycle is shown?
 - direct development
 - seasonal development
 - complete metamorphosis
 - incomplete metamorphosis
- What is the term for the developmental stage labeled C?
 - pupa
 - larva
 - nymph
 - caterpillar

DIRECTIONS: Complete the following analogy.

- queen factor : queen bee :: royal jelly :
 - drone
 - worker
 - queen bee
 - pheromone

INTERPRETING GRAPHICS: The diagram below shows the external structure of a grasshopper. Use the diagram to answer the questions that follow.



- What is the term for the structure(s) labeled D?
 - thorax
 - labrum
 - abdomen
 - antennae
- Which of the following structures is part of the structure labeled B?
 - ovipositor
 - mandibles
 - tympanum
 - malpighian tubules

SHORT RESPONSE

Defensive adaptations in insects increase the chances of insects' survival.

Distinguish between passive defenses and aggressive defenses, and give two examples of each.

EXTENDED RESPONSE

Farmers often try to limit the number of insects on crops by applying chemical insecticides to the crops. Sometimes, an insecticide that had previously been effective no longer affects certain types of insects.

Part A Why do farmers try to control insects?

Part B How can the effect of insecticides on certain insect populations change?

Test TIP

For a question about an illustration that has labels, read the labels carefully and then check that the answer you choose matches your interpretation of the labels.

Observing Grasshopper Anatomy

OBJECTIVES

- Examine the external and internal anatomy of a grasshopper.
- Infer function from observation of structures.

PROCESS SKILLS

- relating structure and function
- observing

MATERIALS

- safety goggles
- gloves
- preserved grasshopper (1 for each student)
- dissection tray
- forceps
- fine dissection scissors, with pointed blades
- hand lens or dissecting microscope
- blunt probe
- sharp probe
- dissection pins



Background

1. List the distinguishing characteristics of insects.
2. In this lab, you will dissect a grasshopper to observe its external and internal structure. To which order of insects do grasshoppers belong?
3. What characteristics of the grasshopper place it into this order?

PART A Observing the External Anatomy of the Grasshopper

1.   **CAUTION** Wear safety goggles and gloves during this lab. Keep your hands away from your eyes and face when working with preserved specimens. Using forceps, hold a preserved grasshopper under running water to gently but thoroughly remove excess preservative. Then place the grasshopper in a dissection tray.
2. Use a hand lens to observe the grasshopper's parts. While referring to Figure 37-3, identify the head, thorax, and abdomen. Note how the thorax and abdomen are divided into segments.
3. Use forceps to spread out and examine both pairs of wings. Notice that the forewings are narrow and the hindwings are wide. Observe how the hindwings fold fanlike against the body.
4. Observe the legs. Grasp one of each pair of legs and notice how the legs are divided into segments. Gently bend the legs to observe their normal range of motion.
5. Examine the 11 segments of the abdomen. On abdominal segment 1, find the tympanum. Then, along each side of abdominal segments 1–8, locate the spiracles, which look like small dots. Gently touch the abdomen with a blunt probe to find the flexible membrane that connects the segments to one another.
6. While referring to Figure 37-4a, examine the grasshopper's head. Find the two antennae, two compound eyes, and three simple eyes. Use a sharp probe to push apart the mouthparts. Locate and identify the mandibles, maxillae, labium, and labrum. Note that each maxilla has a segmented feeler called a palpus and that the labium has two palpi.

PART B Observing the Internal Anatomy of the Grasshopper

7.  **CAUTION** Dissecting instruments can cut you. Always cut in a direction away from your face and body. Using scissors, snip off the grasshopper's legs, wings, and antennae at their bases. Pin the body to the dissection tray. Then use the scissors to make a shallow cut just above the spiracles through the exoskeleton along both sides of the thorax and abdomen.
8. As you remove the exoskeleton, look at its underside, where the heart and aorta may be attached. Also look for muscles attached to the inside of the exoskeleton. Find and remove any fatty tissue (which your teacher can help you identify) that may hide other organs.
9. If your grasshopper is a female, look for its ovaries, which may contain elongated eggs. Refer to Figure 37-5 to see what the ovaries look like. If your grasshopper is a male, examine the ovaries and eggs in the female grasshopper of another student. Remove the ovaries and eggs (if present) from one side of the abdomen to uncover the digestive tract.
10. Make a table in your lab report like the one shown. As you observe each of the structures listed in the table, fill in the function of that structure.
11. Referring to Figure 37-5, look for the organs of the digestive tract: esophagus, crop, gizzard, midgut, hindgut, and anus. Find the salivary glands and the gastric ceca, which are also parts of the digestive system.
12. On the surface of the midgut and hindgut, look for the Malpighian tubules, which are tiny tubes that connect to the digestive tract.
13. Locate and identify the brain, ventral nerve cord, and ganglia.
14. Carefully cut away and remove the organs of the digestive system to expose some parts of the respiratory system. Referring to Figure 37-5, locate and identify tracheae that run along the dorsal and ventral parts of the body, as well as other tracheae that connect them. The larger tracheae lead to spiracles in the abdomen. Look also for swollen tracheae, called air sacs, which increase the volume of air drawn into the abdomen when the grasshopper breathes.
15.   Dispose of your specimen according to the directions from your teacher. Then clean up your materials and wash your hands before leaving the lab.

FUNCTION OF GRASSHOPPER STRUCTURES

Structure	Function
Esophagus	
Crop	
Gizzard	
Midgut	
Hindgut	
Salivary glands	
Gastric ceca	
Malpighian tubules	
Tracheae	
Spiracles	

Analysis and Conclusions

1. How do you think the membrane between segments helps the grasshopper in its movements?
2. How does the function of the stiff, leathery forewings differ from that of the more delicate hindwings?
3. Trace the path of food through the grasshopper's digestive tract.
4. To what system do the Malpighian tubules belong?
5. Why is the circulatory system of the grasshopper described as an open circulatory system?
6. Compared with invertebrates such as flatworms and earthworms, grasshoppers are highly responsive to environmental stimuli. What are some structural adaptations of the grasshopper that make this responsiveness possible?

Further Inquiry

1. Prepare an illustrated chart that compares and contrasts the characteristics of grasshoppers, beetles (order Coleoptera), and butterflies (order Lepidoptera). What trait of each kind of insect is reflected in the name of its order? Include other traits in your chart.
2. The fruit fly of the genus *Drosophila* is an insect that, unlike the grasshopper, undergoes complete metamorphosis. Research the life cycles of the grasshopper and the fruit fly, and make a chart that compares and contrasts their life cycles.

ECHINODERMS AND INVERTEBRATE CHORDATES

This blue sea star, *Linckia laevigata*, shows the pentaradial symmetry of most echinoderms.

SECTION 1 *Echinoderms*

SECTION 2 *Invertebrate Chordates*

ECHINODERMS

*The phylum Echinodermata (ee-KIE-noh-duhr-MAH-tuh) is a group of invertebrates that includes sea stars, sand dollars, sea urchins, and sea cucumbers. The members of this phylum, called **echinoderms**, inhabit marine environments ranging from shallow coastal waters to ocean trenches more than 10,000 m deep. They vary in diameter from 1 cm to 1 m and often are brilliantly colored.*

CHARACTERISTICS

Echinoderms are radially symmetrical animals. Like cnidarians and ctenophores, which are also radially symmetrical, echinoderms have no head or any other sign of cephalization. Unlike cnidarians and ctenophores, however, adult echinoderms develop from bilaterally symmetrical larvae. A few examples of echinoderm larvae are illustrated in Figure 38-1. This feature of development indicates that echinoderms probably evolved from bilaterally symmetrical ancestors.

The fossil record of echinoderms dates back to the Cambrian period, more than 500 million years ago. Early echinoderms from this period appear to have been sessile, and biologists believe these animals evolved radial symmetry as an adaptation to a sessile existence. Echinoderms later evolved the ability to move from place to place. Today, the vast majority of echinoderm species can move by crawling slowly along the ocean bottom, and only about 80 species are sessile.

Echinoderms are deuterostomes, which makes them different from the other invertebrates you have studied so far. Recall that *deuterostomes* are coelomates whose embryos have radial cleavage.

OBJECTIVES

- **Discuss** four distinguishing characteristics of echinoderms.
- **Describe** representative species from each of the five classes of echinoderms.
- **Describe** the water-vascular system and other major body systems of echinoderms.
- **Compare** sexual and asexual reproduction in sea stars.

VOCABULARY

echinoderm
ossicle
water-vascular system
tube foot
test
pedicellaria
madreporite
stone canal
ring canal
radial canal
ampulla
cardiac stomach
pyloric stomach
bipinnaria

FIGURE 38-1

Notice the bilateral symmetry in these echinoderm larvae. The larvae develop into radially symmetrical adults.



Word Roots and Origins

ossicle

from the Latin
ossiculum, meaning
“little bone”

Also, in deuterostomes, the anus forms from the blastopore, and the mesoderm arises from outpockets of the endoderm. Because they develop as deuterostomes, echinoderms are more closely related to chordates than they are to other invertebrates. Chordates are also deuterostomes and are discussed in the second part of this chapter.

Echinoderms have four major characteristics that are not shared by any other phylum. (1) Most echinoderms have a type of radial symmetry called *pentaradial symmetry*, in which the body parts extend from the center along five spokes. (2) They have an endoskeleton composed of calcium carbonate plates known as **ossicles**. The ossicles may be attached to spines or spicules that protrude through the skin. The name *echinoderm* means “spiny skin.” (3) They have a **water-vascular system**, which is a network of water-filled canals inside their body. (4) They have many small, movable extensions of the water-vascular system called **tube feet**, which aid in movement, feeding, respiration, and excretion. The water-vascular system and tube feet will be discussed in more detail.

CLASSIFICATION

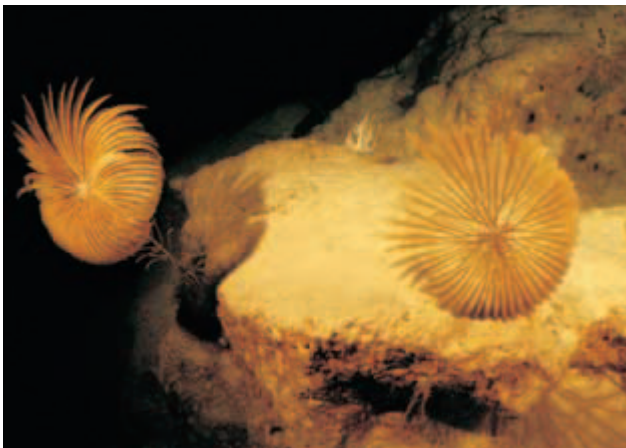
Taxonomists divide the 7,000 species of echinoderms into six classes, five of which will be described here: Crinoidea (kri-NOYD-ee-uh), Ophiuroidea (OH-fee-yoor-OYD-ee-uh), Echinoidea (EK-uh-NOYD-ee-uh), Holothuroidea (HOH-loh-thuh-ROYD-ee-uh), and Asteroidea (AS-tuh-ROYD-ee-uh).

Class Crinoidea

Members of the class Crinoidea, called *crinoids* (KRI-NOYDZ), include the sea lilies and feather stars, which are shown in Figure 38-2. The name *crinoid* means “lily-like.” Sea lilies most closely resemble the fossils of ancestral echinoderms from the Cambrian period. They are sessile as adults, remaining attached to rocks or the sea bottom by means of a long stalk. Feather stars, in contrast, can swim or crawl as adults, although they may stay in one place for long periods.

FIGURE 38-2

This sea lily, *Cenocrinus* (a), and these feather stars, *Oxycomanthus bennetti* (b), are members of the class Crinoidea. Notice their adaptations for filter feeding.



(a)



(b)

In both types of crinoids, five arms extend from the body and branch to form many more arms—up to 200 in some feather star species. Mucus-covered tube feet located on each arm filter small organisms from the water. The tube feet also serve as a respiratory surface across which crinoids exchange oxygen and carbon dioxide with the water. Cilia on the arms transport trapped food to the crinoid's mouth at the base of the arms. The mouth faces up in crinoids, but in most other echinoderms the mouth faces downward.

Class Ophiuroidea

The 2,000 species of basket stars and brittle stars make up the largest echinoderm class, Ophiuroidea, which means “snake-tail.” Members of this class are distinguished by their long, narrow arms, which allow them to move more quickly than other echinoderms. As you can see in Figure 38-3, the thin, flexible arms of basket stars branch repeatedly to form numerous coils that look like tentacles. Brittle stars, so named because parts of their arms break off easily, can regenerate missing parts.

Basket stars and brittle stars live primarily on the bottom of the ocean, often beneath stones or in the crevices and holes of coral reefs. They are so numerous in some locations that they cover the sea floor. Some species feed by raking in food with their arms or gathering it from the ocean bottom with their tube feet. Others trap suspended particles with their tube feet or with mucous strands located between their spines.

Class Echinoidea

The class Echinoidea consists of about 900 species of sea urchins and sand dollars. *Echinoidea* means “spinelike,” a description that applies especially well to many of the sea urchins, such as the ones shown in Figure 38-4. In both sea urchins and sand dollars, the internal organs are enclosed within a fused, rigid endoskeleton called a **test**.

The spherical sea urchins are well adapted to life on hard sea bottoms. They move by means of their tube feet and feed by scraping algae from hard surfaces with the five teeth that surround their mouth. The teeth and the muscles that move them are part of a complex jawlike mechanism called *Aristotle's lantern*. The spines that protrude from the test may be short and flat, long and thin, or wedge shaped, depending on the species. In some sea urchins, the spines are barbed, and in others, they are hollow and contain a venom that is dangerous to predators as well as swimmers.

Sand dollars live along seacoasts. As their name implies, they are usually found in sandy areas and have the flat, round shape of a silver dollar. Their shape is an adaptation for shallow burrowing. The short spines on a sand dollar are used in locomotion and burrowing, and they help clean the surface of the body. Sand dollars use their tube feet to capture food that settles on or passes over their body.

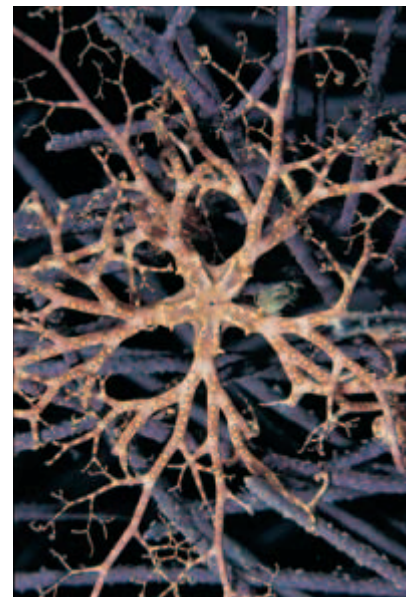


FIGURE 38-3

This basket star, *Astrophyton muricatum*, has long, flexible arms with many coiled branches.

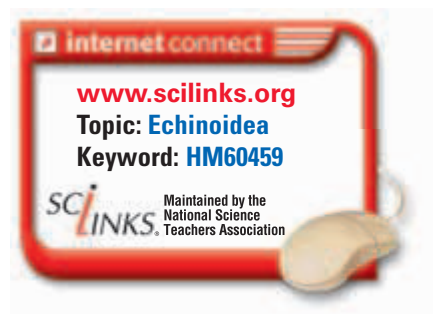


FIGURE 38-4

The long, sharp spines that cover these sea urchins, *Strongylocentrotus*, provide protection against most predators.



FIGURE 38-5

Tentacles around the mouth of this sea cucumber collect food and bring it to the animal's mouth. Five rows of tube feet that run along the body are evidence of the sea cucumber's pentaradial symmetry.



Class Holothuroidea

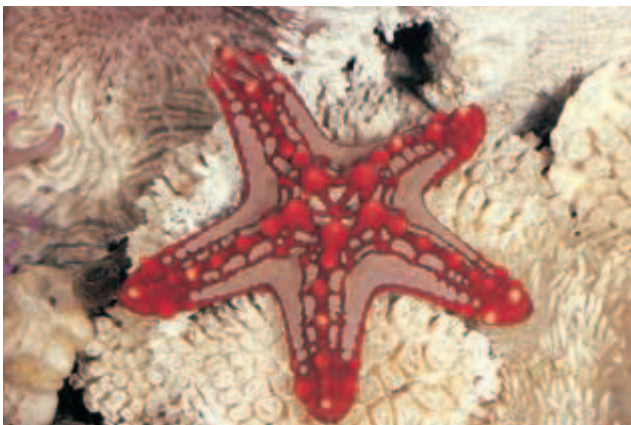
Sea cucumbers belong to the class Holothuroidea. Most of these armless echinoderms live on the sea bottom, where they crawl or burrow into soft sediment by using their tube feet. The ossicles that make up their endoskeleton are very small and are not connected to each other, so their bodies are soft. Modified tube feet form a fringe of tentacles around the mouth. When these tentacles are extended, as shown in Figure 38-5, the animals resemble the polyp form of some cnidarians. This resemblance explains the name of this class, which means “water polyp.” A sea cucumber uses its tentacles to sweep up sediment and water. It then stuffs its tentacles into its mouth and cleans the food off them.

Class Asteroidea

The sea stars, or starfish, belong to the class Asteroidea, which means “starlike.” Sea stars live in coastal waters all over the world. They exist in a variety of colors and shapes, two of which are shown in Figure 38-6. Sea stars are economically important because they prey on oysters, clams, and other organisms that humans use as food.

FIGURE 38-6

Sea stars are found on rocky coastlines worldwide. One of the more colorful varieties is the African sea star, *Proto-reaster linckii* (a). The sunflower star, *Pycnopodia helianthoides* (b), can have up to two dozen arms.



(a)



(b)

STRUCTURE AND FUNCTION OF ECHINODERMS

The sea star will be used to demonstrate some of the details of echinoderm structure and function. As you read about how echinoderms carry out life functions, consider how they differ from the other groups of invertebrates you have studied.

External Structure

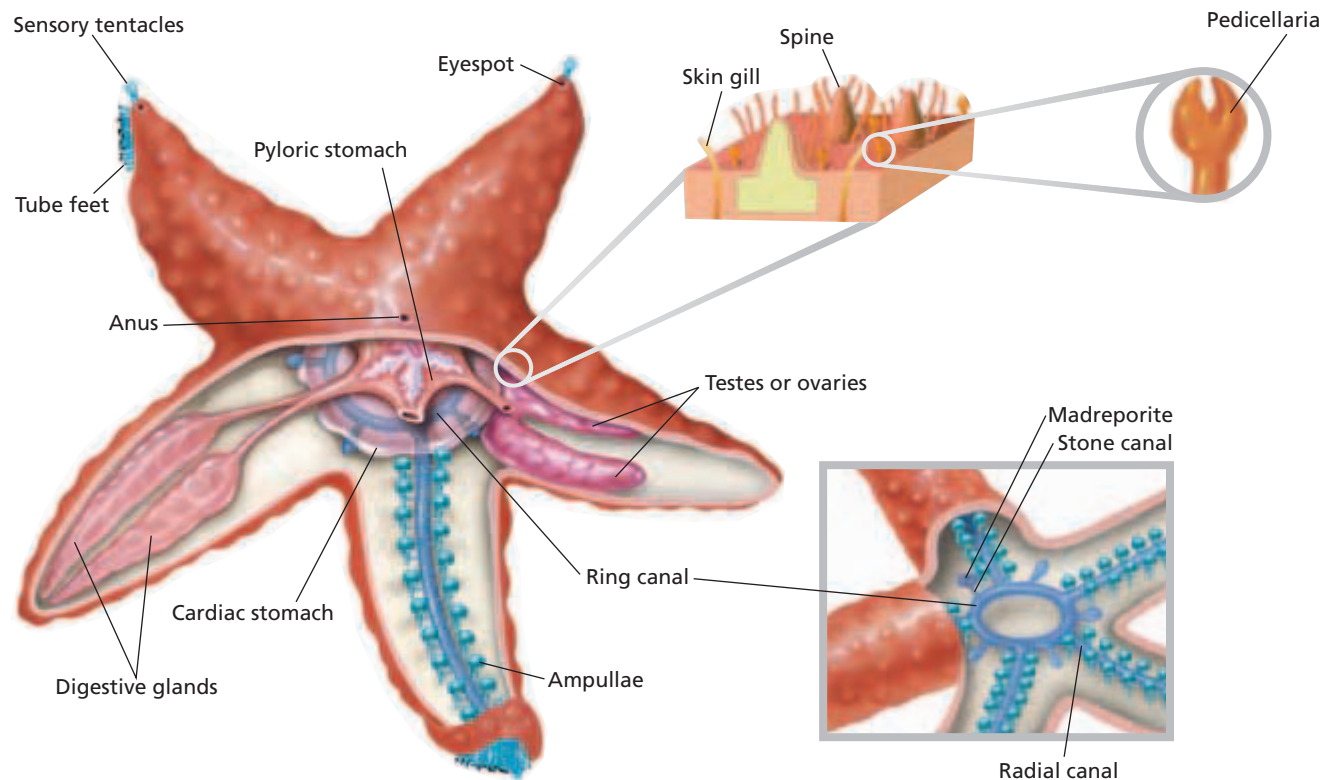
As you can see in Figure 38-7, the body of a sea star is composed of several arms that extend from a central region. Sea stars typically have five arms, but in some species, such as the one shown in Figure 38-6b, there may be as many as 24. Two rows of tube feet run along the underside of each arm. The body is often flattened.

In echinoderms, the side of the body where the mouth is located is referred to as the *oral surface*. The side of the body that is opposite from the mouth is called the *aboral* (A-BOHR-uhl) *surface*. In sea stars, the oral surface is on the underside of the body.

The body of a sea star is usually covered with short spines that give the animal a rough texture. Surrounding each spine in many sea stars are numerous tiny pincers called **pedicellariae** (PED-uh-suh-LAR-ee-ee), which are shown in Figure 38-7. Pedicellariae help keep the body surface free of foreign objects, including algae and small animals that might grow on the sea star or damage its soft tissues. Pedicellariae are found in sea stars and some sea urchins.

FIGURE 38-7

Sea stars have a number of structural features that are unique to the phylum Echinodermata. Their pentaradial symmetry is indicated by their five arms, each of which contains a division of their internal organ systems. The water-vascular system consists of a network of canals connected to hundreds of tube feet. The inset shows that the sea star's exterior is dotted with short spines, pincerlike pedicellariae, and skin gills.



Word Roots and Origins

ampulla

form the Latin *ampulla*,
meaning "flask"

Water-Vascular System

The water-vascular system is a network of water-filled canals that are connected to the tube feet. Use Figure 38-7 to follow the path of water through the water-vascular system. Water enters the system through small pores in the **madreporite** (MA-druh-PAWR-IET), a sieve-like plate on the aboral surface. Water then passes down the **stone canal**, a tube that connects the madreporite to the **ring canal**, which encircles the mouth. Another tube, the **radial canal**, extends from the ring canal to the end of each arm. The radial canals carry water to the hundreds of hollow tube feet. Valves prevent water from flowing back into the radial canals from the tube feet.

The upper end of each tube foot is expanded to form a bulblike sac called an **ampulla** (am-PUHL-uh). Contraction of muscles surrounding the ampullae forces water into the tube feet, causing them to extend. Contraction of muscles lining the tube feet forces water back into the ampullae and shortens the tube feet. In this way, the sea star uses water pressure to extend and withdraw its tube feet. In many species, small muscles raise the center of each tube foot's disklike end, creating suction when the tube feet are pressed against a surface. These coordinated muscular contractions enable sea stars to climb slippery rocks and capture prey.

Feeding and Digestion

The sea star's mouth is connected by a short esophagus to the **cardiac stomach**, which the sea star can turn inside out through its mouth when it feeds. The cardiac stomach transfers food to the **pyloric stomach**, which connects to a pair of digestive glands in each arm. The cardiac stomach, pyloric stomach, and digestive glands break down food with the help of the enzymes they secrete. Nutrients are absorbed into the coelom through the walls of the digestive glands, and undigested material is expelled through the anus on the aboral surface.

Most sea stars are carnivorous, feeding on mollusks, worms, and other slow-moving animals. When a sea star captures a bivalve mollusk, such as a clam, it attaches its tube feet to both halves of the clamshell and exerts a steady pull, as Figure 38-8 shows. Eventually, the clam's muscles tire, and the shell opens slightly. The sea star then inserts its cardiac stomach into the clam and digests the clam's soft tissues while they are still in the shell. The sea star then withdraws the stomach, containing the partially digested food, back into its body, where the digestive process is completed.

Other Body Systems

Like other echinoderms, the sea star has no circulatory, excretory, or respiratory organ systems. Fluid in the coelom bathes the organs and distributes nutrients and oxygen. Gas exchange and waste excretion take place by diffusion through the thin walls of the tube feet and through the *skin gills*, hollow tubes that project from the coelom lining to the exterior. You can see the skin gills in the inset in Figure 38-7, on the previous page.

FIGURE 38-8

This sea star, *Asterias rubens*, is prying open the shell of a clam to feed on the clam's soft tissues.



Because echinoderms have no head, they also have no brain. The nervous system consists mainly of a *nerve ring* that circles the mouth and a *radial nerve* that runs from the nerve ring along each arm. The nerve ring and radial nerves coordinate the movements of the tube feet. If the radial nerve in one arm is cut, the tube feet in that arm lose coordination. If the nerve ring is cut, the tube feet in all arms become uncoordinated, and the sea star cannot move.

Sea stars also have a nerve net near the body surface that controls the movements of the spines, pedicellariae, and skin gills. The end of each arm has an eyespot that responds to light and several tentacles that respond to touch. The tube feet also respond to touch, and other touch-sensitive and chemical-sensitive cells are scattered over the surface of the sea star's body.

Reproduction and Development

Most sea star species have separate sexes, as do most other echinoderms. Each arm of the sea star contains a pair of ovaries or testes. Females produce up to 200 million eggs in one year. Fertilization occurs externally, when the eggs and sperm are shed into the water. Each fertilized egg develops into a bilaterally symmetrical, free-swimming larva called a **bipinnaria** (BIE-pin-AR-ee-uh). After about two months, the larva settles to the bottom, and begins metamorphosis into a pentaradially symmetrical adult.

Echinoderms have remarkable powers of regeneration. Sea stars can regenerate arms from the central region of their body, even if they lose all of their arms. The process of regeneration is very slow, taking as long as a year. Sea stars use their regenerative ability as a defensive mechanism, automatically shedding an arm at its base when the arm is captured by a predator. As you can see in Figure 38-9, some sea stars can even regenerate a complete, new individual from a detached arm, as long as the arm is attached to a portion of the central region. Certain species reproduce asexually when the body splits through the central region. The two parts that are formed then regenerate the missing structures.

FIGURE 38-9

As long as a sea star retains part of its central region, it can regenerate any arms it loses. The sea star shown here, a member of the genus *Echinaster*, is regenerating five new arms.



SECTION 1 REVIEW

1. What are the characteristics that distinguish Echinodermata from other phyla?
2. Give the common names of members of each of the five classes of echinoderms.
3. How does a sea star extend and withdraw its tube feet?
4. Describe how sea stars are able to reproduce asexually.

CRITICAL THINKING

5. **Making Comparisons** Compare development in sea urchins with development in mollusks.
6. **Inferring Relationships** How would a sea star's ability to feed be affected by losing the water in its water-vascular system?
7. **Analyzing Concepts** Sea stars release millions of eggs during reproduction. How is this method of reproduction adaptive?

SECTION 2

OBJECTIVES

- **List** the major characteristics of chordates.
- **Describe** the evolution and classification of invertebrate chordates.
- **Describe** the structure of lancelets.
- **Describe** the structure of tunicates.

VOCABULARY

atriopore

INVERTEBRATE CHORDATES

The phylum Chordata (kawr-DAY-tuh) includes all of the vertebrates, or animals with backbones. It also includes two groups of invertebrates—animals that lack backbones.

CHARACTERISTICS

All animals with a backbone are vertebrates, and they make up one of the subphyla in the phylum Chordata, whose members are called *chordates*. Chordates are so named because they have a *notochord*, a stiff but flexible rod of cells that runs the length of the body near the dorsal surface. Figure 38-10 illustrates the notochord. The stiffness of the notochord provides a resistance against which the body muscles can exert force when they contract. The flexibility of the notochord allows the body to bend from side to side as well as up and down.

Some kinds of chordates retain the notochord throughout their life. In most vertebrates, however, the notochord is present in embryos but becomes greatly reduced when the vertebral column, or backbone, develops. In adult mammals, the notochord persists only as small patches of tissue between the bones of the vertebral column.

Recall that in addition to having a notochord, all chordates have the following three characteristics during some stage of their life: (1) a dorsal nerve cord, (2) pharyngeal pouches, and (3) a postanal tail. These characteristics are also illustrated in Figure 38-10. Unlike the ventral nerve cords of invertebrates such as annelids and arthropods, the dorsal nerve cord of a chordate is a hollow tube.

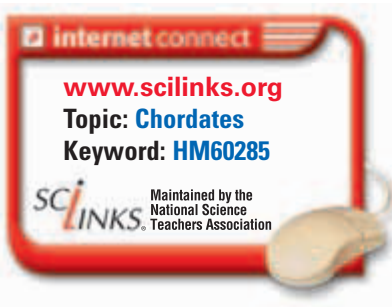
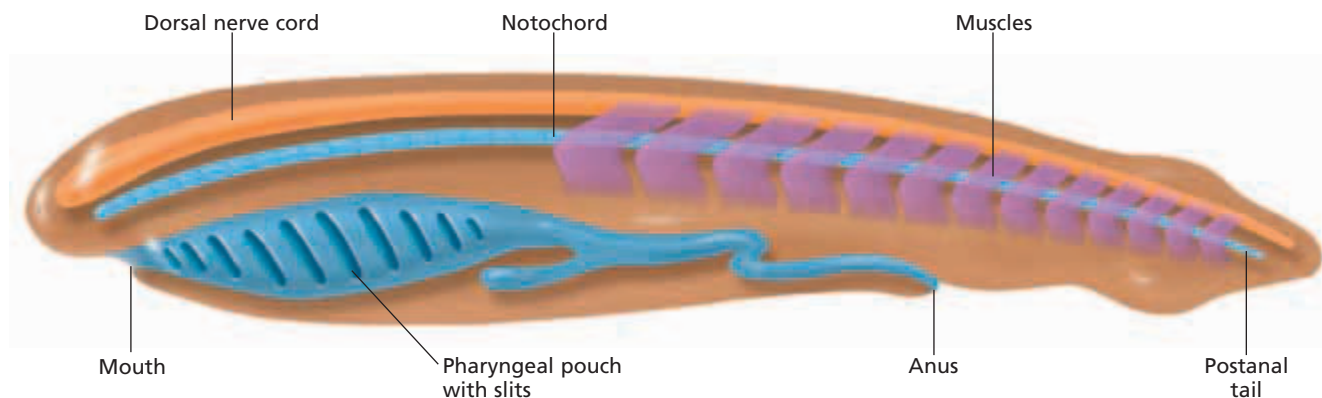


FIGURE 38-10

All chordates have a notochord, a dorsal nerve cord, pharyngeal pouches, and a postanal tail during at least some stage of their life.



In vertebrates, the anterior end of the nerve cord enlarges during development to form the brain, and the posterior end forms the spinal cord. The brain receives information from a variety of complex sensory organs, many of which are concentrated at the anterior end of the body.

The pharyngeal pouches are outpockets in the pharynx, the portion of the digestive tract between the mouth and the esophagus. In aquatic chordates, the pharyngeal pouches have slits and evolved first into filter-feeding structures and later into gill chambers. In terrestrial chordates, the pouches evolved into a variety of structures, including the jaws and inner ear.

The notochord extends into the postanal tail, and muscles in the tail can cause it to bend. The postanal tail provides much of the propulsion in many aquatic chordates. Invertebrates in other phyla lack this form of propulsion, and the anus, if present, is located at the end of the body.

EVOLUTION AND CLASSIFICATION

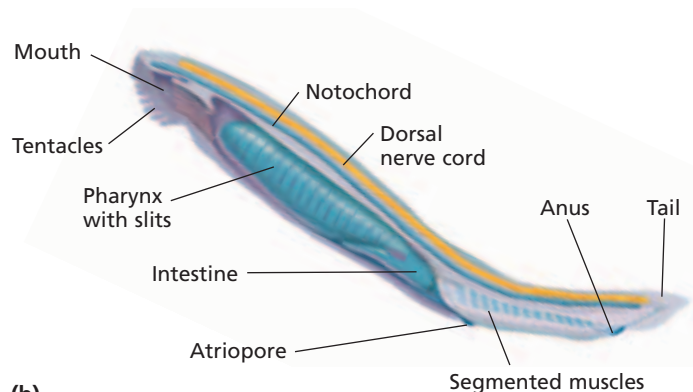
Like echinoderms, chordates are deuterostomes. This similarity provides evidence that echinoderms and chordates likely evolved from a common ancestor. The phylum Chordata is divided into three subphyla: Vertebrata, Cephalochordata (SEF-uh-loh-kawr-DAY-tuh), and Urochordata (YOOR-uh-kawr-DAY-tuh). Members of the subphylum Vertebrata, the vertebrates, constitute more than 95 percent of all chordate species. Members of the other two subphyla live only in the ocean. They are the closest living relatives of the early animals from which all chordates evolved.

Subphylum Cephalochordata

The subphylum Cephalochordata contains about two dozen species of blade-shaped animals known as *lancelets*. Figure 38-11 shows that lancelets look much like the idealized chordate drawn in Figure 38-10. They retain their notochord, dorsal nerve cord, pharyngeal pouches, and postanal tail throughout their life.



(a)



(b)



Quick Lab

Modeling Chordate Characteristics

Materials several colors of clay, toothpicks, masking tape

Procedure Build clay models of a lancelet and an adult tunicate by using different colors of clay for the structures shown in Figures 38-11 and 38-12. Make flags using masking tape attached to toothpicks, and use them to identify any of the four major characteristics of chordates that are found in your models.

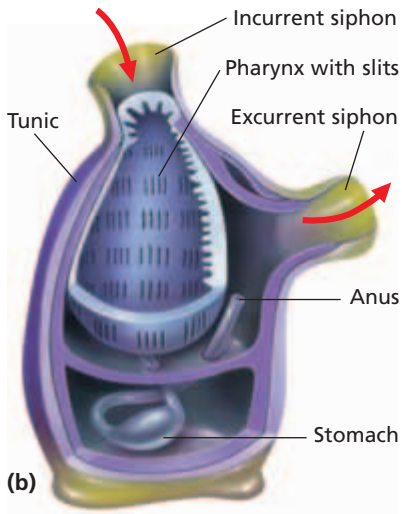
Analysis Which of the major characteristics of chordates are found in the lancelet? the adult tunicate? Which of the four characteristics is shared by both? Why is the tunicate classified as a chordate despite the fact that it has only three of the four chordate characteristics?

FIGURE 38-11

(a) The lancelet *Branchiostoma lanceolatum* lives with most of its body buried in the sand. (b) Even as adults, lancelets clearly show all four chordate characteristics.



(a)



(b)

FIGURE 38-12

Most adult tunicates, such as *Polycarpa aurata* (a), are sessile filter feeders. A drawing of a tunicate's internal structure (b) shows its enlarged pharynx with slits, the only chordate characteristic retained by adult tunicates.

Lancelets live in warm, shallow waters. They wriggle backward into the sand with a muscular tail. Only their anterior end protrudes from the sand. Lancelets have cilia that draw water into the pharynx through the mouth. Food particles in the water are trapped as the water passes through the slits in the pharynx. The food enters the intestine to be digested. The water leaves the body through an opening called the **atriopore** (AY-tree-oh-POHR).

Lancelets can swim weakly, powered by the coordinated contraction of muscles that run the length of their body. If you look closely at Figure 38-11, you can see that these muscles are arranged as a series of repeating segments. Body segmentation is another common feature of chordates. Recall that annelids and arthropods also have segmented bodies. However, animals in those phyla probably evolved body segmentation independently of chordates.

Subphylum Urochordata

The 2,000 species in the subphylum Urochordata are commonly called *tunicates* because their bodies are covered by a tough covering, or tunic. Tunicates are also called *sea squirts* because they squirt out a stream of water when touched. As adults, most tunicates are sessile, barrel-shaped animals that live on the sea bottom. They may be solitary or colonial.

Figure 38-12 shows that tunicates are adapted for filter feeding. Propelled by the beating of cilia, water enters the body through an incurrent siphon, passes through slits in the pharynx, and exits through an excurrent siphon. Food that is filtered by the pharynx moves into the stomach. Undigested material leaves via the anus, which empties into the excurrent siphon.

Tunicates are hermaphrodites. Sperm and eggs are released through the excurrent siphon into the surrounding water, where fertilization occurs.

Adult tunicates bear little resemblance to the idealized chordate shown in Figure 38-10. Although they do have a pouchlike pharynx with slits, they have no notochord, dorsal nerve cord, or postanal tail. Larval tunicates, however, possess all four chordate characteristics, but they lose most of them during metamorphosis.

SECTION 2 REVIEW

1. What are the four major characteristics of chordates?
2. What is the function of pharyngeal pouches in aquatic chordates?
3. What evidence of body segmentation do lancelets display?
4. Which of the chordate characteristics do tunicates retain as adults?

CRITICAL THINKING

5. **Making Comparisons** Compare the function of pharyngeal gill slits with that of gills in a crayfish.
6. **Inferring Relationships** How is burrowing backward, with the head out, adaptive in lancelets?
7. **Relating Concepts** In tunicates, the anus empties into the excurrent siphon. What is the advantage of this anatomical arrangement?

CHAPTER HIGHLIGHTS

SECTION 1

Echinoderms

- Most echinoderms develop from free-swimming, bilaterally symmetrical larvae into bottom-dwelling adults with pentaradial symmetry. The larval stage is evidence that echinoderms may have evolved from bilaterally symmetrical ancestors.
- Echinoderms have an endoskeleton made of ossicles and a water-vascular system, which includes many movable extensions called tube feet.
- The class Crinoidea includes sea lilies and feather stars, which are filter feeders that catch small organisms with their mucus-covered tube feet.
- The class Ophiuroidea consists of basket stars and brittle stars, fast-moving echinoderms with long, flexible arms.
- The class Echinoidea includes sea urchins and sand dollars, whose internal organs are enclosed inside a rigid endoskeleton called a test. Many sea urchins have long spines.
- The class Holothuroidea is made of sea cucumbers, armless echinoderms with soft bodies.
- The class Asteroidea consists of sea stars, which have from 5 to 24 arms. Two rows of tube feet run along the underside of each arm.
- The water-vascular system of a sea star consists of a network of canals that connect to bulblike ampullae. Contraction of muscles surrounding the ampullae extends the tube feet, and contraction of muscles lining the tube feet makes the tube feet retract.
- Sea stars can turn one of their stomachs inside out through their mouth to feed on prey they have captured. After the food is partially digested outside the body, it is brought inside, where digestion is completed.
- Sea stars lack circulatory, excretory, and respiratory organ systems, and they have no head or brain. They use skin gills for gas exchange and waste excretion.
- Most sea stars have separate sexes, and fertilization is external. Sea stars can also reproduce asexually by regeneration.

Vocabulary

echinoderm (p. 761)

ossicle (p. 762)

water-vascular system (p. 762)

tube foot (p. 762)

test (p. 763)

pedicellaria (p. 765)

madreporite (p. 766)

stone canal (p. 766)

ring canal (p. 766)

radial canal (p. 766)

ampulla (p. 766)

cardiac stomach (p. 766)

pyloric stomach (p. 766)

bipinnaria (p. 767)

SECTION 2

Invertebrate Chordates

- Chordates have a notochord, a stiff but flexible rod that runs the length of the body. In one group of chordates, the vertebrates, the notochord is largely replaced by the vertebral column, or backbone.
- Other common characteristics of chordates are a dorsal nerve cord, pharyngeal pouches, and a postanal tail. These characteristics are not present at all life stages in all chordates.
- Like echinoderms, chordates are deuterostomes, which suggests that echinoderms and chordates evolved from a common ancestor.
- Lancelets are animals in the subphylum Cephalochordata. These blade-shaped animals live partially buried in the sand, but they can swim from place to place. They retain all of the major chordate characteristics throughout their life.
- Tunicates are animals in the subphylum Urochordata. Tunicate larvae have all of the major chordate characteristics, but they lose most of them when they develop into adults. Most tunicate adults are sessile.

Vocabulary

atriopore (p. 770)

CHAPTER REVIEW

USING VOCABULARY

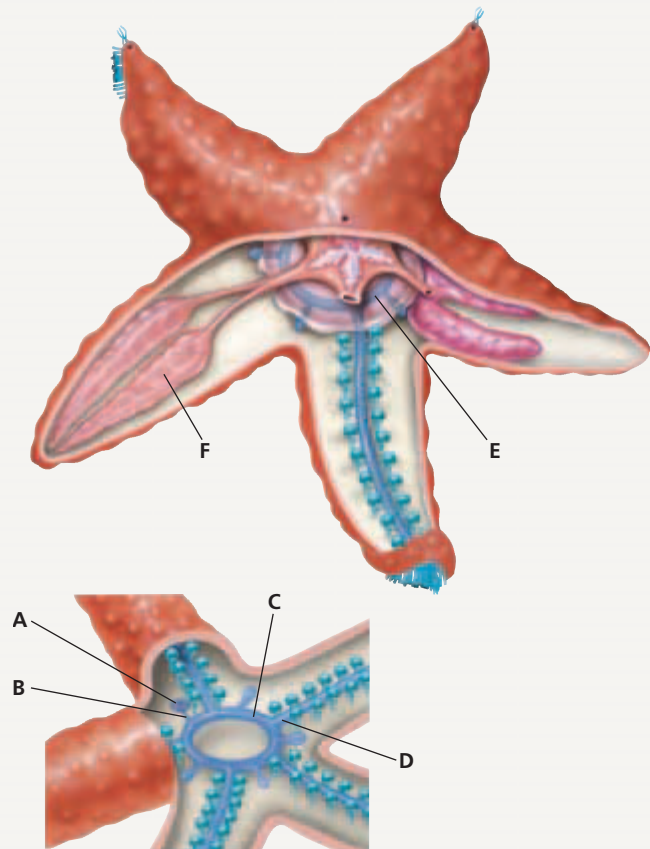
- For each pair of terms, explain how the meanings of the terms differ.
 - ossicle* and *test*
 - tube foot* and *pedicellaria*
 - stone canal* and *ring canal*
 - madreporite* and *atriopore*
- Explain the relationship between the *cardiac stomach* and the *pyloric stomach* in sea stars.
- Choose the term that does not belong in the following group, and explain why it does not belong: *tube foot*, *bipinnaria*, *radial canal*, and *ampulla*.
- Word Roots and Origins** The word *pyloric* is derived from the Greek *pyloros*, which means “gatekeeper.” Using this information, explain why the term *pyloric stomach* is a good name for the structure that the term describes.

UNDERSTANDING KEY CONCEPTS

- Explain** why echinoderms are thought to have evolved from bilaterally symmetrical ancestors.
- Identify** the functions for which echinoderms use their tube feet.
- Compare** feeding in crinoids and basket stars.
- Compare** the ways that sea urchins and sand dollars are adapted to their environment.
- Describe** how the sea cucumber transports food to its mouth.
- Explain** why sea stars are of economic importance.
- Summarize** the process of feeding and digestion in the sea star.
- Compare** sexual and asexual reproduction in sea stars.
- Summarize** how pharyngeal pouches have become modified through evolution in aquatic chordates and in terrestrial chordates.
- Interpret** the significance of the notochord and the postanal tail to aquatic chordates, such as the lancelet.
- Explain** why members of the subphylum Urochordata are called tunicates.
- CONCEPT MAPPING** Use the following terms to create a concept map that sequences the path of water through the water-vascular system of a sea star: *water*, *radial canal*, *madreporite*, *ring canal*, *tube foot*, and *stone canal*.

CRITICAL THINKING

- Forming Reasoned Opinions** Scientists have found many echinoderm fossils from the Cambrian period, but they have found few fossils of other species from this period. What might explain the large number of fossilized echinoderms?
- Analyzing Concepts** Sea lilies and sea cucumbers are mostly sessile animals as adults. Their larvae, however, can swim. What adaptive advantage do swimming larvae provide?
- Inferring Relationships** Basket stars are active at night. During the day, basket stars curl up their arms and become a compact mass. What are possible explanations for this behavior?
- Interpreting Graphics** Identify the structures labeled “A–F” in the diagram below.



- Applying Information** Commercial oyster farmers usually want to prevent sea stars from feeding in the oyster beds. In the past, the farmers would control the sea stars by chopping the sea stars in half and throwing them back into the water. Was this a good way to protect the oysters from predation by the sea stars? Why or why not? If not, describe a better method.

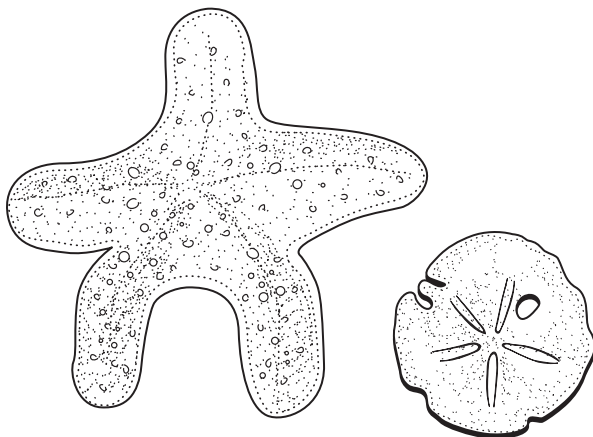


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- In a sea star, gas exchange and excretion of wastes take place by diffusion through which of the following structures?
 - pharynx
 - skin gills
 - atriopore
 - radial canals
- Which of the following types of symmetry is characteristic of echinoderms?
 - biradial
 - bilateral
 - pentaradial
 - pentalateral
- Which of the following classes of echinoderms most closely resembles the fossils of ancient echinoderms?
 - Crinoidea
 - Asteroidea
 - Echinoidea
 - Holothuroidea
- Which of the following is found in adult tunicates, or sea squirts?
 - eyespots
 - notochord
 - spinal cord
 - pharynx with slits

INTERPRETING GRAPHICS: The illustration below shows a sea star and a sand dollar. Use the illustration to answer the question that follows.

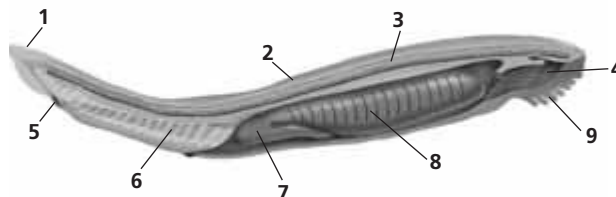


- To which phylum do these animals belong?
 - Chordata
 - Vertebrata
 - Arthropoda
 - Echinodermata

DIRECTIONS: Complete the following analogy.

6. Cardiac stomach : digestion :: atriopore :
- excretion
 - respiration
 - circulation
 - reproduction

INTERPRETING GRAPHICS: The diagram below shows a lancelet. Use the diagram to answer the question that follows.



7. Which four structures are the main distinguishing characteristics of chordates?
- 1, 2, 3, 4
 - 1, 2, 3, 8
 - 2, 3, 4, 6
 - 2, 3, 4, 9

SHORT RESPONSE

Echinoderms and chordates are believed to have evolved from a common ancestor.

Describe the evidence that supports this view.

EXTENDED RESPONSE

The fossil record of echinoderms dates to the Cambrian period, more than 500 million years ago. Scientists have observed many well-preserved echinoderm species from this period.

Part A Based on the fossil evidence, which characteristics did the earliest adult echinoderms have, in terms of symmetry and ability to move?

Part B Based on your answer to Part A, describe the evolution of echinoderms.

Test TIP

If you are unsure of the answers to particular questions, put a mark beside them (on your answer sheet or test booklet), and go on to other questions. If you have time, go back and reconsider the questions you skipped. (Do not write in this book.)

Comparing Echinoderms

OBJECTIVES

- Compare the structure of two types of echinoderms.
- Infer function from the observation of structures.

PROCESS SKILLS

- using dissection instruments and techniques
- observing

MATERIALS

- preserved sea star
- preserved sea urchin
- sea urchin test
- dissection tray
- fine dissection scissors with pointed blades
- hand lens or dissecting microscope
- forceps
- blunt probe
- sharp probe
- dissection pins
- gloves (optional)
- 100 mL beaker
- household bleach
- Petri dish

Background

1. List the major characteristics of echinoderms.
2. Which feature of echinoderms gives their phylum its name?
3. To which classes of echinoderms do sea stars and sea urchins belong?
4. What is pentaradial symmetry?



PART A Observing the External Anatomy of a Sea Star

1.  **CAUTION** Put on safety goggles, gloves, and a lab apron. Using forceps, hold a preserved sea star under running water to gently but thoroughly remove excess preservative. Then, place the sea star in a dissection tray.
2. What evidence can you find that indicates that the sea star has pentaradial symmetry?
3. Make a table in your lab report like the one shown on the next page. As you observe each of the structures listed in the table, fill in the function of that structure.
4. While referring to Figure 38-7, find the madreporite on the aboral surface of your sea star. Record the madreporite's position and appearance.
5. Use a hand lens to examine the sea star's spines. Describe their size and shape. Are they distributed in any recognizable pattern on the surface of the sea star? Are they covered by tissue or exposed? Are they movable or fixed?
6. Are pedicellariae present? What is their location and arrangement? Draw one as it appears under the hand lens or dissecting microscope.
7. Use the dissecting microscope to look for small skin gills. If any are present, describe their location and structure.
8. Now, examine the sea star's oral surface. Find the mouth, and describe its location and structure. Use forceps or a probe to gently move aside any soft tissues. What structures are found around the mouth?
9. Locate the tube feet. Describe their distribution. Using a dissecting microscope, observe and then draw a single tube foot.

PART B Observing the Internal Anatomy of a Sea Star

10.  Using scissors and forceps, carefully cut the body wall away from the aboral surface of one of the sea star's arms. Start near the end of the arm, and work toward the center. The internal

organs may stick to the inside of the body wall, so use a sharp probe to gently separate them from the body wall as you cut. Be careful not to damage the madreporite.

11. While referring to Figure 38-7, find the digestive glands in the arm you have opened. Describe their appearance. If you have dissected carefully, you should be able to find a short, branched tube that connects the digestive glands to the pyloric stomach. If you cannot find the digestive glands or this tube, repeat step 10 on one of the other arms, and look for them there.
12. Cut the tube that connects the digestive glands to the pyloric stomach, and move the digestive glands out of the arm. Look for the testes or ovaries. If your specimen is an immature animal, these organs may be small and difficult to find.
13. Locate the two rows of ampullae that run the length of the arm. What is the relationship between the ampullae and the tube feet, which you observed on the oral surface?
14. Carefully remove the body wall from the aboral surface of the central region of the sea star. Try to avoid damaging the underlying structures. Locate the pyloric stomach and the cardiac stomach. How does a sea star use its cardiac stomach during feeding?
15. Remove the stomachs, and find the canals of the water-vascular system: stone canal, ring canal, and radial canals. In which direction does water move through these canals?
16.  **CAUTION** Bleach is a highly corrosive agent. If you get it on your skin or clothing, wash it off at the sink while calling to your teacher. If you get it in your eyes, immediately flush it out at the eyewash station while calling to your teacher. Cut a 1 cm cross section out of the middle of one of the arms. Using forceps, transfer the section to a small beaker containing enough bleach to cover it. The bleach will eat away the soft tissues, exposing the endoskeleton.
17. After about 10–15 minutes, use the forceps to carefully transfer the endoskeleton to a Petri dish containing tap water. Observe the endoskeleton under a dissecting microscope. Can you find individual ossicles?

FUNCTION OF SEA STAR STRUCTURES

Structure	Function
Madreporite	
Pedicellaria	
Skin gill	
Tube foot	
Digestive gland	
Ampulla	
Ossicle	

PART C Observing the Anatomy of a Sea Urchin

18. Using forceps, hold a preserved sea urchin under running water to remove excess preservative. Then, place the sea star in a dissection tray. What evidence can you find that indicates that the sea urchin has pentaradial symmetry?
19. Observe the sea urchin's spines. Answer the same questions for the sea urchin's spines that you answered for the sea star's spines in step 5.
20. Examine the sea urchin's oral surface. Find the mouth, and use a sharp probe to explore the structures around the mouth. How does the sea urchin's mouth differ from the sea star's?
21. Examine the sea urchin test. What might be the function of the rows of small pores on the test? What might be the function of the small bumps on the test?
22.   Dispose of the specimens according to the directions from your teacher. Then, clean up your materials, and wash your hands before leaving the lab.

Analysis and Conclusions

1. Which features are shared by sea stars and sea urchins?
2. What are some of the structural differences between sea stars and sea urchins?

Further Inquiry

Observe how living sea stars move in a saltwater aquarium. Add some live mussels, and observe the feeding behavior of the sea stars.

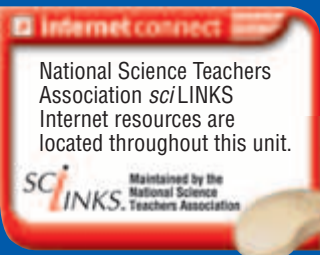
UNIT 9

CHAPTERS

- 39 *Fishes*
- 40 *Amphibians*
- 41 *Reptiles*
- 42 *Birds*
- 43 *Mammals*
- 44 *Animal Behavior*



References to *Scientific American* project ideas are located throughout this unit.



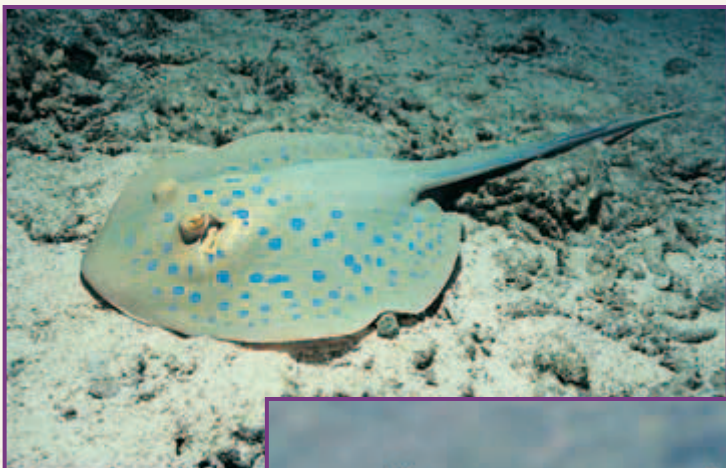
VERTEBRATES

“Nature discloses the secrets of her past with the greatest reluctance. We paleontologists weave our tales from fossil fragments poorly preserved in incomplete sequences of sedimentary rocks. Most fossil mammals are known only from teeth—the hardest substance in our bodies—and a few scattered bones.”

From “History of the Vertebrate Brain,” from *Ever Since Darwin: Reflections in Natural History*, by Stephen Jay Gould. Copyright © 1973, 1974, 1975, 1976, 1977 by the American Museum of Natural History; copyright © 1977 by Stephen Jay Gould. Reprinted by permission of *W. W. Norton & Company, Inc.*, www.norton.com



Opossums are the only North American marsupial mammals.



*This blue-spotted stingray, *Taeniura lymna*, is one of about 100 species of stingrays that belong in the class Chondrichthyes.*



Antlers, such as those on this caribou, are bony outgrowths that are shed each winter.



These colorful rainbow lorikeets, also known as brush-tongued parrots, can be found in eastern Australia, where they feed on eucalyptus flowers.



*Flap-necked chameleon, *Chamaeleo delipis**



The whale shark lives in a saltwater environment, for which it has special organs and biochemical adaptations. The whale shark is the largest fish, growing up to 46 feet long. Notice the size of the whale shark compared to the diver above.

SECTION 1 *Introduction to Vertebrates*

SECTION 2 *Jawless and Cartilaginous Fishes*

SECTION 3 *Bony Fishes*

INTRODUCTION TO VERTEBRATES

Although the vertebrates are not the most diverse or numerous group of animals, they are the most familiar to us. Vertebrates are an important part of our diet, and many are pets.

CHARACTERISTICS

Members of the subphylum Vertebrata, within the phylum Chordata (kawr-DAY-tuh) have, at some stage of life, a notochord (a rod-shaped supporting axis below the nerve cord), a dorsal hollow nerve cord, pharyngeal pouches (paired structures in the throat region), and a post-anal tail (a tail that extends beyond the anus). Vertebrates have three characteristics that distinguish them from other chordates. First, vertebrates have **vertebrae** (singular, *vertebra*), bones or cartilage that surround the dorsal nerve cord and form the spine. Second, vertebrates have a **cranium**, or skull, that protects the brain. Third, all vertebrates have an endoskeleton (an internal skeleton) composed of bone or cartilage.

Vertebrate Classes

Today, there are about 45,000 species of vertebrates. They occupy all but the most extreme terrestrial habitats. The nine classes of vertebrates are summarized below.

- **Hagfishes** (class Myxini)—These fishes have elongated, eel-like bodies. They lack jaws, paired fins, and bone. The notochord remains throughout life. Hagfishes do not have vertebrae. Many scientists do not consider them vertebrates. They are included with vertebrates because they do have a cranium and an endoskeleton.
- **Lampreys** (class Cephalaspidomorphi)—Lampreys lack jaws, paired fins and bone, and retain a notochord throughout life. However, unlike hagfishes, lampreys have a primitive vertebral column composed of cartilage that surrounds the notochord.
- **Sharks, Rays, Skates, and Ratfishes** (class Chondrichthyes)—These predatory fishes have jaws and paired fins. Their skeleton is made of cartilage, not bone, and many have skin covered by a unique kind of scale.
- **Ray-finned Fishes** (class Actinopterygii)—Most familiar fishes are ray-finned fishes. All have jaws and paired fins, most have a skeleton composed of bone. These fish have fins supported by rays of bone that fan out from a central bony axis.

OBJECTIVES

- **Identify** the distinguishing characteristics of vertebrates.
- **List** an example for each of the nine classes of vertebrates.
- **Describe** the characteristics of the early vertebrates.
- **Explain** the importance of jaws and paired fins for fishes.

VOCABULARY

vertebrae
cranium
gill arches

Word Roots and Origins

vertebra

from the Latin *vertebra*,
meaning "a joint"



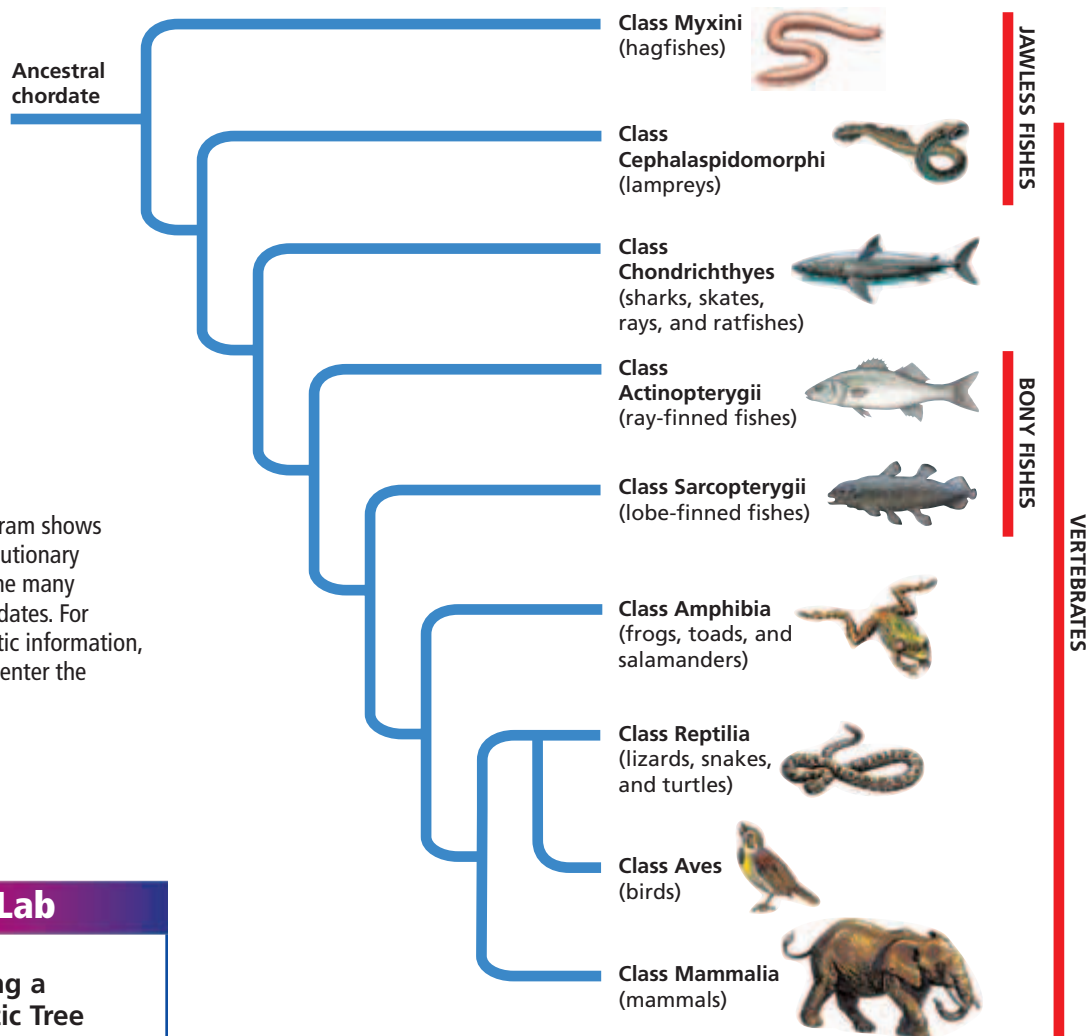


FIGURE 39-1

This phylogenetic diagram shows hypotheses of the evolutionary relationships among the many diverse groups of chordates. For updates on phylogenetic information, visit go.hrw.com and enter the keyword **HM6 Phylo**.



Quick Lab

Analyzing a Phylogenetic Tree

Materials paper, pencil

Procedure

1. Draw the phylogenetic tree shown on this page on your paper.
2. Using the information on pp. 779 and 780, determine the key characteristics that distinguish each vertebrate group. Indicate these evolutionary changes on the branches of the tree to make a diagram of the relationship that exists among vertebrates. Begin at the bottom of the tree with the key characteristics that distinguish vertebrates from other chordates.

Analysis Which characteristics are shared by all vertebrates? Which key characteristic separates the classes Chondrichthyes and Actinopterygii? Which adaptations led to the divergence of mammals?

- **Lobe-finned Fishes** (class Sarcopterygii)—These fishes have fins that are supported by a main axis of bone. There are two living groups of lobe-finned fishes: lungfishes and the coelacanth. Extinct lobe-finned fishes are thought to be the ancestors of amphibians.
 - **Amphibians** (class Amphibia)—About 4,880 species of frogs, toads, and salamanders belong to this group. Their skin is thin and is permeable to gases and water. Most species lay their eggs in water and pass through an aquatic larval stage.
 - **Reptiles** (class Reptilia)—This group includes turtles, crocodiles, alligators, lizards, and snakes. The skin of reptiles is dry and scaly. The eggs of reptiles protect the embryo from drying out and can be laid on land. There are about 8,000 species.
 - **Birds** (class Aves)—Birds are characterized by adaptations that enable flight, including feathers, hollow bones, and a unique respiratory system. There are over 10,000 species.
 - **Mammals** (class Mammalia)—Humans, cats, mice, and horses are among the members of this group. All mammals have hair and nurse their young with milk. There are about 4,400 species.
- Figure 39-1 shows the relationships among the nine classes of living vertebrates.

VERTEBRATE EVOLUTION

Most biologists think that vertebrates originated about 560 million years ago, shortly after the first chordates appear in the fossil record. The oldest known vertebrate fossils are those of tadpole-like jawless fishes. They appear in the fossil record about 560 million years ago. Figure 39-2 shows an artist's reconstruction of one of these fishes. Jawless fishes were the only vertebrates for more than 50 million years. The survivors became the ancestors of today's jawless fishes.

Origin of Jaws and Paired Fins

Almost 450 million years ago, the first fishes with jaws and paired fins appeared. Paired fins increased fishes' stability and maneuverability, and jaws allowed them to seize and manipulate prey. Jaws are thought to have evolved from the first pair of **gill arches**, the skeletal elements that support the pharynx. Figure 39-3 shows three possible stages in this transformation. Modern fishes—the sharks and rays and the bony fishes—make their first appearance in the fossil record about 400 million years ago.

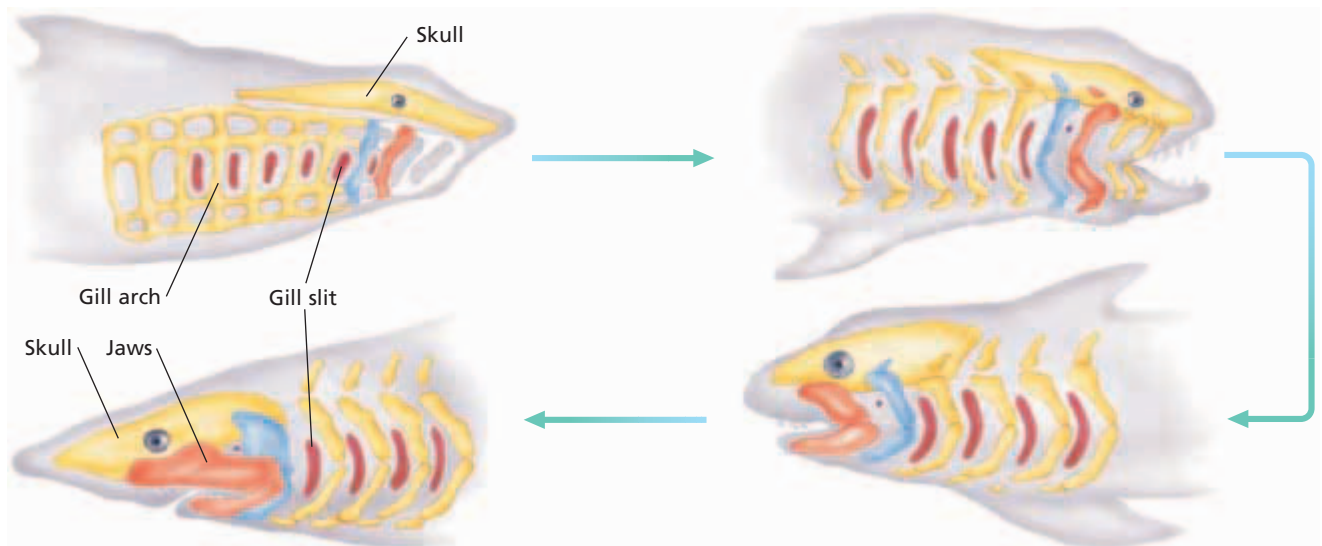


FIGURE 39-2

Early jawless fishes, such as this *Pharyngolepis*, lacked paired fins and probably fed on small invertebrates. Most species of early jawless fishes were less than 15 cm (6 in.) in length.

FIGURE 39-3

Jaws are thought to have developed from the anterior gill arches of early jawless fishes. These figures represent hypothesized stages of evolution.



SECTION 1 REVIEW

1. List three distinguishing characteristics of vertebrates.
2. List the nine classes of vertebrates, using both their scientific and common names.
3. State the characteristics of early jawless fishes.
4. Identify the advantages for fish of having jaws and paired fins over not having these structures.

CRITICAL THINKING

5. **Analyzing Information** Why is it important to use unique characteristics to classify animals?
6. **Applying Information** Explain why early jawless fishes might have been awkward swimmers.
7. **Recognizing Relationships** Explain why scientists think that vertebrates evolved from chordates in the sea.

SECTION 2

OBJECTIVES

- **Identify** three characteristics that make fishes well suited to aquatic life.
- **Describe** three sensory systems in fishes.
- **Evaluate** the similarities between jawless fishes and early vertebrates.
- **Identify** two characteristics of cartilaginous fishes.
- **Contrast** reproduction in lampreys with reproduction in cartilaginous fishes.

VOCABULARY

chemoreception
lateral line
external fertilization
cartilage
placoid scale
internal fertilization

JAWLESS AND CARTILAGINOUS FISHES

The term fish generally refers to three distinct groups of living vertebrates: jawless fishes; cartilaginous fishes; and bony fishes. Fishes are the most numerous and widespread of all vertebrates.

FISH ADAPTATIONS

The body plan of a fish makes it well suited to live in water. A streamlined shape and a muscular tail enable most fishes to move rapidly through the water. Paired fins allow fishes to maneuver right or left, up or down, and backward or forward. Unpaired fins on the back and belly increase stability. In addition, most fishes secrete a mucus that reduces friction as they swim, and that helps protect them from infections.

Most of the tissues in a fish's body are denser than water. By controlling the amount of gas in their bodies, many fishes can regulate their vertical position in the water. Some fishes also store lipids, which are less dense than water and therefore add buoyancy.

Fishes need to absorb oxygen and rid themselves of carbon dioxide. However, scales on fishes limit diffusion through the skin. Instead, most exchanges between water and blood take place across the membranes of gills—the internal respiratory organs of fishes.

Homeostasis

The concentration of solutes in a fish's body usually differs from the concentration of solutes in the water in which the fish swims. The body of a freshwater fish is *hypertonic*. It has a higher concentration of solutes than the surrounding water does, so the fish tends to gain water and lose ions, such as sodium and chloride ions, through diffusion. Most saltwater fishes are *hypotonic*; they contain lower concentrations of solutes than their surroundings do. Thus, saltwater fishes tend to lose water and gain ions.

Like all organisms, fishes must also rid themselves of the waste products produced by metabolism. The kidneys and gills play important roles in maintaining homeostasis in the tissues and in getting rid of metabolic wastes. The kidneys filter the blood and help regulate the concentration of ions in the body. The gills release wastes, such as carbon dioxide and ammonia, and either absorb or release ions, depending on whether the fish lives in fresh water or in salt water.

Eco Connection

Hagfish Depletion

Hagfishes are economically important. Most "eelskin" products, such as wallets, are actually made from the tanned skin of hagfishes. The demand for these products is so high that hagfish populations in some parts of the world have been almost wiped out by overfishing.

Sensory Functions

Fishes have a variety of organs that allow them to sense the world around them. Fish can sense light, chemicals, and sound. Some can also sense electrical and magnetic fields. Fish eyes are similar to eyes of land vertebrates. Many bony fishes have color vision, but most cartilaginous fishes do not.

Chemoreception is the ability to detect chemicals in the environment. Chemoreception includes the senses of smell and taste. Most fish have one or two nostrils, which lead to sensory cells located in olfactory sacs on both sides of the head. Sharks and salmon are examples of fish with a well-developed sense of smell. Fishes have taste buds located in their mouths. They may also have taste buds on their lips, fins, and skin and on whisker-like organs near their mouths called *barbels*.

Fishes have a unique organ called the **lateral line**, which allows them to sense vibration in the water. The lateral line is made of a system of small canals in the skin. The canals are lined with cells that are sensitive to vibration. Fishes also perceive sound waves with their inner ears, which contain a fluid-filled set of canals that contain sensory cells similar to those in the lateral line canals.

Cartilaginous fishes also have sense organs called the *ampullae of Lorenzini* that can detect weak electrical fields, such as those given off by muscles when they contract. This system has been shown to help them locate prey.

JAWLESS FISHES

The only existing jawless fishes are the 80 species of hagfishes and lampreys. These fish were formerly grouped together in the class Agnatha. They are now divided into two classes: Myxini (hagfishes) and Cephalaspidomorphi (lampreys). Their skin has neither plates nor scales. Hagfishes and lampreys have an eel-like body, a cartilaginous skeleton, and unpaired fins. The notochord remains throughout life. Hagfishes live only in the oceans. Many lampreys live permanently in fresh water, and all lamprey species reproduce in fresh water.

Hagfishes

Hagfishes, shown in Figure 39-4, are bottom dwellers in cold marine waters. Hagfishes are unique in that they do not have vertebrae. Hagfishes are also unique because they are *isotonic*, which means that their body fluids have nearly the same ion concentration as sea water. They feed on small invertebrates or on dead and dying fish. Because the hagfish lacks jaws, it cannot bite, but within its mouth are two movable plates and a rough tonguelike structure that it uses to pinch off chunks of flesh. Hagfishes often burrow into the body of a dead fish through the gills, skin, or anus. Once inside, they eat the internal organs.

Word Roots and Origins

agnatha

from the Greek *gnathus*, meaning "jaws," and *a*, meaning "without"

FIGURE 39-4

Hagfishes are modern jawless fishes. Hagfishes lack paired fins, which allows them to burrow into the bodies of the dead fish on which they feed.





FIGURE 39-5

A lamprey's mouth is adapted for latching onto prey and feeding on body fluids of other fishes.

Lampreys

About half the species of lampreys are free-living (non-parasitic). The other half are parasites as adults and feed on the blood and body fluids of other fishes. Once a suitable host is located, a lamprey uses its disk-shaped mouth, shown in Figure 39-5, to attach to the host. Then, it scrapes a hole in the host with its rough tongue and secretes a chemical that keeps the host's blood from clotting. After feeding, the lamprey drops off. The host may recover, bleed to death, or die from an infection.

Some lamprey species spend most of their adult lives in the ocean. Others live in rivers or lakes and never enter salt water. All lampreys breed in fresh water. Fertilization occurs outside the body of either parent, a process known as **external fertilization**. The eggs hatch into larvae that resemble an amphioxus, an invertebrate chordate. The larvae eventually transform into adults.

CARTILAGINOUS FISHES

Sharks, skates, and rays belong to the class Chondrichthyes. Because the fishes in this class have skeletons composed of cartilage, they are also called *cartilaginous fishes*. **Cartilage** is a flexible, lightweight material made of cells surrounded by tough fibers of protein. Sharks, skates, rays, and ratfishes differ from lampreys and hagfishes in that they have movable jaws, skeletons, and paired fins. Almost all of the approximately 800 species of sharks, skates, rays, and ratfishes live in salt water. All species are carnivores, and some are scavengers. The skin of cartilaginous fishes is covered with **placoid** (PLAK-OYD) **scales**—small, toothlike spines that feel like sandpaper. Placoid scales, shown in Figure 39-6, probably reduce turbulence of the water flow and thus increase swimming efficiency.

Sharks

Sharks have smooth, torpedo-shaped bodies that reduce turbulence when swimming. This shape is called a *fusiform* body shape. Figure 39-7 shows the smooth, fusiform body of a shark.

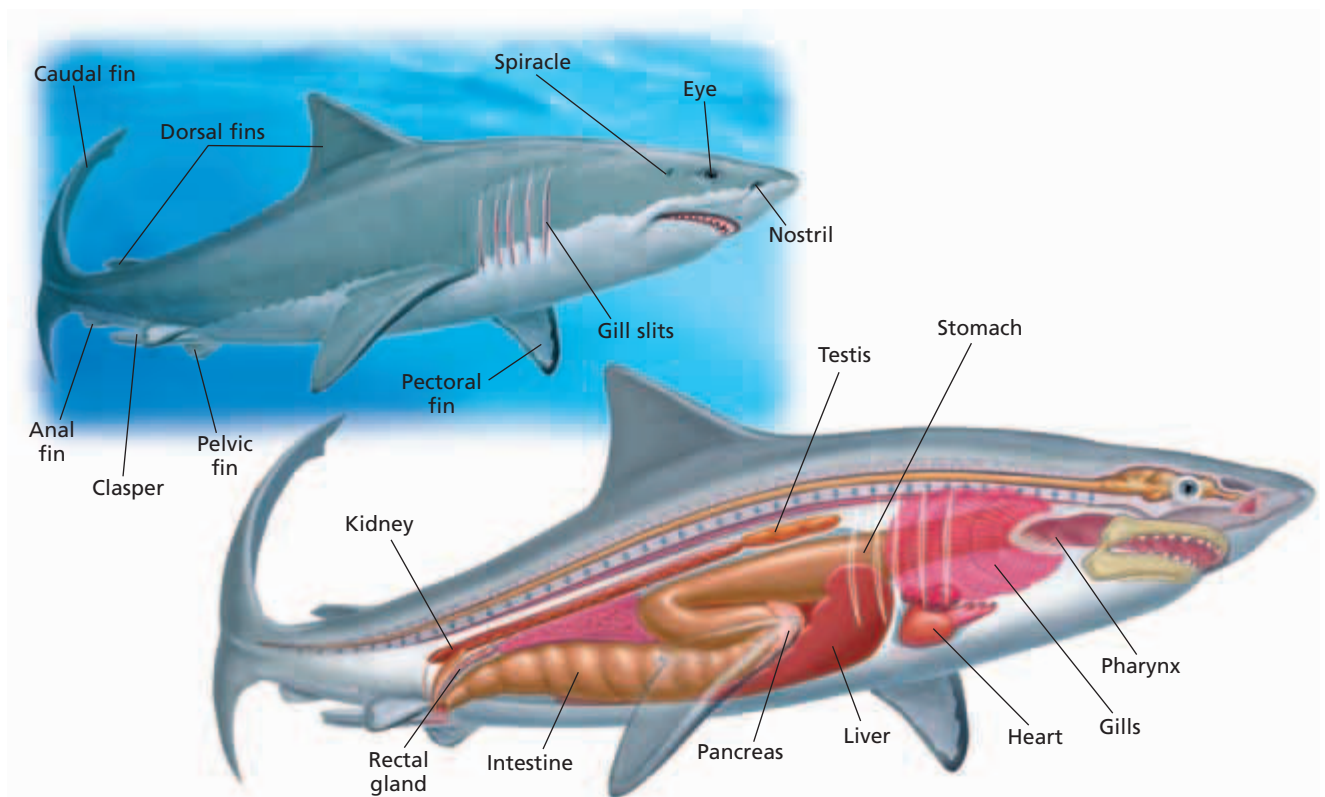
The largest sharks, the whale shark (up to 18 m, or 59 ft, long) and the basking shark (up to 15 m, or 49 ft, long), feed on plankton, floating plants and animals. Like other filter-feeding fishes, the whale and basking sharks filter the water with slender projections on the inner surface of the gills, called *gill rakers*.

The mouth of a typical shark has 6 to 20 rows of teeth that point inward. When a tooth in one of the front rows breaks or wears down, a replacement moves forward to take its place. One shark may use more than 20,000 teeth over its lifetime. The structure of each species' teeth is adapted to that species' feeding habits. Sharks that feed primarily on large fish or mammals have big, triangular teeth with sawlike edges that hook and tear flesh.

FIGURE 39-6

These toothlike placoid scales are found on the skin of cartilaginous fishes. What advantage might they give a shark in swimming?





Rays and Skates

Rays and skates have flattened bodies with paired winglike pectoral fins and, in some species, whiplike tails. Rays have diamond- or disk-shaped bodies, but most skates have triangular bodies. Most rays and skates are less than 1 m (3.3 ft) long. Rays and skates are primarily bottom dwellers. Their flat shape and coloration camouflage them against the floor of their habitat. Most rays and skates feed on mollusks and crustaceans. Figure 39-8 shows an example of a ray.

Ratfishes

There are about 25 species of strange-looking, mostly deep-water cartilaginous fishes that are grouped separately because of their unique features. The ratfishes, or *chimaeras*, have gill slits covered by a flap of skin. Some have a long, ratlike tail. They feed on crustaceans and mollusks.

Adaptations of Cartilaginous Fishes

In cartilaginous fishes, gas exchange occurs in the gills, which lie behind the head. Efficient gas exchange requires a continuous flow of water across the gills. Some fast-swimming sharks are able to push water through their mouth, over their gills, and out of their gill slits by swimming. However, most cartilaginous fishes pump water over their gills by expanding and contracting their mouth cavity and pharynx. When lying on the bottom, rays and skates cannot bring in water through their mouth, which is located on the ventral surface of their body; instead, they draw water in through their spiracles, which are two large openings on the top of the head, just behind the eyes.

FIGURE 39-7

A fusiform body shape allows sharks to slip through the water with little resistance. The dorsal fins provide stability. The pectoral fins provide lift similar to the way airplane wings function in the air. The shark's internal anatomy includes organs for digestion, reproduction, and maintaining homeostasis.

FIGURE 39-8

This blue-spotted stingray, *Taeniura lymna*, is an example of a bottom dweller. This stingray was photographed in the Red Sea near Egypt.





Quick Lab

Modeling a Shark Adaptation

Materials 8 cm dialysis tubing, 8 cm length of string (2), 100 mL salt solution (5 percent), 250 mL beaker, 10 mL distilled water, scale, graduated cylinder

Procedure

1. Tightly tie one end of the dialysis tubing with string. Place 10 mL of distilled water inside the dialysis tubing, and tie the other end of the tube tightly with string.
2. Record the initial weight of the filled tube.
3. Add the filled dialysis tubing to a beaker filled with 100 mL of salt solution.
4. After 10 minutes, remove the tubing from the beaker, blot the outside, and reweigh it. Record your observations.

Analysis Explain the reason for the change in the weight of the tube, if any. Are sharks likely to lose or gain ions? What physical structures of a shark play a key role in maintaining ionic homeostasis in sharks?

Instead of releasing ammonia, cartilaginous fishes use energy to convert ammonia into a compound called *urea*, which is much less toxic. Sharks retain large amounts of urea in their blood and tissues, thus raising the concentration of solutes in their body to at least the same level as that found in sea water. Because the concentration of sodium and chloride in the body of a shark is less than the concentration found in sea water, these ions still diffuse into the body across the gills and are absorbed with food. The *rectal gland*, located in the posterior portion of the intestine, removes excess sodium and chloride ions from the blood and releases them into the rectum for elimination. Sharks that periodically migrate to fresh water excrete sodium, chloride, and urea along with any excess water that enters their bodies.

Cartilaginous fishes maintain their position in the water in two ways. First, because the caudal and pectoral fins generate lift, or upward force, as a fish swims, it can remain at the same level in the water, counteracting the tendency to sink, as long as it keeps moving. Second, many cartilaginous fishes store large amounts of low-density lipids, usually in the liver. A shark's lipid-filled liver may account for 25 percent of its mass. Lipids give these sharks buoyancy by reducing the overall density of the body.

Reproduction in Cartilaginous Fishes

Cartilaginous fishes differ from jawless fishes in that fertilization occurs inside the body of the female. This type of fertilization is called **internal fertilization**. During mating, the male transfers sperm into the female's body using modified pelvic fins called *claspers*. In a few species of sharks and rays, the females lay large yolky eggs right after fertilization. The young develop within the egg, are nourished by the yolk, and hatch as miniature versions of the adults. The eggs of many species develop in the female's body, and the young are born live. In some of these species, the mother nourishes the developing sharks while they are in her body. No cartilaginous fishes provide parental care for their young after birth or hatching.

SECTION 2 REVIEW

1. Identify each characteristic of fishes that makes them well suited to aquatic life.
2. Describe two unique sensory systems that can be found in fishes.
3. Contrast the feeding behavior of hagfish to the feeding behavior of lampreys.
4. List three characteristics that distinguish cartilaginous fishes from living jawless fishes.
5. Identify the advantages of internal fertilization versus external fertilization.

CRITICAL THINKING

6. **Forming a Hypothesis** A student takes fish A from a saltwater tank and fish B from a freshwater tank. The student returns each fish to the wrong aquarium, and the next day both fish are dead. Form a hypothesis that explains why.
7. **Evaluating Differences** Organisms that use external fertilization usually produce more eggs at one time than organisms that use internal fertilization. What might explain this difference?
8. **Inferring Relationships** Explain why all scales point toward a fish's tail.

BONY FISHES

Of the 25,000 known species of fishes, about 95 percent are bony fishes, formerly grouped in the class Osteichthyes. Bony fishes account for most of the vertebrates living in fresh water and in salt water. In this section, you will study some of the adaptations of this group.

CHARACTERISTICS

The bony fishes are characterized by three key features:

- **Bone**—This material is typically harder and heavier than cartilage. The skeletons of most bony fishes contain bone.
- **Lungs or swim bladder**—Early bony fishes had *lungs*, internal respiratory organs in which gas is exchanged between the air and blood. Only a few species of bony fishes have lungs today. Most bony fishes have a **swim bladder**, a gas-filled sac that is used to control buoyancy. The swim bladder is thought to have evolved from the lungs of the early bony fishes.
- **Scales**—The body of a bony fish is usually covered with scales. Scales protect the fish and reduce friction when swimming. There are two main groups of bony fishes. These are the lobe-finned fishes and the ray-finned fishes.

Characteristics of Lobe-Finned Fishes

The **lobe-finned fishes** have fleshy fins that are supported by a series of bones. Two groups of lobe-finned fishes exist today, six species of lungfishes and one species of coelacanth. Lungfishes, shown in Figure 39-9a, exchange gases through both lungs and gills. They live in shallow tropical ponds that periodically dry up. The coelacanth, shown in Figure 39-9b, lives deep in the ocean and was thought to be extinct until 1938. Extinct lobe-finned fishes are ancestors of amphibians.

OBJECTIVES

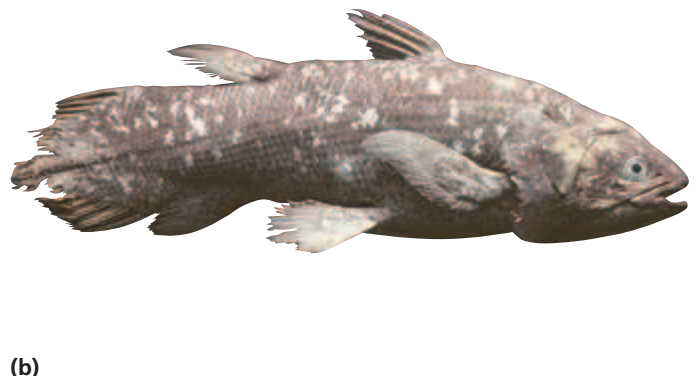
- **List** three characteristics of bony fishes.
- **Distinguish** between lobe-finned fishes and ray-finned fishes.
- **Describe** three key features of bony fishes' external anatomy.
- **Summarize** the major body systems in bony fishes.
- **Describe** the function of the swim bladder.
- **Discuss** reproduction in bony fishes.

VOCABULARY

swim bladder
lobe-finned fish
ray-finned fish
operculum
countercurrent flow
optic tectum
spawning

FIGURE 39-9

Lobe-finned fishes have fleshy fins that are supported by a series of bones. Lungfishes (a) have the ability to gulp air into lungs as an oxygen source. Coelacanths (b) have muscular fins with stout bones.



Characteristics of Ray-Finned Fishes

Ray-finned fishes do not have fins with a central bony axis—they have fins that are supported by long, segmented, flexible bony elements called *rays*. Rays probably evolved from scales. Ray-finned fishes are diverse in appearance, behavior, and habitat. Ray-finned fishes include most familiar fishes, such as, yellow perch, trout, salmon, guppies, bass, herring, goldfish, and eels.

EXTERNAL ANATOMY

Word Roots and Origins

operculum

from the Latin *operculum*,
meaning "cover"

Figure 39-10 shows the external anatomy of a yellow perch, a bony fish common in freshwater lakes of the eastern United States and Canada. The yellow perch, like all bony fishes, has distinct head, trunk, and tail regions. On each side of the head is the **operculum** (oh-PUHR-kyoo-LUHM), a hard plate that opens at the rear and covers and protects the gills.

Fins

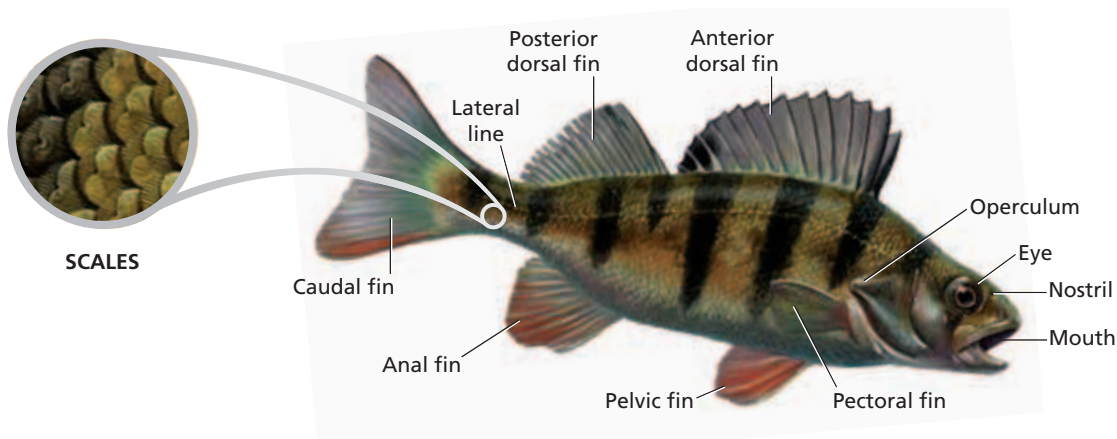
The fins of the yellow perch are adapted for swimming and navigating through the water. The caudal fin extends from the tail. It moves from side to side and amplifies the swimming motion of the body. Two dorsal fins, one anterior and one posterior, and a ventral anal fin help keep the fish upright and moving in a straight line. The fish uses paired pelvic fins and pectoral fins to navigate, stop, move up and down, and even back up. The pelvic fins also orient the body when the fish is at rest. The fins are supported by either rays or spines. Rays are bony yet flexible, while spines are bony and rigid.

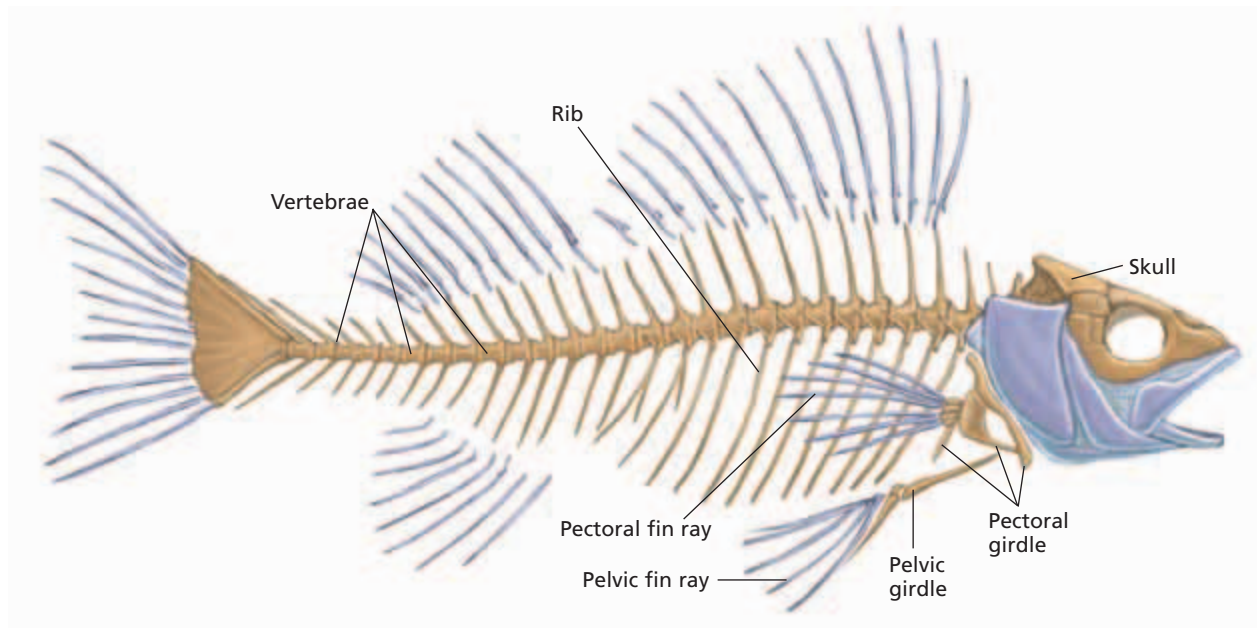
Skin

The skin of the yellow perch is covered with scales. Scales are thin, round disks of a bonelike material that grow from pockets in the skin. As Figure 39-10 shows, scales overlap like roof shingles. They all point toward the tail to minimize friction as the fish swims. Scales grow throughout the life of the fish, adjusting their growth pattern to the food supply. The scales grow quickly when food is abundant and slowly when it is scarce.

FIGURE 39-10

The external features of the yellow perch, *Perca flavescens*, are representative of bony fishes. Note the growth rings on the scales shown in the inset. They indicate the fish's approximate age.





INTERNAL ANATOMY

The major parts of a fish's skeleton, shown in Figure 39-11, are the skull, spinal column, pectoral girdle, pelvic girdle, and ribs. The spinal column is made up of many bones, called vertebrae, with cartilage pads between each. The spinal column also partly encloses and protects the spinal cord. The *pectoral girdle* supports the pectoral fins, and the *pelvic girdle* supports the pelvic fins. In a human skeleton, the pectoral girdle is the shoulder and its supporting bones, and the pelvic girdle is the hips. A fish's skull is composed of a large number of bones (far more than are in the human skull) and is capable of a wide range of movements. Note the pectoral fin rays and pelvic fin rays that are key characteristics of ray-finned fishes.

Digestive System

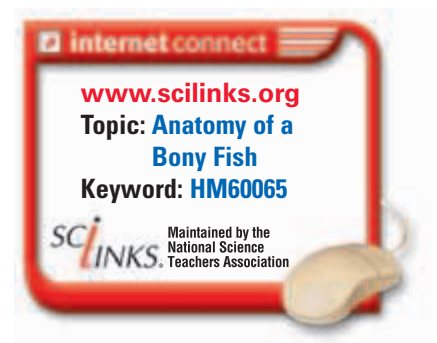
Bony fishes have diverse diets but commonly are carnivores. The jaws of predatory fishes are lined with many sharp teeth that point inward to keep prey from escaping. Strong muscles operate the jaws, which are hinged to allow the mouth to open wide.

Figure 39-12 on the next page shows the internal anatomy of a bony fish. Food passes from the mouth into the pharynx, or throat cavity, and then moves through the *esophagus* to the stomach. The *stomach* secretes acid and digestive enzymes that begin to break down food. From the stomach, food passes into the *intestine*, where digestion is completed and nutrients are absorbed.

The *liver*, located near the stomach, secretes *bile*, which helps break down fats. The *gallbladder* stores bile and releases it into the intestine. The *pancreas*, also located near the stomach, releases digestive enzymes into the intestine. The lining of the intestine is covered with fingerlike extensions called villi that increase the surface area for absorption of digested foods. Undigested material is then eliminated through the *anus*.

FIGURE 39-11

The skeleton of *Perca flavescens* is similar to that of other bony fishes. The general structure of the vertebrae, rib cage, and fins is found in many fishes.



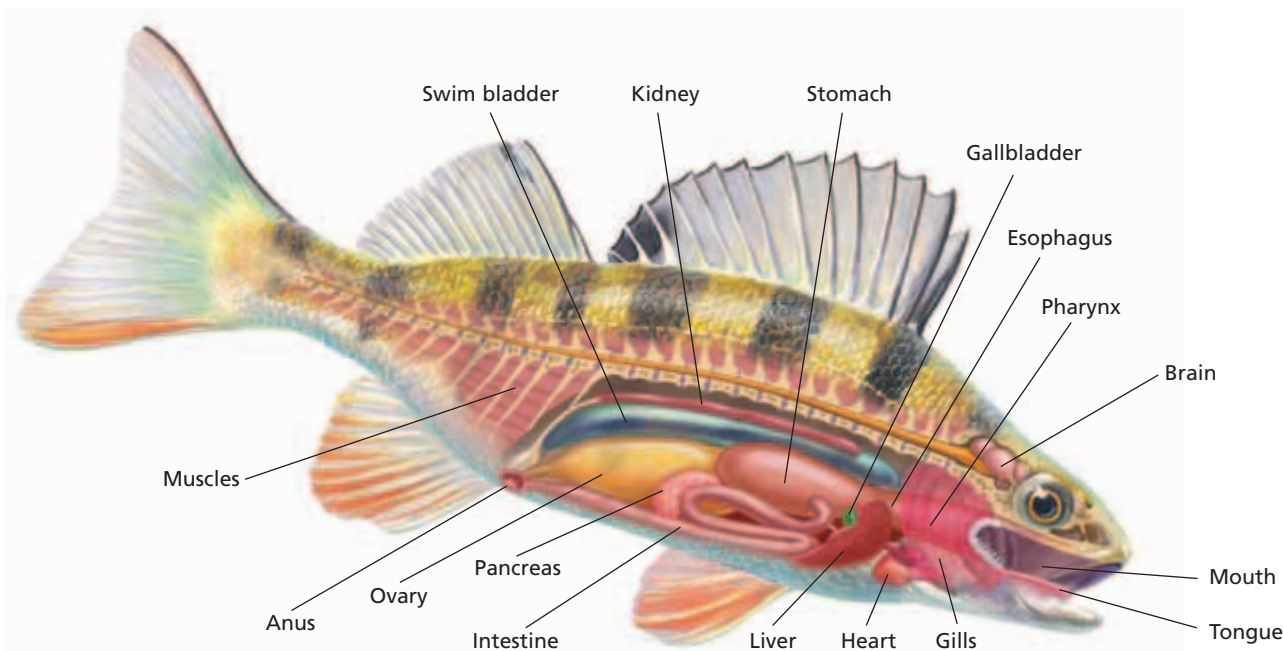


FIGURE 39-12

The internal anatomy of a bony fish, such as this perch, is a model for the arrangement of organs in all vertebrate descendants of fish. Food passes first from the mouth through the esophagus, then to the stomach and intestines. Finally, undigested waste is eliminated through the anus. Digestion of protein occurs in the stomach, and absorption of nutrients occurs in the intestine.

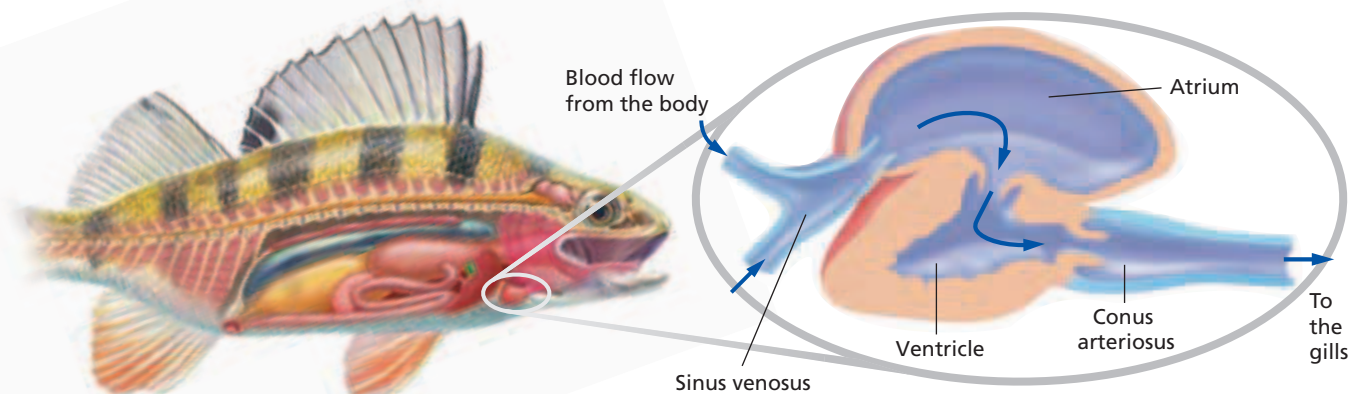
Circulatory System

The circulatory system of a fish delivers oxygen and nutrients to the cells of the body. It also transports wastes produced by metabolism—carbon dioxide and ammonia—to the gills and kidneys for elimination. The circulatory system consists of a heart, blood vessels, and blood. The heart pumps blood through *arteries* to small, thin-walled vessels, called *capillaries*, in the gills. There the blood picks up oxygen and releases carbon dioxide. From the gills, the blood then travels to the body tissues, where nutrients and wastes are exchanged. The blood returns to the heart through *veins*.

The heart of a bony fish has two chambers in a row, as you can see in Figure 39-13. Deoxygenated blood from the body empties into a collecting chamber called the *sinus venosus*. Next, blood moves into the larger *atrium*. Contraction of the atrium speeds up the blood and drives it into the muscular *ventricle*, the main pumping chamber of the heart. Contraction of the ventricle provides most of the force that drives the blood through the circulatory system. The *conus arteriosus* is a thickened, muscular part of the main artery leaving the heart. It has an elastic wall and usually contains valves to prevent blood from flowing back into the ventricle. The conus arteriosus smooths the flow of blood from the heart to the gills.

FIGURE 39-13

A fish's heart is a series of two chambers that act in sequence to move blood through the body, transporting oxygen to the cells and wastes to organs for elimination. Note the thickness of the muscle in the ventricle.



Respiratory and Excretory Systems

The large surface area of a fish's gills allows for rapid gas exchange. Gills are supported by four sets of curved bones on each side of the fish's head. Each gill has a double row of thin projections, called *gill filaments*. In most bony fishes, water is taken into the mouth and pumped over the gills, where it flows across the gill filaments before exiting behind the operculum. As you can see in Figure 39-14, water flows across the gill filaments in a direction opposite to blood flow. This arrangement is known as **countercurrent flow**. Countercurrent flow causes more oxygen to diffuse into the blood than would be possible if blood and water flowed in the same direction.

A fish's kidneys filter dissolved chemical wastes from the blood. The resulting solution, called *urine*, contains ammonia, ions such as sodium and chloride, and water. Urine is carried from the kidneys through a system of ducts to the *urinary bladder*, where it is stored and later expelled. By varying the amount of water and salts in the urine, the kidneys help regulate the water and ion balance in fresh and saltwater fishes.

As blood flows through the gill filaments, ammonia generated by metabolism diffuses from the blood into the water passing over the gills and is removed from the body. The gills also regulate the concentration of ions in the body. Recall that saltwater fishes have lower ion concentrations than sea water has. Therefore, they lose water through osmosis and gain ions, such as sodium and chloride ions. Saltwater fishes make up for this water loss by excreting small amounts of concentrated urine and by drinking sea water, but this increases their internal concentration of sodium and chloride ions. Both kinds of ions are actively transported out through the gills. Freshwater fishes tend to gain water and lose ions. They respond by producing large amounts of dilute urine and actively transporting sodium and chloride ions in through the gills.

Swim Bladder

Most bony fishes have a swim bladder. This thin-walled sac in the abdominal cavity contains a mixture of oxygen, carbon dioxide, and nitrogen obtained from the bloodstream. Fish adjust their overall density by regulating the amount of gas in the swim bladder, enabling them to move up or down in the water.

Swim bladders evolved from balloonlike lungs, which ancestral bony fishes may have used to supplement the oxygen absorbed by the gills. In some fishes, the swim bladder is known to amplify sound by vibrating and transmitting sound to the inner ear.

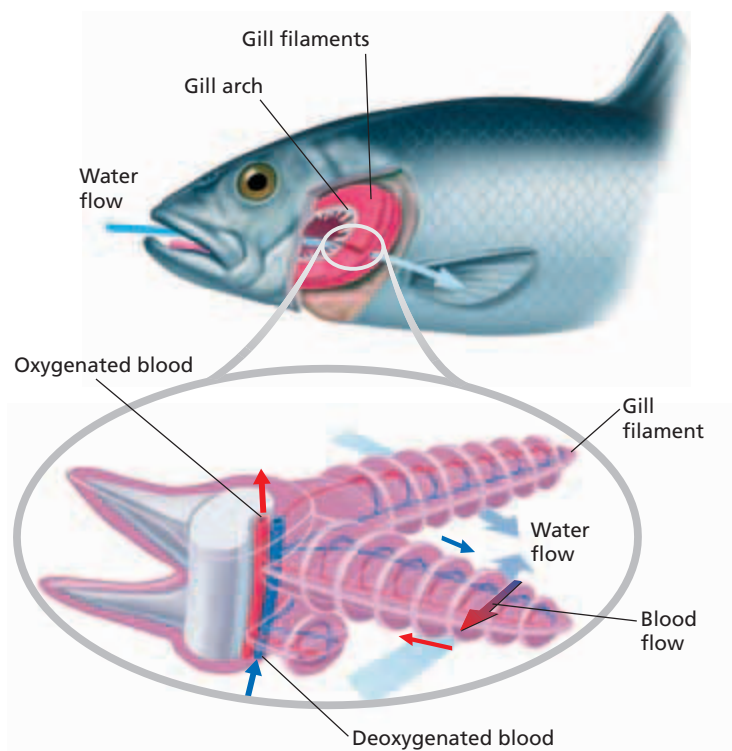


FIGURE 39-14

The gills are located directly behind the head and interior to the operculum. The gill filaments provide the organism with a large surface area, thus enabling gas exchange to occur quickly.

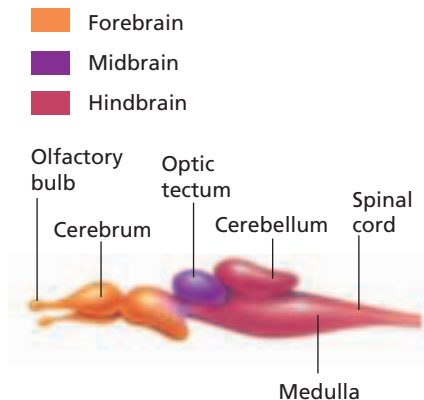


FIGURE 39-15

The fish brain, like the shark brain, has a well-developed medulla to coordinate muscle control.

FIGURE 39-16

This male ringtailed cardinal fish, *Apogon aureus*, is carrying fertilized eggs in its mouth. This behavior lowers losses of eggs to predators and contributes to the success of the species.



Nervous System

The nervous system of a bony fish includes the brain, spinal cord, nerves, and various sensory organs. The fish brain is illustrated in Figure 39-15. The most anterior part of the brain, the forebrain, contains the olfactory bulbs, which process information on smell. The forebrain also includes the *cerebrum*, which has areas that integrate information from other parts of the brain. Behind the forebrain lies the midbrain, which is dominated by the **optic tectum**. The optic tectum receives and processes information from the fish's visual, auditory, and lateral-line systems.

The most posterior division of the brain is the hindbrain, which contains the *cerebellum* (SER-uh-BEL-uhm) and the *medulla oblongata* (mi-DUL-uh AHB-lahng-GAHT-uh). The cerebellum helps coordinate muscles, movement, and balance. The medulla oblongata helps control some body functions and acts as a relay station for stimuli from sensory receptors throughout the fish's body. From the medulla oblongata, the spinal cord extends the length of the body and carries nerve impulses to and from the brain.

REPRODUCTION

Eggs are produced by ovaries in the female, and sperm are produced by the testes in the male. Eggs and sperm are released through an opening behind the anus. Fertilization in most species takes place externally. Mortality among the eggs and young fishes is often very high. Many species of fishes lay large numbers of eggs, which ensures that at least a few individuals survive to become adult fish.

Some bony fishes bear live young. Using a modified anal fin, the male inserts sperm into the female, and fertilization is internal. The female carries the eggs in her body until the young are born. Other species care for the eggs as shown in Figure 39-16.

The reproductive, or **spawning**, behavior of bony fishes varies widely. Some species build crude nests from plants, sticks, and shells. Many species migrate to warm, protected shallow water to spawn.

SECTION 3 REVIEW

1. List three key features that characterize bony fishes.
2. Contrast ray-finned and lobe-finned fishes.
3. Describe the external anatomy of a bony fish.
4. Identify the organs of four internal organ systems found in fishes.
5. Name two functions of the swim bladder.
6. Describe two methods of reproduction that can be found in bony fishes.

CRITICAL THINKING

7. **Applying Information** Which is the most muscular chamber of the fish heart? Explain why.
8. **Forming a Hypothesis** Bottom-dwelling fish often lack a swim bladder. Explain why lack of a swim bladder is an adaptive advantage.
9. **Recognizing Relationships** Explain how countercurrent flow allows the diffusion of more oxygen into the blood than would be possible if blood and water flowed in the same direction.

CHAPTER HIGHLIGHTS

SECTION 1

Introduction to Vertebrates

- Vertebrates are chordates and have a hollow dorsal nerve cord, notochord, pharyngeal pouches, and a post-anal tail at some stage of life.
- The characteristics that distinguish vertebrates from other chordates are vertebrae, a cranium, and an endoskeleton.
- Vertebrates are classified into nine classes: hagfish (class Myxini); lampreys (class Cephalaspidomorphi); sharks and rays (class Chondrichthyes); lobe-finned fishes (class Sarcopterygii); ray-finned fishes (class Actinopterygii); amphibians (class Amphibia); reptiles (class Reptilia); birds (class Aves); and mammals (class Mammalia).
- Early vertebrates were jawless fishes, which had bony scales, cartilaginous skeletons, and no paired fins.
- Paired fins increased stability and maneuverability in fishes; jaws allowed fish to seize and manipulate prey.

Vocabulary

vertebrae (p. 779)

cranium (p. 779)

gill arches (p. 781)

SECTION 2

Jawless and Cartilaginous Fishes

- Fishes have streamlined bodies, paired fins, and secrete mucus that reduces friction when swimming. The fish body plan makes them well suited to aquatic life.
- Fishes have a variety of organs that allow them to sense their environment. Fish can see, smell, and taste the world around them. They can also sense vibration, and some can sense electrical fields.
- Like early vertebrates, living agnathans lack jaws and paired fins and retain a notochord throughout life.
- Lampreys and hagfish are two types of jawless fishes that are alive today. Hagfishes burrow into and eat dead fish. Parasitic lampreys attach themselves to their host with their disc-shaped mouths and feed on the host's blood.
- Cartilaginous fishes have internal skeletons made of cartilage, and their skin is covered with placoid scales.
- Jawless fishes reproduce by external fertilization (fertilization occurs outside of the body of either parent). Most cartilaginous fishes reproduce by internal fertilization (the male inserts sperm into the female's body, and the young develop in an egg inside the female).

Vocabulary

chemoreception (p. 783)
lateral line (p. 783)

external fertilization (p. 784)
cartilage (p. 784)

placoid scale (p. 784)
internal fertilization (p. 786)

SECTION 3

Bony Fishes

- Bony fishes are characterized by three key features: scales on the body, lungs or a swim bladder, and bone in the skeleton.
- Lobe-finned fishes have fleshy fins with a central bony axis, while ray finned fishes have non-fleshy fins supported by long flexible bones called rays.
- The external anatomy of a bony fish has several distinct characteristics—an operculum, fins, and scales.
- A fish's heart has two chambers that work together to move blood through the body.
- Water flows over fishes gills in a direction opposite to blood flow. Oxygen diffuses from the water into the blood very efficiently as a result of this process, which is called *countercurrent flow*.
- Fish adjust their overall density by regulating the amount of gas in the swim bladder, enabling them to move up or down in the water.
- Unlike cartilaginous fishes, most bony fishes reproduce by external fertilization in a process called *spawning*.

Vocabulary

swim bladder (p. 787)
lobe-finned fish (p. 787)

ray-finned fish (p. 787)
operculum (p. 788)

countercurrent flow (p. 791)
optic tectum (p. 792)

spawning (p. 792)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
 - atrium* and *ventricle*
 - caudal fin* and *anal fin*
 - internal fertilization* and *external fertilization*
 - operculum* and *gill*
- Use the following key terms in the same sentence: *cerebrum*, *optic tectum*, *cerebellum*, and *medulla oblongata*.
- Explain the relationship between olfactory bulbs and lateral line system.
- Word Roots and Origins** The word *Actinopterygii* comes from the Greek *actinos*, meaning “ray,” and *pteryx*, meaning “wing” or “fin.” Using this information, explain why *Actinopterygii* is a good name for these fish.

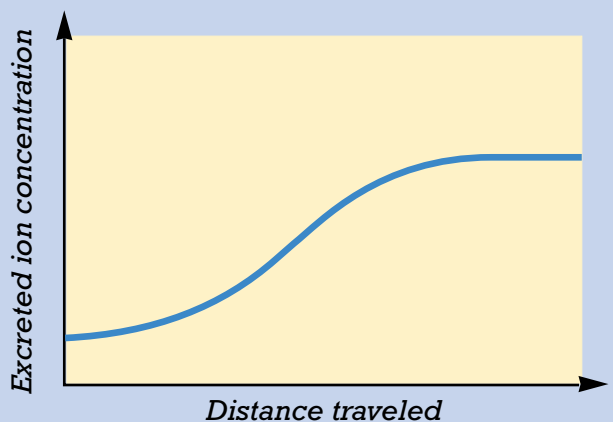
UNDERSTANDING KEY CONCEPTS

- List** the characteristics that would be found in a typical vertebrate.
- Identify** the class of vertebrates to which each of the following organisms belong: goldfish, sand sharks, pigeons, dogs, Pacific lamprey, bullfrog.
- Name** the characteristics that distinguish early vertebrates from modern vertebrates.
- Describe** two major changes in fish anatomy over millions of years.
- Evaluate** the relationship between fishes’ body form and function in an aquatic environment.
- Identify** the structures that are involved in sensory systems in fishes.
- Name** one characteristic of early chordates that distinguishes hagfishes from lampreys.
- Describe** the key characteristics that distinguish cartilaginous fishes.
- Contrast** the reproductive strategy of lampreys with the reproductive strategy of sharks.
- Identify** the characteristics that distinguish bony fishes from cartilaginous fishes.
- Explain** the differences between the two main groups of bony fishes.
- State** how the caudal, dorsal, pectoral, and pelvic fins function in helping fish swim.
- Explain** how the heart and gills function in fish.
- Propose** a possible explanation for the evolution of primitive lungs into swim bladders.
- Compare** the advantages and disadvantages of internal fertilization and spawning.
- CONCEPT MAPPING** Use the following terms to create a concept map that describes the characteristics of jawless, cartilaginous, and bony fishes: *bony fishes*, *gills*, *cartilage*, *operculum*, and *countercurrent flow*.

CRITICAL THINKING

- Applying Information** Sometimes fish develop a swim bladder disorder when they are overfed and food displaces the swim bladder. How might displacement of the swim bladder by food impair the function of this organ?
- Analyzing Research** A famous study showed that a shark could locate and capture a stationary fish buried in the sand at the bottom of its tank. When the fish was enclosed in electrical insulation and then buried, however, the shark could not locate it. From this information can you conclude that the shark is using only its electrical sense to locate the buried fish?
- Recognizing Relationships** Cod fishes lay eggs near the surface of the water. In contrast, the male largemouth bass scoops out a nest in a river bottom and waits for a female to deposit her eggs. What hypothesis would you make regarding the relative number of cod and bass eggs?
- Analyzing Graphics** Study the chart below, which shows the excreted ion concentration of a fish as it travels from one body of water to another. Is the fish traveling from fresh water to salt water or salt water to fresh water?

Effect of Environment on Ion Excretion





Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- Which of the following is true of sharks and rays?
 - They have lungs.
 - They have placoid scales.
 - Most species live in fresh water.
 - They do not have a lateral line system.
- Which of the following is not involved in controlling buoyancy?
 - a fat-filled liver
 - the rectal gland
 - the swim bladder
 - continuous swimming
- What is the function of the lateral line system?
 - initiates migration
 - detects vibrations
 - acts as camouflage
 - keeps fish moving in a straight line
- What do sharks use claspers for?
 - startle other fish
 - increase maneuverability
 - transfer sperm while mating
 - hold on to prey while feeding

INTERPRETING GRAPHICS: The table below shows the salinity of fresh water, salt water, and the body fluids of fish. Use the table to answer the questions that follow.

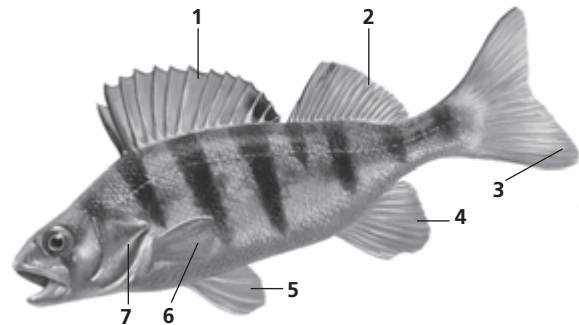
Salinity of Water and Fish Body Fluids			
	Salinity of fresh water	Salinity of salt water	Salinity of body fluids
Freshwater fish	<0.1%	—	0.8 – 1%
Saltwater fish	—	3.5%	0.8 – 1.4%

- What tendency do freshwater fish have in a freshwater environment?
 - lose water and salts
 - take on water and salts
 - take on water and lose salts
 - lose water and take on salts
- What tendency do saltwater fish have in a saltwater environment?
 - lose water and salts
 - take on water and salts
 - lose water and take on salts
 - take on water and lose salts

DIRECTIONS: Complete the following analogy.

7. Hagfish : Myxini :: Sharks :
- Aves
 - Agnatha
 - Mammalia
 - Chondrichthyes

INTERPRETING GRAPHICS: The figure below shows the external anatomy of a bony fish. Use the figure to answer the following question.



8. Which of the fins shown on the fish are dorsal fins?
- 1 & 2
 - 2 & 4
 - 5 & 6
 - 5 & 4

SHORT RESPONSE

Countercurrent flow enhances the diffusion of oxygen from the water into the bloodstream of fishes.

Explain the meaning of countercurrent flow in gills.

EXTENDED RESPONSE

Humans have a four chambered heart with two ventricles. One ventricle pumps blood to the lungs and the other pumps blood that returns from the lungs to the body.

Part A Compare the structure and blood flow of the fish heart to the human heart.

Part B Which heart is able to pump blood more forcefully around the body? Why?

Test TIP

For a question involving experimental data, determine the constants, variables, and control **before** answering the questions.

Observing Structure and Behavior in Fishes

OBJECTIVES

- Observe body shapes, camouflage, and feeding behavior in various species of fish.
- Relate observed adaptations to the survival of each species.

PROCESS SKILLS

- making observations
- collecting data
- identifying and recognizing patterns
- interpreting data
- predicting
- analyzing results
- drawing conclusions

MATERIALS

- clipboard and paper or notebook
- live fish in an aquarium tank
- pencil or pen

Background

1. Identify the three major categories of modern fish.
2. State some key characteristics of bony fish.
3. Define an adaptation.
4. How are fish adapted to life in water?

PART A Observing Body Shapes

1.  **CAUTION** To avoid injuring live fish do not touch or frighten them. Find a tank that contains several types of tropical fish.
2. Identify the species of fish in the tank. In Table 1, record the common and scientific names of the species.
3. Describe the shape of the fish's body. In Table 1, record your description.
4. Observe the activity of the fish. How does it spend most of its time? In Table 1, record your observations.
5. Repeat steps 2–4 with another species of fish.

PART B Observing Body Color and Patterns

6. Find three examples of fish that use body colors, markings, or postures to blend with their surroundings.
7. For each species, record its common and scientific names in Table 2, and describe the type of camouflage it uses.

TABLE 1 BODY SHAPES AND ACTIVITY

Species	Body shape	Activity

TABLE 2 CAMOUFLAGE

Species	Type of camouflage

PART C Observing Feeding Behaviors

- Find three examples of fish that use different methods for feeding.
- For each species, record its common and scientific names in Table 3, and describe its feeding method.

Analysis and Conclusions

- How are the body shapes of the fishes you observed related to the types of activities that those fishes engage in?
- What structural adaptations are associated with the different feeding behaviors that you observed?
- How does camouflage help fish to survive in the cases where you observed it?
- Would you expect camouflage to be more important for fish that live in shallow water or for those that live in very deep water? Explain your reasoning.



Further Inquiry

Design an experiment about fish behavior using fish sold at a local pet store and materials available in your classroom. Report the results of your experiment in an oral presentation.

TABLE 3 FEEDING BEHAVIOR

Species	Type of feeding behavior

AMPHIBIANS



Amphibians are thought to have been the first vertebrates on land. Many amphibians, such as this southern leopard frog (*Rana pipiens*), still live part or all of their life in water.

SECTION 1 *Origin and Evolution of Amphibians*

SECTION 2 *Characteristics of Amphibians*

SECTION 3 *Reproduction in Amphibians*